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Uganda Climate Promise Project

Technical Report on Enhanced Mitigation and Adaptation Targets

Updating Uganda's Nationally Determined Contribution (NDC)

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1 Chapter 1 Introduction

1.1 Background and Objectives

The world is facing environmental challenges due to unprecedented anthropogenic climate change. It is universally agreed that climate change is caused by global warming as a result of increasing concentrations of greenhouse gases (GHG) in the atmosphere. Temperature increases in Africa are projected to be above global average. Drying will intensify because of reduced precipitation and increased evaporation in the continent. As a result, it is widely discussed that climate change will have devastating impacts throughout the continent and other parts of the world. For developing countries, climate change presents additional layers to poverty and other socio-economic challenges.

Countries of the world agreed in 1992, to a global framework to address challenges brought by climate change, termed the United Nations Convention on Climate Change (UNFCCC). It facilitates global cooperation to arrest and prevent progress of climate change to levels that will be unbearable to humans and ecosystems. It focusses on building resilience of systems to climate change and GHG mitigation efforts. However, implementation of the Convention has been sluggish, concentrations of the GHGs in the atmosphere are increasing and climate change continuing as though unabated. The UNFCCC process formulated the Paris Agreement in 2015 to enhance implementation of the Convention. The Agreement responds to the requirements of science that indicate that to limit the average global temperature increase 2°C or preferably less than 1.5°C, compared to pre-industrial levels, net global emissions must reach net zero by 2050. The Paris Agreement requires that countries develop and communicate their Nationally Determined Contributions (NDCs) to the UNFCCC every five years. The NDCs contain national targets to the global requirement.

Uganda produced its first NDC in 2015 and the country is in the process of updating its adaptation and mitigation contributions. The country outlined that its priority to climate change is to develop policies that will enhance national resilience. The government of Uganda also committed to GHG reductions of 22% compared to business-as-usual in 2030. This revision considers a bottom-up approach to review this target.

This report presents the technical inputs, assumptions and methodologies used in revising those targets and determining enhanced adaptation and mitigation targets for the NDC.

1.2 Overall approach to updating the NDC

An update to Uganda's Nationally Determined Contribution (NDC) was prepared with support from Zutari as part of a contract funded by the UNDP and was carried out in parallel with the development of a long-term climate change strategy (LTS) for Uganda, led by Ricardo. To ensure a consistent approach to both the updated NDC and the LTS, the worked collaboratively on both the adaptation and mitigation components.

A six-step process, composed of technical processes and stakeholder processes that inform and feed off each other was used in updating the NDC as shown in the figure below:

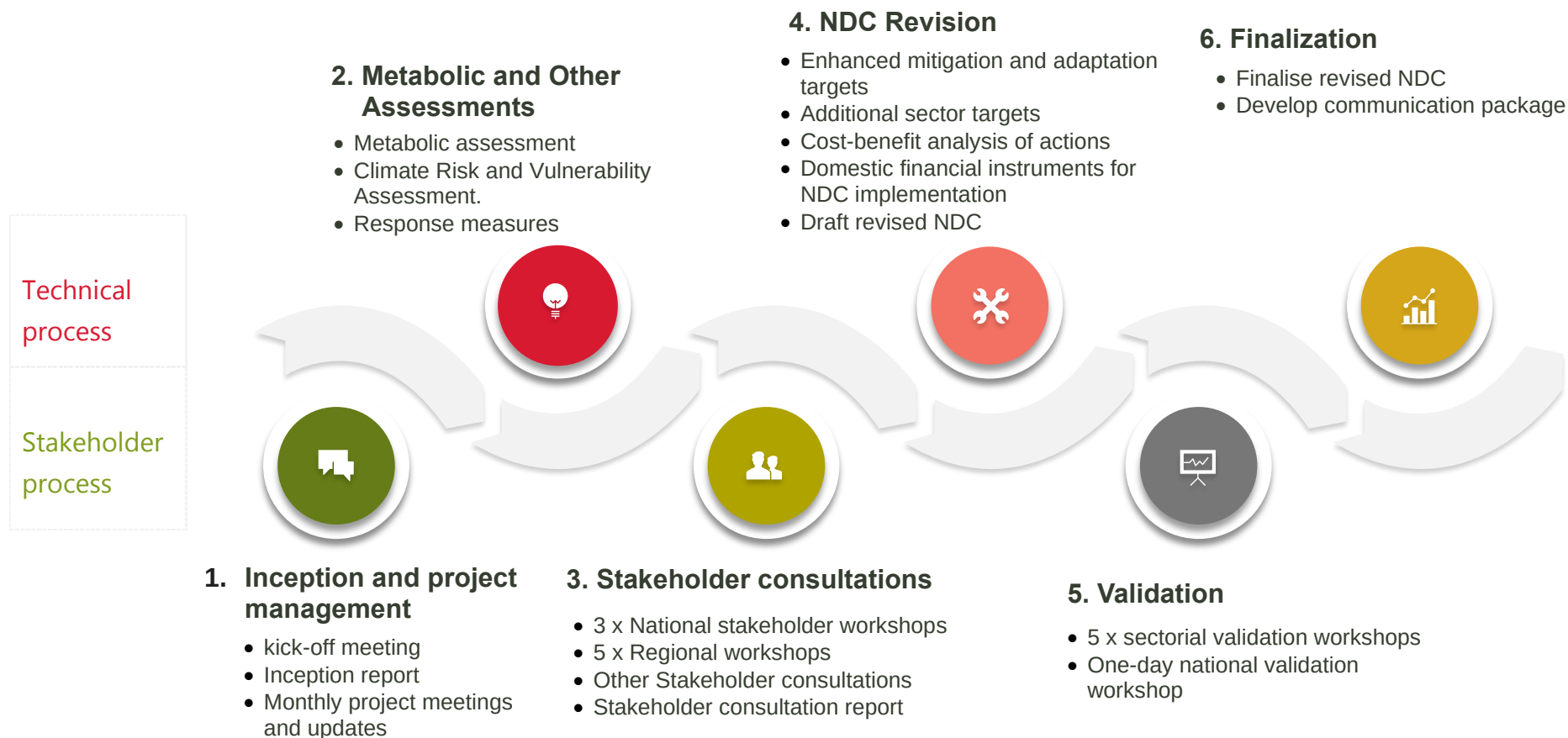


Figure 1-1: Overall approach to updating the NDC

1.2.1 Key outcomes of the metabolic assessment

A metabolic assessment is a life-cycle consumption and production assessment that was carried out to identify Uganda's areas of priority for the updated NDC, particularly the mitigation component. It identified priority sectors in terms of resource use, emissions and socio-economic impacts, and connected them with circular economy opportunities. The opportunities that bear greenhouse gas mitigation potential were prioritised for purposes of the NDC update process. The circular economy analysis aimed to provide additional aspirations for Uganda, beyond the emission reductions which Uganda has already committed to.

The assessment identified seven sectors of Uganda's economy where the enhancement of Uganda's NDC targets should focus. These are presented in the table below, together with the reasons for prioritization.

Table 1-1: Identified priority sectors for circular economy actions.

Sector	Reason for prioritisation
Agriculture	High resource use, GHG emissions and employment
Mining and quarrying	High resource and emission trends, and high mineral depletion
Food and beverage	High consumption emissions
Petroleum, chemical and non-metallic mineral products	High production emissions and trends
Construction	High resource use and emission trends
Education, health and other services	High production and consumption emissions, and high employment
Households	High resource use, production and consumption emissions, air pollution and species loss

As such the report proposed 10 main circular economy activities that need to be prioritized, and these are shown in the figure below where they have also been mapped onto the relevant IPCC sectors.

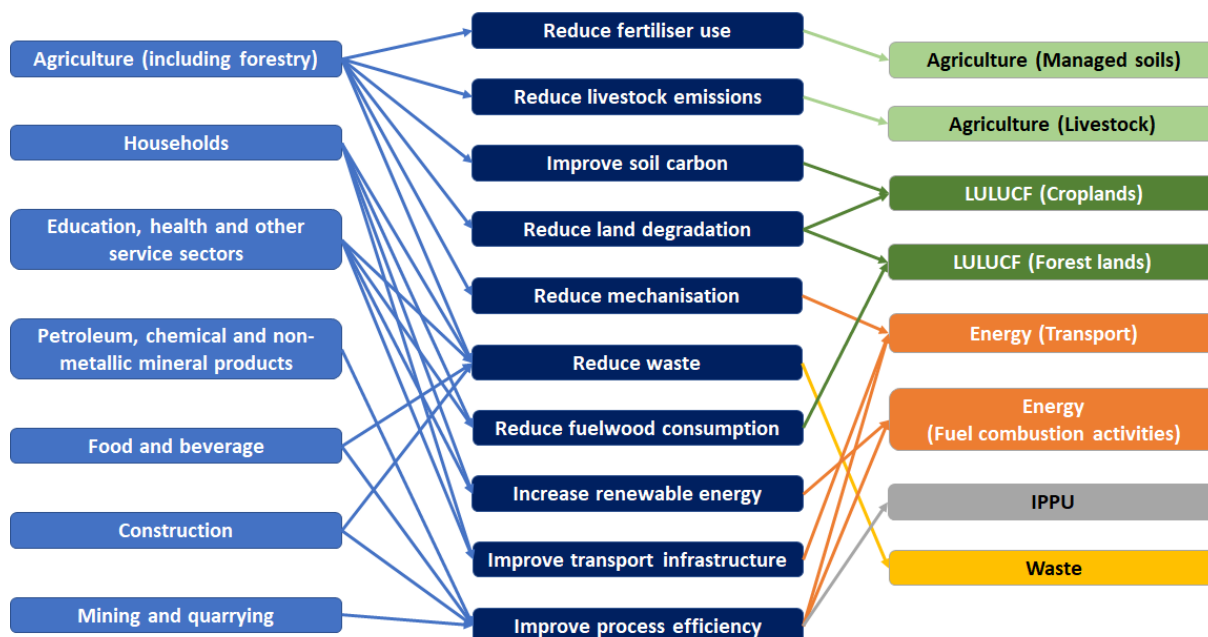


Figure 1-2: Links between proposed circular economy activities and mitigation actions for the NDC.

These recommendations formed the basis for identifying additional sectors and actions to be included in the updated NDC. For the mitigation component specifically, these recommendations show that the

transport sector, Waste sector and IPPU sector, which were not part of the country's first NDC, should be considered for inclusion in the updated NDC.

The full metabolic assessment report is attached hereto as Annex A to this report

1.2.2 Stakeholder engagements

The work of the revision of the NDC included several consultations with a wide spectrum of stakeholders. The Climate Change Department of the Ministry of Water and Environment coordinated the consultations. These consultations were done with Government officials at the national level and district levels, civil society organisations, youth formations and bilateral engagements. There were a total of three national and five regional workshops conducted as shown in the table below:

Table 1-2: Stakeholder Consultations Schedule

STAKEHOLDER CONSULTATIONS	DATE
National Consultations	
National Youth Consultative Meeting	25 th February, 2021
Civil Society (CSO) Consultative Meeting	26 th February 2021
National Climate Change Advisory Committee (NCCAC) Consultative Meeting	23 rd March, 2021
Regional Consultations	
Central Region Stakeholder Consultative Workshop, Entebbe	25 th to 26 th March, 2021
Western Region Stakeholder Consultative Workshop, Mbarara	30 th to 31 st March, 2021
Eastern Region Stakeholder Consultative Workshop, Mbale	7 th to 8 th April, 2021
Northern Region Stakeholder Consultative Workshop, Gulu	12 th to 13 th April, 2021
West Nile Region Stakeholder Consultative Workshop, Arua	15 th to 16 th April, 2021
Validation workshops	
Agriculture, Forestry, Land use (AFOLU) & Wetlands	14 th September 2021
Energy & Transport	15 th September 2021
Waste & Industrial Processes and Productive use (IPPU)	16 th September 2021
Water, Disaster Risk, Health & Urban Environment (Cities)	17 th September 2021
National Validation workshop	21 st October 2021

Due to restrictions made by the Government in order to manage the Covid-19 pandemic, the meetings were both physical and virtual so that traveling was minimised and large crowds avoided.

The main objectives of the consultations were to discuss the proposed mitigation and adaptation actions, identify appropriate indicators Multicriteria Assessment (MCA) methodology to be used to prioritise the proposed measures and actions, and to get feedback on the assumptions used to determine baseline emissions in various sectors. The consultations were also used to collect climate change adaptation and mitigation measures that may exist at the regional level.

While the full stakeholder report is annexed to this technical report (Annex B), summaries of the outcomes from the stakeholder engagements are presented in the sections below.

1.1.1.1 Summary of Stakeholder feedback on Mitigation

Countrywide, stakeholders emphasized the following should be incorporated in the updated NDC;

- Include tax measures that limit importation of used vehicles greater than 5 years old and encourage importation of new vehicles (as opposed to the current norm where the environment levy that is half the value of the car is currently levied. The Ministry of Finance Planning and Economic Development (MoFPED) should consider staggered tax levies (lower taxes on new vehicles and higher taxes on old vehicles)

- Include measures to promote the use of public transport that include bus rapid transport in Great Kampala Metropolitan Area, rail, water and non-motorised transport through provision of cycling and pedestrian lanes during design of new road infrastructure.
- Include strategies that promote fuel efficiency. Actions and measures for periodic data collection and storage on fuel conservation data, fuel efficiency issues, number of vehicles in the country (disaggregated by age, location in rural or urban areas and by size/type of vehicle) need to be included to increase the data availability for estimation of emission factors and projection of greenhouse gas emissions.
- Include strategies that promote green building and infrastructure technologies for roads and buildings for example, greening of road reserves, use of solar rooftops, provision of solar energy for social service centres such as health centres, prisons.
- The updated NDC should emphasize measures and actions that promote access to affordable energy in communities and increase overall electrification rates in Uganda.
- The updated NDC should include measures to promote alternative renewable energy sources that include: biogas technology in institutions, briquette production, wind and solar energy.
- The updated NDC should advocate for the creation of a special authority and special fund for climate change mitigation to facilitate implementation of mitigation measures and taxing gas emitters.
- The updated NDC should promote cooperatives (e.g., green villages) that can support energy generation for productive use and include measures to promote reforestation and urban greening to create carbon sinks
- The updated NDC should include promotion of research in feed formulation as a mitigation measure under the agriculture sector.

1.1.1.2 Summary of Stakeholder feedback on Adaptation

Countrywide, stakeholders pointed out that the following should be emphasized in the update of the NDC

- Prioritise adaptation in infrastructure such as the transport sector for example through building resilience of road and bridge infrastructure to emerging climate change threats, grass planting and greening of highway and urban road reserves and avoidance of construction of roads in fragile ecosystems.
- Mainstream gender, persons with disability, youth and other vulnerable groups in the adaptation and mitigation measures.
- Include a measure on mainstreaming climate change in education. This aligns with feedback from the youth.
- Include measures to increase human disease control and surveillance. There are some diseases which have come as a result of due to the effects of climate change. Good waste management practices for solid and liquid wastes should be adopted.
- Include tax measures that limit importation of used vehicles greater than 5 years old and encourage importation of new vehicles. i.e. provide higher tax levies on old vehicles and lower tax levies of new vehicles.
- Enforce wetlands protection laws in both rural and urban centres.
- Promote agroforestry (focused on indigenous trees), use of early-maturing and disease-resistant crop varieties at community level. Seed banks for indigenous trees research geared towards nurturing of indigenous trees should be promoted.

- Consider development of improved climate responsive infrastructure for value addition and post-harvest handling of fisheries, and increase marketing of the produce.
- Enforcement of the land use policy, standards and guidelines to avoid unplanned developments. Greenbelts should also be included in the design of infrastructure and should be conserved and preserved.
- Include strategies for reduction in the cost of electricity and use of alternative renewable energy sources such as solar and biogas. as gas.
- Promote cleaner industrial production mechanisms to reduce pollution and improved solid waste management for industrial, electronic and medical waste.
- Include measures of updating technology for rainfall forecasting, disaster risk detection (e.g. flooding and landslides) and early warning systems and vulnerability risk mapping.

The detailed technical approaches and methods used in enhancing the mitigation and adaptation targets on Uganda's NDC, based on the outcomes of the metabolic assessment and the stakeholder engagements, are presented in the chapters that follow. Information on enhanced mitigation targets is presented first, followed by details of the enhanced adaption targets.

2 Chapter 2 Enhanced Mitigation Targets

2.1 Overall Mitigation Approach and Methodology

Mitigation assessments were conducted by building different scenarios as indicated in the Figure below.

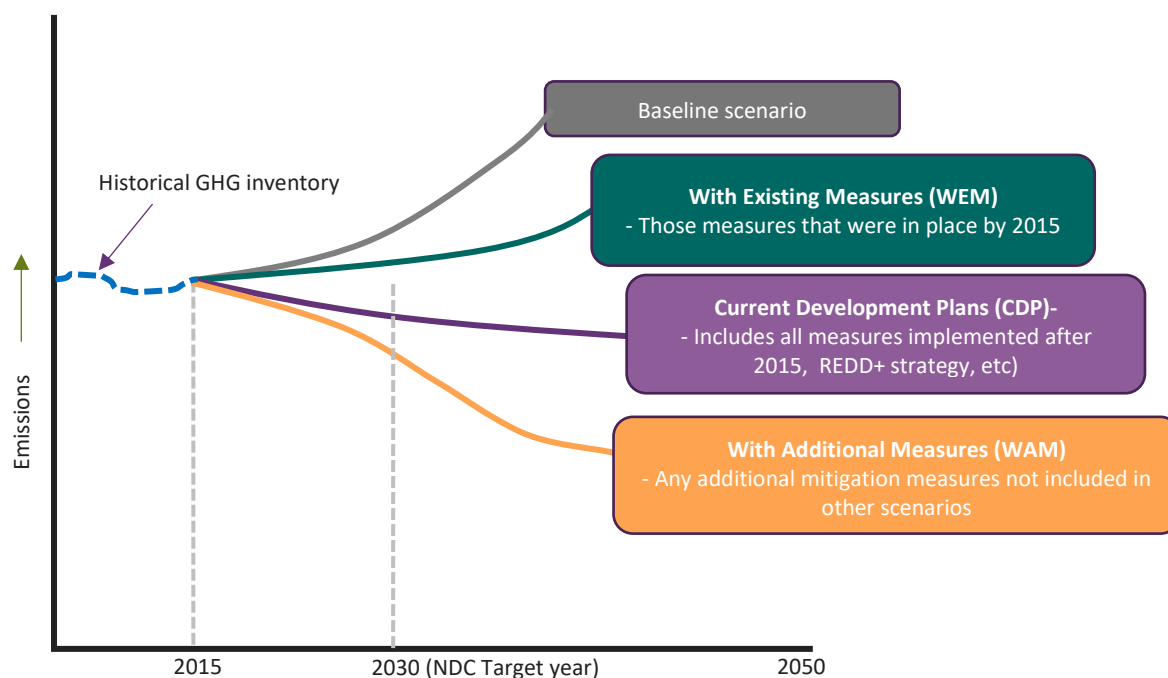


Figure 2-1: Modelled mitigation scenarios

These four emission scenarios were modelled from the 2015 base year, and were defined as follows:

- **Baseline / Business-as-usual:** a scenario that ignores existing measures, and just forecasts future GHG emissions based on expected trends.
- **'With Existing Measures' (WEM):** includes mitigation measures that were in place up to 2015 (the base year).
- **Current Development Policy (CDP):** includes mitigation measures that have been in place since 2015, but also include planned measures and/or targets and commitments that have not yet been implemented.
- **With Additional Measures (WAM):** includes additional mitigation measures or measures in earlier scenarios implemented to a higher degree of ambition.

While the NDC target year is 2030, all the above scenarios were modelled up to 2050 so that they can also respond to the modelling needs of the LTS process.

The Low Emissions Analysis Platform (LEAP¹) model was used to develop future emissions scenarios for the energy sectors (both energy use, and energy supply) and the industrial processes and product use (IPPU) sector. LEAP is an integrated, scenario-based modelling tool that can be used to track energy consumption, production and resource extraction in all sectors of an economy. The benefits of using LEAP are:

¹ Previously known as the Long-range Energy Alternatives Planning model

- It is a model that is familiar to key stakeholders in Uganda and has been used for previous modelling exercises, so will allow for greater comparability with previous GHG scenarios.
- The LEAP model was used for the INDC. Using it for the LTS therefore will help ensure strong linkages between the NDC and the LTS.
- The model has recently been updated with increased functionality, but it remains relatively simple to use.
- The model is free for the Government to use, increasing the chances that they will continue to engage with the model beyond the duration of this project.
- Its low initial data requirements are well suited to a country like Uganda where accessing robust data has been, and will continue to be, a challenge.
- It presents outputs in a transparent and intuitive way.

LEAP can also be used for modelling future emissions in the waste and AFOLU sectors, but it was decided that it would be more appropriate to develop bespoke Excel-based models for these sectors and for the results to be then incorporated into LEAP. This is because LEAP is less well suited to modelling emissions from non-energy sectors, so it makes more sense to use bespoke models which are better able to capture the specific characteristics of these sectors.

Zutari and Ricardo split the work by sector – Ricardo led the development of the energy and IPPU sectors (with Zutari providing a review/quality check) while Zutari led the development the AFOLU and waste sectors (with Ricardo providing a review/quality check).

2.1.1 Sectoral interactions

A key consideration when developing any projections is the sectoral interactions. Actions taken in one sector may have consequences in other sectors, so when developing economy wide projections these interactions needed to be accounted for. This is most relevant when looking at energy use. Demand for energy in the end-use sectors (e.g. residential, commercial, industry) will affect activities in the energy supply sectors (e.g. charcoal production, power generation). Moreover, since a large proportion of the energy use in Uganda is associated with firewood and charcoal, there are important interactions with the land-use sector through the harvesting of forests for fuel.

To ensure these sectoral interactions are addressed, the projections firstly consider the expected demand for energy in the 'end use' sectors (i.e. residential, commercial, industry), and then the energy supply sector projections are developed to ensure alignment (i.e. supply meets demand).

2.1.2 Accounting for CO₂ emissions from biogenic sources

The sector briefing notes are provided for each of the sectors within Uganda's national greenhouse gas inventory (GHGI). These sectors are defined from a production basis i.e. in terms of the sources where the emissions originate from. This means that emissions from electricity generation are allocated to the sector 1A1ai, as reported in the inventory, even though the electricity consumption may be by households, for example.

Likewise, emissions associated with the combustion of biogenic sources (e.g. firewood, charcoal) are accounted for in accordance with the GHGI accounting principles. This means that emissions of carbon dioxide (CO₂) are not accounted for in the sectors where the fuel is combusted (e.g. residential) but are accounted for in the land use sector (e.g. forests), where the biomass originates from. In contrast, emissions of methane (CH₄) and nitrous oxide (N₂O) are accounted for in the sector where the fuel is combusted.

The implication of this accounting approach is that while GHG emissions in Uganda are currently largely driven by demand for energy in the residential, commercial and industrial sectors, these emissions are

'accounted for' in the land use sector. Thus, the land use sector is by far the largest source of greenhouse gas emissions in Uganda. However, some of the largest opportunities to mitigate these emissions is from actions taken in the end-use (e.g. more efficient stoves) or energy supply/transformation (e.g. more efficient charcoal production) sectors.

2.2 Information on sectoral projections

The following sections of the chapter provide an assessment of the GHG projections for each of the main sources in the GHG inventory. These are organised as follows:

- Energy demand
 - Section 2.3 – industrial energy use (1A2)
 - Section 2.4 – commercial/institutional energy use (1A4a)
 - Section 2.5 – residential energy use (1A4b)
 - Section 2.6 – energy use in agriculture / forestry / fishing (1A4c)
 - Section 2.7 – transport (1A3)
- Energy supply
 - Section 2.8 – electricity generation (1A1a)
 - Section 2.9 – petroleum refining (1A1b)
 - Section 2.10 – charcoal production (1A1c)
- Section 2.11 – industrial processes and product use (2)
- Section 2.12 – livestock (3A)
- Section 2.13 – forestry and land use (3B)
- Section 2.14 – emissions from managed soils (3C)
- Section 2.15 – waste (4)

2.3 Industrial energy use (1A2)

IPCC sector 1A2 is manufacturing industries and construction. This chapter covers energy use in industry in Uganda. Emissions from construction are not disaggregated from manufacturing industries in the national GHG inventory, and so are not considered separately here. Emissions from industrial processes (as opposed to from energy use in industry) are covered in Chapter **Error! Reference source not found..**

2.3.1 Baseline situation

Characteristics of the sector

Uganda's industrial sector can be split into: Manufacturing, Construction, Mining and Quarrying, Electricity and water production (Ministry of Water and Environment, 2019)².

Manufacturing is currently the largest component of the industry sector, contributing 41% of the industrial sector output in 2017/18. It is largely comprised of agro-processing and mineral processing. Agro-processing includes coffee, tea, cotton, tobacco, grains and cereals, meat, dairy, leather and fish. The mineral processing industry is comprised of cement, ceramics, marble, sand, iron and steel, fertiliser, and lead acid batteries among others (Ministry of Water and Environment, 2019)³.

² Ministry of Water and Environment (2019), Uganda's First Biennial Update Report (BUR) to the United Nations Framework Convention on Climate Change. Accessed from: https://unfccc.int/sites/default/files/resource/FBUR%20Final_2019.pdf

³ Ibid.

Most firms are small and medium enterprises concentrated in Kampala and Central region” (Calabrese *et al.*, 2019⁴).

Energy use

In 2015, direct combustion of fuel for energy in the industry sector was around 116,000 TJ (Ministry of Water and Environment, 2019⁵), of which the vast majority (89%) was from biomass sources (wood fuel and agricultural residues and waste). The remainder (10%) was from liquid fuels (e.g. gas/diesel oil) and solid fuels (1%). In addition to energy from direct combustion, the industry sector also consumed ~5700 TJ of electricity (data from 2015 energy balance).

With respect to bioenergy demand, the Biomass Energy Strategy (Ministry of Energy and Mineral Development, 2013)⁶ estimated the annual wood needs for key industry sub-sectors as follows:

- brick industry – 6,000,000 tonnes of wood
- tea industry – 71,000 tonnes of wood
- small scale lime production – 270,000 tonnes of wood and 75 tonnes of charcoal
- tobacco – 200,000 tonnes of wood
- sugar jaggeries – 500 tonnes of wood and 2,000 tonnes of bagasse
- vegetable oil processes – 170,000 tonnes of agricultural waste and 75,000 tonnes of wood
- fish smoking – 22,400 tonnes wood
- cement production – 80,000 tonnes of agricultural waste
- textile production – 17,000 tonnes of agricultural waste

The emissions associated with land use change, including deforestation are discussed further in Section 2.14.

GHG emissions

Fuel combustion in the industry sector was responsible for 1,253 Gg CO_{2eq} in 2015 (Ministry of Water and Environment, 2019⁷). This value relates to combustion of liquid and solid fossil fuels (all greenhouse gases), and the methane and nitrous oxide emissions from the combustion of biomass sources.

Carbon dioxide emissions from biomass sources are not included in the value, as these emissions are accounted for the land sector (5). Likewise, emissions from the generation of electricity consumed in the industry sector are accounted for in the sector (1A1a)⁸.

⁴ Calabrese *et al.* (2019), Industrial Development in Uganda: An Assessment of the Policy Framework. Accessed from: https://www.researchgate.net/publication/341297907_Industrial_development_in_Uganda_An_assessment_of_the_policy_framework

⁵ Ministry of Water and Environment (2020), Republic of Uganda National Greenhouse Gas Inventory (NIR) Report for 2005-2015. Draft. June 2020.

⁶ Ministry of Energy and Mineral Development (2018), Biomass Energy Strategy (BEST) Uganda. Accessed from: <https://data2.unhcr.org/en/documents/details/64163>

⁷ Ministry of Water and Environment (2019), Uganda's First Biennial Update Report (BUR) to the United Nations Framework Convention on Climate Change. Accessed from: https://unfccc.int/sites/default/files/resource/FBUR%20Final_2019.pdf

⁸ The latest inventory does not distinguish between generation for supply to the grid and auto generation by end users.

2.3.2 Business as usual scenario

Drivers and assumptions

2.3.2.1.1 GDP growth rates

The key drivers of emissions in this sector are population growth and economic development (Ministry of Water and Environment, 2014)⁹. For the purposes of the BAU scenario emissions were initially projected forward based on the GDP growth rates in Vision 2040 (National Planning Authority (2010))¹⁰. However, following stakeholder consultation, these values were then adjusted downwards as the growth rates were considered too high, and inconsistent with assumption used elsewhere (such as the electricity demand forecasts used by Electricity Regulatory Authority). This is also consistent with analysis of historical trends in energy demand in Africa by the IEA (2019)¹¹ who found that the rate of growth lower compared to GDP growth. A growth rate of 4.5% was therefore applied across all years in the final analysis.

Table 2-1: Growth rates for Industry sector

Year	2015	2020	2025	2030	2035	2040
Industry	4.5	4.5	4.5	4.5	4.5	4.5

2.3.2.1.2 Fuel mix

The future fuel mix is assumed to remain unchanged from the current average fuel mix in 2015, with one exception. In accordance with the NDPIII goal, LPG is assumed to have an increasing share of the fuel mix. By 2025, the LPG share in the industry sector is assumed to increase to 8%, rising to 12% by 2050.

Electrification

The Rural Electrification Strategy and Plan (RESP ¹²) set out the target to achieve a rural electrification access of 26% (i.e. consumers who will be utilizing electricity in their homes, businesses, or institutions) by 2022 from the current level of about 7%. The RESP aims for access to be at 51% by 2030 and 100% by 2040. Access to electricity will support switching from bioenergy to cleaner energy sources. These targets are assumed to be achieved across all scenarios. It is important to note that 100% access to electricity does not equate to 100% use of electricity – and in the BAU scenario the use of electricity is assumed to remain unchanged from the current mix, as described above.

Sector disaggregation

The entire industry sector is handled together rather than splitting into sub-sectors, due to lack of activity data at sub-sector level.

⁹ Ministry of Water and Environment (2014), Uganda's Second National Communication to the United Nations Framework Convention on Climate Change. Accessed from: <https://unfccc.int/resource/docs/natc/uganc2.pdf>

¹⁰ National Planning Authority (2010). Uganda Vision 2040. National Planning Authority, Kampala Uganda. Accessed from: <http://www.npa.go.ug/uganda-vision-2040/>

¹¹ IEA (2019). Africa energy outlook 2019. Accessed from: https://iea.blob.core.windows.net/assets/2f7b6170-d616-4dd7-a7ca-a65a3a332fc1/Africa_Energy_Outlook_2019.pdf

¹² Ministry of Energy and Mineral Development (2013), Rural Electrification Strategy and Plan 2013-2022. Accessed from: <https://s3-eu-west-1.amazonaws.com/s3.sourceafrica.net/documents/119217/Strategy-and-Plan-2013-2022.pdf>

Efficiency of technologies

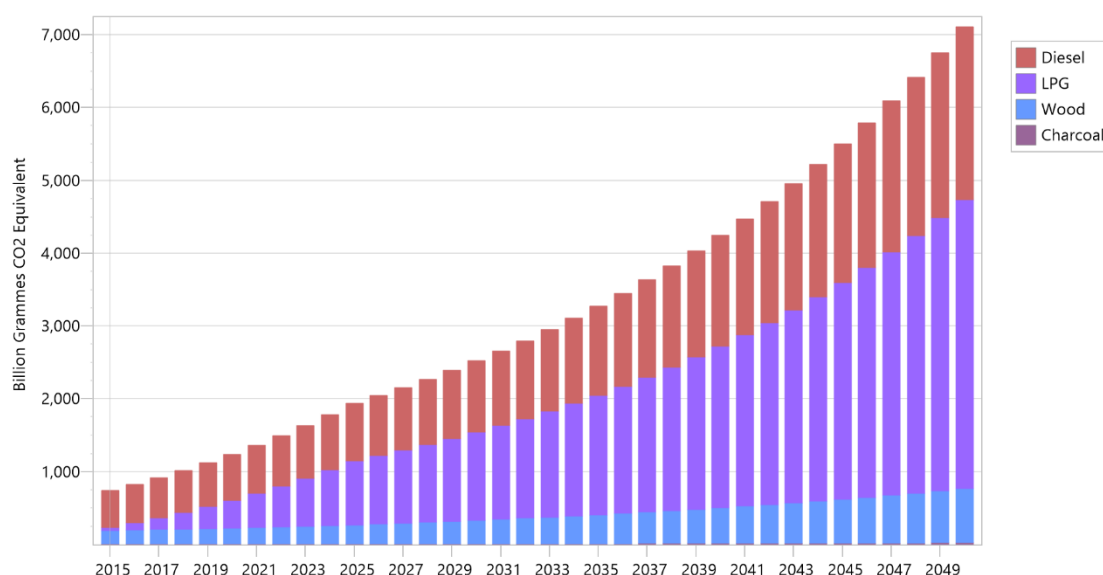
The projections assume that the main energy technologies remain the same, with no improvements in efficiency.

GHG emission projections

The projected GHG emissions from fuel combustion in the BAU scenario are shown below. It is important to note that all GHG emissions from fossil fuel combustion are included, but only CH₄ and N₂O emissions from biogenic energy sources (i.e., wood and charcoal) as the CO₂ emissions are accounted for the land sector (3B).

It can be seen in **Error! Reference source not found.** that a large increase is expected under the BAU scenario of all GHG emissions from fossil fuels, such as diesel and LPG, and of methane and nitrous oxide emissions from combustion of wood. Methane and nitrous oxide emissions from charcoal have not increased by much because charcoal accounted for a

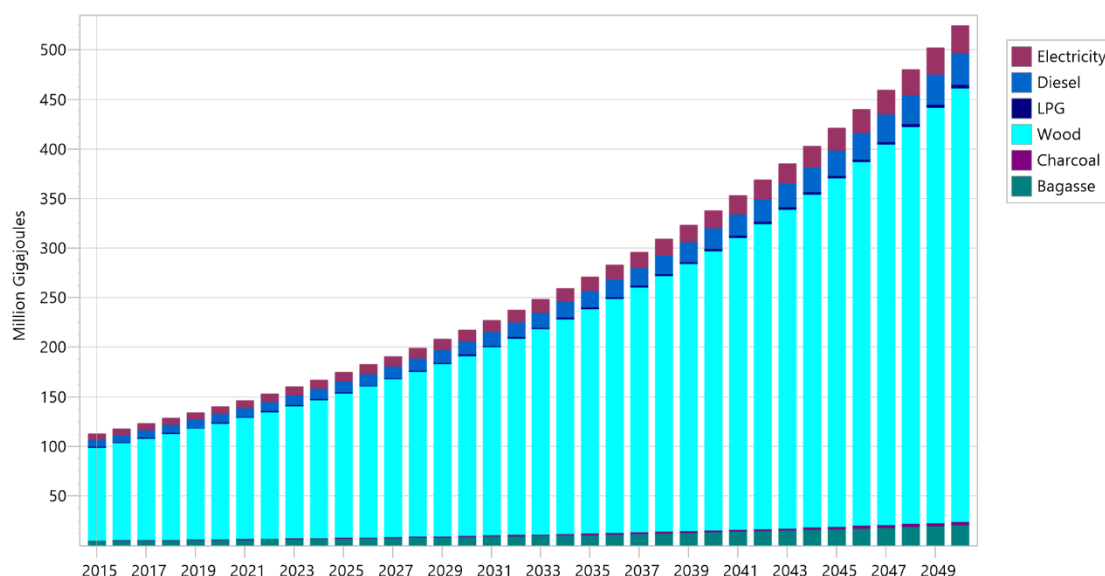
Figure 2-2: GHG emissions (GgCO₂e) from industrial energy use in Uganda under the business-as-usual scenario



far smaller share of the 2015 fuel mix than the other sources.

The emissions projections shown above are driven by energy demand, but for some sources (e.g. electricity, bioenergy) accounting principles mean that some or all of the associated emissions are accounted for elsewhere. The relative importance of these fuels is reflected in the energy demand projections, as shown in Figure 2-3 below, with wood continuing to dominate energy demand in the BAU scenario.

Figure 2-3: Energy demand projections (million GJ) by fuel, from industrial energy use in Uganda under the business-as-usual scenario



Key uncertainties

A key uncertainty for industrial energy use is whether any new industries are likely to arise in Uganda, or alternative manufacturing processes adopted – the BAU projection assumes only the existing industries will exist in the future (i.e., no new activities). A further uncertainty is how the existing sectors will develop as the economy grows. We have assumed a linear trend, but for industries this can sometime involve a step-changes in emissions new sites come into operation. The assumption that the fuel mix will remain unchanged is a simplification from what is likely to happen in practice, likewise the assumption that there will be no improvement in the efficiency of technologies. The impact of changes in fuel mix and technologies are explored in the mitigation scenarios below.

2.3.3 Mitigation options

The main mitigation options relevant to the industry sector are improvements in energy efficiency and fuel switching.

Industrial energy efficiency

Uganda's Second National Communication (Ministry of Water and Environment, 2014)¹³ notes that industries suffer from poor energy efficiency, and the Energy and Mineral Sector Development Plan (ESDP) (Ministry of Energy and Mineral Development, 2015)¹⁴ notes that a 10% reduction in industrial energy use is possible simply by employing best practice measures.

Pilot projects such as 'Promoting sustainable product innovation and energy efficient practices among small scale industries in Uganda' UNEP (2020)¹⁵ have demonstrated the potential energy savings in the sector, with the micro, small and medium enterprises involved in the project achieving a 35.7 per

¹³ Ibid.

¹⁴ Ministry of Energy and Mineral Development (2015) Energy and Mineral Sector Development Plan. Accessed from: <http://npa.go.ug/wp-content/uploads/2018/01/Energy-Sector-Development-plan-Final.pdf>

¹⁵ UNEP (2017). Promoting sustainable product innovation and energy efficient practices among small scale industries in Uganda. Impact Sheet. Accessed at: <https://wedocs.unep.org/bitstream/handle/20.500.11822/34890/UGSP.pdf?sequence=1&isAllowed=y>

cent reduction in energy and water consumption. Likewise, one of the tea factories in the project “Demand-side management of energy use in micro, small and medium sized enterprises (MSMEs) in Uganda through promotion of energy efficiency techniques and practices’ UNEP (2020)¹⁶ achieved an energy productivity improvement of more than 17%.

Further improvements in energy efficiency were identified in the Biomass Energy Strategy (Ministry of Energy and Mineral Development, 2013)¹⁷ such as more efficient kilns in the lime making, brick making and fish smoking industries, use of wood pellets rather than round wood logs in tea making and tobacco, use of industrial briquettes in vegetable oil processing. In particular, the use of more efficient kilns in brick making was estimated to deliver saving equivalent to a 45% reduction in demand for fuelwood by the sector in 2040, relative to a business-as-usual scenario. Given the high level of energy use in brickmaking the savings were estimated at 175 PJ in 2040.

With respect to electricity consumption, a detailed assessment of the potential energy savings from energy efficiency measures in Uganda was prepared by de la Rue du Can et al (2017)¹⁸. The study estimated that in a business-as-usual scenario electricity consumption in industry would double by 2030 to 3,167 GWh (~11,400 TJ), with over 50% of the 2030 electricity demand associated with motors. Process heating and process cooling, and refrigeration, were other important sources of demand. A series of energy efficiency measures were identified to reduce electricity consumption by 2030, including energy management systems, high-efficiency motors, high-efficiency chillers, and more efficient LED lighting. In combination these technologies were estimated to be able to reduce industrial electricity consumption by 948 GWh (~3,400 TJ) in 2030 – or 30% of the total electricity consumption. Moreover, the large majority of these savings were deemed economic i.e. the value of the lifetime energy savings is greater than the initial cost of investment.

Table 2-2: Cost of energy efficiency measures in industry sector (electricity use)

Mitigation Measure	Cost description	Cost unit	Cost value	Source of data
Electrical energy efficiency	Calculated weighted cost of conserved energy for key industrial categories, based on projected consumption savings in 2030	USD c/kWh	4.0	Energy Efficiency Roadmap for Uganda Making Energy-Efficiency Count (de la Rue du Can et al, 2017) ¹⁹ .

Industrial fuel switching

Uganda’s industries are highly reliant on firewood as a fuel source. The Biomass Energy Strategy (Ministry of Energy and Mineral Development, 2013)²⁰ notes that non-traditional forms of biomass (e.g. bagasse, vegetal wastes) could act as a replacement for traditional biomass sources. It was estimated that vegetal waste, shrubs, and bushes could sustainably supply industrial briquette making with

¹⁶ UNEP (2020). Promoting sustainable product innovation and energy efficient practices among small scale industries in Uganda’ Impact Sheet. Accessed at: <https://wedocs.unep.org/bitstream/handle/20.500.11822/35339/SWDSM.pdf>

¹⁷ Ministry of Energy and Mineral Development (2018), Biomass Energy Strategy (BEST) Uganda. Accessed from: <https://data2.unhcr.org/en/documents/details/64163>

¹⁸ de la Rue du Can, Stephane, David Pudleiner, David Jones, and Aleisha Khan (2017). Energy Efficiency Roadmap for Uganda, Making Energy Efficiency Count. Accessed at: [Energy Efficiency Roadmap for Uganda, Making Energy Efficiency Count | LBL International Energy Study Group](#)

¹⁹ de la Rue du Can, Stephane, David Pudleiner, David Jones, and Aleisha Khan (2017). Energy Efficiency Roadmap for Uganda, Making Energy Efficiency Count. Accessed at: [Energy Efficiency Roadmap for Uganda, Making Energy Efficiency Count | LBL International Energy Study Group](#)

²⁰ Ministry of Energy and Mineral Development (2013), Biomass Energy Strategy (BEST) Uganda. Accessed from: <https://data2.unhcr.org/en/documents/details/64163>

capacity to meet 40% of the country's biomass needs in 2040. Based on the estimates in the strategy this is equivalent to 780 PJ in 2040²¹.

In addition, some industrial energy needs could be met through of alternative fuels including electricity. Likewise, fuel switching to liquified petroleum gas (LPG) would reduce demand for fuelwood. This would also be consistent with the third NDP which has a goal of increasing national LPG uptake from 1% to 8% by 2025.

Table 2-3: Cost of briquettes from agricultural waste

Mitigation Measure	Cost description	Cost unit	Cost value	Source of data
Sustainable sourced biomass	Cost of a tonne of briquettes from agricultural waste	USD/tonne	282	Based on Kampala Jellitone Supplies report from 2017

2.3.4 Mitigation scenarios

Key policies/strategies

The following policies/measures include activities relevant for mitigating emissions in the industry sector:

Policy/measures	Mitigation action	Status
Energy and Mineral Sector Development Plan	Notes that industrial energy use can be reduced by 10% simply by employing best practice measures	Planned
Biomass Energy Strategy	Estimated that improved kilns in brickmaking and lime production could reduce energy demand by 45% in 2040 relative to current efficiency levels. Estimated that 40% of biomass needs could come from sustainable sources by 2040.	Planned
Third National Development Plan (NDP III)	Has a goal to increase access and consumption of clean energy. One of the key results associated with this goal is <i>'increase national LPG uptake from the current 1 percent to 8 percent on the energy balance'</i>	Planned
Minimum Energy Performance Standards for products	Voluntary programme in place to increase the efficiency of products entering the market. 12 standards in place, 1 testing equipment	Implemented (voluntary)

With existing measures (WEM)

Apart from the Minimum Energy Performance Standards for products, which is voluntary and only relates to a small proportion of overall energy use, no other implemented policies have been identified for industrial energy use. Therefore, the WEM scenario is unchanged from the BAU.

Current development plans (CDP)

Industrial energy use is assumed to be reduced in line with suggestions in the ESDP²², which notes that industrial energy use can be reduced by 10% simply by employing best practice measures. This is 10% reduction from BAU is assumed to be fully met in 2025 (the ESDP timeline is to 2020, so the target year

²¹ The strategy assumed a 5 fold increase in energy demand between 2005 and 2040, which is greater than the growth assumptions in the current study

²² Ibid.

has been extended slightly). After this, a 10% efficiency improvement is assumed in all future years to 2050.

With respect to fuel switching, 25% of biomass is assumed to come from sustainable sources (e.g. bagasse) by 2030, rising to 40% by 2040 (in line with suggestion from Biomass Energy Strategy²³). This is assumed to be the maximum potential, and the share doesn't increase further beyond 2040.

The LPG share is assumed to be the same as in the BAU scenario i.e. an 8% share in 2025, rising to 12% by 2050.

Under the CDP scenario, LPG and sustainable biomass primarily act to reduce wood consumption but also replace some of the diesel consumption.

No fuel switching to electricity is assumed in this scenario.

With additional measures (WAM)

An increase level of energy efficiency is assumed in the WAM scenario, based on energy savings identified in the energy efficiency roadmap and biomass energy strategy. By 2030, industrial energy use is assumed to be 20% lower than the BAU scenario, and 30% lower by 2040.

With respect to fuel switching, the same levels of sustainable biomass and LPG use are assumed as in the CDP scenario, but due to the additional efficiency savings described above will represent a slightly different % share. For LPG, this represents a 10% share in 2030 (compared to 8% under CDP), and a 16% share by 2050 (compared to 12% under CDP). For sustainable biomass, this represents a 28% share in 2030 (compared to 25% under CDP) and a 52% share in 2040 (compared to 40% under CDP).

Under WAM, it is also assumed that the remaining industrial energy consumption in 2050 (32%) will be met by electricity, meaning diesel, wood and charcoal are phased out in the industrial demand sector by 2050.

Summary of scenarios

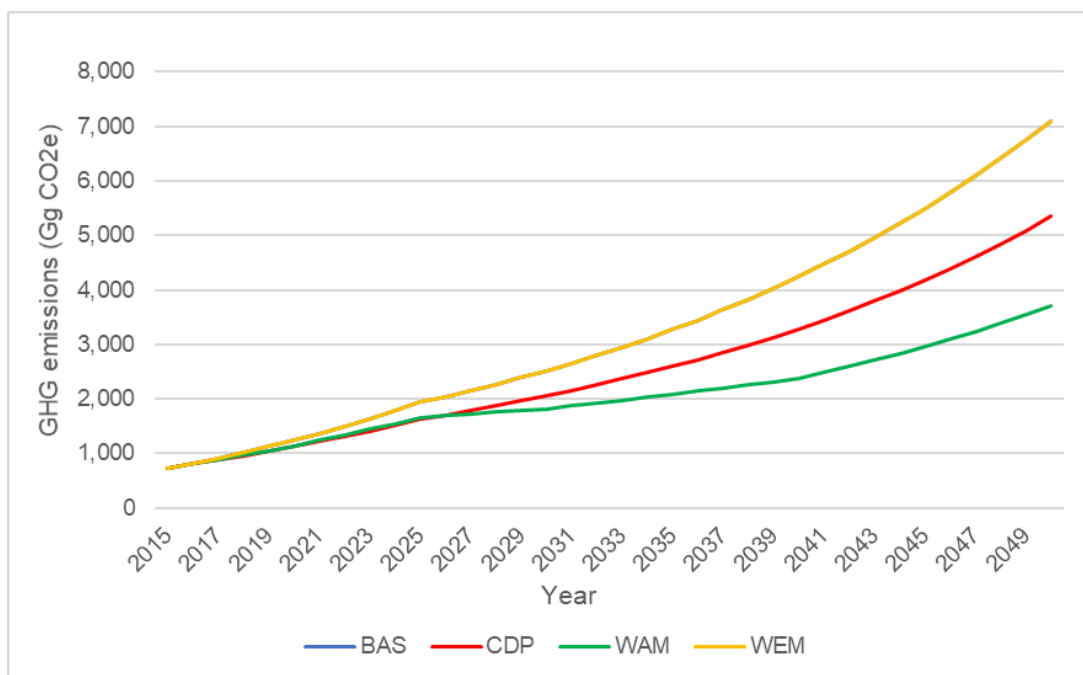
Measure	BAU	WEM	CDP	WAM	Notes
Energy efficiency	Static efficiency	Unchanged from BAU	10% reduction from BAU by 2025	20% reduction from BAU by 2030 30% reduction from BAU by 2040	WEM is unchanged from BAU because no further <i>implemented</i> measures were identified. The BAU assumptions are based on projections rather than targets.
Fuel switching	Same fuel mix as 2015, but with 8% of energy needs met by LPG in 2025, rising to 12% by 2050	Unchanged from BAU	25% of energy met by biomass from sustainable sources by 2030, rising to 40% by 2040 8% of energy needs met by LPG in 2025, rising to 12% by 2050	28% of energy met by biomass from sustainable sources by 2030, rising to 52% by 2040 10% of energy met by LPG in 2025, rising to 16% by 2050 Remaining 32% met by electricity	WEM is unchanged from BAU because no further <i>implemented</i> measures were identified. The BAU assumptions are based on projections rather than targets.

²³ Ibid.

GHG emission projections

The GHG emissions arising for the different mitigation scenarios is shown below. This shows that the direct emissions from the sector are actually greater in the CDP and WAM scenarios as a result of the fuel switching to LPG. This is because the direct combustion emissions are accounted for here, whereas the CO₂ emissions associated with reduced firewood and charcoal use (e.g. reduced deforestation) are accounted for in the forestry sector.

Figure 2-4: GHG emissions (GgCO₂e) from industrial energy use in Uganda under the three mitigation scenarios



Key uncertainties

The mitigation potentials for the measures are based on previous detailed estimates for Uganda, so should be relatively robust. However, in practice, there will be technical, behavioural and financial barriers to be overcome to deliver this level of ambition. For example, increased uptake of sustainable biomass may require investment in industrial production of briquettes, and increased electricity consumption requires upgrades to the electricity transmission and distribution infrastructure.

2.4 Commercial/Institutional energy use (1A4a)

2.4.1 Baseline situation

Characteristics of the sector

The commercial sector comprises a range of commercial and institutional activities. In terms of key energy users, the Biomass Energy Strategy (Ministry of Energy and Mineral Development, 2013)²⁴ splits commercial demand into the following categories:

²⁴ Ministry of Energy and Mineral Development (2013), Biomass Energy Strategy (BEST) Uganda. Accessed from: <https://data2.unhcr.org/en/documents/details/64163>

- Bakeries
- Hotels, restaurants and bars
- Breweries and distilleries

Institutions are split as follows:

- Schools
- Governmental organisations

The mitigation analysis presented here assumes a total of ~40,000 institutional settings in Uganda.

Energy use

In 2015, direct combustion of fuel for energy in the commercial sector was around 19,000 TJ (Ministry of Water and Environment, 2019²⁵), of which the vast majority (almost 91.5%) was from biomass sources²⁶. The remainder (8.5%) was from liquid fuels (e.g. gas/diesel oil). In addition to energy from direct combustion, the commercial sector also consumed 1,172 TJ of electricity (data from 2015 energy balance).

GHG emissions

Fuel combustion in the commercial sector was responsible for 250.8 ktCO_{2eq} in 2015 (Ministry of Water and Environment, 2019²⁷). This value relates to combustion of liquid fuels (all greenhouse gases), and methane emissions and nitrous oxide emissions from combustion of biomass sources (firewood and charcoal).

Carbon dioxide emissions from biomass sources are not included in the value, as these emissions are accounted for in the land sector (3B). Likewise, emissions associated with electricity consumption are accounted for in the sector (1A1a).

2.4.2 Business as usual scenario

Drivers and assumptions

2.4.2.1.1 GDP growth rates

The key drivers of emissions in this sector are population growth and economic development (Ministry of Water and Environment, 2014)²⁸. For the purposes of the business as usual scenario emissions have been projected forward based on UBOS population projections²⁹.

Table 2-4: Population growth assumptions (based on UBOS population projections) used to drive growth in the commercial/institutional sector

Year	2016	2020	2025	2030	2035	2040	2045	2050
Population (Millions)	36.63	41.59	48.50	55.98	63.68	71.45	79.15	86.59

²⁵ Ibid.

²⁶ According to the energy balance, just over half was from charcoal, and just under half from firewood

²⁷ Ibid.

²⁸ Ibid.

²⁹ UBOS (no date), Population and censuses. Accessed from: <https://www.ubos.org/explore-statistics/20/>

2.4.2.1.2 Fuel mix

The future fuel mix is assumed to remain unchanged from the current average fuel mix in 2015, with one exception. In accordance with the NDPIII goal, LPG is assumed to have an increasing share of the fuel mix. By 2030, the LPG share in the industry sector is assumed to increase to 15%.

2.4.2.1.3 Electrification

The Rural Electrification Strategy and Plan (RESP³⁰) set out the target to achieve a rural electrification access of 26% (i.e. consumers who will be utilizing electricity in their homes, businesses, or institutions) by 2022 from the current level of about 7%. The RESP aims for access to be at 51% by 2030 and 100% by 2040. Access to electricity will support switching from bioenergy to cleaner energy sources. These targets are assumed to be achieved under BAU. It is important to note that 100% access to electricity does not equate to 100% use of electricity – the contribution of electricity to the fuel mix is dictated by fuel switch measures in the mitigation scenarios.

2.4.2.1.4 Sector disaggregation

The entire commercial/institutional sector is handled together rather than splitting into sub-sectors, due to lack of activity data at sub-sector level.

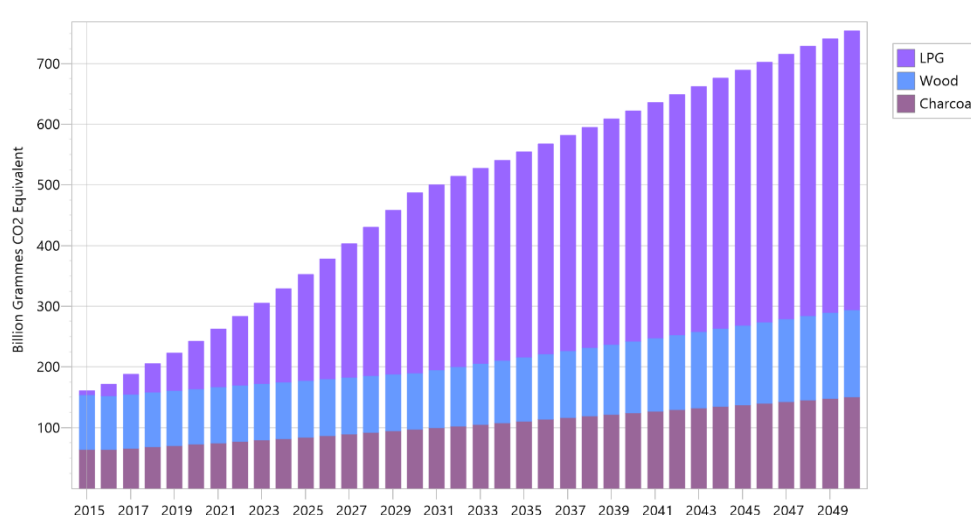
2.4.2.1.5 Efficiency of technologies

The projections assume that the main energy technologies remain the same, with no improvements in efficiency.

2.4.2.1.6 GHG emissions projections

The projected GHG emissions from fuel combustion in the BAU scenario are shown below. Emissions are dominated by non-CO₂ emissions firewood and charcoal; the contribution from LPG is small.

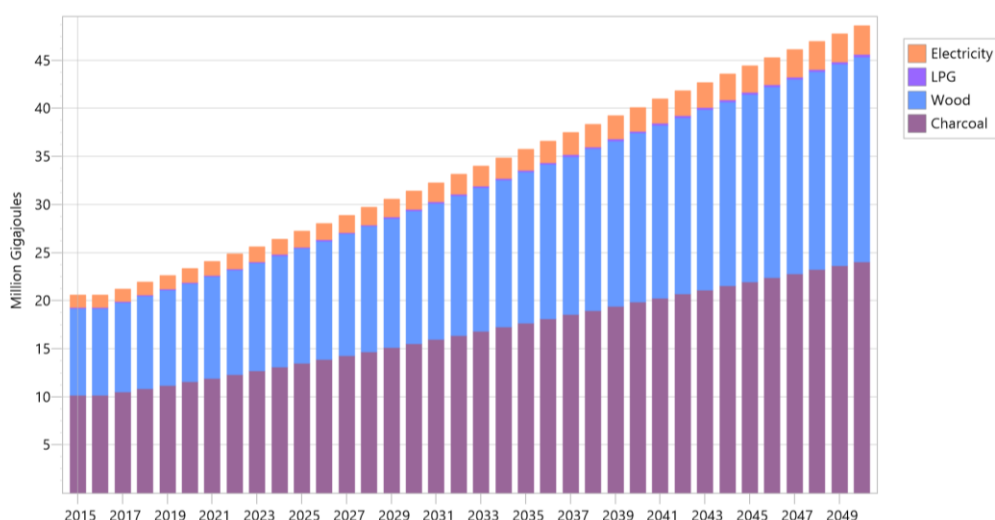
Figure 2-5: GHG emissions (GgCO₂e) from commercial/institutional energy use in Uganda under the business-as-usual scenario



³⁰ Ministry of Energy and Mineral Development (2013), Rural Electrification Strategy and Plan 2013-2022. Accessed from: <https://s3-eu-west-1.amazonaws.com/s3.sourceafrica.net/documents/119217/Strategy-and-Plan-2013-2022.pdf>

The emissions projections shown above are driven by energy demand, but for some sources (e.g. electricity, bioenergy) accounting principles mean that some or all of the associated emissions are accounted for elsewhere. Energy demand projections are shown in Figure 2-6 below, which illustrates that electricity is another important energy source.

Figure 2-6: Energy demand by fuel, from commercial/institutional energy use in Uganda under the business-as-usual scenario



2.4.2.1.7 Key uncertainties

There are a number of uncertainties with the projections in this sector. Firstly, it assumes that the fuel mix remains the same in the future as it is now, with the exception of LPG – so doesn't account for increased electricity use (likely as access to electricity increases under the RESP). It also assumes that all activities within the commercial and institutional sector are the same, when in practice the characteristics of e.g. hotels, will be different to e.g. schools. Finally, it assumes no change in the efficiency of the technologies that are used, or any increases in consumption as the economy grows. Some of these factors may lead to reductions in emissions e.g. more efficient technologies and other may lead to increases in emissions e.g. increased demand for new technologies such as air conditioning. The current approach implicitly assumes these effects even out.

2.4.3 Mitigation options

In the Commercial and Institutional sector, the main mitigation options are (Ministry of Water and Environment, 2014)³¹:

- Fuel switching;
- Use of more energy efficiency technologies and/or equipment;
- Adoption of better energy management practices.

As described above, the majority of current energy needs are met by charcoal and firewood and the CO₂ emissions associated with these fuels are 'accounted for' in other sectors. Therefore, improvements in energy efficiency will result in changes in the carbon stock in the land sector (5) rather than in the emissions profile for the commercial/institutional sector.

³¹ Ibid.

Fuel switching

This would include measures such as cleaner fuels (to replace charcoal/wood) for roadside food vendors (Ministry of Energy and Mineral Development, 2013³²). For institutions like educational facilities, prisons and hospitals large quantities of bioenergy are used for cooking. The alternative fuels may include electricity, biogas and LPG.

A key consideration for institutions when considering alternative fuel options is the cost of energy. A recent study by Rhamz International (2020)³³ analysed the performance, availability, affordability and accessibility of different cooking fuels and technologies. The analysis involved a kitchen laboratory test on the different technologies to determine the difference in expenditures of alternative fuel sources. The analysis was based on current market prices, and no estimates were made of how the cost of the technologies and/or fuels may change in the future. The controlled cooking tests (CCTs) performed during the study revealed that briquettes were the cheapest option followed by charcoal. Electricity was more than a factor of 10 more expensive than charcoal; LPG was the most expensive option overall. Traditional woodfuel stoves were also expensive to run, costing more than electricity, but less than LPG. Despite the large variation in costs the authors recommended mass promotion of all of the improved stoves for institutions, both improved installed institutional wood stove & mobile improved charcoal stove and clean fuel stoves i.e. LPG and electric stoves.

Table 2-5 Cost of electric cook stoves

Mitigation Measure	Cost description	Cost unit	Cost value	Source of data
Electric cook stove	Average Cost of electric stoves in 2019/2020	USD/stove	1390	An energy audit experiment to promote renewable energy in large institutions and households. UNDP Uganda Country Report- September 2020.

Use of more efficient technologies

The introduction of more efficient technologies in the commercial/institutional sector would reduce direct fuel use and the associated emissions. This could include measures such as more efficient cookstoves for schools/hotels/restaurants/bars (Ministry of Energy and Mineral Development, 2018³⁴), or the introduction of water filters to reduce the energy needed to boil water (to make it consumable). Institutional biolatrines (using biogas technology) have also been noted as a potential mitigation option.

Table 2-6: Cost of more efficient institutional cook stove

Mitigation Measure	Cost description	Cost unit	Cost value	Source of data
Improved institutional cookstoves	Average Cost of institutional stoves in 2019/2020	USD/stove	764	An energy Audit experiment to promote renewable energy in large institutions and households-UNDP Uganda Country Report- September 2020.

³² Ministry of Energy and Mineral Development (2013), Biomass Energy Strategy (BEST) Uganda. Accessed from: <https://data2.unhcr.org/en/documents/details/64163>

³³ Rhamz International (2020). An energy audit experiment to promote renewable energy in large institutions and households. Draft report to UNDP

³⁴ Ministry of Energy and Mineral Development (2018), Biomass Energy Strategy (BEST) Uganda. Accessed from: <https://data2.unhcr.org/en/documents/details/64163>

With respect to electricity consumption, a detailed assessment of the potential energy savings from energy efficiency measures in Uganda was prepared by de la Rue du Can et al (2017)³⁵. The study estimated that in a business-as-usual scenario electricity consumption in industry would double by 2030 to 1,480 GWh (~5,300 TJ), with around 30% of the 2030 electricity demand associated with lighting, followed by air conditioning (14%) computer equipment (13%) water heating (13%) and cooking (10%). A series of energy efficiency measures were identified to reduce electricity consumption by 2030, including solar water heating, heat pump water heater, LED lighting, and high efficiency computers. In combination these technologies were estimated to be able to reduce commercial electricity consumption by 491 GWh (~1,770 TJ) in 2030 – or 33% of the total electricity consumption. Moreover, the large majority of these savings (90%) were deemed economic i.e. the value of the lifetime energy savings is greater than the initial cost of investment.

Cost of energy efficiency measures in industry sector (electricity use)

Mitigation Measure	Cost description	Cost unit	Cost value	Source of data
Electrical energy efficiency	Calculated weighted cost of conserved energy for key commercial categories, based on projected consumption savings in 2030	USD c/kWh	4.0	Energy Efficiency Roadmap for Uganda Making Energy-Efficiency Count (de la Rue du Can et al, 2017) ³⁶ .

2.2.3.3 Adoption of better energy management practices

Improved energy management practices can improve energy efficiency in the commercial/institutional sector, acting to decrease energy consumption and the associated GHG emissions. This may include measures such as improved energy management in distilleries (Ministry of Energy and Mineral Development, 2018³⁷).

2.4.4 Mitigation scenarios

Key policies/strategies

The following policies/measures are relevant for the commercial/institutional sector:

Policy/measures	Mitigation action	Status
Revolving Loan Facility for the Uptake of Improved Institutional Cook Stoves in Ugandan Schools	The full implementation of this project will see more than 18,000 primary schools, almost 3,000 secondary schools, and about 50 tertiary schools from all over the country provided with improved cookstoves (Ministry of Water and Environment, 2019)	Implemented
Energy Sector Development Plan	Aims to "Promote adoption of improved institutional kilns, oven, and stoves in all educational institutions, hospitals, and prisons" (Ministry of Energy and Mineral Development, 2015)	Planned

³⁵ de la Rue du Can, Stephane, David Pudleiner, David Jones, and Aleisha Khan (2017). Energy Efficiency Roadmap for Uganda, Making Energy Efficiency Count. Accessed at: [Energy Efficiency Roadmap for Uganda, Making Energy Efficiency Count | LBL International Energy Study Group](#)

³⁶ de la Rue du Can, Stephane, David Pudleiner, David Jones, and Aleisha Khan (2017). Energy Efficiency Roadmap for Uganda, Making Energy Efficiency Count. Accessed at: [Energy Efficiency Roadmap for Uganda, Making Energy Efficiency Count | LBL International Energy Study Group](#)

³⁷ Ministry of Energy and Mineral Development (2018), Biomass Energy Strategy (BEST) Uganda. Accessed from: <https://data2.unhcr.org/en/documents/details/64163>

Policy/measures	Mitigation action	Status
Third National Development Plan (NDPIII)	The NDPIII intends to “promote uptake of alternative and efficient cooking technologies”. It includes a target to reduce share of biomass energy used for cooking from 85% in 2017/18 to 50% in 2024/25 (and increase share of clean energy used for cooking from 15% in 2017/18 to 50% in 2025/25). It also includes the target to ‘increase national LPG uptake from the current 1 percent to 8 percent on the energy balance’	Planned
REDD+ Strategy and Action Plan	Strategy Option 5: Energy efficient cooking stoves, is concerned with the replacement of traditional firewood and charcoal stoves with more efficient technologies. It is aimed at both households and institutional users (educational institutions, restaurants and cafeterias, hospitals, prisons, industries and other similar entities) (Ministry of Water and Environment, 2017 ³⁸).	Planned
Charcoal to Power initiative	This initiative aims to replace the use of traditional biomass for cooking with low-carbon electricity in households and institutions (ERA, 2021 ³⁹)	Planned
Minimum Energy Performance Standards for products	Voluntary programme in place to increase the efficiency of products entering the market. 12 standards in place, 1 testing equipment	Implemented (voluntary)

With existing measures (WEM)

The WEM scenario assumes the full implementation of the ‘Revolving Loan Facility for the Uptake of Improved Institutional Cook Stoves in Ugandan Schools’ CDM project. This project will provide 21,000 (out of approx. 40,000 schools/institutions; i.e., ~50%) with improved cookstoves. This is assumed to be fully implemented by 2030. After this, no further dissemination of cookstoves is assumed.

Current development plans (CDP)

The NDPIII goal to increase use of clean fuels/electricity to 50% by 2025, as well as to increase national LPG uptake to 8% of the energy balance by 2025. It is assumed that LPG stoves are included within the 50% clean fuels target. As described above the cost of cooking for both electric and LPG stoves is currently more than 10 times more expensive than the cost for more efficient charcoal stoves (Rhamz International, 2020)⁴⁰. LPG and electric stoves also have high upfront costs. This cost differential is likely to limit the uptake of clean stoves in the short term, and financial incentives are likely to be needed to promote switching. This will make the delivery of the NDPIII goals challenging.

Achieving a high level of uptake of more efficient charcoal and firewood stoves appears more plausible in the short-term given the lower upfront cost of these stoves, and the better cost-effectiveness overall. However, these stoves won’t contribute towards the relevant NDPIII goals.

Despite the high costs of cleaner cookstoves and for consistency with the approach taken in other sectors, the CDP scenario assumes that the NDPIII goals are achieved. It is therefore assumed that by 2030, improved charcoal stoves will be used by 50% of institutions and the remaining 50% of institutions will use clean cookstoves, with LPG being used by 15% of institutions (as in BAU), and electric stoves being used by the remaining 35%.

³⁸ Ministry of Water and Environment (2017). REDD+ Strategy and Action Plan. Accessed at: <https://www.mwe.go.ug/sites/default/files/library/Final%20-%20Uganda%20REDD%2B%20Strategy%20and%20Action%20Plan-October%202017.pdf>

³⁹ Ibid.

⁴⁰ Rhamz International (2020). An energy audit experiment to promote renewable energy in large institutions and households. Draft report to UNDP

With additional measures (WAM)

The CDP scenario is already very challenging, and the WAM scenario assumed the same level of uptake until 2030 (50% improved charcoal stoves, 15% LPG and 35% electric stoves). After 2030, it is assumed that electric cookstoves continue to increase in uptake (in line with NDPIII/ERA Least Cost Generation Plan targets to increase use of clean fuels/electricity, and in line with the goals of Uganda's electrification strategy), replacing charcoal stoves, but also some of the LPG stoves. Therefore, by 2050 70% of the institutional cookstoves are assumed to be electric.

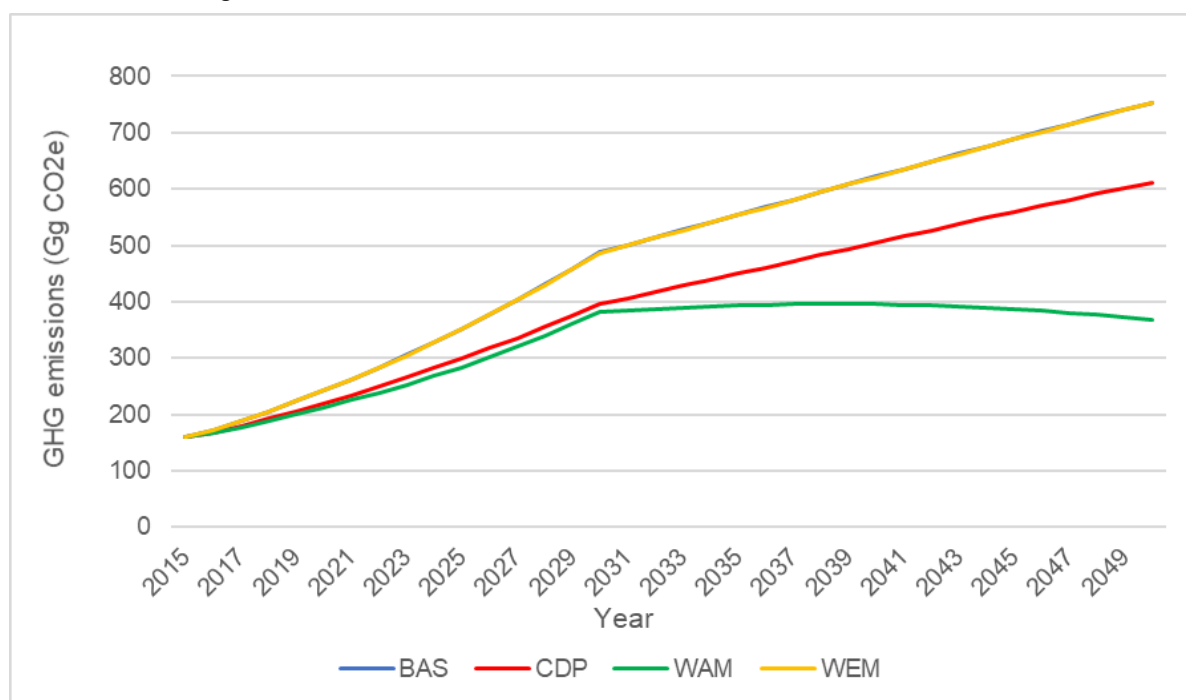
2.2.4.5 Summary of scenarios

Measure	BAU	WEM	CDP	WAM	Notes
Improved cookstove efficiency	Same efficiencies as 2015	50% of schools/institutions to be using improved charcoal cookstoves by 2030 (replacing traditional charcoal stoves)	50% of schools/institutions to be using improved charcoal cookstoves by 2030	Same as CDP up to 2030 (then gradual decrease in improved charcoal cookstoves from 2030-2050 as fuel switch to electricity occurs – see below)	WEM is unchanged from BAU because no further <i>implemented</i> measures were identified. The BAU assumptions are based on projections rather than targets.
Switching the cleaner fuels	Same fuel mix as 2015, but with 15% of energy needs met by LPG in 2030	Unchanged from BAU	15% LPG stoves and 35% electric stoves by 2030	Same as CDP up to 2030. After 2030, electricity use gradually increases, reaching 70% in 2050.	WEM is unchanged from BAU because no further <i>implemented</i> measures were identified. The BAU assumptions are based on projections rather than targets.

GHG emission projections

The projected GHG emissions from fuel combustion in the mitigation scenario are shown below. It is important to note that, as these scenarios deal with switches away from biogenic sources, much of the emissions reduction will be allocated to other sectors in the inventory (e.g. land sector, 3B). The CDP and WAM scenarios see significant emissions reductions due to the measure targeting a fuel switch to electricity – this allows emissions in the WAM scenario to reach 0 by 2050.

Figure 2-7: GHG emissions (GgCO₂e) from commercial/institutional energy use in Uganda under the three mitigation scenarios.



Key uncertainties

A key uncertainty for these scenarios is the timeliness of implementation – in many cases, the target year specified in the reference documents has been delayed slightly based on judgement of current progress. Another key uncertainty surrounds exactly how many schools/institutions/commercial settings there are in Uganda – we have assumed there to be 40,000 based on the best available data from desk research, but this is uncertain. Further uncertainties relate to the level of take up of clean fuels, since technical and behaviour barriers may limit full take up.

2.5 Residential energy use (1A4b)

2.5.1 Baseline situation

Characteristics of the sector

The most recent Uganda National Household Survey (UBOS, 2018⁴¹) provides information on some of the key characteristics of the residential sector in Uganda as of 2016/2017. These include:

- 81.1% of rural dwellings, and 84% of urban dwellings, have only 1 sleeping room / 2.2% and 1.6% of rural and urban dwellings respectively have more than 2 rooms
- Most dwellings have iron sheet roofing (68.6% rural / 92.6% urban) and brick walls (60.8% rural / 81.4% urban). Around a third of rural dwellings have thatched roofing (31.1%) and/or walls made of mud and poles (34.8%) Earth floors are most common in rural dwellings (73.1%) while cement floors are most common in urban dwellings (70.1%).

Access to electricity

Access to electricity is another important factor influencing consumption levels. Currently, 61.5% of rural households and 21.9% of urban households have no electricity (Ministry of Energy and Mineral Development, 2020⁴²). Of those households that have access to at least one source of electricity (51% of total households) approximately half have access through the grid (National or mini-grid), while the other half only have access through off-grid solutions like generators, rechargeable batteries and Solar - mostly Solar Lighting System (SLS) and Solar lanterns.

Moreover, even where households have access to at least one source of electricity the availability, quality and reliability of supply varies greatly. Therefore, overall, almost 30% of urban and 78% of rural households have less than reasonable access to electricity even for basic lighting use (Ministry of Energy and Mineral Development, 2020⁴³).

Energy use

In 2015, direct combustion of fuel for energy in the residential sector energy was approximately 369,000 TJ (Ministry of Water and Environment, 2019⁴⁴), of which the vast majority (almost 99.5%) was from biomass sources. The remainder (0.5%) was from liquid fuels (e.g. kerosene). In addition to energy from direct combustion, the residential sector also consumed 2,177 TJ of electricity (data from 2015 energy balance).

For most households, the majority (>80%) of energy-use is associated with cooking (UBOS, 2018)⁴⁵.

⁴¹ Uganda Bureau of Statistics (UBOS), 2018. Uganda National Household Survey 2016/2017. Kampala, Uganda; UBOS https://www.ubos.org/wp-content/uploads/publications/03_20182016_UNHS_FINAL_REPORT.pdf

⁴² Ministry of Energy and Mineral Development (2020), National Electrification Report. Accessed from: http://www.energyandminerals.go.ug/site/assets/files/1081/memd_national_electrification_survey_-_2018_printed_2020.pdf

⁴³ Ministry of Energy and Mineral Development (2020), National Electrification Report. Accessed from: http://www.energyandminerals.go.ug/site/assets/files/1081/memd_national_electrification_survey_-_2018_printed_2020.pdf

⁴⁴ Ibid

⁴⁵ Ibid.

GHG emissions

In 2015 direct fuel combustion in the residential sector was responsible for 3,549 Gg CO_{2eq}. (Ministry of Water and Environment, 2019⁴⁶). This value relates to combustion of liquid fuels (all greenhouse gases), and methane emissions and nitrous oxide emissions from combustion of biomass sources (firewood and charcoal).

Carbon dioxide emissions from biomass sources are not included in the value, as these emissions are accounted for the land sector (3B). Likewise, emissions associated with electricity consumption are accounted for in the sector (1A1a).

2.5.2 Business as usual scenario

Drivers and assumptions

Population growth and urbanisation

Energy use is expected to grow in line with population growth and economic development, with population the key driver. Population is assumed to grow in line with the UBOS household projections⁴⁷. Urbanisation will also be a key driver of emissions in the residential energy sector – projections for this variable are informed by the 2040 target (60% urban) from Vision 2040 (National Planning Authority (2010)⁴⁸ with the base year (2015) value informed by World Bank data⁴⁹. From 2040-2050, a lower rate of urbanisation compared to previous years is assumed.

Table 2-7: UBOS household projections used to drive growth in the residential sector.

Year	2016	2020	2025	2030	2035	2040	2045	2050
Households (million)	8.2	9.6	11.8	14.4	17.7	21.3	25.0	28.6
Household size (persons)	4.5	4.3	4.1	3.9	3.6	3.3	3.1	2.9

Table 2-8: Urbanisation assumptions used for the residential sector.

Year	2015	2020	2025	2030	2035	2040	2045	2050
Urban	22%	30%	37%	45%	52%	60%	65%	70%
Rural	78%	70%	63%	55%	48%	40%	35%	30%

Fuel mix

The future fuel mix is based on the current (2015) mix, with future demand projected forward using population numbers. As described below, increased electricity access will mean a greater proportion of household will have access to electricity, and therefore a change in fuel mix is assumed to follow for those households.

A further change in the fuel mix that has been assumed related to LPG. The Third NDP has a target to 'increase national LPG uptake from the current 1 percent to 8 percent on the energy balance' by

⁴⁶ Ibid.

⁴⁷ UBOS (no date), Population and censuses. Accessed from: <https://www.ubos.org/explore-statistics/20/>

⁴⁸ National Planning Authority (2010). Uganda Vision 2040. National Planning Authority, Kampala Uganda. Accessed from: <http://www.npa.go.ug/uganda-vision-2040/>

⁴⁹ World Bank (no date), Urban population (% of total population) - Uganda. Accessed from: <https://data.worldbank.org/indicator/SP.URB.TOTL.IN.ZS?locations=UG>

2024/25. It is therefore assumed that 8% of household energy needs will be met by 2025, increasing to 10% by 2030, and remaining at this level until 2050.

Electrification

The Rural Electrification Strategy and Plan (RESP⁵⁰) set out the target to achieve a rural electrification access of 26% (i.e. consumers who will be utilizing electricity in their homes, businesses, or institutions) by 2022 from the current level of about 7%. The RESP aims for access to be at 51% by 2030 and 100% by 2040. Access to electricity will support switching from bioenergy to cleaner energy sources. These targets are assumed to be achieved under BAU, because so much progress has already been made that it is likely to have exerted impacts on emissions prior to 2015 (the base year for this analysis). It is important to note that 100% access to electricity does not equate to 100% use of electricity – the contribution of electricity to the fuel mix is dictated by fuel switch measures in the mitigation scenarios.

Sector disaggregation

Due to the high demand in energy from the residential sector, fuel consumption was split further into 4 segments: 1) Urban-Electrified, Urban Non-electrified, Rural-Electrified, Rural Non-electrified. A further disaggregation between energy use for cooking and lighting has also been made.

Efficiency of technologies

In the BAU scenario, traditional fuel sources are expected to remain important. However, the Biomass Energy Strategy notes that ~65,000 improved residential cookstoves are already distributed each year. Therefore, the BAU scenario assumes the penetration of more efficient cookstoves at this level. Faster rates of uptakes are assumed in the other scenarios.

GHG emissions projections

The projected GHG emissions from fuel combustion in the BAU scenario are shown below.

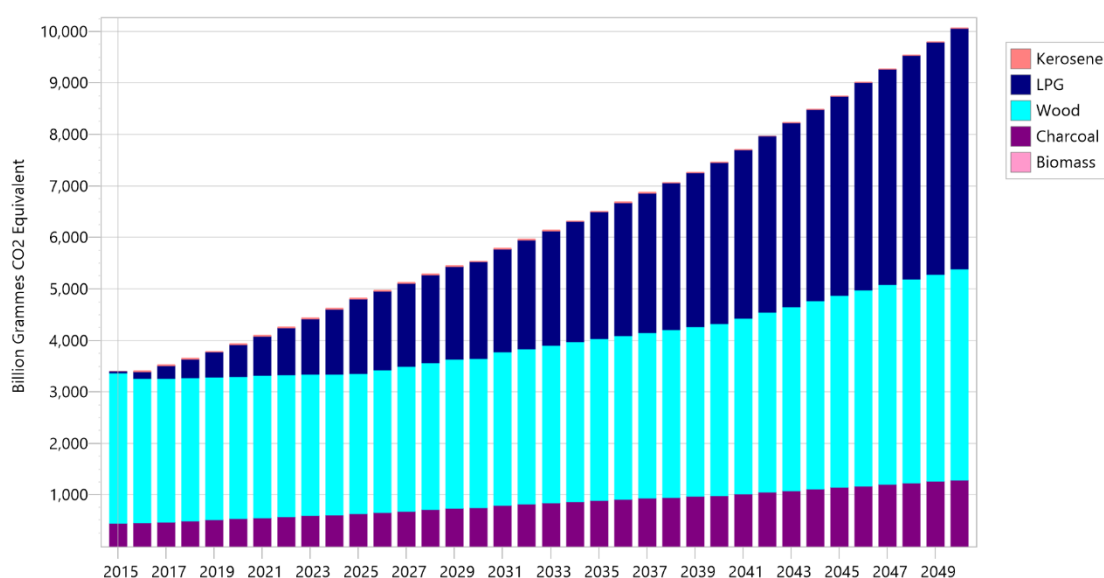
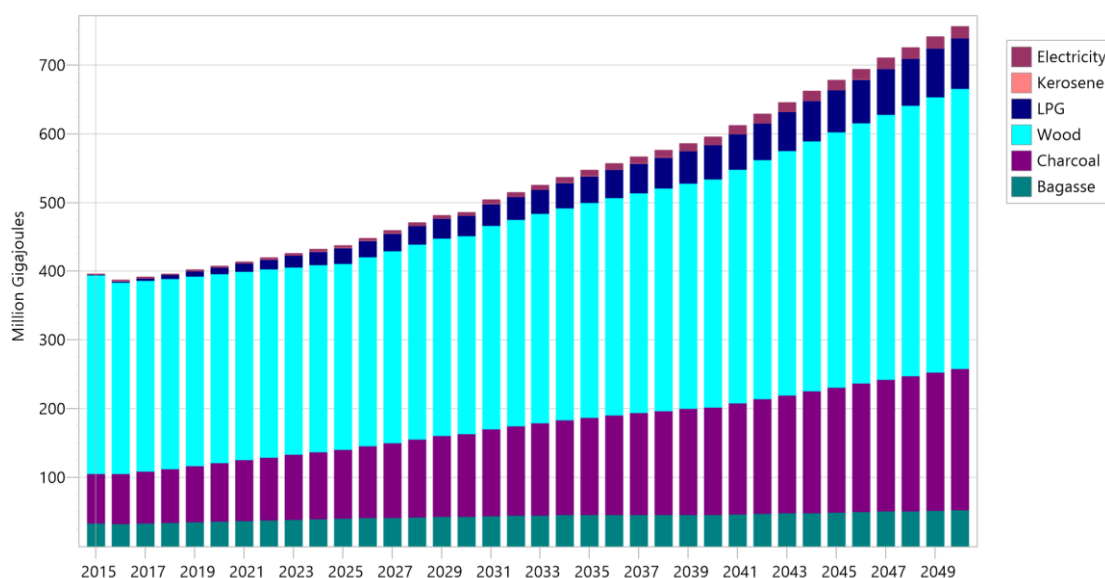


Figure 2-8: GHG emissions (GgCO2e) from residential energy use in Uganda under the business-as-usual scenario.

⁵⁰ Ministry of Energy and Mineral Development (2013), Rural Electrification Strategy and Plan 2013-2022. Accessed from: <https://s3-eu-west-1.amazonaws.com/s3.sourceafrica.net/documents/119217/Strategy-and-Plan-2013-2022.pdf>

The emissions projections shown above are driven by energy demand, but for some sources (e.g. electricity, bioenergy) accounting principles mean that some or all of the associated emissions are accounted for elsewhere. Energy demand projections are shown in Figure 2-9 below, which illustrates that electricity is another important energy source.

Figure 2-9: Energy demand by fuel, from residential energy use in Uganda under the business-as-usual scenario



Key uncertainties

There are a number of uncertainties with the projections in this sector. A key uncertainty for residential energy use is whether any new energy uses are likely to arise in the future – the BAU scenario assumes that only current energy uses (i.e., cooking and lighting) will continue in the future. This means that potential future energy uses such as electrical appliances and space cooling are not considered. There is also uncertainty with respect to the delivery of the NDP goal associated with LPG use, which is included in the BAU, as well as the timing and levels of electrification.

2.5.3 Mitigation options

Overview

The following mitigation measures have been identified for reducing emissions associated with residential energy use.

- More efficient technologies:
 - Efficient cookstoves (improved wood stove, improved charcoal stove etc)
 - Efficient lighting (CFLs, LEDs etc)
- Fuel switching:
 - Reduce use of traditional cooking fuels (charcoal, firewood) and lighting fuels (kerosene) and replace with cleaner fuels – e.g. electric stove, gas stove, electric lighting, solar lighting
 - Rural electrification – enhancing access to the grid in rural areas in order to decrease reliance on biomass

- Solar Home Systems – to decrease reliance on traditional biomass fuel sources.

As described above, the majority of current energy needs are met by charcoal and firewood and the CO₂ emissions associated with these fuels are ‘accounted for’ in other sectors. Therefore, improvements in energy efficiency will mostly result in reduced GHG emissions in the land sector (3B), rather than in the emissions profile for the residential sector.

Efficient cookstoves/lighting

According to the Biomass Energy Strategy⁵¹ Uganda is characterised by high share of traditional and inefficient cooking technologies. This includes the use of three stone fuelwood stove at 90% in rural areas, and traditional metallic charcoal stoves estimated at 80% in urban areas. Emissions from cooking can be reduced through the use of more efficient cookstoves (e.g. improved wood stove, improved charcoal stove) The use of more efficient cookstoves expected to achieve approximately 40% efficiency saving over traditional cooking stoves (Ministry of Water and Environment, 2015a)⁵².

According to the Biomass Energy Strategy⁵³ the adoption of efficient charcoal and fuelwood stoves is about 10% or less in both urban and rural areas. In terms of the penetration of more efficient cookstoves, the Strategy indicates that about 65,000 improved cookstoves are distributed annually through out Uganda. This represents around 0.8% of the total households.

The NDP III⁵⁴ published in 2020, suggested that 15% of households currently (2017/18) use clean energy for cooking; the NDP included a target to increase this to 50% by 2024/25. In the mitigation scenarios presented here, “Clean energy” is assumed to refer to electricity, in line with additional plans from the ERA Least Cost Generation Plan 2020-2030 (ERA 2021⁵⁵) which states Uganda’s intention to swap from biomass for cooking to electricity (the “Charcoal to Power for Cooking” initiative), but also LPG stoves.

With respect to lighting, the ESDP⁵⁶ notes that replacement of incandescent bulbs for lighting can significantly reduce residential energy consumption for this purpose, stating that “electricity consumption for lighting in households and other buildings can be reduced by up to 70- 80% simply by replacing incandescent bulbs by Compact Fluorescent Lamps (CFLs)”.

The opportunities for energy savings associated with electricity use were further explored in a report by de la Rue du Can et al (2017)⁵⁷. The study estimated that in a business-as-usual scenario electricity consumption in the residential sector would more than triple by 2030 to 2,391 GWh (~8,600 TJ), with around 25% of the 2030 electricity demand from on-grid urban households (the main sub-sector) associated with lighting, followed by refrigeration (24%) and TVs (10%). A series of energy efficiency measures were identified to reduce electricity consumption by 2030, including LED lighting, solar water heating, high efficiency refrigerator and more efficient television. In combination these technologies were estimated to be able to reduce commercial electricity consumption by 568 GWh (~2,040 TJ) in 2030 – or 33% of the total electricity consumption. Moreover, the large majority of these savings (84%) were deemed economic i.e. the value of the lifetime energy savings is greater than the initial cost of investment.

⁵¹ Ibid.

⁵² Ministry of Water and Environment (2015a) Uganda’s Intended Nationally Determined Contribution (INDC)

⁵³ Ibid.

⁵⁴ National Planning Authority (2020), Third National Development Plan (NDP III) 2020/21 - 2024/25. Accessed from: http://www.npa.go.ug/wp-content/uploads/2020/08/NDP-III-Finale_Compressed.pdf

⁵⁵ Ibid.

⁵⁶ Ibid.

⁵⁷ de la Rue du Can, Stephane, David Pudleiner, David Jones, and Aleisha Khan (2017). Energy Efficiency Roadmap for Uganda, Making Energy Efficiency Count. Accessed at: [Energy Efficiency Roadmap for Uganda, Making Energy Efficiency Count | LBL International Energy Study Group](#)

Mitigation Measure	Cost description	Cost unit	Cost value	Fuel saving	Source of data
More efficient cookstoves	Average Cost of domestic improved cookstoves- Improved Charcoal stove (ICS) and Installed Improved Wood Stove (IWS)	USD/stove	6.25 ICS 55.6 IWS	20% to 35% thermal efficiency improvement between old and new stove resulting in about 50-70%	An energy Audit experiment to promote renewable energy in large institutions and households-UNDP Uganda Country Report- September 2020. Cost:
More efficient lighting					

Fuel switching to cleaner fuels

A key mitigation option in the residential energy sector is switching to cleaner fuels (e.g. electric, gas) to reduce consumption of biomass as fuel. In particular, the “Charcoal to Power” initiative aims to replace the use of traditional biomass for cooking with low-carbon electricity in households and institutions (ERA, 2021⁵⁸). Uganda’s Rural Electrification Strategy is crucial to this, as rural areas without access to grid electricity currently have high reliance on traditional biomass as fuel. Enhancing access to electricity across Uganda will allow biomass to be replaced with low-carbon electricity as a fuel source. Solar Home Systems also offer a viable alternative to traditional biomass fuel sources.

Charcoal production

Whilst not associated with residential energy sector emissions, interventions targeting inefficient charcoal production can act to reduce the impact of charcoal consumption in the energy sector (i.e., by reducing the associated deforestation). This measure is discussed separately, in section 2.10.

2.5.4 Mitigation scenarios

Key policies/strategies

Policy/measures	Mitigation action	Status
Rural Electrification Strategy and Plan (RESP)	The Rural Electrification Strategy aims to enhance grid access in rural areas (with goals of 100% access to electricity by 2040), which will support actions to reduce the use of biomass as an energy source.	Implemented
Biomass Energy Strategy	65,000 improved cookstoves already disseminated every year	Implemented
Up-Energy CDM Project ⁵⁹	A project to provide an additional 50,000 improved residential cookstoves per year until 2040 (the end of the project duration).	In progress

⁵⁸ Ibid.

⁵⁹ Up-Energy CDM Project PoA:

https://cdm.unfccc.int/ProgrammeOfActivities/poa_db/WRL90IO62SKX4BD5CYJT8E1HVZ7QUM/view

Policy/measures	Mitigation action	Status
Third National Development Plan (NDPIII)	Clean fuels to make up 50% of energy use by 2025. NDPIII also notes an electrification target with a more ambitious timeline than the RESP: "The aspiration of Agenda 2030 is to achieve universal access to electricity by 2030" (NDPIII) It also includes the target to 'increase national LPG uptake from the current 1 percent to 8 percent on the energy balance'	Planned / In Progress
REDD+ Strategy and Action Plan	Strategy Option 5: Energy efficient cooking stoves, is concerned with the replacement of traditional firewood and charcoal stoves with more efficient technologies. It is aimed at both households and institutional users (Ministry of Water and Environment, 2017 ⁶⁰).	Planned
Charcoal to Power initiative	This initiative aims to replace the use of traditional biomass for cooking with low-carbon electricity in households and institutions (Electricity Regulatory Authority, 2021)	Planned

With existing measures (WEM)

Electrification in the WEM scenario is unchanged from the BAU scenario (i.e. following the targets of the Rural Electrification Strategy and Plan), as so much progress has already been made.

With respect to more efficient technologies, in addition to the 65,000 improved residential cookstoves already disseminated under the BAU scenario, a further 50,000 improved residential cookstoves are disseminated per year (until 2040) under the WEM scenario (in line with the Up-Energy CDM project⁶¹ for improved residential cookstoves).

Residential energy use for lighting is not targeted in the WEM scenario (i.e., unchanged from BAU) as no existing/implemented measures have been identified.

Current development plans (CDP)

While the BAU and WEM scenarios assume electrification in line with the Rural Electrification Strategy and Plan, a more ambitious timeline for electrification is proposed in NDPIII, which states: "The aspiration of Agenda 2030 is to achieve universal access to electricity by 2030" (NDPIII, p.144). The CDP scenario therefore assumed 100% electrification by 2030.

On top of the improved cookstove penetration assumed in the WEM scenario, the CDP scenario assumes increased uptake of electric stoves (to facilitate a fuel switch to electricity). This is in line with the NDPIII, which intends to "promote uptake of alternative and efficient cooking technologies" (NDPIII, p.149). It includes a target to increase the share of clean energy used for cooking (assumed here to be electricity – in line with the 'Charcoal to Power' initiative detailed in ERA 2021) to 50% in 2025. After 2025, no further penetration of electric stoves is assumed (though improved biomass cookstove penetration continues to increase up until 2040, in line with WEM assumptions)

Residential energy use for lighting is also targeted in the CDP scenario, in line with the following suggestion from the ESDP: "Electricity consumption for lighting in households and other buildings can be reduced by up to 70- 80% simply by replacing incandescent bulbs by Compact Fluorescent Lamps

⁶⁰ Ministry of Water and Environment (2017). REDD+ Strategy and Action Plan. Accessed at: <https://www.mwe.go.ug/sites/default/files/library/Final%20-%20Uganda%20REDD%2B%20Strategy%20and%20Action%20Plan-October%202017.pdf>

⁶¹ Up-Energy CDM Project PoA:

https://cdm.unfccc.int/ProgrammeOfActivities/poa_db/WRL90IO62SKX4BD5CYJT8E1HVZ7QUM/view

(CFLs)." (p.102). The CDP scenario therefore assumed a 75% reduction from BAU in energy use for lighting by 2030, after which no further improvements are assumed.

.With additional measures (WAM)

The WAM scenario contains targets similar in nature to the CDP scenario, but with enhanced levels of ambition as follows:

- Electrification is assumed to reach 100% by 2025 (i.e., a more ambitious timeline than under CDP)
- Energy use for cooking is assumed to be 100% electric by 2050 (in line with the "Charcoal to Power" initiative detailed in ERA (2021, and in line with the NDPIII goal to "promote uptake of alternative and efficient cooking technologies" (NDPIII, p.149)).
- Energy use for lighting is assumed to be reduced by 75% by 2025 (i.e., a more ambitious timeline than under CDP).

Summary of scenarios

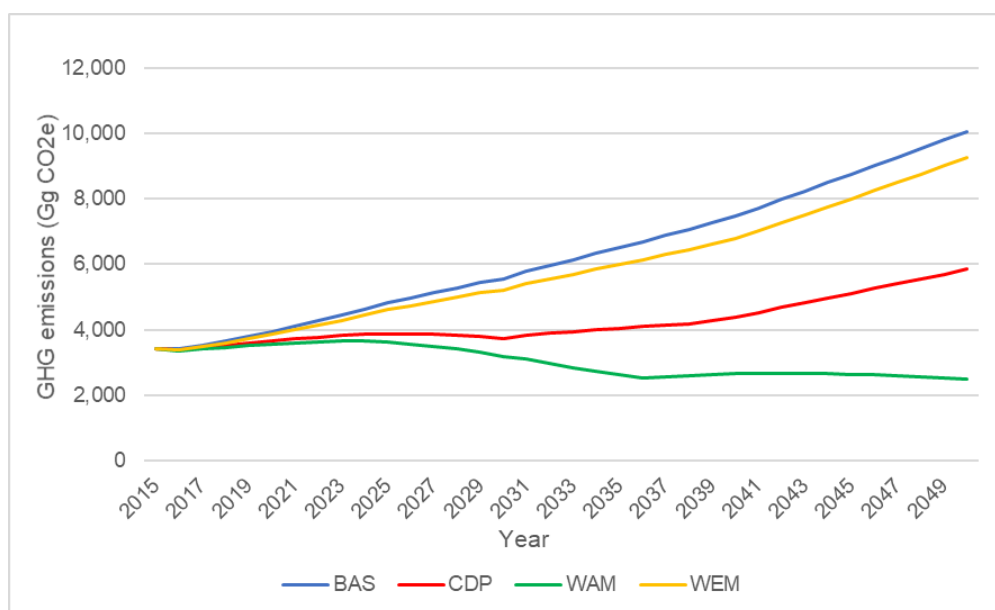
Measure	BAU	WEM	CDP	WAM	Notes
Electricity access	100% by 2040	Unchanged from BAU	100% by 2030	100% by 2025	
Improved cookstove efficiency	65,000 improved cookstoves disseminated per year, growth in energy use driven by UBOS household projections	An additional 50,000 improved cookstoves disseminated per year (115,000 total)	Electric stove penetration (see below)	Electric stove penetration (see below)	
Cooking fuel switch to electricity	Same fuel mix as 2015, growth in energy use driven by UBOS household projections	Unchanged from BAU	Electricity to reach 50% of fuel share by 2025	Electricity to reach 50% of fuel share by 2025 and 100% by 2050.	
Reduced energy use for lighting	Same fuel mix as 2015, growth in energy use driven by UBOS household projections	Unchanged from BAU	75% reduction from BAU in energy use for lighting by 2030	75% reduction from BAU in energy use for lighting by 2025	

GHG emission projections

The projected GHG emissions from fuel combustion in the mitigation scenarios is shown below.

It is also important to note that residential energy demand projections are affected by numerous variables (e.g. rural-urban split changes, electrification measure, fuel switches etc). It is also important to note that the lighting efficiency measure does not produce any significant emissions savings, as it is related to a reduction in electricity consumption (where the electricity being consumed is very low-carbon to begin with). It is clear that measures targeting a fuel switch to electricity have a significant impact on emissions in the CDP and WAM scenarios, with the 100% electricity use assumed by 2050 in the WAM scenario allowing emissions to reduce to 0 by this year.

Figure 2-10: GHG emissions (GgCO₂e) from residential energy use in Uganda under the three mitigation scenarios



2.3.1.7 Key uncertainties

Similar to the BAU scenario, a key uncertainty related to future changes in energy demand. For example, because the BAU does not consider new energy uses such as space cooling, the mitigation scenarios do not consider potential measures to enhance energy efficiency in buildings. Another key uncertainty is the timeliness of implementation of the measures discussed – in many cases, the target year specified in the reference documents has been delayed due to best judgement on current progress.

2.6 Agriculture / Forestry / Fishing (1A4c)

2.6.1 Characteristics of the sector

Agriculture, forestry and fishing are important economic activities in Uganda, contributing 23.6% of Uganda's GDP in 2015 (World Bank⁶²). The characteristics of the agriculture/forestry sector are further detailed in sections **Error! Reference source not found.**, **Error! Reference source not found.**, and **Error! Reference source not found.**

⁶² World Bank (no date), Agriculture, forestry, and fishing, value added (% of GDP) – Uganda. Accessed from: <https://data.worldbank.org/indicator/NV.AGR.TOTL.ZS?locations=UG>

Energy use

Energy use in the agriculture/forestry/fishing sub-sectors arises from fisheries, where fishing boats are responsible for mobile combustion. In 2015, energy use totalled 2,508 TJ, 1254 TJ of which was from mobile combustion in the fishing sub-sector (allocation of the remainder of this energy use is not documented; Ministry of Water and Environment, 2019).

GHG emissions

GHG emissions for sub-sector 1A4c totalled 181.1 Gg CO₂e in 2015.

2.6.2 Business as usual scenario

Drivers and assumptions

The projected GHG emissions from fuel combustion in the agriculture/forestry/fishing sub-sector are assumed to increase in line with the expected growth in GDP. Emissions have been projected forward based on the GDP growth rates in Vision 2040 (National Planning Authority (2010))⁶³.

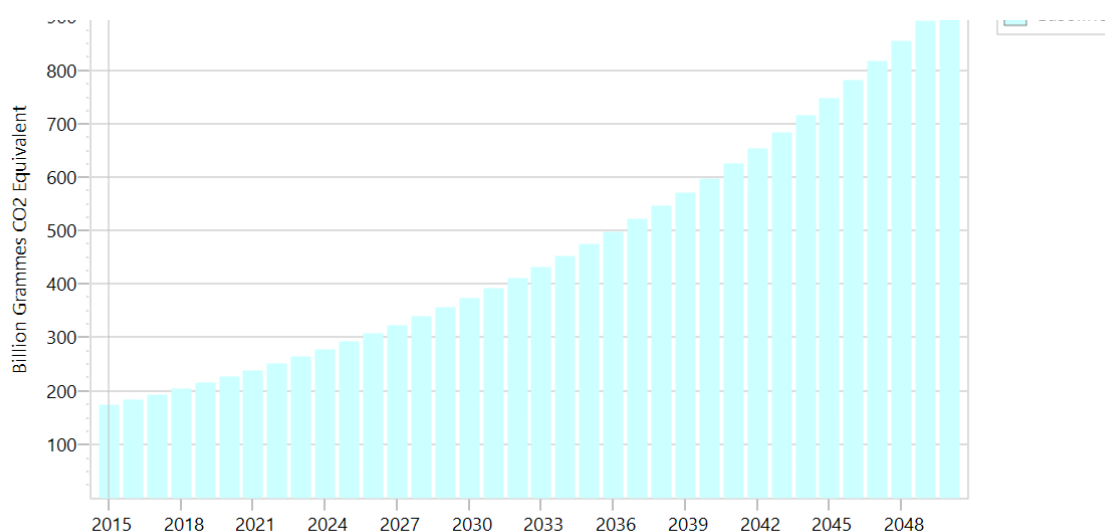
Table 2-9: Real GDP growth rates for Agriculture sector from Vision 2040 (note: 2040-2050 growth factors continue on historic trend)

Year	2015	2020	2025	2030	2035	2040
Agriculture	5.24	5.37	5.15	5.02	4.88	4.65

GHG emissions projections

The projected GHG emissions from fuel combustion in the BAU scenario are shown below.

Figure 2-11: GHG emissions (GgCO₂e) from agriculture/forestry/fishing energy use in Uganda under the business-as-usual scenario.



⁶³ National Planning Authority (2010). Uganda Vision 2040. National Planning Authority, Kampala Uganda. Accessed from: <http://www.npa.go.ug/uganda-vision-2040/>

2.6.3 Mitigation options

As a significant proportion of energy use in this sub-sector can be attributed to mobile combustion from fishing boats, a key intervention could be to reduce fuel consumption from fishing boats (through best practice measures) or to switch to less carbon-intensive fuels.

2.6.4 Mitigation scenarios

With existing measures

No targets for this sub-sector have been identified.

CDP

No targets for this sub-sector have been identified.

With additional measures

No targets for this sub-sector have been identified.

2.7 Transport (1A3)

This sub-sector is detailed to a greater extent than others due to ongoing work under a parallel project to conduct a more detailed mitigation analysis of the transport sector.

2.7.1 Baseline situation

Characteristics of the sector

Uganda's transport system can be divided into five sectors: (1) roads and road transport; (2) rail transport; (3) air transport, (4) inland water transport; and (5) other modes.⁶⁴

Road transport is the dominant mode, carrying over 95% of total traffic.⁶⁵ The public road network, including both classified and unclassified roads, comprises more than 140,000 km. In 2018/2019, about 96 percent of freight cargo and passenger traffic was delivered by road.⁶⁶ The Kampala District in particular accounts for over 50% of the vehicles in Uganda.⁶⁷

The majority of vehicles in Uganda originate from Japan, imported as reconditioned units, usually more than 5 years old. The current motorised vehicle fleet is currently approximately 1,355,090 vehicles, up from 739,036 in 2012, with the average age of vehicles being more than 15 years old. Public passenger transport in Uganda is a mix between private, motorcycles, minibuses and buses.⁶⁸

Civil aviation transport has also expanded rapidly in recent years. International passenger numbers per annum increased during the last 8 years from 781,428 to 1,303,484 in 2016. International traffic dominates flights at Entebbe, up to 97% for passengers and 99% for cargo. Entebbe International

⁶⁴ <https://ledsgp.org/wp-content/uploads/2018/03/GIP01883-LEDS-UgandaFinal.pdf>

⁶⁵ <https://ledsgp.org/wp-content/uploads/2018/03/GIP01883-LEDS-UgandaFinal.pdf>

⁶⁶ http://www.npa.go.ug/wp-content/uploads/2020/08/NDPIII-Finale_Compressed.pdf

⁶⁷ <https://www.mwe.go.ug/sites/default/files/library/National%20Climate%20Change%20Policy%20April%202015%20final.pdf>

⁶⁸ Mitigation assessment of transport sector. MoWT. Unpublished.

Airport is currently undergoing expansion in order to cater for the increasing passenger traffic and cargo freight.⁶⁹

However, rail and waterway services are generally in a run-down state⁷⁰. The current market shares of railways declined from 12% to 5% within the last 8 years.⁷¹ Railway Transport sector has been operating below its capacity due to dilapidation of the railway line (only 26% of the railway line is operational), poor state of locomotives, unavailability of boat ferries to supplement the network, poor state of real estate property and theft of track materials.⁷²

Energy use

Energy use in transport is set out in the 2015 Energy Balance, reproduced in **Error! Reference source not found.** below.

Table 2-10: Uganda 2015 energy balance⁷³

1.1: UGANDA ENERGY BALANCE 2015														
Unit:ktoe	Liquefied Petroleum Gases	Motor Gasoline (Petrol)	Aviation Gasoline	Kerosene	Gas/Diesel Oil	Fuel Oil	Lubricants	Bitumen	Fuelwood, Residues and Bagasse	Charcoal	Hydro	Solar Photovoltaics	Electricity	Total of All Energy Sources
Production									16,625.11		265.83	0.34		17,066.25
Imports	6.78	597.08	109.14	51.26	728.08	44.17	17.05	2.79					4.13	1,560.48
Exports													-10.41	-10.41
Total Primary Energy Supply (TPES)	6.78	597.08	109.14	51.26	728.08	44.17	17.05	2.79	16,625.11		265.83	0.34	-6.28	18,616.33
Statistical differences														
Transformation Processes					-1.03	-17.28			-328.47	1,976.68	-265.83	-0.34	303.41	1,492.16
Main activity producer electricity plants					-1.03	-17.28					-257.48		264.28	-11.52
Autoproducer electricity plants									-328.47		-8.34	-0.34	39.13	-298.03
Charcoal production plants										1,976.68				1,801.71
Energy Industry Own Use													-19.01	-19.01
Own use in electricity plants													-19.01	-19.01
Losses									-4,743.91	-652.52			-62.09	-5,458.51
Final Consumption	6.78	597.08	109.14	51.26	727.04	26.88	17.05	2.79	11,552.74	1,324.16			216.03	14,630.97
Industry		59.92		40.80	218.22	26.88			2,449.37				135.97	2,931.16
Non-specified (industry)		59.92		40.80	218.22	26.88			2,449.37				135.97	2,931.16
Transport		477.23	109.14		436.43									1,022.81
Road		447.27			407.48									854.75
Domestic aviation			109.14											109.14
Rail					28.96									28.96
Water Transport		29.96												29.96
Other	6.78	59.92		10.46	72.39				9,103.37	1,324.16			80.07	10,657.15
Residential	4.52			10.46	36.20				8,254.00	1,103.47			52.37	9,461.02
Commercial and public services	2.26				36.20				196.01	220.69			27.52	482.68
Street Lighting													0.17	
Agriculture/forestry		29.96												29.96
Fishing		29.96												29.96
Non-specified (other)									653.36					653.53
Non-Energy Use							17.05	2.79						19.85
Non-energy use industry							3.01							3.01
Non-energy use in transport							9.03	2.79						11.82
Non-energy use in other							5.02							5.02
Electricity Output in GWh					1.00	78.00			354.00		3,091.00	4.00		3,528.00
Elec output-main activity producer electricity plants					1.00	78.00					2,994.00			3,073.00
Elec output-autoproducer electricity plants									354.00		97.00	4.00		455.00

GHG emissions

CO2 emissions from the transport sector in Uganda have fluctuated since 2005 with a general upward trend, as shown in Figure 2-12 below.

69 Mitigation assessment of transport sector. MoWT. Unpublished.

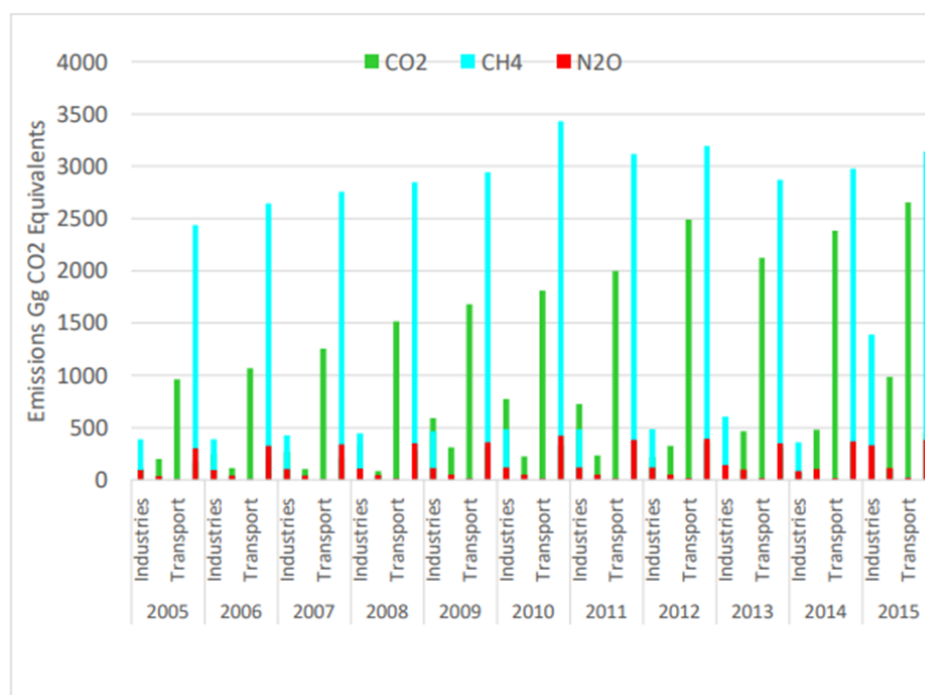
70 <https://works.go.ug/wp-content/uploads/2015/08/National-Transport-Master-Plan-2008-2023.pdf>

71 Mitigation assessment of transport sector. MoWT. Unpublished.

72 Mitigation assessment of transport sector. MoWT. Unpublished.

73 https://www.energyandminerals.go.ug/site/assets/files/1017/2015_statistical_abstract.pdf

Figure 2-12: Emission trends in the transport and the manufacturing industries and construction sectors, 2005 to 2015⁷⁴



2.7.2 Business as usual emissions projection

Drivers and assumptions

Key drivers

The main drivers for transport emissions are population and economic growth. As Uganda's population grows and becomes more wealthy, the overall demand for transport will increase and there may be a shift from non-motorised transport and public transport, to private car use. Vehicle ownership (excluding motor cycles) increased from only 2/1000 in 2008 to 22/1000 in 2016. It is important to note that around 40% of registered vehicles currently are motorcycles. Whilst these are key drivers, these are not the available datapoints used to build the model.

Assumptions

The key long-term strategy document for Uganda, Vision 2040 has emphasised the need for an integrated transport infrastructure network to spur its own economic growth. This will entail development of a highly interconnected transport network and services optimizing the use of rail, road, water and air transport modes.

This has been reinforced by the Second National Development Plan which has recognised infrastructure as one of the development fundamentals required to attain the Middle income status. As a result, a sizeable share of commitments is being directed to infrastructure investments with a focus on reducing travel times between regions, integrating the national market and connecting it to other markets in the East African Community.

⁷⁴ https://unfccc.int/sites/default/files/resource/FBUR%20Final_2019.pdf

As outlined in the National Transport Master Plan, the dominant growth will be in the road sector, where traffic is expected to reflect economic growth and lead to a growth in traffic of about 7% until 2025 when growth reduces to 6%⁷⁵.

The rail sector should also be able to increase its market share substantially however substantial investment is needed in order to overcome the challenges presented in Section 0. Ambitious goals were set-out in Vision 2040, of at least 80% of Uganda freight transport will be carried by rail and the standard gauge rail will transport at least 10% of all persons for interurban and international trips by year 2040. However, project timelines do not currently align with these goals.

In the air sector, stagnation in the international passenger market over the years 1997- 2002 was followed by rapid growth in 2002-07. International air freight has grown rapidly in recent years, particularly in the export market, and the market as a whole may expand at about 9% annually over the next 15 years. Domestic air traffic should be capable of growing at between 5 and 10% annually, with strong growth in general aviation a possibility as the tourist sector expands. Domestic air freight will remain at a low level. Upgrade and expansion of Entebbe International Airport is ongoing.

In the water sector, traffic by wagon ferry will be determined by the development of rail traffic, and also by the relative use of the Southern Route through Tanzania for imports and exports. Similarly, utilisation of the 'road bridge' ferries will follow growth of road traffic. Growth of other lake and river traffic will follow that of local economies in the districts concerned, where the waterways will remain a lifeline for some rural communities.

Modelling approach

Two modelling approaches have been used to model the GHG emission projections here, these are:

- Top-down: This approach takes fuel consumption, with some limited disaggregation, as the primary activity data and applies an emissions factor in order to calculate the GHG emissions
- Bottom-up: This approach takes information on the activity of each mode of transport, i.e. the annual vehicle kilometres travelled (VKM), it then multiplies this by the average load factor the vehicle type e.g. the number of people or tonnes of cargo on-board for an average trip to get passenger kilometres travelled (PKM) or tonnes kilometres travelled (TKM). This is then multiplied by a fuel economy value to estimate fuel consumption. An emissions factor is then applied in order to calculate the GHG emissions.

A top-down methodology is a simpler approach, and provides more limited disaggregation by mode, and therefore does not provide the flexibility to model as many mitigation measures and their impact. A bottom-up approach does allow this but has significantly higher data requirements and therefore can introduce more uncertainty.

The modelling approach used for each sub-sector is listed in Table 2-11.

Table 2-11: Methodology selected for each sub-sector

Sub-sector	Methodology
Road transport	Bottom-up: (Activity (VKM) * load factor = PKM/TKM) * fuel economy * emission factor
Civil aviation	Top-down: Fuel consumption * emission factor
Railways	Top-down / bottom up: (Fuel consumption / fuel economy = VKM) * emission factor

⁷⁵ https://www.researchgate.net/publication/285766191_Energy_Report_for_Uganda_A_100_Renewable_Energy_Future_by_2050

Sub-sector	Methodology
Water-borne navigation	Top-down: Fuel consumption * emission factor

The other key methodological point is that each transport sub-sector was split by passenger and freight transportation in order to allow better targeting of mitigation measures.

For the modelling exercise, the BAU scenario was developed using the following assumptions.

Road transport

Modelling approach: Bottom-up (VKM)

Table 2-12: Bottom-up calculation methodology used for road transport

Activity		Load factor		Fuel economy		Emissions factor
Vehicle kilometres travelled (VKM)	X	Persons per vehicle Tonnes per vehicle	X	Litres of fuel/100 km	X	CO2 produced per energy consumed

Activity

Activity data was split by passenger and freight and then by mode of transport.

To calibrate the modelling, statistics on petrol and diesel consumption published in the energy balances were compared against the estimated petrol and diesel consumption of the model, as calculated bottom up based on the VKM data and estimated vehicle fuel economy. It was found that the fuel consumption in the model was significantly higher than the energy balance. Therefore, the VKM values in the base year (2019) were then scaled down to better reflect the fuel consumption statistics in the energy balance.

VKM were extrapolated forward in BAU under growth rate of 7% to 2025 and 6% from 2025 to 2050.⁷⁶

Energy consumption and GHG emissions by mode of transport were determined based on the estimated split of VKM by mode.

Table 2-13: VKM split - Passenger

Mode	2019 proportion of total passenger VKM
Motorcycles	57%
Cars	29%
Small Bus (Matatu)	13%
Medium and large bus	2%

⁷⁶ https://www.researchgate.net/publication/285766191_Energy_Report_for_Uganda_A_100_Renewable_Energy_Future_by_2050

Table 2-14: VKM split - Freight

Mode	2019 proportion of total freight VKM
Light Trucks (2 Axles)	62%
Medium Trucks (3 Axles)	9%
Large Trucks (4 Axles)	5%
Heavy Trucks (5+ Axles) ⁷⁷	24%

Load factor

Historical vehicle load factors are presented in **Error! Reference source not found.** and **Error! Reference source not found.**, the 2019 values are assumed to remain constant across the modelling period in the baseline.

Table 2-15: Load factor - Passenger

Mode	2019 Load factor ⁷⁸ (People/VKM)
Motorcycles	1.6
Cars	3.0
Small Bus (Matatu)	15.0
Medium and large bus	37.5

Table 2-16: Load factor - Freight

Mode	2019 Load factor (Tonne/VKM)
Light Trucks (2 Axles)	10
Medium Trucks (3 Axles)	15
Large Trucks (4 Axles)	20
Heavy Trucks (5+ Axles)	30

Fuel economy

Data on fuel split of total number of vehicles is publicly available,⁷⁹ i.e. proportion of total vehicles (new and old) that are petrol and diesel fuelled. This data is also disaggregated by engine capacity, however, the fuel split by VKM travelled by mode of vehicle was not available. Therefore an expert judgement was made about the fuel split by vehicle type, for petrol assuming a higher proportion (70-100%) of VKM for vehicles with smaller engine capacity and a passenger transport focus and the reverse for vehicles with larger engine capacity and a freight transport focus.

Table 2-17: Fuel split assumptions - Passenger

⁷⁷ Heavy Trucks combines VKM data for 2019 for 5, 6, 7 and >7 axle vehicles into one category

⁷⁸ Gathered directly from the Government by MEIR under TraCS programme

⁷⁹ https://www.fiafoundation.org/media/461028/africa_vehicle-fuel-economy-baseline-for-uganda.pdf

Mode	Petrol	Diesel
Motorcycles	100%	0%
Cars	70%	30%
Small bus (Matatu)	50%	50%
Medium and large bus	30%	70%

Table 2-18: Fuel split assumptions - Freight

Mode	Petrol	Diesel
Light Trucks (2 Axles)	50%	50%
Medium Trucks (3 Axles)	30%	70%
Large Trucks (4 Axles)	0%	100%
Heavy Trucks (5+ Axles)	0%	100%

Fuel economy figures were taken from Vehicle Fuel Economy Baseline for Uganda.⁸⁰ This document provided fuel economy by engine capacity, these categories were aggregated and averaged to provide 2 distinct categories of light duty vehicles (LDV) and heavy duty vehicles (HDV) as well as including separate calculations from the study of motorcycle fuel economy.

One strange trend in the data is that the fuel economy for petrol and diesel vehicles (except motorcycles) has actually decreased over time, meaning the Ugandan vehicle fleet is consuming more fuel per kilometre. The reason for this has been explored in the literature, as observed in the Vehicle Fuel Economy Baseline for Uganda, “we notice that despite the imposition of the environmental tax by URA in 2008, the average age has been worsening for all categories of engine capacity...The relatively high age of vehicle at registration is of concern to promoting fuel economy national policy, since older cars for any engine capacity and vehicle technology consume more fuel and hence emit more CO₂ per kilometre...One major weakness of a flat environmental tax for instance the current 20% on the value of the vehicle irrespective of age, is that older vehicles have a lower value hence carry a lower charge, accordingly their purchase price is low; and the reverse is true for newer versions; they are of a higher value and they fetch a higher environmental fee hence making their purchase price to rise. For that matter most of the consumers would opt to purchase older cheaper vehicles instead of the newer ones.”

Figure 2-13: Average age of vehicles and category by fuel type and year of registration⁸¹

Weight Category	Diesel				Petrol			
	2005	2008	2011	2014	2005	2008	2011	2014
Heavy Duty	9.8	12.2	12.3	17.0	7.0	8.2	8.7	11.5
Light Duty	6.5	8.4	8.7	15.0	10.4	11.8	12.9	15.5
Grand Average	8.1	10.2	10.5	16.4	10.4	11.7	12.8	15.4

Table 2-19: Petrol fuel economy (L/100km)⁸²

Mode	2005	2008	2011	2014
Moto	2.7	2.5	2.4	1.9
LDV	10.6	10.8	11.1	11.8

⁸⁰ https://www.fiafoundation.org/media/461028/africa_vehicle-fuel-economy-baseline-for-uganda.pdf

⁸¹ https://www.fiafoundation.org/media/461028/africa_vehicle-fuel-economy-baseline-for-uganda.pdf

⁸² https://www.fiafoundation.org/media/461028/africa_vehicle-fuel-economy-baseline-for-uganda.pdf

Mode	2005	2008	2011	2014
HDV	22.2	21.4	21.5	22.9

Table 2-20: Diesel fuel economy (L/100km)

Diesel	2005	2008	2011	2014
LDV	11.6	11.7	11.9	13.4
HDV	24.2	24.9	27.7	29.3

It is assumed that there is a limit to this historical downward trend in vehicle fuel economy continuing into the future and therefore this has not been extrapolated forwards. These limits are that:

- As the fuel economy baseline explains⁸³, as the age of a vehicle increases there is a decrease in its fuel economy/efficiency. However, this decrease is not linear but exponential. Meaning that the difference in fuel efficiency between vehicles that are 0-5 years old is much larger than the difference in fuel efficiency between vehicles that are 10-15 years old and so on until the difference is negligible.⁸⁴
- Also, it is assumed that there are limits to the effects of the environmental levy on purchasing behaviour, costs of maintenance increase with a vehicle's age and therefore it will not always be the most cost-effective decision to buy an older car.

The Global Fuel Economy Initiative (GFEI)'s 50by50 campaign⁸⁵ sets targets for the global vehicle fleet to achieve, as detailed below:

- By 2020: 20% improvement in stock average (on-road) efficiency (against 2005 baseline), reflecting both the improvements in new car fuel economy (with some lag time for stock-turnover) and additional measures such as eco-driving, improved aftermarket components, better vehicle maintenance, etc.
- By 2030: 35% improvement in stock (against 2005 baseline), roughly trailing new car improvements plus on-road improvement measures.
- By 2050: 50% (50 by 50: the ultimate goal) improvement in global stock average fuel economy (against 2005 baseline), following the new car improvement in 2030 and with in-use improvement measures.

As Uganda is reliant on vehicle imports, these timescales can be used as a benchmark to gauge future fuel efficiency in Uganda in the baseline based on the average age of the vehicle fleet. Extrapolating the trend shown in **Error! Reference source not found.** to 2019, across both LDVs and HDVs, for petrol and diesel, the average age of the fleet would be approximately 20 years old. This would mean that updates to vehicle technology have a lag time of roughly 20 years before they penetrate the Ugandan vehicle fleet.

It is assumed that this average fleet age of 20 years old is a peak after which incentives to purchase an older vehicle will diminish due to costs of maintenance. The fuel economy will also therefore peak in 2019. After this point, the BAU assumes the Uganda vehicle fleet moves towards the GFEI targets with the 20-year lag time. The exact assumptions are presented below in **Error! Reference source not found.** and the fuel economy trajectory for cars in the baseline scenario is presented Figure 2-14.

Table 2-21: Baseline fuel economy assumptions

Scenario	20% improvement	35% improvement	50% improvement
GFEI Targets	2020	2030	2050

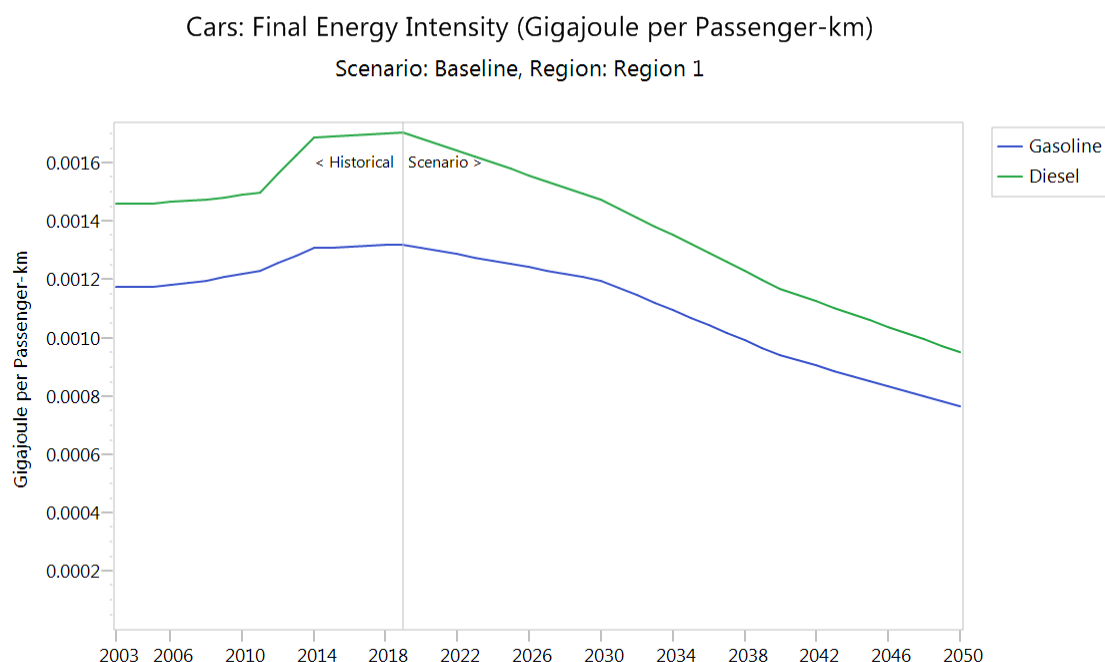
⁸³ https://www.fiafoundation.org/media/461028/africa_vehicle-fuel-economy-baseline-for-uganda.pdf

⁸⁴ This effect is presented very clearly in Figure 6: Relation between Fuel Efficiency and Age of vehicle, of the fuel economy baseline: https://www.fiafoundation.org/media/461028/africa_vehicle-fuel-economy-baseline-for-uganda.pdf

⁸⁵ <https://www.globalfuelconomy.org/media/46127/50by50-report-2009-lr.pdf>

Uganda BAU	2040	2050	2070
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Figure 2-14: Car fuel economy in baseline scenario



Civil Aviation

Modelling approach: Top-down (fuel consumption)

Table 2-22: Top-down calculation methodology used for civil aviation

Fuel consumption		Emissions factor
Total fuel sold	X	CO2 produced per energy consumed

Current levels of jet kerosene consumption, taken from energy balance, are extrapolated forward based on expected growth rates. Upgrade and expansion of Entebbe International Airport is included in BAU. This assumes an increase in passenger capacity from 1.9 million to 3.0 million and increased freight capacity of 100,000 tonnes per year. This data was taken and used to scale jet kerosene consumption. A linear historical growth was assumed across the modelling period and as no baseline fuel economy was provided, no change to vehicle technologies or efficiencies across the modelling period have been assumed.

No other airport infrastructure projects were included at this point either due to lack of data or due to the relatively small impact they would have on total VKM.

Rail

Modelling approach: Top-down (fuel consumption) validated with bottom-up (VKM)

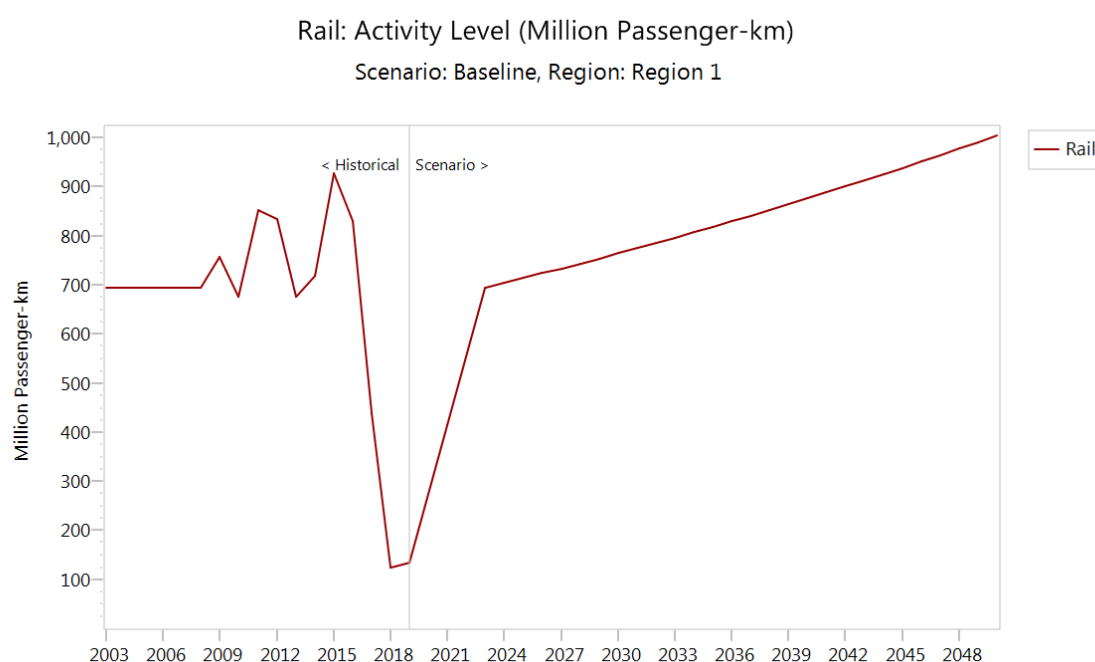
Table 2-23: Hybrid calculation methodology used for rail transport

Fuel consumption		Fuel economy		Activity		Activity type		Fuel economy		Emissions factor
Total fuel sold	/	Litres of fuel/100 km	=	Vehicle kilometres travelled (VKM)	X	Passenger / freight proportion	X	Litres of fuel/100 km	X	CO2 produced per energy consumed

A full bottom-up methodology was not possible due to a lack of passenger VKM data and no fuel economy data provided for rail locomotives. Therefore, the fuel consumption was taken, as the datapoint with the lowest uncertainty, and a reference fuel economy value was used to infer VKM. This VKM figure was then scaled by the proportion of PKM and TKM in the data provided to get the passenger and freight shares. Having the model in this format allows measures to be modelled that directly affect VKM travelled and vehicle fuel economy or fuel use with much greater sensitivity.

Whilst the rail network has experienced some significant dilapidation (“the current market shares of railways declined from 12% to 5% within the last 8 years”⁸⁶), it is assumed that the state of the railway remains roughly stable and it is assumed that the historic growth in usage experienced between 2008 and 2015 continues following recovery from the extreme lows in activity experienced during the past few years.

Figure 2-15: Baseline rail passenger KM travelled



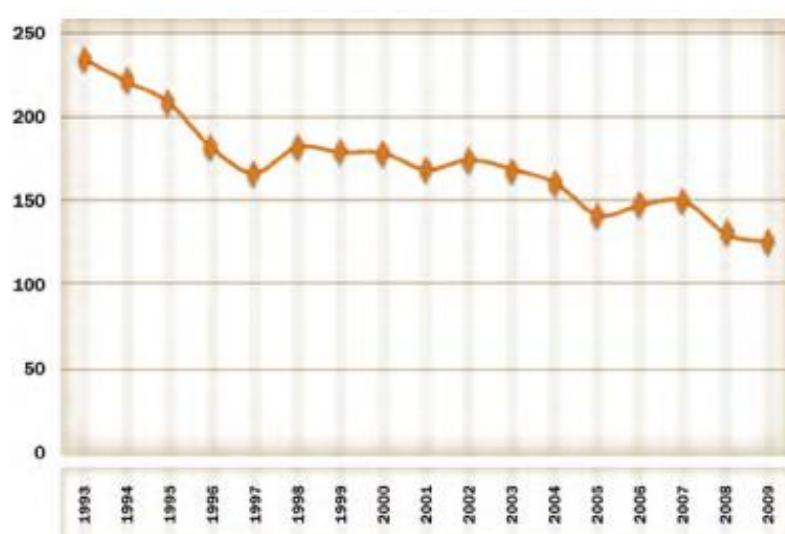
It is assumed that no infrastructure development occurs in the baseline as the two primary projects planned to develop the rail network (meter gauge and SGR) are included under the CDP scenario and therefore the BAU assumes that no further investment is provided.

As no baseline fuel economy was provided, a reference fuel economy from South Africa was used. Assuming an aging vehicle stock, the fuel economy for 1994 was taken and used across the historic

⁸⁶ Mitigation assessment of transport sector. MoWT. Unpublished.

time-series. Beyond this. It is assumed that the fuel economy improves in-line with the trend presented in Figure 2-16 however still delayed, reaching the 2009 levels in 2060.

Figure 2-16: South Africa railway specific energy consumption (KJ/transport unit)⁸⁷



Water-borne

Modelling approach: Top-down (fuel consumption)

Table 2-24: Top-down calculation methodology used for water-borne transport

Fuel consumption		Emissions factor
Total fuel sold	X	CO2 produced per energy consumed

Very little data available for mode, only fuel consumed however no unit or fuel type was provided. Assumed 100% diesel consumption. Assumptions on units made and consumption extrapolated across historical time-series.

Consumption extrapolated forward in BAU under growth rate of 7% to 2025 and 6% from 2025 to 2050 (Energy Report for Uganda).

⁸⁷ https://www.uic.org/com/IMG/pdf/iea-uic_2012final-lr.pdf

GHG emissions projections

Figure 2-17: Total GHG emissions for transport in baseline scenario

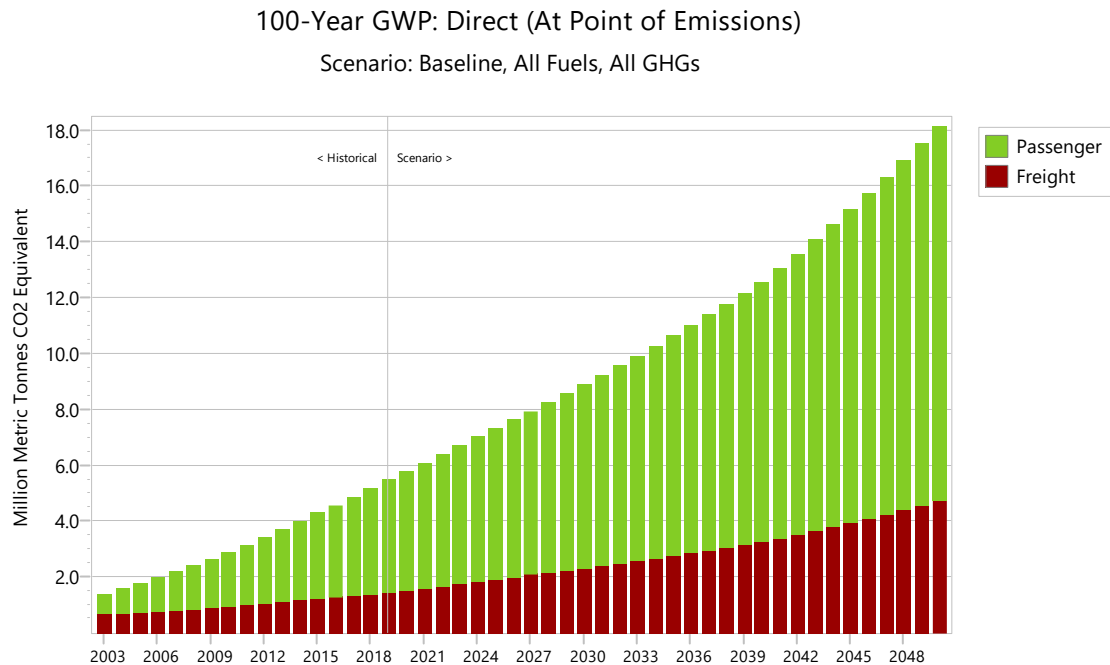


Figure 2-18: GHG emissions by mode for road transport in baseline scenario

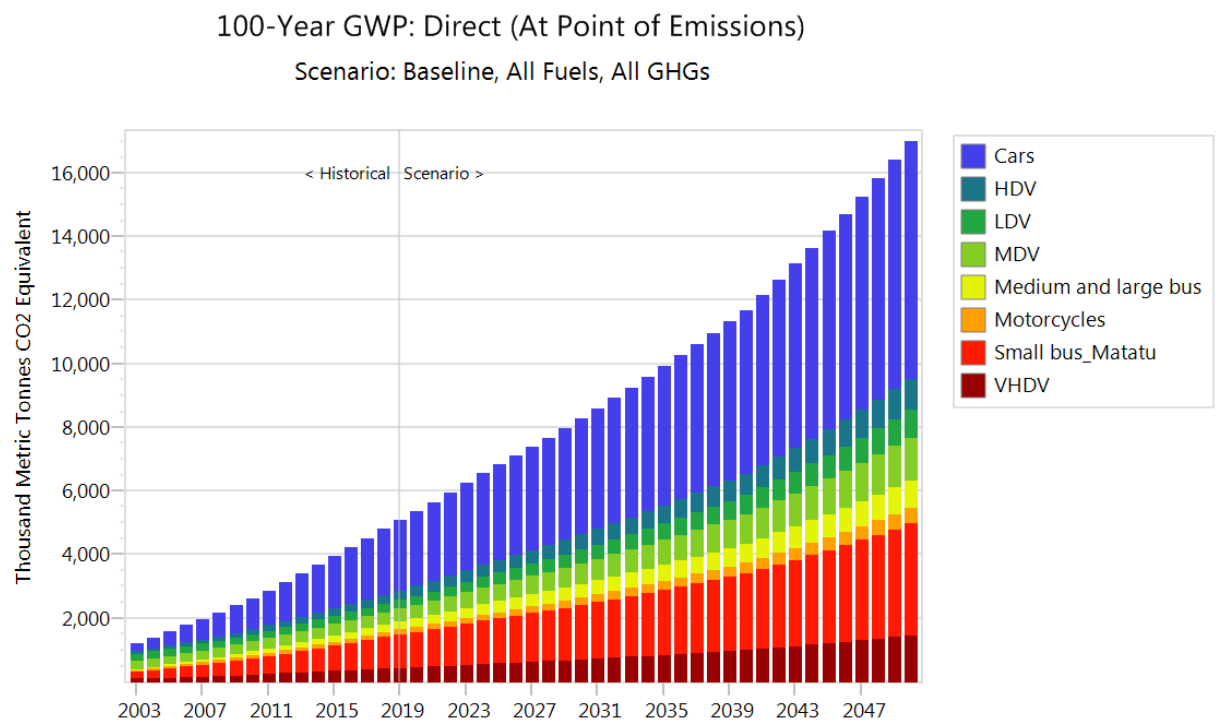


Figure 2-19: Total GHG emissions for civil aviation in baseline scenario

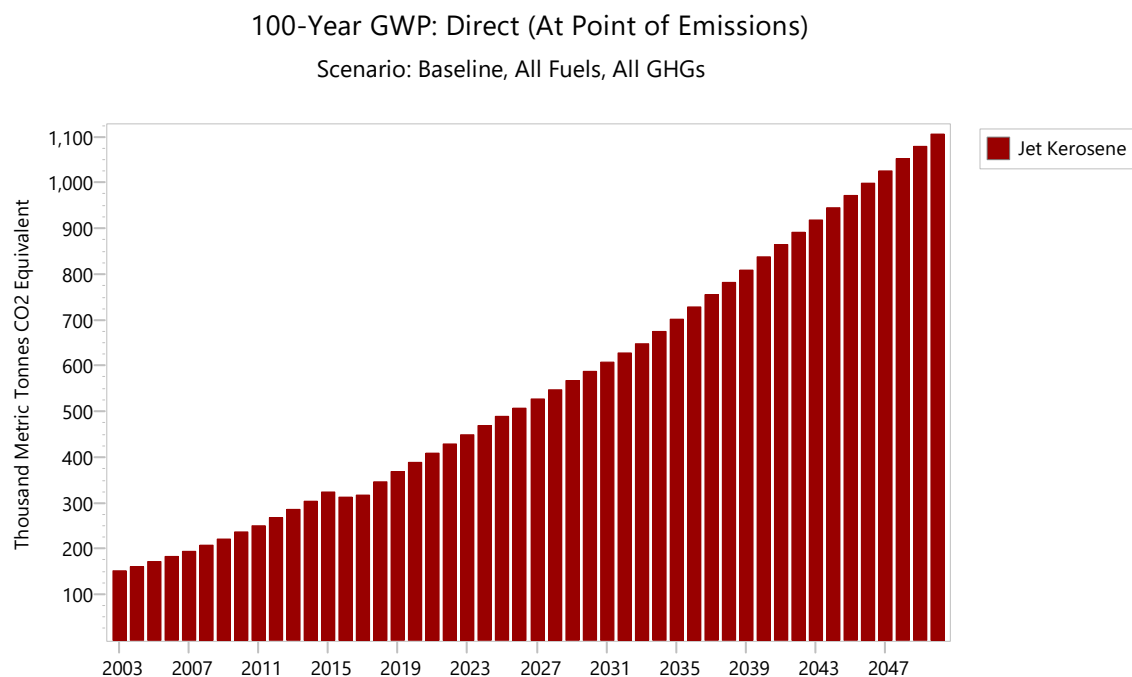


Figure 2-20: Total GHG emissions for rail in baseline scenario

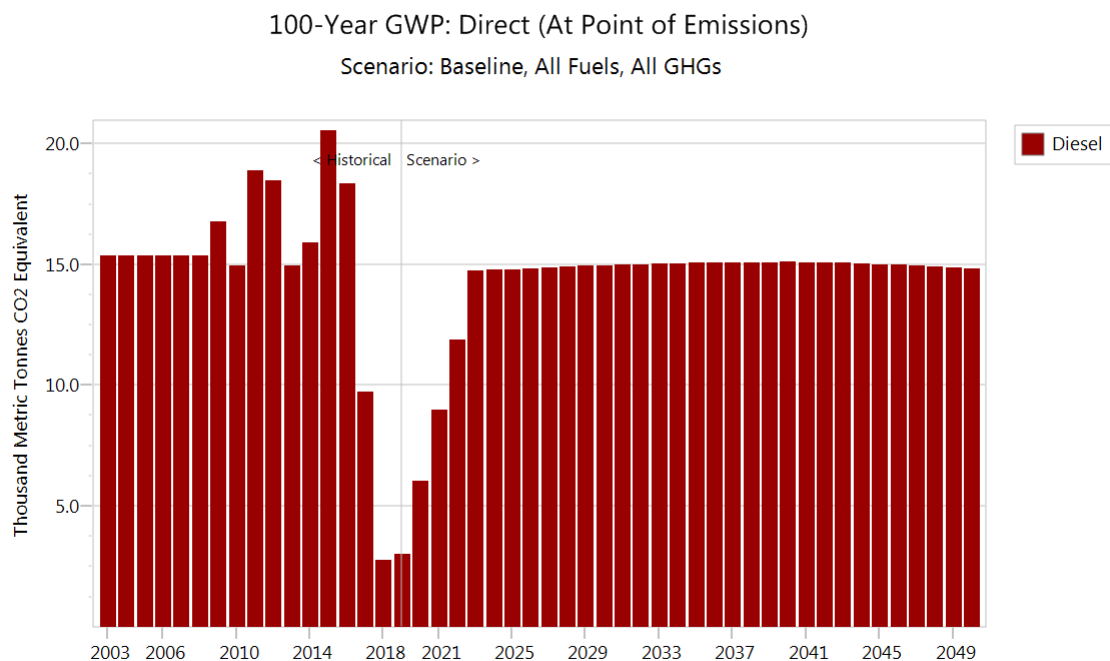
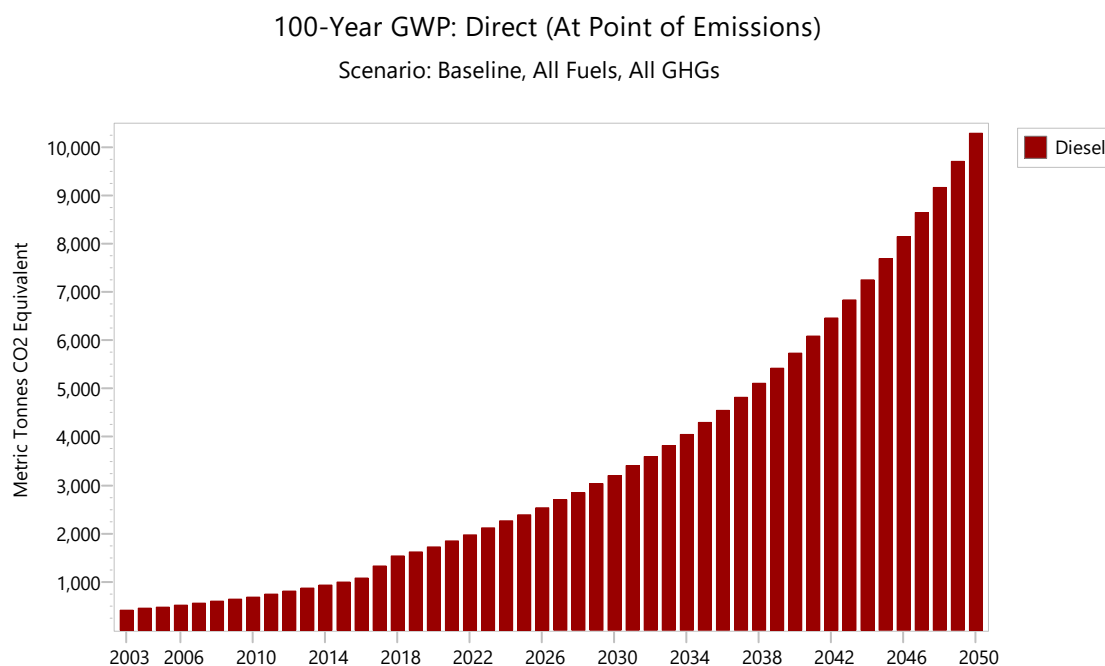


Figure 2-21: Total GHG emissions for water-borne navigation in baseline scenario



Key uncertainties

Road transport

Modelling approach: Bottom-up (VKM)

All assumptions detailed in section 0 above for this sub-sector are key uncertainties however there are some priority focus areas within these datapoints.

As the modelling approach is bottom-up, the key uncertainties for the road transport sector are the split by mode that make-up the historic emissions and the growth assumptions going forward.

As already explained, when validating the road transport numbers against the energy balance values and the 2015 NIR, there were significant differences in the fuel consumed when modelled from a top-down perspective and from a bottom-up perspective.

A disaggregated sub-sectoral breakdown of the 2015 NIR for transport would allow bottom-up values to be scaled to more closely match the energy balance and inventory.

However, as much disaggregation possible will be required as the energy balance simply provides the total fuel consumed (e.g. petrol), which comes from the activity of several different vehicle types all with very different driving behaviour. Vehicle numbers are useful but not sufficient, as you could have equal numbers of two vehicle types (car and boda boda) yet one travel significantly further than the other. The validation of the VKM travelled by mode is therefore key.

It is also assumed that under the BAU, the fuel economy of the fleet will achieve the GFEI average stock targets with a 20 year lag time, if there are any Ugandan specific assumptions that can be shared about how fuel economy would develop under the baseline as well as load factor for both passenger and freight then these would be incredibly useful.

Rail

Modelling approach: Top-down (fuel consumption) validated with bottom-up (VKM)

The BAU for the rail transport sector assumes the existing network is maintained and roughly the same level of activity can be assumed across the modelling period, assuming slight growth. This is a key assumption that must be validated.

The VKM for rail transport has been inferred from the fuel consumption divided by the average fuel economy of the rail transport fleet in South African fleet, assuming a lag time. It is assumed this trend continues going forward. Considering the impact this has on the estimates of VKM, specific fuel economy values for the Ugandan rail fleet, or alternatively comprehensive PKM data for the rail sector and their expected change over time under the baseline would be very useful.

2.7.3 Mitigation options

Road transport fuel efficiency

The Vehicle Fuel Efficiency NAMA⁸⁸ outlines a bundle of possible activities which could be implemented independently:

- Development of a national database on vehicle fleet, fuel consumption and efficiency (set up to eventually include inspection data)
- Development of fuel efficiency policy and standards have benchmark/target consumption for light, medium and heavy duty vehicles being imported into the country
- Development of fuel standards for sulphur. In addition, standards should specify grades of motor oil appropriate for the weather. Use of appropriate oil makes the engine more fuel efficient.
- Development of testing facilities and inspection programme for oil products (as another strategy to promote fuel efficiency): testing quality of oil, fuel once it is already at respective storage facilities to curb cases of adulteration.
- Development of a regulation limiting the age of imported vehicles (compliance linked to pre-shipment inspection)
- Promotion of cleaner fuels and setting of fuel standards
- Design of a vehicle inspection and maintenance system, including certification programme, standards for inspections and establishment of vehicle inspection centres.
- Development of tax incentives to encourage acquisition of more fuel efficient vehicles
- Establishment of a financial incentives scheme for vehicle replacement
- Public information and awareness campaign

WEM

No existing fuel efficiency policy is in place and therefore the WEM scenario includes no difference from the BAU for this measure.

CDP

⁸⁸

https://www4.unfccc.int/sites/PublicNAMA/_layouts/un/fccc/nama/NamaSeekingSupportForPreparation.aspx?ID=139&viewOnly=1

As explained in section 0, the BAU assumes the Uganda vehicle fleet moves towards the GFEI 50by50 targets with the 20-year lag time due to the average age of the fleet. Whilst no interventions are currently explicitly planned, it is assumed that on the basis of the Vehicle Fuel Efficiency NAMA⁸⁹:

The CDP scenario for this measure therefore assumes a reduction in the average age of the vehicle fleet to 10-years and an achievement of the GFEI targets with a 10-year lag time.

Table 2-25: Baseline fuel economy assumptions

Scenario	20% improvement	35% improvement	50% improvement
GFEI Targets	2020	2030	2050
Uganda CDP	2030	2040	2060

WAM

Building upon the CDP scenario, the WAM scenario for this measure assumes the full achievement of the GFEI 50by50 targets (for 2030 and 2050) in-line with their outlined timeframe, assuming additional measures to further reduce the vehicle fleet average age and improve the associated fuel economy.

Table 2-26: Baseline fuel economy assumptions

Scenario	20% improvement	35% improvement	50% improvement
GFEI Targets	2020	2030	2050
Uganda WAM	N/A	2030	2050

Alternative fuel switch

The Vehicle Fuel Efficiency NAMA⁹⁰ refers to “promotion of cleaner fuels and setting of fuel standards,” however no specific interventions are currently planned. The Energy Efficiency Strategy of Uganda for 2010 – 2020 however describes specific aspirations on the use of alternative fuels (LNG and biodiesel), therefore this measure has been modelled separately to explore the effect independently. The strategy describes a possible aim of increasing the use of blended fuels by 1% per year. This is used as the basis for the measure here except this goal is applied to alternative fuels, both biofuels and LNG.

WEM

No existing alternative fuel policy is in place and therefore the WEM scenario includes no difference from the BAU for this measure.

CDP

As no specific interventions are currently planned, the indicative goals from the Energy Efficiency Strategy are used as the basis for this scenario. It assumes the increase of alternative fuels by 1% per year as stated in the Energy Efficiency Strategy, starting in 2021. This 1% increase is divided evenly between liquefied natural gas (LNG) 60%, ethanol (E10) 20% and biodiesel 20%.

Currently, as all transportation fuels are imported,⁹¹ the model does not account for emissions from LNG, ethanol and biodiesel production.

⁸⁹

https://www4.unfccc.int/sites/PublicNAMA/_layouts/un/fccc/nama/NamaSeekingSupportForPreparation.aspx?ID=139&viewOnly=1

⁹⁰

https://www4.unfccc.int/sites/PublicNAMA/_layouts/un/fccc/nama/NamaSeekingSupportForPreparation.aspx?ID=139&viewOnly=1

⁹¹ https://www.energyandminerals.go.ug/site/assets/files/1017/2015_statistical_abstract.pdf

WAM

The WAM scenario for this measure takes the assumptions of the CDP scenario and assumes an increase in ambition to a 2% annual increase.

Modal shift to Non-Motorised Transport (NMT)

This measure uses the planned NMT corridors in the GKMA as well as in other urban areas across Uganda as its basis. However, as no data on VKM avoided through installation, assumptions must be made about related reduction of private car use.

The NMT corridors included are based on the planned implementation of NMT corridors across the country, the CDP scenario presents a more realistic, as validated by the Government, implementation schedule for the plans articulated in Kampala's Multi-Modal Urban Transport Master Plan and the WAM scenario accounts for the full ambition of the plan. Additional assumptions about corridors implemented outside of Kampala are also included.

Category	CDP	WAM	Unit
Walking proportion of trips	46% ⁹²	46%	
Kampala city council total roads	1,030 ⁹³	1,030	KM
Kampala city council paved roads	330	330	KM
Total GKMA NMT corridors implemented by 2030	100	407	KM
Proportion total KM	9.7%	39.5%	
Kampala population proportion Uganda	7.4%	7.4%	
Total motorised transport trips	66,715	66,715	Mil PKM
Total Kampala motorised transport trips	8,317	8,317	Mil PKM
Kampala project NMT proportion	9%	28%	
Kampala project NMT PKM	736	2,356	Mil PKM
Trips rearranged from motorised (NMT+)	11% ⁹⁴	11%	
Trips rearranged from motorised (NMT+)	77	247	Mil PKM
Non-Kampala project PKM	100	200	KM
Non-Kampala project NMT proportion	9%	16%	
Non-Kampala project NMT PKM	736	1,352	Mil PKM
Trips rearranged from motorised (NMT+)	77	142	Mil PKM

Efficient operation of public transportation

Several proposed interventions are described in the Energy Efficiency Strategy of Uganda for 2010 – 2020, including the improved organisation of urban public transport leading to reduced congestion and increased efficiency in the service and a reduction in associated VKM and increase in load factor, increasing the utilisation of the mode.

This would be achieved through interventions such as: the introduction of fixed bus routes with defined frequency, the reorganisation of transport in urban centres, the introduction of traffic lights as much as possible and the introduction of fixed routes with defined frequency.

WEM

⁹² Kampala Capital City Authority (KCCA). (2018). Multi-Modal Urban Transport Master Plan for Greater Kampala Metropolitan Area (GKMA): Final Report. Section 2.3.

⁹³ https://openjicareport.jica.go.jp/pdf/12013025_01.pdf

⁹⁴ <https://www.unil.ch/files/live/sites/ouvdd/files/shared/Colloque%202006/Communications/Mobilite/Theorie/R.%20Frick.pdf>

No existing policy is in place and therefore the WEM scenario includes no difference from the BAU for this measure.

CDP

The CDP scenario for this measure assumes the associated interventions described above are implemented however as no quantified impact is provided, a default assumption of a 5% reduction in PKM is included (including an increase in the load factor and a decrease in the distance travelled).

WAM

The WAM scenario for this measure assumes increased interventions that would achieve an associated 10% reduction in PKM.

Freight avoidance through planning and best practice

No specific interventions are known of in strategy documents or otherwise however this is an equivalent measure to the previous one targeting public transport, only this measure addresses similar inefficiencies in urban freight transport.

WEM and CDP

No existing policy is in place and therefore the WEM and CDP scenarios include no difference from the BAU for this measure.

WAM

The WAM scenario for this measure assumes an achievement of a 10% reduction in TKM (including an increase in the load factor and a decrease in the distance travelled) from improved efficiency in planning.

Residential trip avoidance through town planning and transport orientated development

Uganda's Third National Development Plan (NDP III) 2020/21 – 2024/25⁹⁵ articulates several infrastructure interventions including the development of Transit-Oriented developments along transport infrastructure corridors (such as roadside stations).

WEM

No existing policy is in place and therefore the WEM scenario includes no difference from the BAU for this measure.

CDP

Develop land use and transport master plans incorporating transit-oriented land use and measures to incentivise well-located affordable housing along planned mass rapid transit lines.⁹⁶

Adopt TOD friendly building control rules by 2025.

This results in the achievement of a 5% increase in NMT and mass transit and a 5% decrease in private vehicles.

WAM

⁹⁵ <http://extwprlegs1.fao.org/docs/pdf/uga199743.pdf>

⁹⁶ Kampala Capital City Authority (KCCA). (2018). Multi-Modal Urban Transport Master Plan for Greater Kampala Metropolitan Area (GKMA): Final Report. Section A-5.

Develop land use and transport master plans incorporating transit-oriented land use and measures to incentivise well-located affordable housing along planned mass rapid transit lines.⁹⁷

Adopt TOD friendly building control rules by 2025.

This results in the achievement of a 5% increase in NMT and mass transit and a 5% decrease in private vehicles.

The WAM scenario for this measure uses an example from South Africa as a basis which used the assumption that TOD reduces motorised travel demand by 10% in 2050⁹⁸, assuming additional measures were implemented beyond the CDP scenario.

Develop mass transit solutions to facilitate modal shift

Uganda's Third National Development Plan (NDP III) 2020/21 – 2024/25⁹⁹ articulates several infrastructure interventions including the implementation of an inclusive mass rapid transport system (including intercity rail, light rail transport (LRT), metro and bus rapid transit (BRT)). This measure takes planned infrastructure projects and models the projected increase in capacity to the associated modes. This increase is then offset in part by reduction in other modes (particularly private car use).

The infrastructure projects included in the modelling:

2.7.3.1.1 Development of GKMA-BRT system

This measure assumes the implementation of 101km of BRT in GKMA by 2030.

Table 2-27: GKMA-BRT system general assumptions¹⁰⁰

BRT Line	Length (km)	Daily ridership	Daily PKM	Yearly PKM	Operational
BRT 1	22.2	462,418	10265680	3.75E+09	2030
BRT 2	38.5	170,735	6573298	2.4E+09	
BRT 3	19.9	291,258	5796034	2.12E+09	
BRT 4a	9.8	161,331	1581044	5.77E+08	
BRT 4b	10.6	234,602	2486781	9.08E+08	

2.7.3.1.2 Redevelopment and extension/expansion of GKMA passenger service

Table 2-28: GKMA passenger service general assumptions

Rail line	Length (km)	Annual PKM	Operation
Bujjuko-Mukono	53	2315585	2030
Kampala-Port	8	349522.3	2030

It is assumed this measure is diesel powered.

⁹⁷ Kampala Capital City Authority (KCCA). (2018). Multi-Modal Urban Transport Master Plan for Greater Kampala Metropolitan Area (GKMA): Final Report. Section A-5.

⁹⁸ <https://www.climate-transparency.org/wp-content/uploads/2020/08/CT-Low-Carbon-Transport-SA-DIGITAL.pdf>

⁹⁹ <http://extwprlegs1.fao.org/docs/pdf/uga199743.pdf>

¹⁰⁰ Data collected as part of project. Unpublished.

2.7.3.1.3 Development of the LRT system

This measure assumes the implementation of 100km of LRT in GKMA by 2036.

Table 2-29: LRT system general assumptions

LRT line	Length (km)	Daily ridership	Daily PKM	Yearly PKM	Operational
LRT 4	21	421,353	8848413	3.23E+09	2031
LRT 5	24	453,015	10872360	3.97E+09	2035
LRT 7	54.9	415,603	22816605	8.33E+09	2040

It is assumed this measure is electrically powered.

2.7.3.1.4 Development of metro infrastructure

This measure assumes the implementation of 75km of Metro in GKMA by 2040.

Table 2-30: Metro system general assumptions

Metro line	Length (km)	Daily ridership	Daily PKM	Yearly PKM	Operational
Metro 1	26	544,842	14165892	5.17E+09	2031
Metro 2	26	506,647	13172822	4.81E+09	2031
Metro 3	23	493,531	11351213	4.14E+09	2036

It is assumed this measure is electrically powered.

WEM

The infrastructure projects are planned and therefore the WEM scenario includes no difference from the BAU for this measure.

CDP

The CDP scenario for this measure assumes full implementation of the projects on the timelines outlined above. The computed yearly PKM data is used to scale fuel consumption data for rail to input into the model, for bus travel the PKM data is inputted directly, associated emissions are then calculated.

It is assumed that 100% of the additional PKM comes from modal shift from road transportation, i.e. the difference is divided across motorcycle, private car and matatu use proportionally and decreased in line with the project timeline

For the GKMA project, fuel is assumed to remain as diesel throughout the modelling period. Fuel economy is assumed to improve faster than in the baseline scenario, achieving the 2009 levels outlined in section 0 by 2040 as opposed to 2060 in the baseline scenario.

Bus fuel efficiency increases in-line with the road transport fuel efficiency measure.

WAM

The WAM scenario for this measure assumes significant technological improvement in vehicle technologies or efficiencies for the rail fleet.

Fuel is assumed to remain as diesel throughout the modelling period for all rail projects. Fuel economy is assumed to improve faster than in the baseline scenario, achieving the 2009 levels outlined in section 0 by 2030 as opposed to 2060 in the baseline scenario and achieving a 50% reduction on the baseline historic fuel economy by 2050.

No electrical power has been assumed at this point.

Freight modal shift to rail

The expansion of rail services and the associated modal shift are key priorities for Uganda. Vision 2040¹⁰¹ outlines that at least 80% of Uganda freight transport will be carried by rail.

Current estimated project timelines forecast project completion and operation after 2040 meaning these timelines might not be feasible.

Uganda's Third National Development Plan (NDPIII) 2020/21 – 2024/25¹⁰² also articulates the aim

The infrastructure projects included in the modelling:

2.7.3.1.5 Development of the standard gauge railway system

Table 2-31: Standard gauge railway general assumptions

Standard Gauge Railway	Length (km)	Annual TKM	Annual PKM	Operation
Eastern line	273	3.82E+07	1.04E+09	2040
Western line	377	5.26E+09	1.43E+09	2049
Southern line	430	2619	2.62E+09	2052 ¹⁰³
Northern line	762	3.09E+10	2.89E+09	2044

It is assumed this measure is electrically powered.

2.7.3.1.6 Rehabilitation of the Meter Gauge Railway System

Table 2-32: Metre gauge railway general assumptions

Metre Gauge Railway	Length (km)	Annual TKM	Operation
Northern Line (Tororo-Gulu)	342	451440	2024
Eastern Line (Kampala-Malaba)	292	385440	2026

It is assumed this measure is diesel powered.

WEM

The infrastructure projects are planned and therefore the WEM scenario includes no difference from the BAU for this measure.

CDP

¹⁰¹ <https://www.greengrowthknowledge.org/sites/default/files/downloads/policy-database/UGANDA%29%20Vision%202040.pdf>

¹⁰² <http://extwprlegs1.fao.org/docs/pdf/uga199743.pdf>

¹⁰³ It is estimated that the Southern Line will not be operational by 2050.

The CDP scenario for this measure assumes full implementation of the projects on the timelines outlined above. The computed yearly TKM data is used to scale fuel consumption data for rail to input into the model, associated emissions are then calculated.

As no specific assumptions for the impact of the infrastructure projects on modal shift have been shared, some basic assumptions have been made at this stage of the effect of the measure.

- It is assumed that 100% of the increased TKM from the infrastructure projects is achieved as a result of a modal shift
- This increase in TKM is assumed will be achieved through a modal shift from all road freight vehicles.

Fuel is assumed to remain as diesel for all rail projects throughout the modelling period. Fuel economy is assumed to improve faster than in the baseline scenario, achieving the 2009 levels outlined in section 0 by 2040 as opposed to 2060 in the baseline scenario.

WAM

The WAM scenario for this measure assumes the delivery of the standard gauge railway project 5 years ahead of current predictions as a reasonable acceleration of the project in order to achieve the goals of Vision 2040 sooner.

Standard Gauge Railway	Predicted Operation	WAM Operation
Eastern line	2040	2035
Western line	2049	2044
Southern line	2052	2047
Northern line	2044	2039

The WAM scenario for this measure also assumes significant technological improvement in vehicle technologies or efficiencies for the rail fleet.

Fuel is assumed to remain as diesel throughout the modelling period for all rail projects. Fuel economy is assumed to improve faster than in the baseline scenario, achieving the 2009 levels outlined in section 0 by 2030 as opposed to 2060 in the baseline scenario and achieving a 50% reduction on the baseline historic fuel economy by 2050.

No electrical power has been assumed at this point.

Measures not included currently

Below is a list of measures currently not included due to a lack of detail on the specifics of the measure to model that would mean assumptions made would be too uncertain. **Error! Reference source not found.** presents measures that would be included in the CDP scenario as Uganda has already expressed some level of plans to introduce these measures. Uganda should also consider possible additional measures for the WAM scenario that Uganda could consider modelling the impact of if they feel it is realistic they would be implemented effectively in the timeframe, for example, the administration has the authority to implement the measure.

Table 2-33: Possible additional measures

Measure	Description	Data required
Road improvement projects	Included in the assumed growth rates of the road transport sector and the improved fuel economy. Not included explicitly as a measure.	Expected VKM reduction, energy intensity reduction
Electric road vehicles	Plans for electric boda-boda or buses	Expected VKM/vehicle numbers, energy intensity of vehicles

Measure	Description	Data required
Electrification of rail	Plans for electric traction (e.g. SGR)	Expected energy intensity of vehicles, proportion of VKM/network
Oil pipeline	Plans for pipeline to transfer oil internally in Uganda, replacing road freight VKM	Road freight VKM avoided, emissions associated with pipeline
Utilisation of water transport	Plans for increased water-borne transport	Road PKM and TKM avoided, fuel consumed by craft, energy intensity of craft
What other measures could/should be included? Including new measures not considered before.		

2.7.4 Mitigation scenarios

Key policies and strategies

Uganda currently lacks an up to date, comprehensive, “approved” national transport policy which “is a disadvantage to the country in general and to Kampala District in particular, which accounts for over 50% of the vehicles in Uganda.”¹⁰⁴

However, several key policies form the general vision for the transport sector in Uganda:

- National Transport Master Plan 2008-2023¹⁰⁵
- Uganda Vision 2040¹⁰⁶
- Uganda’s Third National Development Plan (NDP III) 2020/21 – 2024/25¹⁰⁷
- Uganda National Climate Change Policy¹⁰⁸

The National Climate Change neatly outlines the high-level general priorities for GHG reduction in the transport sector:

- Promote the development, approval and effective implementation of a long-term national transport policy and plan that will take GHG mitigation concerns into account.
- Effect a gradual shift to the use of less carbon-intensive fuels (including compressed natural gas, ethanol and LPG) in vehicles instead of relying heavily on gasoline and diesel fuels.
- Promote modes of transport that take into account GHG emission reduction.

As well as outlining some more specific points that require attention in Uganda, it is around this framework that the mitigation modelling for this sector has been constructed:

- Critical issues such as urban mobility are not being given the attention they deserve, as efforts focus instead on main roads and highways.
- The infrastructure in Kampala can no longer handle the traffic, resulting in traffic congestion, high fuel consumption and consequently high GHG emissions.¹⁰⁹
- Mass transit in the city was reintroduced after an absence of more than 20 years. There is currently a very limited number of buses operating in urban centres.
- There is heavy dependence on roads for freight and cargo transport, which leads to fast deterioration of roads.

¹⁰⁴ <https://www.mwe.go.ug/sites/default/files/library/National%20Climate%20Change%20Policy%20April%202015%20final.pdf>

¹⁰⁵ <https://works.go.ug/wp-content/uploads/2015/08/National-Transport-Master-Plan-2008-2023.pdf>

¹⁰⁶ <https://www.greengrowthknowledge.org/sites/default/files/downloads/policy-database/UGANDA%29%20Vision%202040.pdf>

¹⁰⁷ http://www.npa.go.ug/wp-content/uploads/2020/08/NDP-III-Finale_Compressed.pdf

¹⁰⁸ <https://www.mwe.go.ug/sites/default/files/library/National%20Climate%20Change%20Policy%20April%202015%20final.pdf>

¹⁰⁹ Road rehabilitation has not been included in the mitigation analysis due to a lack of data provided on the effect of the measure including negative feedback loops. If assumptions can be provided about the impact of the measure on activity then this measure could be included in the modelling.

- Uganda once had 1,232 kilometres of rail network. At present, only 251 kilometres is operational, on the Kampala-Jinja-Malaba line.

Assumptions for the mitigation scenarios

The table below describes the assumptions made for each transport measure for the three mitigation scenarios:

Measure	BAU	WEM	CDP	WAM
Road transport fuel efficiency	Global Fuel Economy Initiative (GFEI) 50by50 targets, ¹¹⁰ improvement of fuel economy with 20 year time-lag: 2040: 20% 2050: -35% 2070: -50%	N/A	Global Fuel Economy Initiative (GFEI) 50by50 targets, improvement of fuel economy with 10 year time-lag: 2030: 20% 2040: -35% 2060: -50%	Global Fuel Economy Initiative (GFEI) 50by50 targets, improvement of fuel economy with no time-lag: 2030: -35% 2050: -50%
Alternative fuel switch	Road: Assumes no switching to alternative fuels (LNG, ethanol or LNG). All traditional fuels (petrol/gasoline and diesel).	N/A	1% per year increase in alternative fuel use for all road vehicles. ³ - 60% of the increase comes from natural gas. - 20% from ethanol (E10). - 20% from biodiesel. Introduction of at least 200 e-buses in GKMA by 2030. ¹¹¹	2% per year increase in alternative fuel use for all road vehicles. ³ - 60% of the increase comes from natural gas. - 20% from ethanol (E10). - 20% from biodiesel. Introduction of at least 200 e-buses in GKMA by 2030. ¹¹²
Development of NMT infrastructure	Road: Walking accounts for 46% of daily trips. ¹¹³ 2015 passenger KM (PKM) Motorised: 66,715 million Non-motorised: 78,318 million Total road PKM: 145,033 million	N/A	100 km of complete streets or dedicated NMT corridors, constructed in greater Kampala area by 2030 leading to 10% shift in PKM by mode from other passenger modes. Construct 100 km of NMT facilities in secondary cities by 2030.	407 km of complete streets or dedicated NMT corridors, constructed in greater Kampala area by 2030 leading to shift in PKM by mode from other passenger modes. Construct 200 km of NMT facilities in secondary cities by 2030.

¹¹⁰ <https://www.globalfueleconomy.org/media/46127/50by50-report-2009-lr.pdf>

¹¹¹ Transformative Urban Mobility Initiative (TUMI) E-bus Mission.

¹¹² Transformative Urban Mobility Initiative (TUMI) E-bus Mission.

¹¹³ Kampala Capital City Authority (KCCA). (2018). Multi-Modal Urban Transport Master Plan for Greater Kampala Metropolitan Area (GKMA): Final Report. Section 2.3.

Measure	BAU	WEM	CDP	WAM
			Policies introduced to manage the use of personal motor vehicles to reduce congestion and encourage a shift to public transport, walking, and cycling. Including parking management and secure cycle parking.	Policies introduced to manage the use of personal motor vehicles to reduce congestion and encourage a shift to public transport, walking, and cycling. Including parking management and secure cycle parking.
Efficient operation of public transportation	Standard growth rate of PKM of 7% to 2025 and 6% to 2050. No change in load factor.	N/A	Work with the taxi industry to achieve service quality improvements and operator consolidation. Introduce 1,000 high-quality city buses in GKMA ¹¹⁴ together with bus shelters, terminals, and depots. ¹¹⁵ Introduce ITS systems, automatic fare collection, and control centre for public transport in GKMA 2025. ¹¹⁶ 5% reduction in VKM and 5% increase in load factor from improved organisation of urban public transport.	Work with the taxi industry to achieve service quality improvements and operator consolidation. Introduce 1,000 high-quality city buses in GKMA together with bus shelters, terminals, and depots. ¹¹⁷ Introduce ITS systems, automatic fare collection, and control centre for public transport in GKMA 2025. 10% reduction in VKM and 10% increase in load factor from improved organisation of urban public transport.
Freight avoidance through planning and best practice	Standard growth rate of PKM of 7% to 2025 and 6% to 2050 for road freight. No change in load factor.	N/A	N/A	10% reduction* in PKM of mode
Residential trip avoidance through town planning and transport orientated development	Standard growth rate of PKM of 7% to 2025 and 6% to 2050. No change in load factor.	N/A	Develop land use and transport master plans incorporating transit-oriented land use and measures to incentivise well-located affordable housing along planned mass rapid transit lines. Adopt TOD friendly building control rules by 2025.	Develop land use and transport master plans incorporating transit-oriented land use and measures to incentivise well-located affordable housing along planned mass rapid transit lines. Adopt TOD friendly building control rules by 2025.

¹¹⁴ Bus system planned by KCCA in collaboration with Metu Zhongtong.

¹¹⁵ Kampala Capital City Authority (KCCA). (2018). Multi-Modal Urban Transport Master Plan for Greater Kampala Metropolitan Area (GKMA): Final Report. Section C-3.

¹¹⁶ Kampala Capital City Authority (KCCA). (2018). Multi-Modal Urban Transport Master Plan for Greater Kampala Metropolitan Area (GKMA): Final Report. Section C-5.

¹¹⁷ Kampala Capital City Authority (KCCA). (2018). Multi-Modal Urban Transport Master Plan for Greater Kampala Metropolitan Area (GKMA): Final Report. Section C-3.

Measure	BAU	WEM	CDP	WAM
			TOD reduces motorised travel demand by 5% in 2050.	TOD reduces motorised travel demand by 10% in 2050 ¹¹⁸
BRT – Bus Rapid Transit	Road: Standard growth rate of PKM of 7% to 2025 and 6% to 2050. No change in load factor. Minimal policies to encourage mass transit uptake over private vehicles.	N/A	Implement 101 km of BRT in GKMA by 2030.¹¹⁹ Policies introduced to manage the use of personal motor vehicles to reduce congestion and encourage a shift to public transport, walking, and cycling. Including parking management and secure cycle parking.	N/A
GKMA Passenger service	Rail: Historical growth rate continues across period. 12% fuel economy improvement of diesel locomotives achieved by 2030 relative to 2015. Minimal policies to encourage mass transit uptake over private vehicles.	N/A	Implement 61km of passenger MGR rail by 2030. 22% fuel economy improvement of diesel locomotives achieved by 2030 relative to 2015. Policies introduced to manage the use of personal motor vehicles to reduce congestion and encourage a shift to public transport, walking, and cycling. Including parking management and secure cycle parking.	Fuel economy improvement of diesel locomotives achieved 10 years sooner.
Metro	Rail: Historical growth rate continues across period. 12% fuel economy improvement of diesel locomotives achieved by 2030 relative to 2015. Minimal policies to encourage mass transit uptake over private vehicles.	N/A	Implement 75km of fully electrified passenger metro rail by 2040. Policies introduced to manage the use of personal motor vehicles to reduce congestion and encourage a shift to public transport, walking, and cycling. Including parking management and secure cycle parking.	N/A
LRT – Light Rail Transit	Rail: Historical growth rate continues across period. 12% fuel economy improvement of	N/A	Implement 100km of fully electrified passenger LRT rail by 2040.	N/A

¹¹⁸ <https://www.climate-transparency.org/wp-content/uploads/2020/08/CT-Low-Carbon-Transport-SA-DIGITAL.pdf>

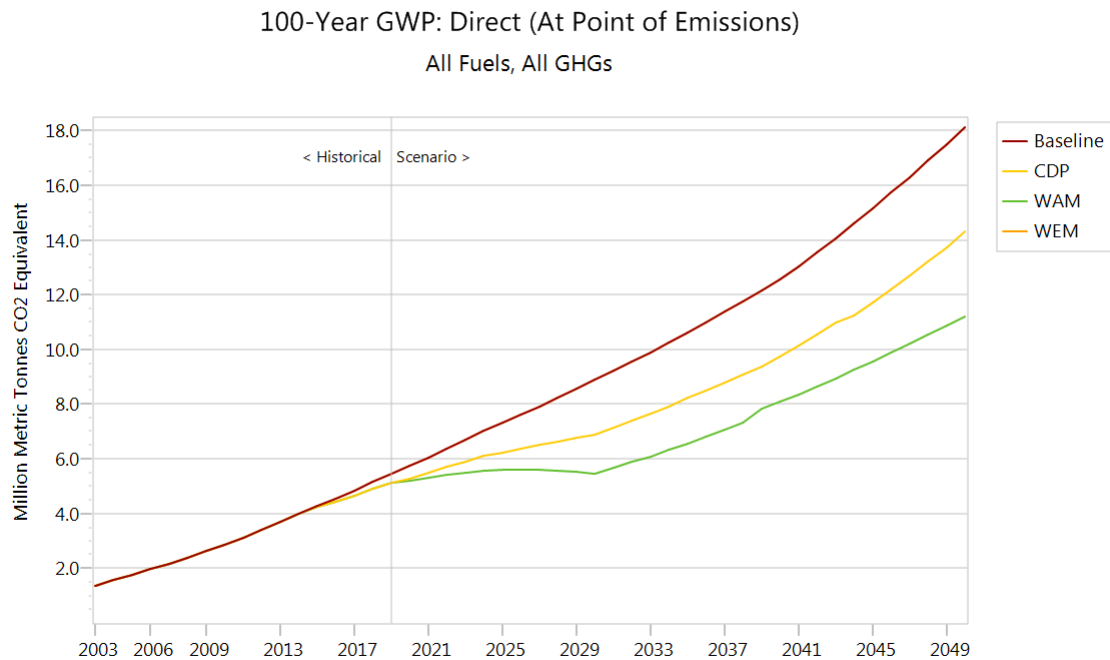
¹¹⁹ Four corridors identified in the MMUTMP. Section 10.5.

Measure	BAU	WEM	CDP	WAM
	diesel locomotives achieved by 2030 relative to 2015. Minimal policies to encourage mass transit uptake over private vehicles.		Policies introduced to manage the use of personal motor vehicles to reduce congestion and encourage a shift to public transport, walking, and cycling. Including parking management and secure cycle parking.	
MGR – Meter Gauge Railway rehabilitation for freight transit	Rail: Historical growth rate continues across period. 12% fuel economy improvement of diesel locomotives achieved by 2030 relative to 2015.	N/A	Rehabilitation of 634km of meter gauge railway by 2026 to facilitate modal shift of freight from road to rail. 22% fuel economy improvement of diesel locomotives achieved by 2030 relative to 2015.	Fuel economy improvement of diesel locomotives achieved 10 years sooner. ¹²⁰
SGR – Standard Gauge Railway	Rail: Historical growth rate continues across period. 12% fuel economy improvement of diesel locomotives achieved by 2030 relative to 2015.	N/A	Implement 1412km of fully electrified standard gauge rail by 2050.	N/A

¹²⁰ Diesel traction assumed currently due to lack of data availability on electric vehicle energy consumption.

GHG mitigation projections

Figure 2-22: Total GHG emissions for transport in all scenarios



The steps in the CDP and WAM curves show the introduction of mass transit and rail infrastructure and the associated increase in rail and bus activity as well as the resulting decrease in road transport from the associated modal shift.

Figure 2-23: Total GHG emissions for passenger road transport in all scenarios

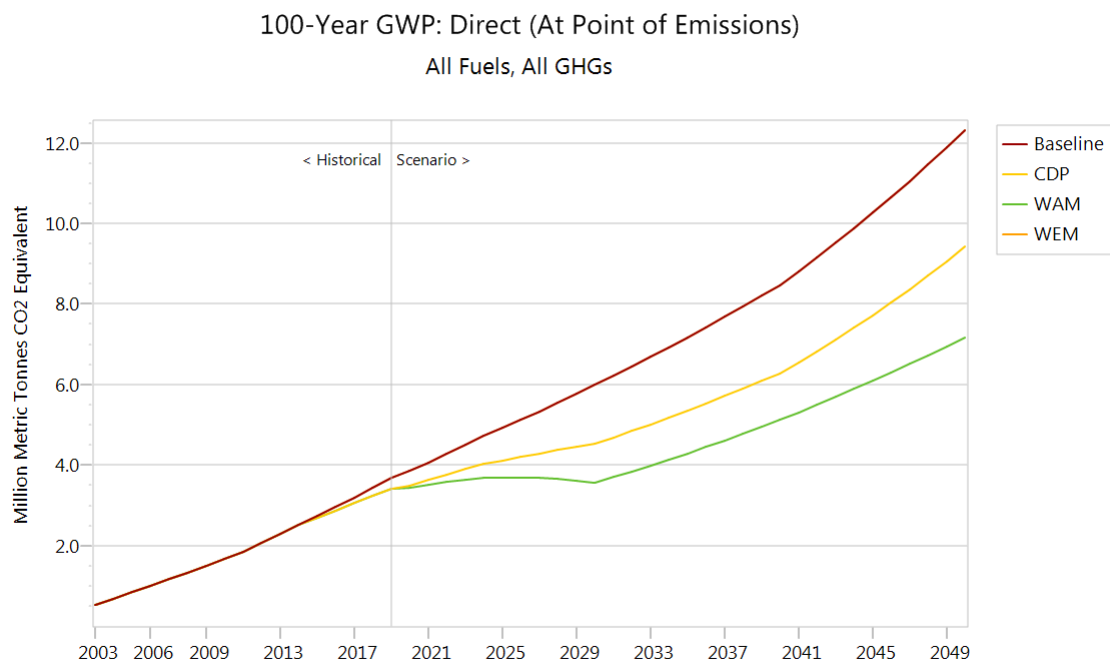


Figure 2-24: Total GHG emissions for freight road transport in all scenarios

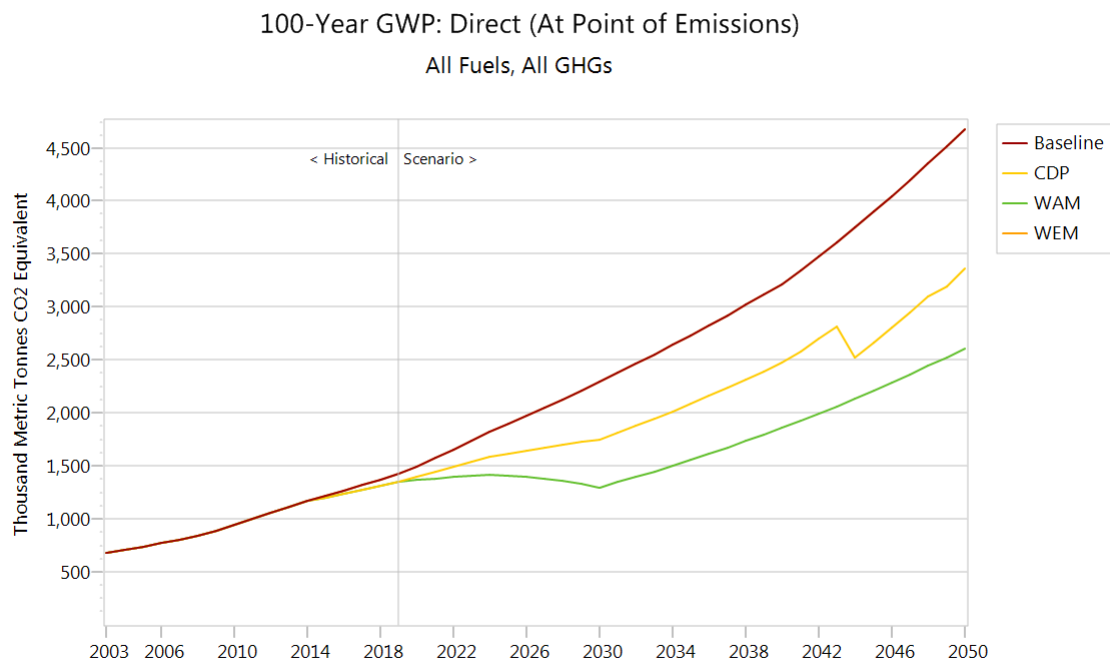
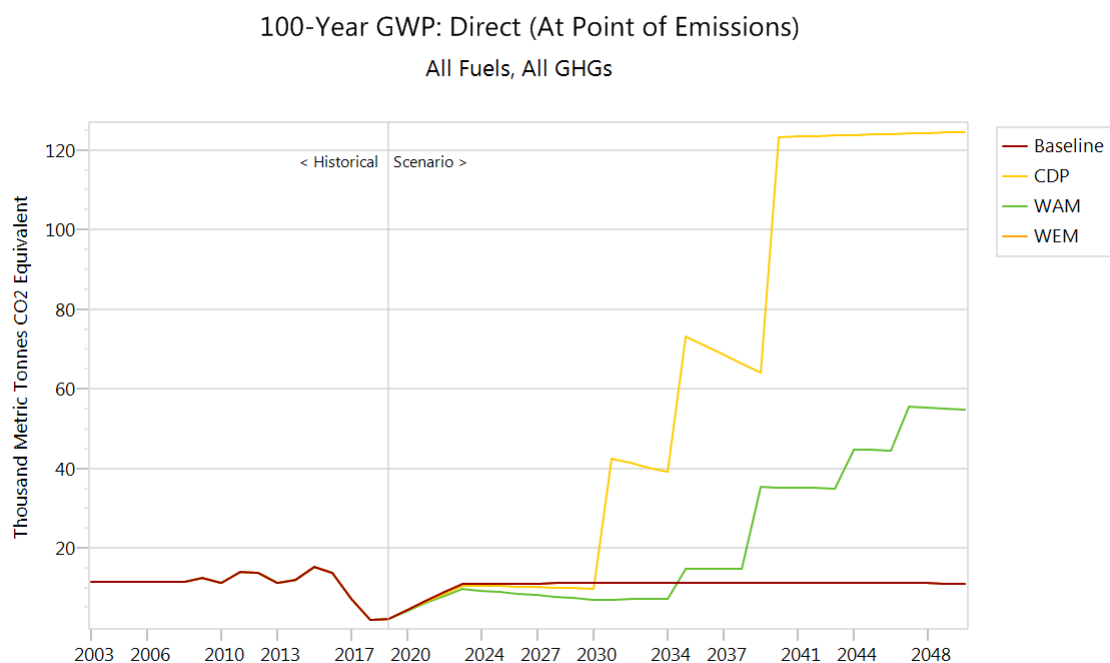


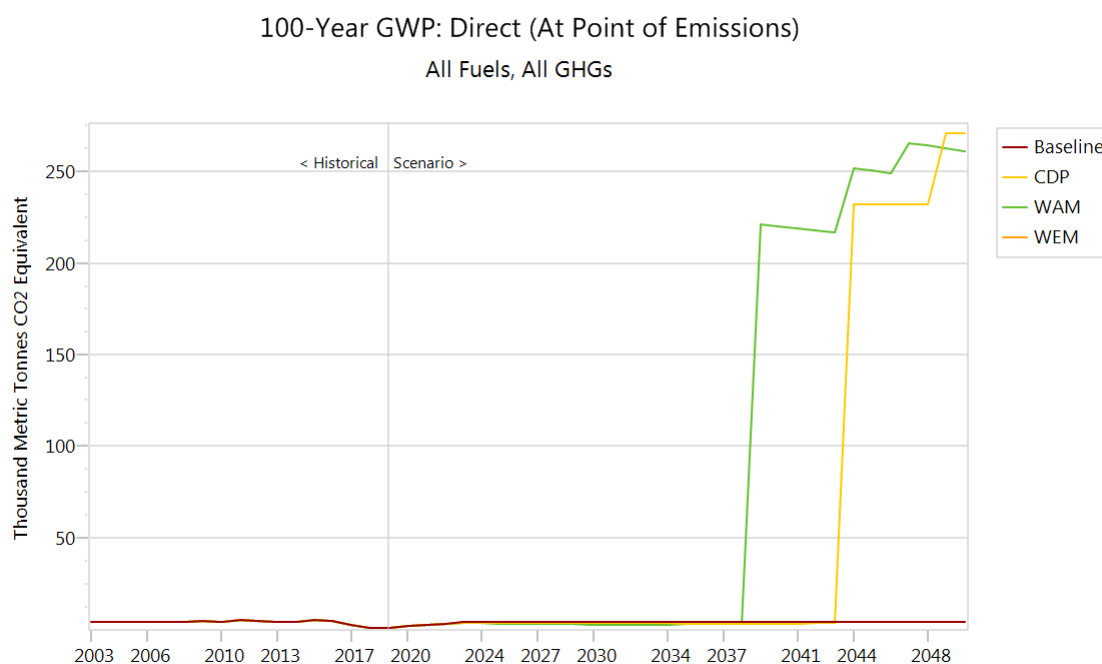
Figure 2-25: Total GHG emissions for passenger rail transport in all scenarios



Emissions in the rail sector increase under the mitigation scenarios (CDP and WAM), as shown in **Error! Reference source not found.** and Figure 2-26. This is because the rail infrastructure projects (MGR, SGR, Metro and LRT) are only included in the mitigation scenarios. They are not included in the BAU as they are classified as “mitigation measures” due to the modal shift that occurs as a result of their construction, therefore under the BAU, rail activity remains unchanged at its present level. The

construction of these infrastructure projects adds significant capacity to the existing dilapidated system and causes an increase in rail activity and therefore emissions from the rail sector. However, this increase in activity in the rail sector is offset by an equivalent reduction in activity in the road transport sector as people and freight shift from road to rail. Due to the efficiencies associated with mass transit and rail freight in comparison to road transport, total emissions are therefore reduced. The jagged profile in emissions is a result of the step-change in activity that occurs as each line of each rail project becomes operational.

Figure 2-26: Total GHG emissions for freight rail transport in all scenarios



Key uncertainties

The mitigation measures that make the most significant contribution to the GHG reductions in the transport sector are those that focus directly on the road transport sector, as this is where the vast majority of emissions come from in the sector. In particular, the key measures include:

- Road transport fuel efficiency
- Alternative fuel switch
- Transit Orientated Development (TOD)

The key uncertainties are therefore the assumptions underpinning these measures:

- Feasible fuel economy improvements
 - o Timeline
- Rate of penetration of alternative fuels
- Uganda specific assumptions about the impact of TOD and freight and public transport on transport activity (efficiency, distance travelled etc).

Whilst the measures listed above have the most significant impact on emissions, the infrastructure project-based modal shift measures are also key as they are critically important for Uganda's national and regional development and will come with very high costs. These measures are:

- Mass transit
- Modal shift to rail

Key uncertainties are therefore the assumptions that underpin these measures, in particular:

- Total expected capacity (passenger/freight)
- Assumed potential modal shift from other modes to rail/mass transit
- Fuel used for traction (diesel or electric)
- Energy intensity (fuel economy) of vehicles

Other uncertainties include the impact of NMT infrastructure on modal shift, the assumed VKM change between modes (e.g. from private transport to NMT).

2.8 Electricity generation (1A1ai)

2.8.1 Baseline situation

Characteristics of the sector

Uganda's Electricity Generation Capacity has grown over the years from about 872.2 MW in 2012 to the current 1,254.2 MW (with 5.9 MW on Off-Grid) by December 2019. This capacity is projected to increase by over 1,000 MW in the next 2 years following the commissioning of the Karuma Hydro Power Project (600 MW) and other projects implemented by Independent Power Producers (IPPs) (ERA, 2021¹²¹).

Current (2019) capacity is dominated by hydro - 1,004 MW (80%), with much smaller contributions from thermal - 100 MW (8%), Bagasse/Cogeneration - 96 MW (8%) and grid-connected Solar - 50 MW (4%) (ERA, 2021¹²²). The thermal heavy fuel oil (HFO) plants are used to provide standby generation in case of shortage of supply from other technologies due to key factors such as drought.

To meet peaks in demand, additional generation capacity is also available from Uganda's neighbouring countries through the regional transmission network. In practice, in 2019, <0.5% of Uganda's electricity mix was met by imports (ERA, 2021¹²³) indicated that at the current time Uganda does not need to import power to meet peak demand.

In 2019, transmission losses from electricity generation were approximately 4% and distribution losses were around 17% (ERA, 2021¹²⁴).

GHG emissions

GHG emissions arise from the combustion of heavy fuel oil in the thermal power plant. In 2015, the GHG emissions from power generation were 56.13 GgCO_{2eq}, or 0.06% of Uganda's total GHG emissions in the year. Emissions of CO₂ from the bagasse are not included in this total as they originate from biogenic sources, but other greenhouse gas emissions are included.

¹²¹ ERA (2021), Least cost electricity expansion Plan 2020-2030. Electricity Regulatory Authority, LN/83-120/021/008. March 2021. Accessed from: <https://www.era.go.ug/index.php/resource-centre/publications/plans/626-least-cost-electricity-expansion-plan-2020-2030/download>

¹²² Ibid

¹²³ Ibid

¹²⁴ Ibid

2.8.2 Business as usual scenario

Drivers and assumptions

2.8.2.1.1 Economic and population growth

Energy sector developments in Uganda are driven by economic development, population growth rates and changes in household size (Ministry of Water and Environment, 2014¹²⁵). Therefore, as Uganda's economy grows, along with its population, so does the demand for electricity.

2.8.2.1.2 Rate of electrification

Access to electricity is another important factor influencing consumption levels. Currently, 61.5% of rural households and 21.9% of urban households have no electricity (Ministry of Energy and Mineral Development, 2020¹²⁶). Of those households that have access to at least one source of electricity (51% of total households) approximately half have access through the grid (National or mini-grid), while the other half only have access through off-grid solutions like generators, rechargeable batteries and Solar - mostly Solar Lighting System (SLS) and Solar lanterns.

Moreover, even where households have access to at least one source of electricity the availability, quality and reliability of supply varies greatly. Therefore, overall, almost 30% of urban and 78% of rural households have less than reasonable access to electricity even for basic lighting use (Ministry of Energy and Mineral Development, 2020¹²⁷).

There are plans to increase grid access significantly, and different plans/strategies have set different targets in this area, although, it is not always clear as to whether they refer to grid access or access to electricity in whatever form:

- The Uganda Vision 2040 (National Planning Authority, 2010¹²⁸) set a target of electricity access to be 80% by 2040
- Rural Electrification Strategy and Plan (Ministry of Energy and Mineral Development, 2013¹²⁹) aims for rural electrified: 7% in 2015 to 100% by 2040; Urban electrified: 55% in 2015 to 100% by 2040.
- The Second National Development Plan (National Planning Authority, 2015¹³⁰) has the objective of 30% grid connection access by 2020, and it referenced the Vision 2040 document as a long-term target.
- The Third National Development Plan (National Planning Authority, 2020¹³¹) has the goal of increasing the proportion of the population with access to electricity to 60% by 2025.

¹²⁵ Ministry of Water and Environment (2014), Uganda's Second National Communication to the United Nations Framework Convention on Climate Change. Accessed from: <https://unfccc.int/resource/docs/natc/uqanc2.pdf>

¹²⁶ Ministry of Energy and Mineral Development (2020), National Electrification Report. Accessed from: http://www.energyandminerals.go.ug/site/assets/files/1081/memd_national_electrification_survey_2018_printed_2020.pdf

¹²⁷ Ministry of Energy and Mineral Development (2020), National Electrification Report. Accessed from: http://www.energyandminerals.go.ug/site/assets/files/1081/memd_national_electrification_survey_-_2018_printed_2020.pdf

¹²⁸ National Planning Authority (2010). Uganda Vision 2040. National Planning Authority, Kampala Uganda. Accessed from: <http://www.npa.go.ug/uganda-vision-2040/>

¹²⁹ Ministry of Energy and Mineral Development (2013), Rural Electrification Strategy and Plan 2013-2022. Accessed from: <https://s3-eu-west-1.amazonaws.com/s3.sourceafrica.net/documents/119217/Strategy-and-Plan-2013-2022.pdf>

¹³⁰ National Planning Authority (2020), Third National Development Plan (NDPII) 2015/16 – 2019/20. Accessed from: <http://npa.go.ug/wp-content/uploads/NDPII-Final.pdf>

¹³¹ National Planning Authority (2020), Third National Development Plan (NDPIII) 2020/21 - 2024/25. Accessed from: [http://www.npa.co.ug/wp-content/uploads/2020/08/NDPIII-Finale Compressed.pdf](http://www.npa.co.ug/wp-content/uploads/2020/08/NDPIII-Finale%20Compressed.pdf)

- The SE4ALL document¹³² sets an ambitious target for universal electricity access for 2030, it provides policy recommendations and suggests a 2:1 split between on-grid and off-grid solutions to achieve it.
- The Electricity Connection Policy (Electricity Regulatory Authority, 2018¹³³) sets an objective of 60% grid connection by 2027 by providing subsidised grid connections to households. It further references the Vision 2040 as a goal out of scope of the ECP's objectives but notes it as still being a national objective for the future.

It can be observed that although there are some differences in the medium-term trajectory for access rates, most policy targets seem to converge on the 80% target for 2040 – although the alignment of definitions is yet to be confirmed.

While there is consensus on the long-term objective the BAU scenario needs to reflect what the expected development might actually look like. Analysis by Ricardo (2019)¹³⁴ for the World Bank concluded¹³⁵ that access to electricity from the grid would grow to no more than 40% by 2030, and would fall back to 30% by 2040 (unless there was continuing investment) as the current pace of efforts would soon be offset by demographic growth. Moreover, total access to electricity of reasonable quality (including off-grid) would increase to no more than 60% by 2030 and would fall to 50% by 2040 (again driven by demographic changes and unless balanced by continuing investment).

Therefore, under the BAU scenario the following is assumed with respect to electricity access:

100% by 2040 as per rral elec.

in the BAU scenario.

2.8.2.1.3 Increase in electricity generation capacity

In the BAU scenario future electricity generation capacity provided by different technologies follows electricity supply projections from the ERA Least Cost Generation Plan 2020-2030 (ERA, 2021¹³⁶). The BAU scenario considers installed capacity (**Error! Reference source not found.**) plus all energy projects that were fully operational by 2020 (i.e., planned/constructed around base year; **Error! Reference source not found.**), as well as the expected future nuclear capacity expected by 2040 (24,000 MW, Vision 2040¹³⁷) as this was indicated as a highly important measure for dealing with future additional demand.

Table 2-34: Installed capacity (MW) included in the BAU scenario (from Table 11 of ERA 2021¹³⁸).

Fuel	2015	2016	2017	2018	2019
Large hydro	630.0	630.0	630.0	630.0	855.0
Small hydro	65.3	65.3	82.3	114.0	149.3
Thermal (HFO)	101.6	101.6	101.6	101.6	100
Bagasse/Cogeneration	96.2	96.2	96.2	96.2	96.2
Solar	0.6	10.8	20.8	40.8	50.8

¹³² Sustainable Energy for All (SEforALL) is an International Organization working with leaders in government, the private sector and civil society to drive further, faster action toward achievement of Sustainable Development Goal 7, which calls for universal access to sustainable energy by 2030 (Source: SE4ALL)

¹³³ Electricity Regulatory Authority (2018) Electricity Connection Policy Accessed from: <https://www.era.go.ug/index.php/resource-centre/regulatory-instruments/policies/318-electricity-connections-policy/download>

¹³⁴ Ricardo (2019) Uganda Distribution Sector Diagnostic Review and Directions for Future Reforms for Long-term Sector Development and Acceleration of Electricity Access Expansion. Report for The World Bank Group

¹³⁵ Based on previous performance – together with the REA 2017-2027 Master Plans and the Umeme Limited 2017-2025 Business Plan

¹³⁶ Ibid

¹³⁷ Ibid

¹³⁸ ERA (2021), Least cost electricity expansion Plan 2020-2030. Electricity Regulatory Authority, LN/83-120/021/008. March 2021. Accessed from: <https://www.era.go.ug/index.php/resource-centre/publications/plans/626-least-cost-electricity-expansion-plan-2020-2030/download>

Table 2-35: Additional power generation projects (online in 2019) included in the BAU scenario (from Table 10 of ERA 2021¹³⁹).

Project Name	Technology	Installed capacity (MW)	Year online
Isimba HPP	Hydro	183	2019
Sindila (Butama)	Hydro	5.3	2019
Siti II Small Hydropower Plant	Hydro	16.5	2019
Ziba Limited (Kyambura) Hydropower Plant	Hydro	7.6	2019
Ndugutu Hydropower Plant	Hydro	4.8	2019
Achwa I Hydropower Plant	Hydro	42	2019
Emerging Power U Ltd (Mayuge/Bufulubi)	Solar	10	2019

2.8.2.1.4 Mix of generation technologies

The fuel mix for future electricity generation capacity has been taken from the ERA Least Cost Generation Plan 2020-2030 (ERA, 2021¹⁴⁰) and other documents, such as the ESDP¹⁴¹ and Vision 2040¹⁴².

The availability of each fuel type (detailed in **Error! Reference source not found.**) has been kept the same across the timeseries. The efficiency of different fuel types has been taken from international sources for that fuel type. No improvement to efficiency has been considered across the timeseries.

The capacity is consistent with a reserve margin of 15% and assumes transmission and distribution losses to decrease from in line with the projections to 2025 detailed in the ERA Least Cost Generation Plan 2016-2025 (ERA, 2016¹⁴³ p.46) – transmission losses are projected to decrease from ~3.8% to ~3.2% between 2015 and 2025, while distribution losses are projected to decrease from ~19% to ~12% between 2015 and 2025.

Expert judgement has been used to determine whether the power plant is used for base or peak (fossil fuels are used for peak power). The base year demand curve is used across all years.

Table 2-36: BAU electricity generation capacity (MW)

Fuel	2015	2020	2030	2040	Efficiency	Availability	Base/Peak
HFO	102	100	100	100	35%	70%	Peak
Bagasse	96	96	96	96	40%	50%	Base
Hydro	695	1004	1004	1264	-	50%	Base
Solar	0	50	50	50	-	20%	Base
Geothermal	0	0	0	0	-	70%	Base
Nuclear	0	0	12571	24000	-	100%	Base
Wind	0	0	0	0	-	30%	Base
Peat	0	0	0	0	45%	70%	Peak

¹³⁹ ERA (2021), Least cost electricity expansion Plan 2020-2030. Electricity Regulatory Authority, LN/83-120/021/008. March 2021. Accessed from: <https://www.era.go.ug/index.php/resource-centre/publications/plans/626-least-cost-electricity-expansion-plan-2020-2030/download>

¹⁴⁰ Ibid

¹⁴¹ Ministry of Energy and Mineral Development (2015) Energy and Mineral Sector Development Plan. Accessed from: <http://npa.go.ug/wp-content/uploads/2018/01/Energy-Sector-Development-plan-Final.pdf>

¹⁴² Ibid

¹⁴³ Ibid

2.8.2.1.5 Demand-supply balance

The BAU scenario assumes that demand for electricity increases in response to population, economic growth and increased electrification, and additional generation capacity is brought online to respond to this demand. However, as noted in the ERA Least Cost Generation Plan 2016-2025 (ERA, 2016¹⁴⁴) if all planned generation projects are commissioned, then Uganda will face a challenge of excess generation capacity if no additional mitigation measures are undertaken to grow demand in the short term to medium.

GHG emission projections

GHG emissions arise from the combustion of heavy fuel oil in the thermal power plant. The BAU projection does not allow for the increased use of fossil fuel generation beyond the current 100MW of HFO capacity. As Uganda exploits its petroleum resources there may be a stronger economic case to increase this capacity. Such a strategy may lead to a significant increase in the GHG emissions from those described above.

The CO₂ emissions arising from the bagasse plant are not included in the projection (although emissions of other greenhouse gasses are) as these emissions are accounted for in the FOLU sector.

It is also important to note that the GHG emissions projections for electricity generation are influenced by many factors beyond just the fuel mix – for example, we can see that there are 0 emissions for many years under BAU (due to the use of renewables to meet projected demand). Where this is not the case (i.e., 2015) it is due to demand exceeding the capacity provided by renewables, meaning heavy fuel oil reserve capacity has to be used to meet the excess. In essence, the graph below represents emissions from the heavy fuel oil plant (and methane/nitrous oxide emissions from biogenic sources) which only occur when demand exceeds renewable capacity.

Figure 2-27: GHG emissions (GgCO₂e) from electricity generation in Uganda under the business-as-usual scenario.



¹⁴⁴ Ibid

Key uncertainties

The source data for the electricity supply projections is the ERA. This data is believed to be of good quality. The plans that are available explain how the electricity grid is expected to develop in the future, taking into account increased demand for electricity. The plans are expected to be more certain for the next 5-10 years, and less certain for years that are further into the future.

A key uncertainty is the introduction of 24GW of capacity of new nuclear energy by 2040. The country has limited experience with nuclear energy, so the development of new capacity is likely to be challenging. Whilst the introduction of nuclear generation will not affect Uganda's GHG emissions greatly as nuclear is low carbon, if it does not go ahead the capacity will need to be met by other sources, which may include thermal generation plant.

A further uncertainty is how demand will increase in the future, and whether the targets for increase electrification will be achieved. This may lead to excess generation capacity in the short to medium term.

2.8.3 Mitigation options

GHG missions from electricity generation in Uganda are currently low due to the high levels of hydro power and low levels of fossil fuel-based generation, although emissions do arise from the combustion of heavy fuel oil in thermal plants. There are also some emissions from bagasse. As a result, there is limited scope for additional abatement, and the BAU scenario already envisages zero emissions from electricity generation from 2016 onwards (Figure 2-27).

However, in the other mitigation scenarios there will be the need for additional electricity generation sources to meet increased electricity demand (e.g. from increasing rates of connection to the main grid, and fuel switching to electricity). Indeed, the Vision 2040¹⁴⁵ plan envisages the need for over 41,000 MW of generation capacity by 2040.

Therefore, renewables (e.g. solar, hydro, wind, bagasse and geothermal) and nuclear are all low carbon generation technologies that can be used to ensure that increased demand for electricity is met in a way that does not increase CO₂ emissions (i.e. instead of additional fossil-based generation). The more renewables that can be added to the mix, the less the reliance on nuclear. Further details on these technologies are provided below.

In addition, further reductions in T&D losses, whilst not delivering additional emissions reductions, will improve the efficiency of the T&D system and will mean that the additional future demand can be met with a lower level of generation capacity.

Bagasse cogeneration

The installed capacity for bagasse-cogeneration is expected to increase in the near future, with an additional 25MW bagasse plant in the pipeline to 2030. The level of generation of the respective plants is affected by the availability of cane and general weather conditions to grow cane in the country (ERA, 2021¹⁴⁶).

Table 2-37: Cost of bagasse cogeneration

Technology	Installed cost (USD per kW)	Operation and maintenance cost
Bagasse cogeneration	1,751	2% and 6% of the total installed costs per year.

¹⁴⁵ Republic of Uganda (2013), Uganda Vision 2040. Accessed from: <https://www.greengrowthknowledge.org/sites/default/files/downloads/policy-database/UGANDA%29%20Vision%202040.pdf>

¹⁴⁶ Ibid

Source: IRENA (2020)¹⁴⁷ Installed cost is based on weighted average costs for the Rest of the World region for 2010-2020. Values in 2020 prices.

Solar PV

Uganda has significant potential for solar PV, with the possibility of harvesting it in almost any part of the country (ERA, 2021¹⁴⁸). The intermittent nature of the technology is one of the main limitations, but as battery technology matures and becomes price competitive this limitation will be reduced.

Table 2-38: Cost of Solar PV

Technology	Installed cost (USD per kW)	Operation and maintenance cost
Solar PV	1,148	9 USD per kW per year for non-OECD countries

Source: IRENA (2020)¹⁴⁹ Installed cost is based on weighted average costs for South Africa, in 2020. Value in 2020 prices. Cost excludes any use of battery technology in combination.

Geothermal

In 2012, the Ministry of Energy estimated that the country had a potential of about 450 MW of Geothermal Capacity. This technology is expected to provide generation capacity to the national grid by end of 2027 (ERA, 2021¹⁵⁰).

Table 2-39: Cost of geothermal

Technology	Installed cost (USD per kW)	Operation and maintenance cost
Geothermal	4,468	115 USD per kW per year

Source: IRENA (2020)¹⁵¹ Installed cost is based on global weighted average costs in 2020. Values in 2020 prices.

Wind

By the end of 2019, one developer was in advanced stages in the feasibility study with possibility of developing a 50 MW Wind Power Plant. Additional generation of 150 MW with the current technology is expected by 2027 on a need arises basis (ERA, 2021¹⁵²). However, only one wind project (Rupa Wind Power Project - 20 MW capacity) is actually in the pipeline.

Table 2-40: Cost of onshore wind

Technology	Installed cost (USD per kW)	Operation and maintenance cost
Onshore wind	1,873	Up to 30% of the total levelized cost.

Source: IRENA (2020)¹⁵³ Installed cost is based on weighted average costs for the Africa region in 2020. Values in 2020 prices.

¹⁴⁷ IRENA (2020). Renewable Power Generation Costs in 2020. Accessed from: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2021/Jun/IRENA_Power_Generation_Costs_2020.pdf

¹⁴⁸ Ibid

¹⁴⁹ IRENA (2020). Renewable Power Generation Costs in 2020. Accessed from: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2021/Jun/IRENA_Power_Generation_Costs_2020.pdf

¹⁵⁰ Ibid

¹⁵¹ IRENA (2020). Renewable Power Generation Costs in 2020. Accessed from: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2021/Jun/IRENA_Power_Generation_Costs_2020.pdf

¹⁵² Ibid

¹⁵³ IRENA (2020). Renewable Power Generation Costs in 2020. Accessed from: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2021/Jun/IRENA_Power_Generation_Costs_2020.pdf

Hydroelectric power

While 80% of Uganda's electricity is generated from hydropower already, there is still additional hydropower potential that can be exploited – ESDP¹⁵⁴ notes 4500MW of hydropower potential (which is over 4x the hydropower capacity currently installed).

Table 2-41: Cost of hydroelectric

Technology	Installed cost (USD per kW)	Operation and maintenance cost
Large hydro (>10MW)	2,417	2% and 6% of the total installed costs per year.
Small hydro (<10MW)	3,256	

Source: IRENA (2020)¹⁵⁵ Installed cost is based on weighted average costs for the Africa region from 2016 - 2020. Values in 2020 prices.

Nuclear

Nuclear Energy is one the technologies expected to support the delivery of the goals in Vision 2040. The outlook for the development of Nuclear Energy is estimated to be 2032 to 2035 (ERA, 2021¹⁵⁶).

Table 2-42: Cost of nuclear

Technology	Overnight cost (USD per kW)
Nuclear (LWR)	3,606

Source: IEA (2020)¹⁵⁷

Overnight costs include: 1. direct construction costs plus pre-construction costs, such as site licensing, including the environmental testing; 2. the indirect costs such as engineering and administrative costs that cannot be associated with a specific direct construction cost category, as well as capitalised indirect costs; 3. owners' costs that include any additional expenses incurred by the owners associated with the plant and plant site, but excluding offsite, "beyond the busbar", transmission costs; and 4. contingency to account for changes in overnight cost during construction.

Reduced losses from transmission and distribution

Reducing transmission and distribution (T&D) losses from electricity generation acts to reduce the amount of electricity generation needed to meet demand (as less is lost in the process).

2.8.4 Mitigation scenarios

Key policies/strategies

To meet the increased demand, the future electricity generation capacity provided by different technologies has been taken from the ERA Least Cost Generation Plan 2020-2030 (ERA, 2021¹⁵⁸) and

¹⁵⁴ Ibid

¹⁵⁵ IRENA (2020). Renewable Power Generation Costs in 2020. Accessed from: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2021/Jun/IRENA_Power_Generation_Costs_2020.pdf

¹⁵⁶ Ibid

¹⁵⁷ IEA (2020). Projected Cost of Generating Electricity. Accessed from: <https://iea.blob.core.windows.net/assets/ae17da3d-e8a5-4163-a3ec-2e6fb0b5677d/Projected-Costs-of-Generating-Electricity-2020.pdf>

¹⁵⁸ Ibid

other documents, such as the ESDP¹⁵⁹ and Vision 2040¹⁶⁰. The following strategies/plans were used to inform assumptions about Uganda's future electricity generation.

Policy/measures	Mitigation action	Status
ERA Least Cost Generation Plan 2020-2030	Increased grid capacity, and increased renewable energy generation.	Some projects implemented, others planned
Energy Sector Development Plan	Increased grid capacity, and increased renewable energy generation.	
Vision 2040	Increased grid capacity, and increased renewable energy generation.	Planned

As described above, Uganda already has a low carbon power sector, and in the short term if all planned generation projects are commissioned, then Uganda will face a challenge of excess generation capacity if no additional mitigation measures are undertaken to grow demand in the short term to medium. However, as a result of measures in the end use sector, including fuel switching, additional capacity will be required in the long terms.

Therefore, the mitigation analysis that was carried out for the electricity sector considered two main measures – additional renewable power generation to meet any electricity demand, and reduction of transmission and distribution losses. Both measures will help ensure that increasing electricity demand in future is met in a low carbon manner.

Increasing access to grid electricity would also displace fossil fuel use (e.g. kerosene for lighting, wood for cooking etc), however, the emissions reductions from this are modelled in the relevant sectors (e.g. households, commercial) where the fuel is consumed.

With existing measures (WEM)

Electricity generation in the WEM scenario includes the energy projects in the BAU scenario, plus others (detailed in the ERA Least Cost Generation Plan 2020-2030) that are expected to come online between 2020 and 2025 (detailed in the table below).

Table 2-43: Additional energy projects (in pipeline to 2030) included in the WEM scenario (from Appendix 2 of ERA 2021¹⁶¹).

Project Name	Technology	Installed capacity (MW)	Year online
Nyamagasani 1 HPP	Hydro	15	2021
Nyamagasani 2 HPP	Hydro	6.1	2021
Muyembe HPP	Hydro	6.9	2021
Achwa 2 HPP	Hydro	41	2021
Karuma HPP	Hydro	600	2021
Kikagati HPP	Hydro	14	2021
Kakaka HPP	Hydro	5	2021
Atari HPP	Hydro	3.3	2022
Kabeywa 1 HPP	Hydro	6.5	2022

¹⁵⁹ Ministry of Energy and Mineral Development (2015) Energy and Mineral Sector Development Plan. Accessed from: <http://npa.go.ug/wp-content/uploads/2018/01/Energy-Sector-Development-plan-Final.pdf>

¹⁶⁰ Republic of Uganda (2013), Uganda Vision 2040. Accessed from: <https://www.greengrowthknowledge.org/sites/default/files/downloads/policy-database/UGANDA%29%20Vision%202040.pdf>

¹⁶¹ ERA (2021), Least cost electricity expansion Plan 2020-2030. Electricity Regulatory Authority, LN/83-120/021/008. March 2021. Accessed from: <http://www.era.go.ug/index.php/resource-centre/publications/plans/626-least-cost-electricity-expansion-plan-2020-2030/download>

Project Name	Technology	Installed capacity (MW)	Year online
Kabeywa 2 HPP	Hydro	2	2023
Nyabuhaka HPP	Hydro	3.2	2023
Simu HPP	Hydro	9.5	2024
Sisi HPP	Hydro	7	2024
Hoimo HPP	Hydro	3.3	2024
Igassa HPP	Hydro	0.3	2024
Kabasanja HPP	Hydro	0.4	2024
Tokwe HPP	Hydro	0.3	2024
Nsongya HPP	Hydro	0.7	2024
Katooke HPP	Hydro	0.3	2024
Nchwera HPP	Hydro	0.5	2024
Warugo HPP	Hydro	0.5	2024
Nyagak III HPP	Hydro	5.5	2025
Nyamwamba II HPP	Hydro	7.3	2025
Nyamabuye HPP	Hydro	7	2025
Sironko HPP	Hydro	7	2025
Kigwabaya HPP	Hydro	4.2	2025
SCOUL Bagasse Plant	Bagasse	25	2021
Xsabo Nkoge Solar	Solar	20	2023
Rupa Wind Power Project	Wind	20	2023

Renewable Energy Feed-in Tariffs (REFIT) support the development of Small Renewables Technologies. The current scheme is in its 4th phase, and able to support hydro and bagasse schemes.

Regarding grid access, we have reflected the Government's strategy, which is to achieve a rural electrification access of 26% (i.e. consumers who will be utilizing electricity in their homes, businesses, or institutions) by 2022 from the current level of about 7%. The pace of acceleration anticipated in the RESP also projects access to be at 51% by 2030 and 100% by 2040 (RESP¹⁶², p.ii). These are the same assumptions that we've made in the BAU scenario.

T&D losses are assumed to follow the same trajectory (to 2025) as in the BAU scenario (based on the ERA Least Cost Generation Plan 2016-2025) as no further implemented measures for this have been identified.

Current development plans (CDP)

For the CDP scenario, we have included the projects from the ERA Least Cost Generation Plan 2020-2030 that are in feasibility study stage (see table below). Where information on expected opening years could not be found, the year 2040 was assumed as the target year. This additional capacity is intended to meet additional demand.

Table 2-44: Additional energy projects (in feasibility study) considered under the CDP scenario (from Appendix 2 of ERA 2021¹⁶³).

Project Name	Technology	Installed capacity (MW)	Year online (assumed, based on best available information)
Mukoki HPP	Hydro	3.4	2040

¹⁶² Ibid

¹⁶³ ERA (2021), Least cost electricity expansion Plan 2020-2030. Electricity Regulatory Authority, LN/83-120/021/008. March 2021. Accessed from: <https://www.era.go.ug/index.php/resource-centre/publications/plans/626-least-cost-electricity-expansion-plan-2020-2030/download>

Project Name	Technology	Installed capacity (MW)	Year online (assumed, based on best available information)
Mitano HPP	Hydro	13.6	2040
Kisinga HPP	Hydro	2.5	2040
Ayago HPP	Hydro	840	2030
Kiiba HPP	Hydro	400	2040
Oriang HPP	Hydro	392	2040
Muzizi HPP	Hydro	48	2030
Nsongi HPP	Hydro	7	2040
Achwa-Aber Multipurpose HPP	Hydro	135	2040
Kiraboha HPP	Hydro	5	2040
Latoro HPP on R Aswa	Hydro	4.2	2040
Buwangani HPP	Hydro	7	2040
Nyakinengo SHP	Hydro	5.2	2040
Lower Achwa HPP	Hydro	17.4	2040
Awere HPP	Hydro	18	2040
Okollo SHPP	Hydro	5	2040
Rwembya SHPP	Hydro	0.4	2040
Lwakhakha HPP	Hydro	6.7	2040
Maziba HPP	Hydro	1.2	2040
Ngenge HPP	Hydro	13.8	2040
Nengo Bridge HPP	Hydro	7.5	2040
Atari 2 HPP	Hydro	2	2040
Pece Biomass Power Project	Biomass	20	2030
Unergy Biomass Power Project	Biomass	20	2030
Jinja Waste to Energy Power Project	Biomass	2.5	2030
Panyimur Geothermal Power Project	Geothermal	10	2030
Ulepi Solar	Solar	10	2040
Kamuli Sugar Bagasse Power Plant	Cogeneration	3	2040
Pramukh Steel Bagasse Power Plant	Cogeneration	8	2040
Excess Associated Gas Thermal Power Project	Thermal	146	2040
Kingfisher Gas to Power Project	Thermal	39.1	2030

It is assumed that grid access reaches 100% by 2030 (and 50% by 2025), based on the aspiration of Agenda 2030 to achieve universal access to electricity by 2030 (NDPIII¹⁶⁴, p.144).

While the BAU scenario considers T&D loss projections to 2025 from the ERA Least Cost Generation Plan 2016-2025 (ERA, 2016¹⁶⁵), the new ERA Least Cost Generation Plan 2020-2030 (ERA, 2021¹⁶⁶) provides updated T&D loss projections to 2040, which are considered in the CDP (current development plans) scenario. The assumption is that T&D losses will decrease from the BAU 2025 values (3.2% transmission losses, and 13% distribution losses) to reach 2% and 8% respectively in 2040.

With additional measures (WAM)

For the WAM scenario we have assumed further increased renewable energy capacity (in order to meet the higher demand assumed in this scenario). It is assumed that Uganda will fully exploit its energy

¹⁶⁴ National Planning Authority (2020), Third National Development Plan (NDPIII) 2020/21 - 2024/25. Accessed from: http://www.npa.go.ug/wp-content/uploads/2020/08/NDPIII-Finale_Compressed.pdf

¹⁶⁵ Ibid

¹⁶⁶ Ibid

potential, as detailed in the Vision 2040¹⁶⁷: “It is estimated that Uganda will require 41,738 MW by 2040 ... The required capacity will be generated from different energy sources namely: hydro power (4500MW); geo-thermal (1500MW); nuclear (24000MW); solar (5000MW); biomass (1700MW); peat (800MW) and thermal (4300MW)” (p.73-74). The table below details full assumptions (note: nuclear is already included under BAU, due to its high strategic importance). **This Vision 2040 target is included in the WAM scenario, rather than the lower ambition scenarios, because the projects that are needed to achieve this ~41,800MW capacity are not in existing plans or feasibility studies (i.e., the projects that are in the pipeline and in feasibility study are not sufficient to reach this capacity, meaning additional measures are needed).**

Table 2-45: Additional grid capacity assumed under the WAM scenario (from Vision 2040 p.73-74), where Uganda’s full potential for various energy sources is assumed to be met (in 2040).

Energy Source	Full potential (MW) met in 2040
Natural Gas	4300
Geothermal	1500
Hydro	4500
Solar	5000
Biomass	1700
Peat	800

For grid access, 100% access by 2025 has been assumed. Although undoubtedly challenging, this would be consistent with past trends.

For the WAM scenario, we have simply raised the ambition from the CDP scenario – it is assumed that T&D losses will decrease even further by 2050, reaching 1% and 6% respectively (from the 2% and 8% respectively achieved in 2040 under CDP).

Summary of scenarios

The table below describes the assumptions made for each measure for the three mitigation scenarios:

Table 2-46: assumptions for mitigation measures in the electricity generation sector

Measure	BAU	WEM	CDP	WAM	Notes
Renewable generation	Energy projects that have come online prior to 2020 considered. 2040 nuclear potential also included, due to strategic importance.	Energy projects in pipeline to 2030 considered	Energy projects currently in feasibility study stage considered	Full potential met in 2040	For full details on energy projects included in each scenario, see associated tables
Reduction in transmission and distribution losses	T&D loss projections to 2025 (based on ERA Least Cost Generation Plan 2016-2025) are considered	Unchanged from BAU	T&D loss projections to 2040 are considered (based on ERA Least Cost Generation Plan 2020-2030)	T&D losses are assumed to be further reduced (i.e., beyond CDP ambition) by 2050	WEM is unchanged from BAU because no further <i>implemented</i> measures were identified. The BAU assumptions are based on projections rather than targets.

¹⁶⁷ Ibid

Key uncertainties

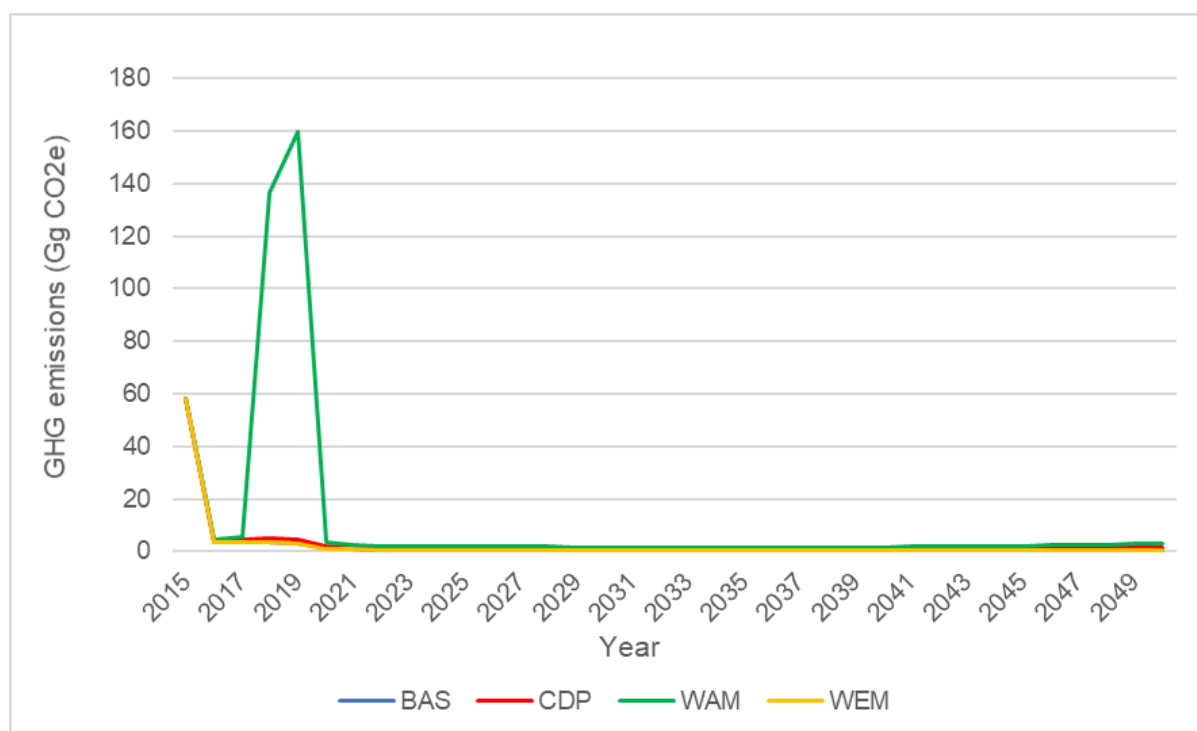
The main uncertainty associated with these projections is surrounding the timeliness of implementation of some of the renewable energy projects. Where possible, desk research has provided up to date estimations of the year that projects will come online. For the WAM scenario, however, it is assumed that Uganda's full energy potential will be met in 2040, which is a date that was decided based on best judgement.

GHG emission projections

It is important to note that CO₂ emissions from biogenic sources (e.g. bagasse) are not counted here, as the emissions belong to AFOLU – only methane and nitrous oxide emissions from biogenic sources are included here.

It is also important to note that the GHG emissions projections for electricity generation are influenced by many factors beyond just the fuel mix - when demand exceeds the renewable supply (as it does occasionally in higher ambition scenarios, as a result of higher ambition electrification targets) the HFO plant must provide the deficit. This is the case in 2015 for all scenarios, though BAU and WEM then see emissions reduce to 0 for the entire period 2016-2050 (indicating that renewable supply (informed by ERA 2021 projections) meets demand for the whole projection period). This is also the case from 2018-2019 in the higher ambition (CDP and WAM) scenarios, but this is potentially a function of uncertainty in demand projections. Most notably, the WAM scenario sees emissions from electricity generation increase significantly from 0 in 2040 to over 600 Gg CO₂e in 2050 – this is due to a mismatch between electricity supply and demand (which is further discussed in section **Error! Reference source not found.** below). Vision 2040 projected that approximately 41,800 MW of capacity was needed in 2040 to meet the increased demand expected by this year – our analysis is consistent with that, with low-carbon energy sources clearly meeting demand in 2040 under our highest ambition (WAM) scenario (where emissions are 0). However, our analysis indicates that longer term planning (i.e., to 2050) needs to consider even more ambitious renewable energy measures in order to prevent significant growth in emissions from electricity generation. It is important to note, though, that while 2050 emissions from electricity generation in the WAM scenario (>600 GgCO₂e) appear large in the graph above, they are actually rather small in the grand scheme of things.

Figure 2-28: GHG emissions (GgCO₂e) from electricity generation in Uganda under the three mitigation scenarios.



Electricity supply versus demand

It is important to assess whether the electricity generation scenarios assumed here align with the electricity demand associated with the mitigation measures included in the energy demand sectors – excess demand may result in the need for (potentially high-carbon) imports, while excess supply may negatively impact financial sustainability of the grid. This analysis also provides important context to the GHG emissions projections for the different scenarios shown above in section **Error! Reference source not found.**

The four graphs below show that, in all scenarios, supply closely follows demand until at least 2040. This means that the spike in emissions seen in 2018-2019 in the CDP/WAM scenarios is due to the reserve capacity of the HFO plant being used to meet additional demand (i.e., renewable capacity is exhausted in those years). As noted above, this is potentially a function of uncertainty in demand projections. Note: electricity demand in the WEM scenario is the same as BAU because no measures targeting a fuel switch to electricity were included in the WEM scenarios (and electricity access assumptions are the same between the two).

In WAM scenario, demand exceeds supply from 2044 onwards. However, GHG emissions in the WAM scenario (shown above) begin to increase from 2042. This means that the spike in emissions from 2042-2044 is due to the use of the HFO plant, but from 2044-2050 it is due to high-carbon energy imports being needed to meet additional demand. The implications of this on emissions (and for longer term planning) are discussed in section **Error! Reference source not found.** above.

Figure 2-30: Electricity supply versus electricity demand in the BAU

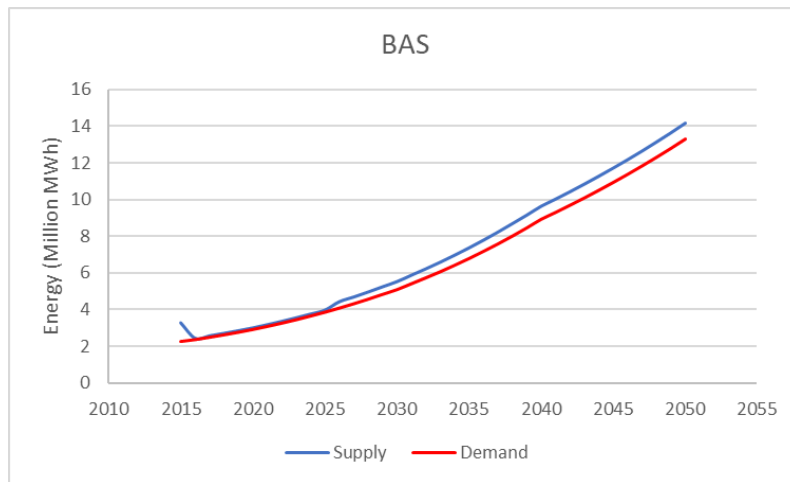


Figure 2-29: Electricity supply versus electricity demand in the WEM scenario.

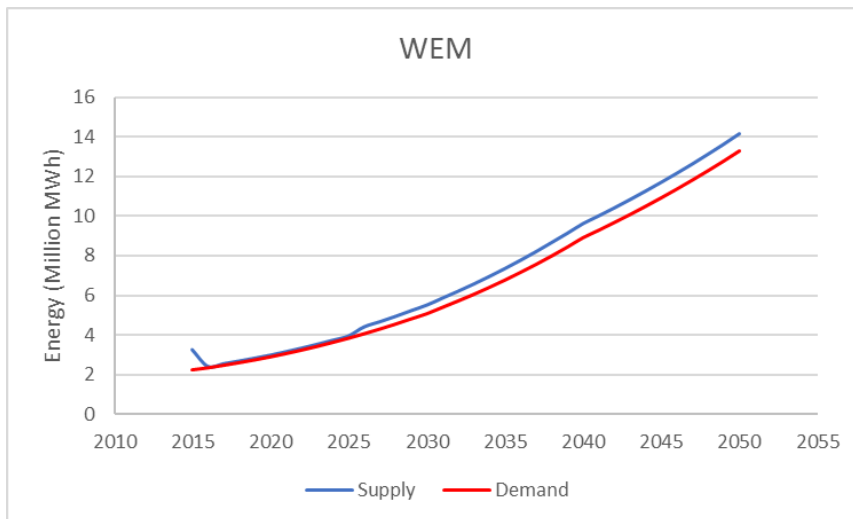


Figure 2-31: Electricity supply versus electricity demand in the CDP scenario

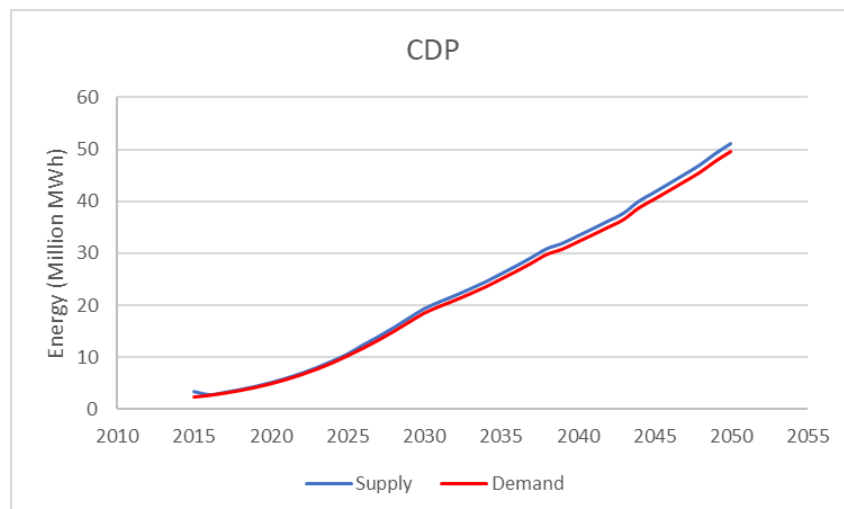
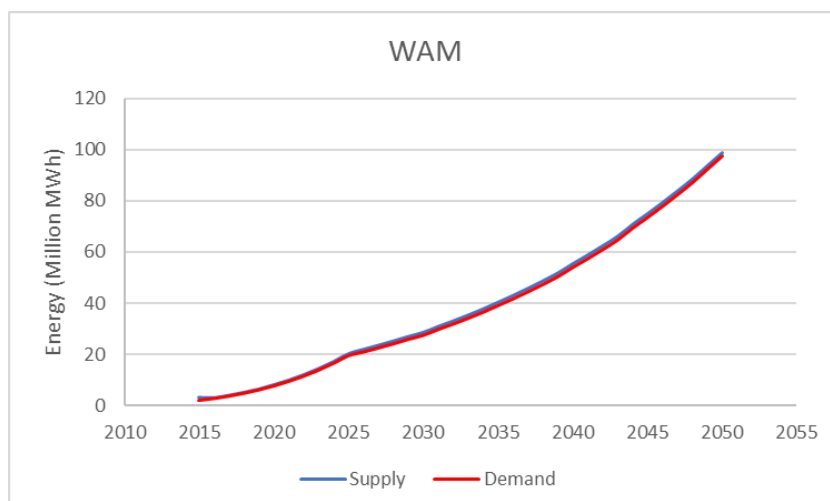


Figure 2-32: Electricity supply versus electricity demand in the WAM scenario.



2.9 Petroleum Refining (1A1b)

2.9.1 Baseline situation

Characteristics of the sector

Uganda does not currently have a mature petroleum industry, although it does have recoverable reserves and plans for exploitation. According to the Ministry of Energy and Mineral Development (2015)¹⁶⁸, Uganda has 1.4 billion barrels of crude oil that are recoverable.

To meet current demand for petroleum products, a total of 1.3 billion litres of petroleum products were imported in 2012, mostly for use in the transport sector. Of these, 41.1%, 6.1% and 52.8% were petrol, kerosene and diesel imports, respectively (Ministry of Energy and Mineral Development, 2015¹⁶⁹).

GHG emissions

There are currently no GHG emissions arising from this source.

2.9.2 Business as usual scenario

A number of activities are planned as part of the development of Uganda's oil and gas sector. The key project activities are discussed further in Section 2.11 and include the development of oil fields (Tilenga and Kingfisher projects), the development of the East African Crude Oil Pipeline (EACOP) and the development of a 60,000 barrels of oil per day refinery at Kabaale. This latter project is considered here.

¹⁶⁸ Ministry of Energy and Mineral Development (2015) Energy and Mineral Sector Development Plan. Accessed from: <http://npa.go.ug/wp-content/uploads/2018/01/Energy-Sector-Development-plan-Final.pdf>

¹⁶⁹ ibid

Drivers and assumption

2.9.2.1.1 Refinery capacity

The Kabaale project include the refinery, a 211 km long multi-products pipeline that will evacuate refined products from the refinery to a storage terminal in Buloba near Kampala city, a Water Intake facility and its corresponding water pipeline to the refinery and, the Buloba storage terminal itself¹⁷⁰. The refinery is assumed to be operational in 2026 with an output of 60,000 BPSD. The refinery may be expanded based on a number of considerations in the future.

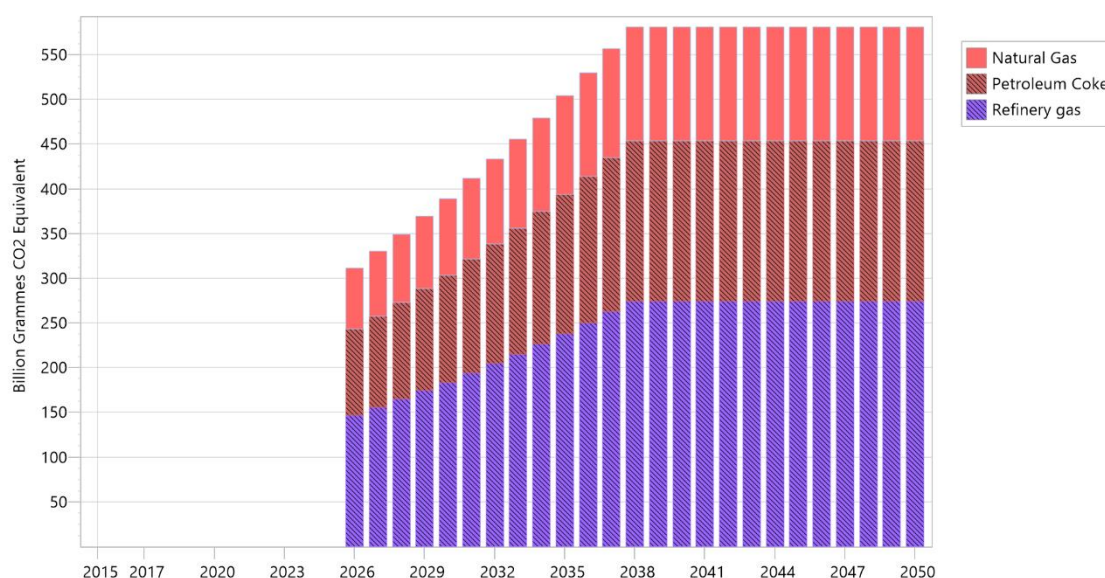
The BAU scenario assumes that oil refinery output increases to meet national demand.

GHG emission projections

The Environmental and Social Impact Assessment (ESIA) of the refinery project is currently ongoing and is planned to be concluded by December 2021. Therefore, the emissions estimates provided below are based on generic assumptions in the LEAP model.

The BAU emissions projections for petroleum refining are shown in the chart below. As noted above, the activities are assumed to start in 2026 and output – and the associated emissions - increase in response to national demand.

Figure 2-33: GHG emissions (GgCO₂e) from petroleum refining in Uganda under the business-as-usual scenario.



Key uncertainties

As there are currently no existing refinery operations in Uganda, there is a high level of uncertainty associated with the projections. This uncertainty relates both to the timing of the activities and the associated emissions. When the ESIA of the project is published it should provide a better understanding of the expected emissions.

¹⁷⁰ <https://www.unoc.co.ug/midstream/the-uganda-refinery-project/>

2.9.3 Mitigation options

Given the uncertainty around the design of the refinery, no specific mitigation options have been identified for this source. However, as a new plant, it might be expected that the design already represents international best-practice for controlling emissions (e.g., process efficiencies, use of co-generation). Therefore, the emissions estimate provided above may overestimate the actual emissions. At the same there may be additional mitigation options that are not included within the current plans which can deliver additional emissions reductions, such as carbon capture and storage.

2.9.4 Mitigation scenarios

With existing measures (WEM)

The WEM scenario is assumed to be the same as the BAU.

Current development plans (CDP)

The CDP scenario is assumed to be the same as the BAU.

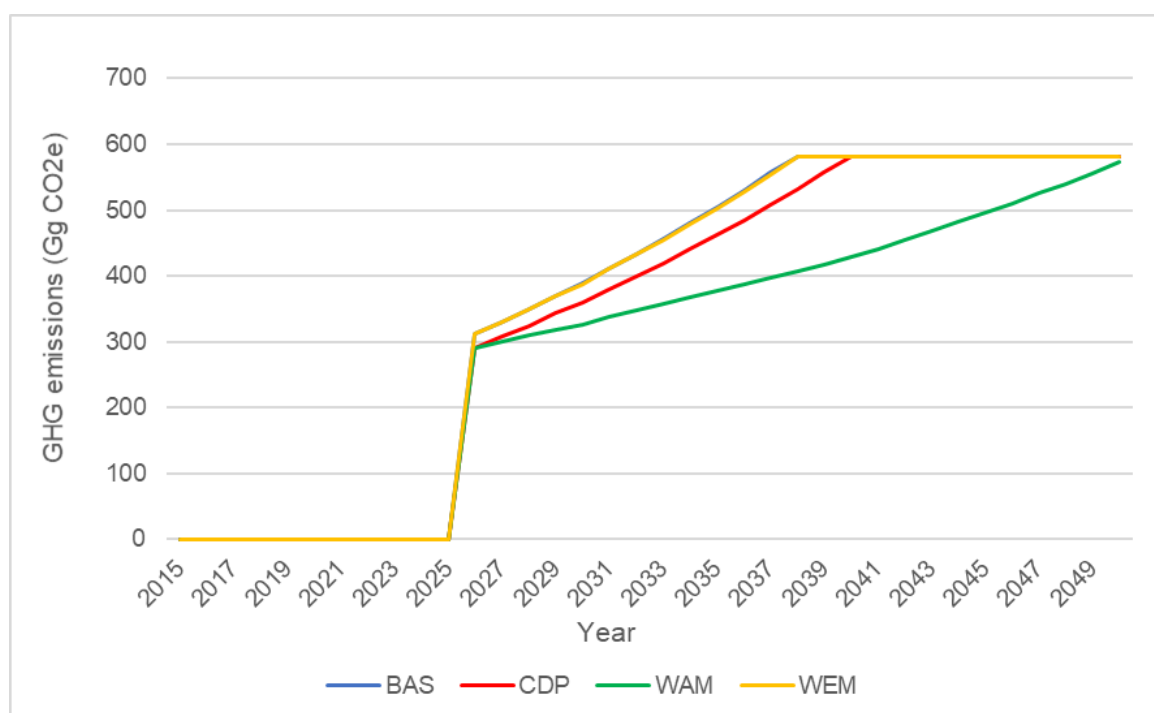
With additional measures (WAM)

The WAM scenario is assumed to be the same as the BAU.

GHG emission projections

The GHG emissions from petroleum refining do vary across the scenarios, but this is due to indirect effects of measures implemented in other sectors (e.g., fuel switching away from liquid fuel in the demand sectors).

Figure 2-34: GHG emissions (GgCO₂e) from petroleum refining in Uganda under the three mitigation scenarios. Note: the differences between scenarios seen here are a function of measures implemented in other sectors.



2.10 Charcoal production (1A1c)

2.10.1 Baseline situation

Characteristics of the sector

Charcoal is created by heating wood in the absence of oxygen in kilns. The majority of these kilns are small scale and very inefficient (~ efficiency 11%, by mass, for traditional earth kiln¹⁷¹) which means that large quantities of wood (~89%) are being wasted during this process.

Another potential issue with charcoal is the transportation that comes with it. Charcoal requires woody inputs and therefore it often needs to be transported to urban demand centres. As urbanisation increases the distances that charcoal needs to be transported would also increase.

Forests are regenerative and so charcoal can be used as part of a sustainable system. However, for the forests to be sustainable, the quantity of wood removed from the forests needs to match the quantity of wood grown by the forest in the year. The Biomass Energy Strategy estimated that the sustainable content of wood in Uganda is 13.7 million tonnes, and that Uganda was using 44.2 million tonnes of wood-equivalent (i.e., including charcoal) per year (Ministry of Energy and Mineral Development, 2013¹⁷²).

Energy use

In 2015, the manufacture of charcoal was associated with 82,760 TJ of energy consumption (Ministry of Water and Environment, 2019¹⁷³).

GHG emissions

Charcoal production accounted for 1,719 Gg CO_{2eq} of emissions in 2015 (Ministry of Water and Environment, 2019¹⁷⁴). Only CH₄ and N₂O emissions from these sources are included in this value, as CO₂ emissions associated with the wood used are accounted for in the land sector (3B).

2.10.2 Business as usual emissions projection

Drivers and assumptions

The production of charcoal is driven by demand for charcoal fuel from other sectors. In particular, charcoal is the most common fuel in urban settings, since its energy density is higher and therefore it can be stored more easily in urban settings.

¹⁷¹ Ministry of Water and Environment (2017). Accessed at: <https://www.mwe.go.ug/sites/default/files/library/Final%20-%20Uganda%20REDD%2B%20Strategy%20and%20Action%20Plan-October%202017.pdf>

¹⁷² Ministry of Energy and Mineral Development (2013), Biomass Energy Strategy (BEST) Uganda. Accessed from: <https://data2.unhcr.org/en/documents/details/64163>

¹⁷³ Republic of Uganda (2020), National Greenhouse Gas Inventory (NIR) Report for 2005-2015 (Draft provided by in-country partners).

¹⁷⁴ Ibid.

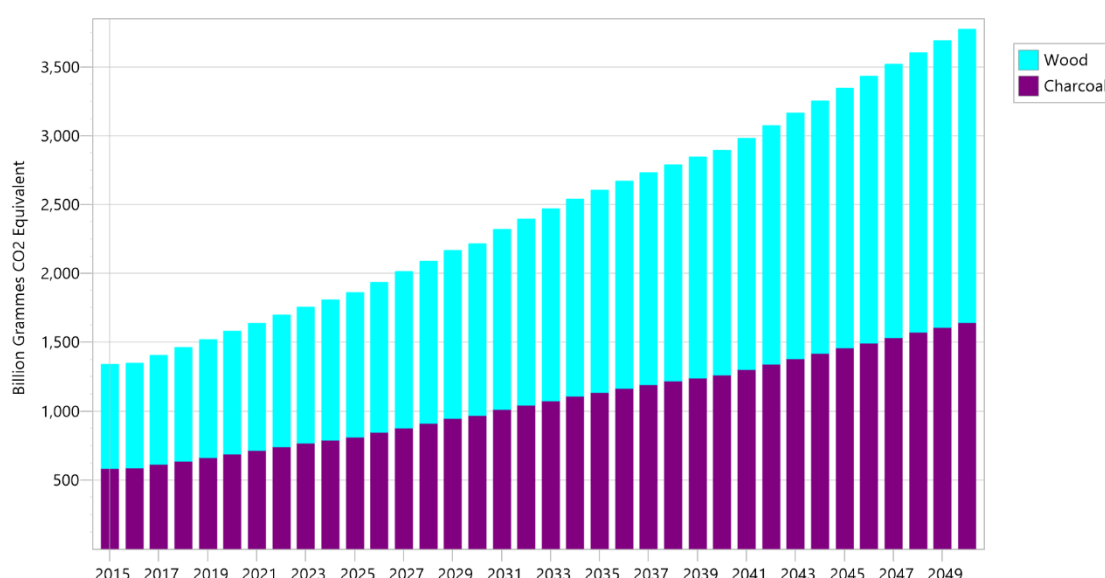
Therefore, the BAU growth in charcoal production is driven by the assumptions that are made with respect to growth in energy demand from end-use sectors. An additional contributing factor is the assumed rate of urbanisation.

The following key assumptions are made in the BAU scenario:

- Input fuel – Wood is assuming to come from indigenous sources (and therefore has interactions with the forestry sector)
- Capacity – Assumed to match indigenous demand in other end-use sectors, so capacity increases to match the BAU level of demand.
- Efficiency - Remains constant (90% of charcoal produced in normal kilns with efficiency of 20% (on an energy basis), 10% in improved with efficiency of 60% (on an energy basis) – Ministry of Energy and Mineral Development (2013)¹⁷⁵)
- Imports/exports – None

GHG emission projections

The projected GHG emissions from charcoal production in the BAU scenario is shown below.



Key uncertainties

The volume of charcoal that is currently being produced in Uganda is well understood, although the efficiency of the production process is more uncertain. Likewise, there is uncertainty about how demand for charcoal might increase in the future. There is also uncertainty with respect to how quickly more efficient kilns might be taken up in the BAU scenario. The current assumptions that kilns remain at the current level of efficiency is simplistic but provides a reasonable basis for assessing the mitigation scenarios.

2.10.3 Mitigation options

It is important to note that measures targeting the efficiency of charcoal production will not reduce CO₂ emissions in the energy sector but may contribute to decreased rates of deforestation in the forestry sector, and any associated emissions from the land use change are accounted for there.

¹⁷⁵ Ibid.

Efficient charcoal kilns

Charcoal used for cooking etc is currently produced in very inefficient kilns. Increasing the efficiency of these kilns would decrease the amount of firewood needed to produce the charcoal. This would involve replacing traditional inefficient kilns (20% efficiency on energy basis) with more efficient ones (60% on energy basis)¹⁷⁶ (Ministry of Energy and Mineral Development, 2013¹⁷⁷).

Table 2-47: Cost of more efficient charcoal kiln

Mitigation Measure	Cost description	Cost unit	Cost value	Source of data
More efficient charcoal kiln	Unit value per kiln (Adam Retort type)	USD	1,400	REDD+ Strategy and Action plan ¹⁷⁸

Alternative fuel sources

In addition to woody biomass, there are other types of biomass that can also be used for heat and as demand for wood decreases these may become more important. Examples include shrubs, grass, vegetal waste and bagasse. In addition to direct use, these non-woody biomass feedstocks can go through the briquetting process and be turned into something similar to charcoal but without the same deforestation that occurs with it. The Biomass Energy Strategy (Ministry of Energy and Mineral Development, 2013)¹⁷⁹ estimated that these sources could sustainably supply industrial briquette making with capacity to meet 40% of the country's biomass needs in 2040.

The Government of Uganda has set up a large-scale briquetting unit at Kampala Jerritone Suppliers to scale up briquette production in the country as an alternative for charcoal (ESDP¹⁸⁰).

Mitigation Measure	Cost description	Cost unit	Cost value	Source of data
Use of alternative fuel sources-non-woody biomass sources e.g briquettes	Cost of a tonne of briquettes from agricultural waste	USD/tonne	282	Based on Kampala Jellitone Supplies report from 2017

2.10.4 Mitigation scenarios

Key policies/strategies

The following policies/measures detail future plans for the sector:

¹⁷⁶ In the REDD+ strategy the efficiency of the Adam Retort kilns is stated to be as high as 66% on an energy basis.

¹⁷⁷ Ibid.

¹⁷⁸ Ministry of Water and Environment (2017). REDD+ Strategy and Action Plan. Accessed at: <https://www.mwe.go.ug/sites/default/files/library/Final%20-%20Uganda%20REDD%2B%20Strategy%20and%20Action%20Plan-October%202017.pdf>

¹⁷⁹ Ministry of Energy and Mineral Development (2013), Biomass Energy Strategy (BEST) Uganda. Accessed from: <https://data2.unhcr.org/en/documents/details/64163>

¹⁸⁰ Ibid.

Policy/measures	Mitigation action	Status
Biomass Energy Strategy	Aims to “Maintain wood demand for charcoal at the 2013 level by increasing per cent of charcoal made in improved technologies to 75% by year 2020”.	In progress
Energy Sector Development Plan	Aims to promote adoption of improved institutional kilns.	Planned
REDD+ Strategy and Action Plan	Strategic option 2 includes the sub-option on Improved charcoal kilns linked to bioenergy woodlots. This assumes 100,000 more efficient kilns. Strategy Option 3 includes the sub-option on Improved charcoal kilns linked to timber plantation sites. This assumed 15,000 more efficient kilns (Ministry of Water and Environment, 2017 ¹⁸¹).	Planned

With existing measures (WEM)

The Biomass Energy Strategy¹⁸² aims to “Maintain wood demand for charcoal at the 2013 level by increasing per cent of charcoal made in improved technologies to 75% by year 2020”. The timeframes in the REDD+ Strategy and Action Plan¹⁸³ are less clear (the analysis is to 2042) along with the relative share of kilns, but it can be inferred from the analysis that it assumes 100% of kilns are more efficient.

Both of the documents are essentially strategies, and they don’t in themselves represent specific policy instruments to deliver the improved kilns. Therefore, for the purposes of the WEM scenario it is difficult to claim that significant changes will be delivered in the short term. It has been assumed that the Biomass Energy Strategy target of 75% of kilns being more efficient will only be delivered by 2040, rather than 2020. It is also assumed that improvements will continue after this target is reached, with charcoal made in improved technologies assumed to reach 100% in 2050.

Current development plans (CDP)

This scenario assumes 100% of charcoal will be made in improved kilns by 2040, in line with the REDD+ Strategy and Action Plan¹⁸⁴. A linear trend is assumed between 2020 and 2040 for the purposes of the analysis.

With additional measures (WAM)

100% of charcoal to be made in improved technologies by 2030 (i.e., enhanced ambition in terms of target year). A linear trend is assumed between 2020 and 2040 for the purposes of the analysis.

Summary of scenarios

The table below describes the assumptions made for each measure for the three mitigation scenarios:

Table 2-48: assumptions for mitigation measures in the electricity generation sector

¹⁸¹ Ministry of Water and Environment (2017). REDD+ Strategy and Action Plan. Accessed at: <https://www.mwe.go.ug/sites/default/files/library/Final%20-%20Uganda%20REDD%2B%20Strategy%20and%20Action%20Plan-October%202017.pdf>

¹⁸² Ministry of Energy and Mineral Development (2013), Biomass Energy Strategy (BEST) Uganda. Accessed from: <https://data2.unhcr.org/en/documents/details/64163>

¹⁸³ Ministry of Water and Environment (2017). REDD+ Strategy and Action Plan. Accessed at: <https://www.mwe.go.ug/sites/default/files/library/Final%20-%20Uganda%20REDD%2B%20Strategy%20and%20Action%20Plan-October%202017.pdf>

¹⁸⁴ Ibid.

Measure	BAU	WEM	CDP	WAM	Notes
Improved efficiency of charcoal production	Remains constant from base year (90% in normal kilns with efficiency of 20%, 10% in improved with efficiency of 60%)	75% of charcoal to be made in improved efficiency kilns by 2040 (and 100% by 2050)	100% of charcoal to be made in improved efficiency kilns by 2040	100% of charcoal to be made in improved efficiency kilns by 2030	Penetration rates based on expert judgement

GHG emission projections

The IPPC's refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC 2019¹⁸⁵) suggest that where country specific emission factors are not available, Tier 1 emission factor should be used for estimating emissions from charcoal production. These calculate emissions based on charcoal produced.

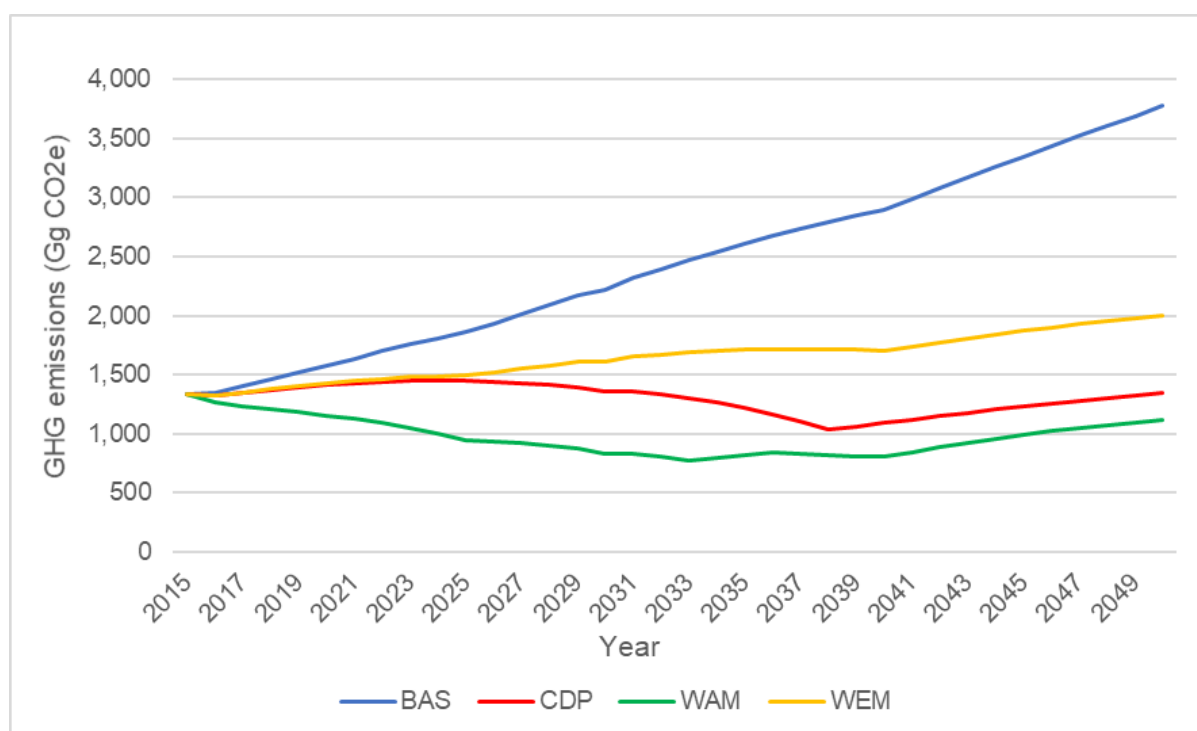
The guidance notes that “kilns with lower efficiency tend to have a higher emission factor, and vice versa” and that “the emission factors depend on the type of kiln and corresponding efficiency”. However, “emission factors are not available by kiln type”. What this means is that in the emissions projections shown below, the changes in emissions arise entirely from changes in charcoal produced and respond to changes in demand as a result of actions taken in other sector (e.g., more efficient charcoal stoves).

The use of more efficient charcoal kilns may also result in change in GHG emissions as a result of the reduced demand for firewood (for the same level of charcoal demand) and these changes in emissions accounted for in the forestry sector.

For example, under the WAM scenario emissions from charcoal reduction are reduced to 0 by 2050, but this is largely a function of the shift from charcoal to electricity as a fuel source.

¹⁸⁵ IPCC 2019, 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Calvo Buendia, E., Tanabe, K., Kranjc, A., Baasansuren, J., Fukuda, M., Ngarize, S., Osako, A., Pyrozhenko, Y., Shermanau, P. and Federici, S. (eds). Published: IPCC, Switzerland. Chapter 4 Fugitive emissions: Accessed from: <https://www.ipcc-nggip.iges.or.jp/public/2019rf/>

Figure 2-36: GHG emissions (GgCO₂e) from charcoal production in Uganda under the three mitigation scenarios.



Key uncertainties

There is uncertainty over the timescales over which new kilns will be introduced. The strategies that have been published have ambitious timescales, but it does not clear these are achievable. Another uncertainty is the extent to which demand for charcoal will change over time, as end user switch to charcoal e.g., from firewood, or switch to alternative fuels such as briquettes, LPG or electricity.

2.11 Oil and natural gas (B2)

2.11.1 Baseline situation

Characteristics of the sector

As described in Section 2.9, Uganda has an emerging oil and gas industry, with recoverable reserves of crude oil and plans for exploitation. These plans will lead to an increase in greenhouse gas emissions in Uganda in the future.

The following activities are planned:

- **Tilenga Project** – Refers to the development of six oil fields in the Albertine Graben¹⁸⁶ via a single Central Processing Facility (CPF) with a capacity to treat some 190,000 barrels of oil per day¹⁸⁷ of produced oil together with associated gas, produced water, injection water and associated utilities

¹⁸⁶ https://totalenergies.com/sites/g/files/nytnzq121/files/documents/2021-03/Tilenga_esia_non-tech-summary_28-02-19.pdf

¹⁸⁷ <https://www.pau.go.ug/the-tilenga-project/>

and camps, A total of about 412 wells will be drilled from 31 well pads, and a network of buried pipelines will collect the oil production from each well pad and transport it to the CPF. All produced water shall be reinjected for oil recovery purposes. From the CPF the treated oil will be sent to the proposed oil export system and refinery at Kabaale via the 96 km 24" Tilenga Feeder Pipeline. The CPF will also be connected to a water abstraction facility on the shores of Lake Albert. There will be no routine flaring during production, and the produced associated gas will be used to generate power and heat to run the oil and gas facilities as well as provide power for EACOP Uganda facilities.

- **Kingfisher Oil project** – Refers to the development of the Kingfisher field in the Albertine Basin, which is part of the western arm of the East African Rift System, via a single Central Processing Facility (CPF) with a capacity to treat some 40,000 barrels of oil per day of produced oil together with associated gas, produced water, injection water and associated utilities and camps. A total of about 31 wells will be drilled from 4 well pads, and a network of buried pipelines will collect the oil production from each well pad and transport it to the CPF. There will be no flaring during normal operation. The excess gas will be utilized for heat and power generation and LPG production.¹⁸⁸
- **East African Crude Oil Pipeline (EACOP)** - The East African Crude Oil Pipeline (EACOP) System is an export pipeline project that will transport oil from an inlet flange at the Kabaale pumping station (PS1), in Hoima district, Uganda, to an export flange at a proposed marine storage terminal (MST) at Chongoleani, Tanga district, on the coast of Tanzania.

GHG emissions

There are currently no estimates available of the GHG emissions currently arising from these activities. Emissions from exploratory activities at the sites are expected to be relatively small.

2.11.2 Business as usual scenario

Drivers and assumption

The drivers and assumptions follow those within the respective ESIA documents, including the well productivity.

GHG emission projections

The design of the oil and gas projects is on-going. Environmental and Social Impact Assessments (ESIA) have been prepared for each of the projects, based on the proposed design at that time. In some cases, more recent emissions estimates are available which reflect more recent plans. For the Tilenga project the BAU emissions are based on the latest estimates for the project¹⁸⁹. For the Kingfisher project a high-level GHG emissions estimated provided by CNOOC Uganda Ltd has been used. For the EACOP the emissions estimated reported in the ESIA¹⁹⁰ has been used.

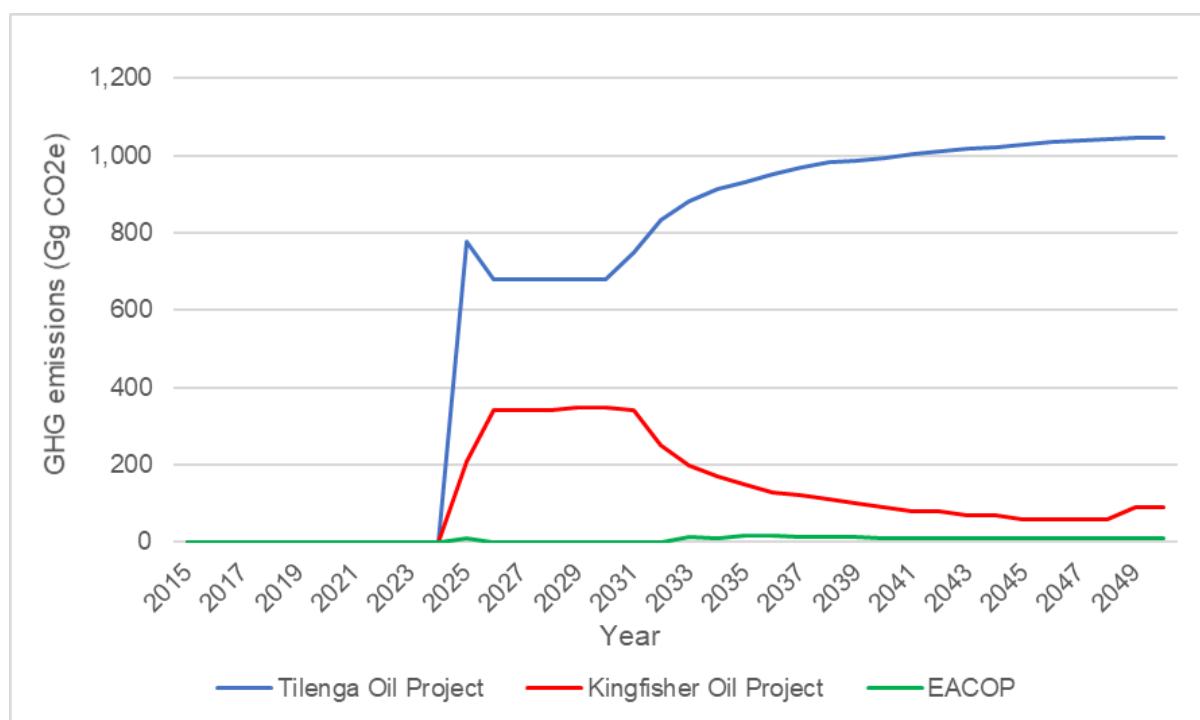
All of the estimated emissions arise from the combustion of fuels.

¹⁸⁸ https://cnoocinternational.com/-/media/cnooc-images-and-files/operations/middle-east-and-north-africa/uganda/esia-documents/vol_0_cnooc_kingfisher_esia_non-technical_summary_final.ashx?la=en&hash=F0BD0C69E4A9B8A9D90E363C3D0A0978

¹⁸⁹ Personal communication, Totalenergies

¹⁹⁰ Appendix G Impact Assessments - Uganda

Figure 2-37: GHG emissions (GgCO₂e) from fuel combustion at the Tilenga, Kingfisher and EACOP projects



Key uncertainties

As the projects are still at the design stage, there is a moderate level of uncertainty associated with the projections. This uncertainty relates both to the timing of the activities and the associated emissions. For all projects the developers have made an estimate of GHG emissions, and all projects have an ESIA. However, as the plans are further developed this may lead to a change in the expected GHG emissions.

2.11.3 Mitigation options

As the projects are subject to further design consideration, no specific mitigation options have been identified for the projects. However, as new projects, it might be expected that the design already represents international best-practice for controlling emissions (e.g., utilisation of excess gas for heat and power). At the same there may be additional mitigation options that are not included within the current plans which can deliver additional emissions reductions, such as the inclusion of an LPG recovery facility Tilenga.

2.11.4 Mitigation scenarios

There are no mitigation scenarios; the WEM, CDP and WAM are the same as in the BAU scenario. The emissions from this scenario are shown above.

2.12 Industrial processes and product use (2)

2.12.1 Baseline situation

Characteristics of the sector

Uganda's NIR includes reported emissions in 2015 for two industrial processes and product use categories:

- 2.A.1 - Cement production
- 2.A.2 - Lime production

Uganda produces two types of cement. The Portland cement and Pozzolana Portland Cement. The clicker fractions the Portland cement and Pozzolana Portland Cement are 95% and 72% respectively (Ministry of Water and Environment, 2019).

GHG emissions

Emissions from the IPPU are mainly from the mineral industry, accounting for 99.5% of the emissions and are dominantly CO₂ (Ministry of Water and Environment, 2019). The overall emissions from mineral industries were 487 Gg of CO₂e in 2015, the majority of which (347 Gg CO₂e) are attributed to cement production (Ministry of Water and Environment, 2019).

2.12.2 Business as usual scenario

Drivers and assumptions

2.12.2.1.1 Population

Since cement and lime production is strongly influenced by domestic demand in the construction sector, such as large infrastructure such as dams, road network and commercial buildings (Ministry of Water and Environment, 2019) population growth is a key driver. The UBOS household projections (**Error! Reference source not found.**) have therefore been used as the primary driver of emissions. Emissions from RAC production are also driven by household projections.

2.12.2.1.2 GDP growth rates

Metal and lubricant production is driven by GDP growth projections for industry (Table 2-1), as these materials are primarily used for industrial purposes.

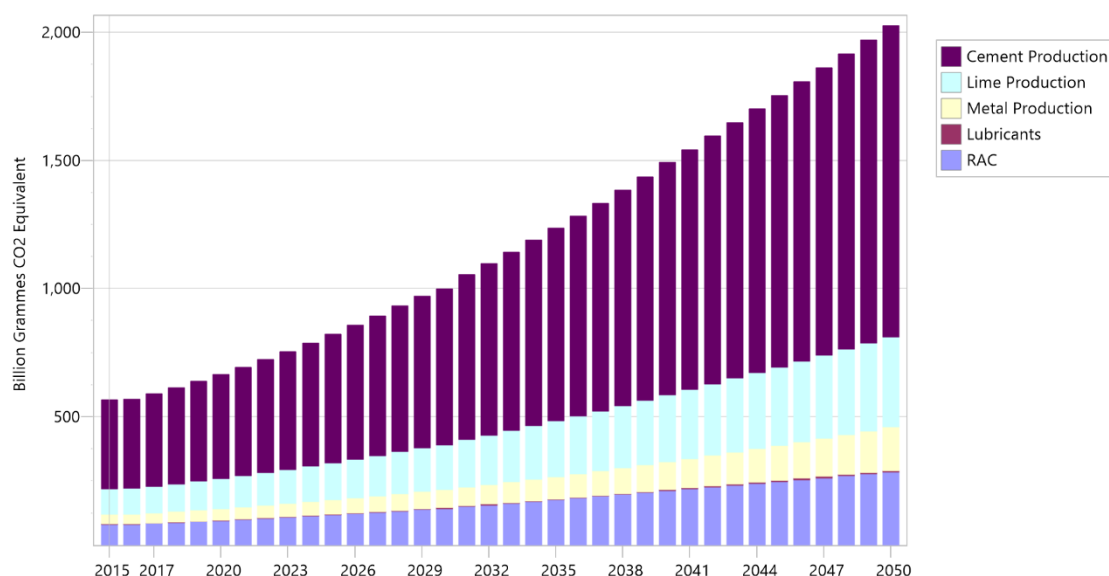
2.12.2.1.3 Technology developments

No changes to the current cement or lime production technologies are assumed.

GHG emission projections

The projected GHG emissions from industrial processes and product use (IPPU) in the business as usual scenario is shown below.

Figure 2-38: GHG emissions (GgCO₂e) from IPPU in Uganda under the business-as-usual scenario.



Key uncertainties

The performance of the current cement plant is relative well established, as are the emissions. However, what is more uncertain is the growth in demand, and how this will affect the operations at the plant. This include demand for the different types of cement that are currently produced. For example, no account has been taken of the need to increase capacity, and whether this would result in the construction of newer and more efficient plants. Likewise, no technology upgrades at existing plant have been assumed.

2.12.3 Mitigation options

Clinker substitution in cement production

Cement production is responsible for the majority of emissions in the IPPU sector – targeting this sub-sector is therefore the best course of action for reducing IPPU emissions. Clinker is a component of cement, and is essentially a mix of limestone and minerals that has been heated and transformed in a kiln. The use of other constituents (i.e., pozzolana, fly-ash or slag to replace clinker) in cement means lower emissions and lower energy use.

Mitigation Measure	Cost description	Cost unit	Cost value	Source of data
Clinker substitution in Cement production		USD/tCO ₂ e	7.6	GACMO database

2.12.4 Mitigation scenarios

With existing measures (WEM)

No existing/implemented measures for cement production have been identified, meaning the WEM scenario is unchanged from BAU.

Current development plans (CDP)

No quantified targets for cement production have been identified, meaning the CDP scenario is unchanged from WEM/BAU.

With additional measures (WAM)

Assumptions for clinker substitution in cement production in the WAM scenario are based on assumptions from Zimbabwe's Mitigation Assessment. The two substitution measures detailed provide approximately 60 thousand tonnes of emissions savings in 2030 (assuming 16% substitution by 2030) – this saving has been scaled to Uganda's cement production (6.8million tonnes, compared to 2 million for Zimbabwe) resulting in proposed emissions savings of 195 Gg CO₂e by 2030, which is 32% below the value here for BAS in 2030). The assumption is therefore that the rate of growth in emissions from cement production in the WAM scenario is 68% of BAS (32% reduction).

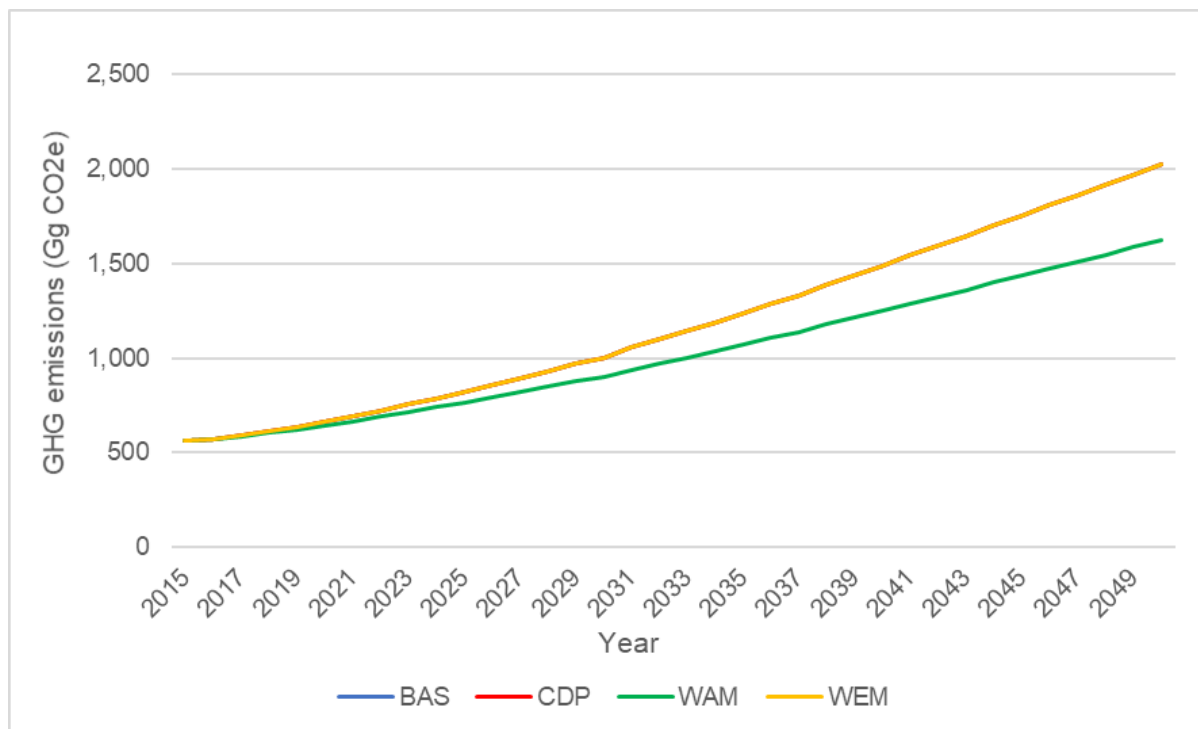
Summary of scenarios

Measure	BAU	WEM	CDP	WAM	Notes
Clinker substitution	Clinker fraction unchanged from 2015; growth in emissions driven by UBOS household projections.	Unchanged from BAU	Unchanged from BAU	32% reduction in emissions from BAU	Based on assumptions from Zimbabwe's Mitigation Assessment

GHG emission projections

Emissions projections for the three mitigation scenarios are shown below. Note: WEM and CDP are the same as each other (and the same as BAU) because no measures are included for IPPU under these scenarios.

Figure 2-39: GHG emissions (GgCO₂e) from IPPU in Uganda under the three mitigation scenarios.



Key uncertainties

This scenario is highly uncertain, as it is based on assumptions from a mitigation assessment conducted for Zimbabwe. While best efforts have been made to scale assumptions to the context of Uganda, significant uncertainty remains. For example, the Zimbabwe assessment assumes 16% substitution by 2030, but data on the current average clinker content of cement for Uganda could not be found. As such, it is intrinsically assumed that the starting point (i.e., baseline characteristics) of clinker in cement production were the same as in Zimbabwe.

2.13 Livestock (3A)

2.13.1 Baseline situation

Characteristics of the sector

In Uganda cattle production can be classified into two distinct types: “traditional” systems (small-scale farmers producing at a small farm or household level) and “commercial” systems (better developed and more mechanised). Traditional dairy consists of small-holder extensive, medium-holder extensive, pastoral (semi-nomadic pastoralists) and aro-pastoral sub-systems. Commercial systems consist of small-holder intensive, medium-holder intensive and large-scale producers. Traditional cattle represent over 90% of total national cattle (MAAIF, undated). The ‘commercial’ and ‘traditional’ classifications

generally correspond to the IPCC (2006) definitions of high-productivity systems and low-productivity systems for dairy systems respectively. However, it should be noted that this description of cattle production differs from the definition used in the draft NIR for 2005 – 2015 (i.e. Uganda, 2020). The cattle population can be disaggregated into different subcategories according to their age groups (see table below).

Figure 2-40: Cattle subcategories according to age groups in Uganda (Data from MAAIF)

Proportion of cattle (%)				
	Age (years)	Local breeds	Exotic breeds	All breeds
Bulls	>2	3,46	6,67	3,70
Cows	>2	77,85	74,67	77,62
Heifers	1-2	4,38	3,56	4,32
Bullocks	1-2	8,77	7,11	8,64
Female calves	<1	2,77	4,00	2,86
Male calves	<1	2,77	4,00	2,86
		100	100	100

Manures decompose and produce methane and nitrous oxide. Formation and quantities of the GHGs depend on the management systems that are practised in the country and environmental conditions. Data on manure management systems in Uganda are based on expert judgements and interpolations. The data show that pasture, range and paddock is the common manure management system used in the country for nearly all the livestock types.

GHG emissions

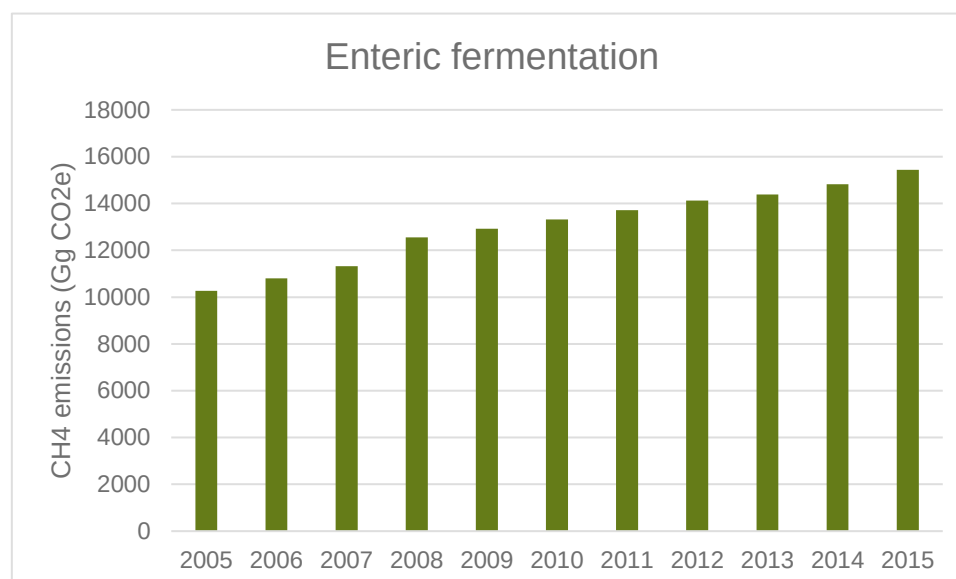
GHG emissions from enteric fermentation are influenced by animal size, type of feed intake and milk produced particularly for dairy systems (IPCC, 2006). With its use of locally derived emission factors, a Tier 2 approach provides an opportunity to conduct mitigation analysis for agricultural systems transitioning from low to high input production systems, unlike a Tier 1 approach that only considers changes in number of animals with time in the country (IPCC, 2006). Tier 2 uses livestock population disaggregated into subcategories and local emission factors. In the absence of local emission factors in Uganda, emission factors from western Kenya (Ndungu et al., 2018) have been assumed. These emission factors are similar to those of Goopy et al. (2018), which are for cattle in similar conditions and location/region. The choice of Ndungu et al factors was based on typical animal mass which although higher, came close to the value for Uganda used in the draft NIR (Uganda, 2020). Moreover, these factors are representative of cattle production systems in East Africa (Ndungu et al., 2018). Default IPCC (2006) emission factors for a dairy cow in Africa is 46 kg CH₄ head⁻¹ year⁻¹ and 31 kg CH₄ head⁻¹ year⁻¹ for other cattle.

Ndungu et al. (2018)			Goopy et al. (2018)	
	Emission factors (kg CH ₄ head ⁻¹ year ⁻¹)	Typical animal mass (kg)	Emission factors (kg CH ₄ head ⁻¹ year ⁻¹)	Typical animal mass (kg)
Males 1-2 yrs	33,2	156,88	30,1	143,5
Males > 2 yrs	45,5	214,14	39,5	216
Females 1-2 yrs	28,5	149,98	23	154,6
Females > 2 yrs	50,6	306,85	28,3	216,3
Calves < 1 yr	29	67,1	15,7	73,4

Enteric fermentation is the largest source of emissions from livestock. It is responsible for 96% of total livestock emissions. For cattle, enteric fermentation accounts for between 62% and 93% of emissions depending on the production and manure management systems (FAO & NZ, 2020). CH₄ emissions

account for 98% of cattle emissions (FAO & NZ, 2019). As a result of their large numbers and high emissions per unit (head), cattle are the major source of the enteric fermentation. They are followed by goats and sheep. Rate of cattle population growth in Uganda (6.3% per annum, FAO & NZ, 2020) has a very influence in the high increase of emissions from enteric fermentation which averages 5% per year (MWE, 2020).

Figure 2-41: Emissions from enteric fermentation in Uganda (MWE, 2020)



GHG emissions from manure management consist of CH₄ and N₂O. These two GHGs from manure management systems produced 713 Gg CO₂e in 2015 (MWE, 2020). Compared to CH₄ from enteric fermentation, manure management systems are responsible for a small fraction of the livestock emissions. For cattle, CH₄ is the main GHG emitted by manure management systems across all cattle production systems in Uganda (FAO & NZ, 2020).

2.13.2 Business as usual scenario

Drivers and assumptions

2.13.2.1.1 Enteric fermentation

The main drivers of emissions from enteric fermentation are the number and type of animals, the size and age of the animals, the type of feed and the amount of milk produced. Of these, the most significant drivers are the number of animals and their feed, which in turn will depend on the size of the population and dietary habits (which could, in turn, be linked to income levels).

Cattle was disaggregated by age groups using information from MAAIF. No age disaggregation was assumed in other livestock types and all animals were assumed to be mature.

Historical data of animal population from 2005-2015 were used to determine annual growth rates of animals. This gave the following growth rates:

- Cattle population increases at a linear rate based on the trend of their population between 2005 and 2015.

- Other livestock growth rates were; sheep (6.68%), goats (5.08%), swine (5.25%), poultry (2.74%) and mules and asses (0.56%) based on historical livestock population data (draft NIR, Uganda, 2020).

Default emission factors were used for all livestock except cattle, where the Ndungu et al. (2018) EFs were used. It was assumed that the emission factors remained constant for the business as usual scenario.

2.13.2.1.2 Manure management

Livestock manure management systems are a source of N₂O and CH₄. Production of the emissions depend on the environment (including climate and soil) and feed given to animals. There is little knowledge and data on livestock manure management in Uganda. Data (**Error! Reference source not found.**) that are used in the national GHG inventories are based on interpolations and expert judgements. Livestock farming in Uganda is primarily traditional and communal, making pasture, range and paddock to be the major management system in the country particularly for the outdoor animals. These fractions were assumed to stay constant through to 2050.

Table 2-49: Manure management systems by animal type in Uganda (adopted from draft NIR 2020)

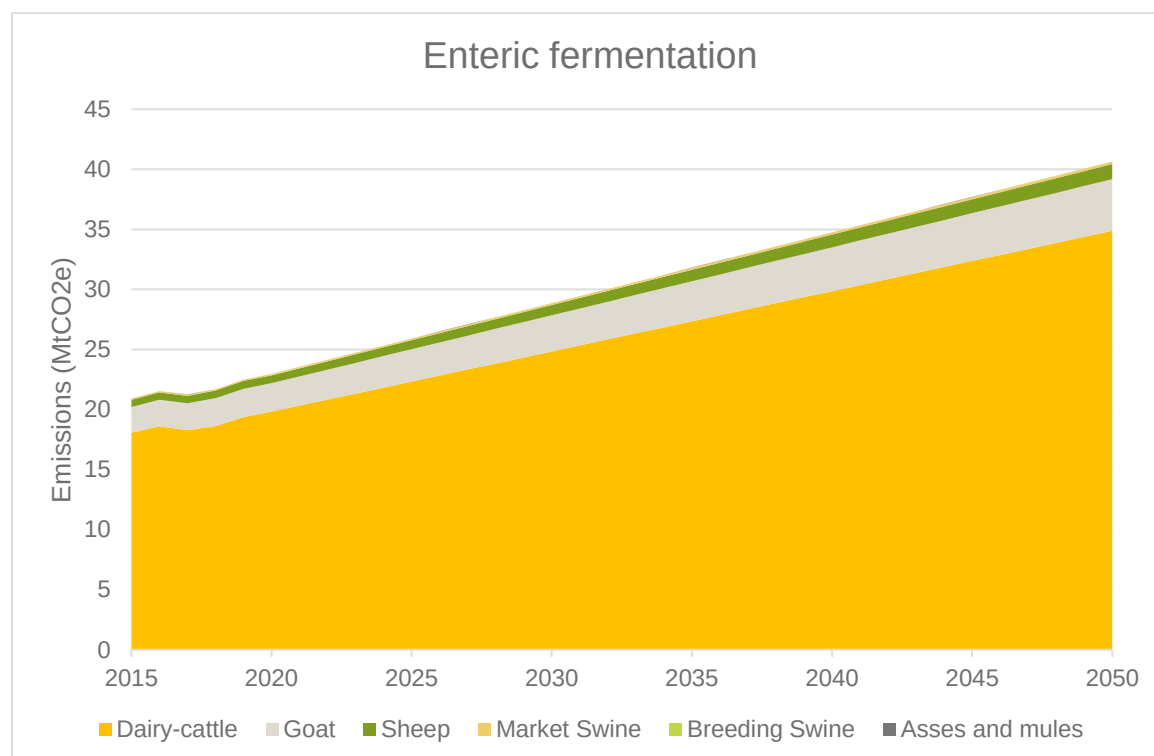
Livestock category	Pasture, Range and Paddock	Solid storage	Pit storage below animal confinement	Anaerobic digesters
Commercial dairy cattle	90%	5%	4%	1%
Traditional dairy cattle	90%	10%	-	-
Sheep	100%	-	-	-
Goats	80%	20%	-	-
Breeding swine	-	90%	10%	-
Market swine	-	30%	70%	-
Poultry	70%	30%	-	-

Default 2006 IPCC emission factors were used for all livestock types to calculate the CH₄ and N₂O from manure management systems. Commercial dairy cattle assumed default emission factors for dairy cattle while traditional cattle used default emission factors for non-dairy cattle.

GHG emissions projections

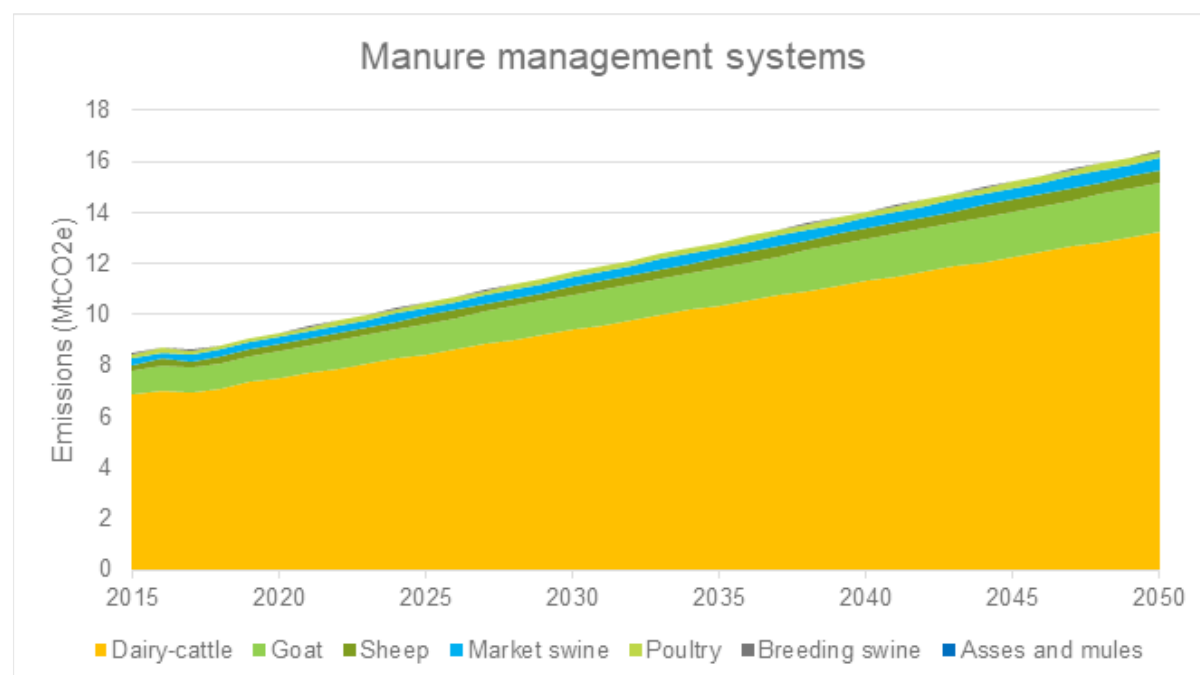
Following the assumptions outlined above, the business as usual emissions projection from the agriculture sector are as shown in **Error! Reference source not found.** below.

Figure 2-42: Projected business as usual GHG emissions from enteric fermentation to 2050 (MtCO₂e)



Due to high livestock population growth rates, emissions from enteric fermentation are projected to increase more than 2-fold by 2050 compared to 2015. Cattle will remain the major source of emissions in 2050, followed by goats. Other livestock types will continue as small sources of emissions in this category. Mature cows (> 2 years old) are the largest source of emissions.

Figure 2-43: Projected business as usual GHG emissions from manure management to 2050 (MtCO₂e)

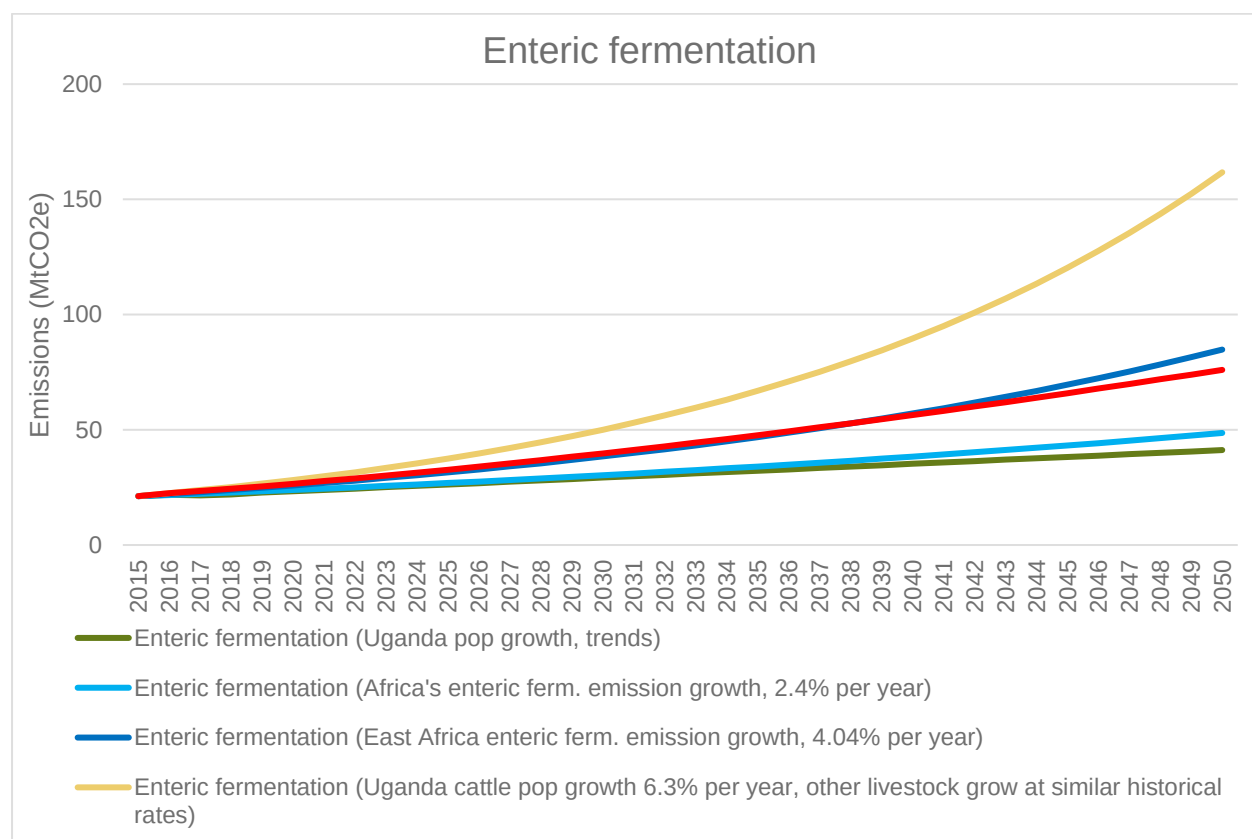


Greenhouse gas emissions from manure management systems are projected to increase further between 2015 and 2050. Dairy cattle are forecasted to remain the major source of the emissions

followed by goats and sheep. The emissions from this source are expected to be driven by population growth of the animals due to the increasing demand of livestock products in the country.

Key uncertainties

The growth rate of livestock is the major uncertainty in the emissions. The rate of 6.3% per year for cattle, reported in (FAO & NZ study; Agric NAMA) was considered too high. The emissions associated to it would grow exponentially fast as shown below (for enteric fermentation, manure management would result with the similar pattern). Other possible growth rates are shown.



Compared to the default factors that have been used in previous calculations, emission factors of Ndungu et al (2018) have slightly increased enteric fermentation emissions. These factors are more representative than the default factors as they are derived using local conditions and primary management systems in East Africa.

Disaggregation of cattle into categories by age groups can be another key uncertainty. Breakdown of cattle population into these categories is always a challenge particularly for traditional production systems that do not keep farm records. Surveys to generate this information are also not held regularly in Africa.

2.13.3 Mitigation options

In the agriculture sector, the main mitigation actions will be those included with Uganda's REDD+ Strategy (Ministry of Water and Environment, 2017). The measures are assessed alongside those for the forestry sector in the Section **Error! Reference source not found.** below.

2.13.4 Mitigation scenarios

Key policies and strategies

The mitigation measures for the agriculture sector are assessed alongside those for the forestry sector in Section **Error! Reference source not found.** below.

GHG mitigation projections

The mitigation projections for the agriculture sector are included with those for the forestry sector in section in Section **Error! Reference source not found.** below.

2.14 Forestry and other land use (3B)

2.14.1 Baseline situation

Characteristics of the sector

Natural forest vegetation in Uganda is categorized into Tropical High Forest Well-stocked and Low-stocked (THF) and the more dominant woodlands (Obua et al., 2010¹⁹¹; MWE, 2018¹⁹²). In addition to natural forests, there are plantations which consist of broadleaved and coniferous tree types (Obua et al., 2010; MWE, 2018). Forests are described as vegetation dominated by trees most of which are at maturity and are more than 5 m tall and form a tree canopy cover of 30% of the land (MWE, 2018). Woodlands are predominantly covered with woody plants, trees over 4 m high, shrubs and grasses (Obua et al., 2010). Differences in climate, soil, altitude, drainage and past human activities have contributed to diverse structures and compositions of forests across the country (Obua et al., 2010). Total economic value of forests in Uganda was 6% of the gross domestic product in 2004 (Obua et al., 2010).

Uganda is experiencing rapid deforestation (MWE, 2018). Overall forest land represented 24% of national land in 1990, decreasing to 15.4% (or 3.2 million ha) of national land area in 2000 and 12.4% (or 2.5 million ha) in 2015 (Obua et al., 2010; MWE, 2018). Forest area decreased by 27% (1.8% per year) between 1990 and 2005 (Nakakaawa et al., 2011¹⁹³). Natural forest reduced from 30% of land area in 1990 to approximately 10% in 2015, an average decline of 1.8% per year (MWE, 2017). However, land area under plantations *increased more than 2-fold* from 32,225 ha in 1990 to 107,608 ha in 2015 (MWE, 2017¹⁹⁴). It is estimated that forest cover may have been around 50% of Uganda total land area during the beginning of the 20th century (Obua et al., 2010). The decrease of forests in Uganda is due to high population growth which increases demand for forest products for both domestic and industrial use. Nearly two-thirds (64%) of forest lands are located outside of the protected areas (Obua et al., 2010).

Deforestation in Uganda is estimated to be between 50 000 ha and 55 000 ha per year which represents 0.9% per year (Obua et al., 2010; MWE, 2018). Woodlands were reduced by 0.47% per year between 1990 and 2015 (Mwanjalolo et al., 2018). Other estimates put the deforestation rate at 1.10-3.15% per year (Obua et al., 2010). Woodlands accounted for 90% (1 196 510 ha) of forest cover decrease between 1990 and 2005 (Nakakaawa et al., 2011).

¹⁹¹ Obua J, Agea JG, Ogwal JJ, 2010. Status of forests in Uganda. African Journal of Ecology, 48, 853-859.

¹⁹² MWE, 2018. Proposed forest reference emission level for Uganda. Ministry of Water and Environment. Kampala, Uganda.

¹⁹³ Nakakaawa CA, Vedeld PO, Aune JB, 2011. Spatial and temporal land use and carbon stock changes in Uganda: implications for a future REDD strategy. Mitig Adapt Strateg Glob Change, 16, 25-62.

¹⁹⁴ MWE, 2017. Forest investment program for Uganda. Ministry of Water and Environment. Kampala, Uganda.

Table 2-50: Change in forest area in Uganda from 1990 to 2015¹⁹⁵

	1990 ^b	2000	2005	2015
Total forest area (million ha) [% of country land]	4.932b,c [20b-24a]	3.1d-3.2a,e [15.4]a	3.6 [14]b,c	2.42d-2.5a [11.8d-12.4a] 1.84f [10]
Forest cover disaggregation million ha [% of forest land]				
Woodlands	3.975b,c [80.6]b,c	2.26e	2.778c	1.62 [64]a,e 1.16f
THF	0.924b,c [18.7]b,c	0.63e	0.792c	0.51 [20]a,e
THF + degraded THF				0.430 +0.136f
Plantations	0.033b,c [0.7]b,c	0.27e	0.0334c	0.42[16]a,e 0.108f
Change of forest cover per year (%)				
Woodlands			-2.20e	
THF			-1.50e	
Plantations			2.96e	

GHG emissions

Deforestation and forest degradation produced CO₂ emissions of 15.9 Mt in 2000 and nearly 40 Mt in 2015 (MWE, 2019a¹⁹⁶, Uganda 2020¹⁹⁷). Total annual losses due to deforestation and forest degradation between 1990 and 2005 were estimated at 2.6×10^6 tC yr⁻¹, resulting in 9.5 Mt CO₂ year⁻¹) (Nakakaawa et al., 2011). GHG emissions from deforestation are 8.3 million tCO₂ per year and forest degradation produced 0.82 million tCO₂ per year (MWE, 2017; MWE, 2018). Annual GHG emissions from deforestation are in the range of 12 to 17 MtCO₂e (MWE, 2017). Although woodlands experienced highest deforestation between 2000 and 2015, THF is the largest source of the emissions with 53% due to its higher emission factor.

Forest degradation (forest remaining forest with a permanent reduction of forest carbon stocks) occurs at a large scale in Uganda (MWE, 2018). Between 24% and 30% of THF have been classified as being degraded or have lost their quality in 2005 compared to 1990 (Nakakaawa et al., 2011; Obua et al., 2010). Between 1990 and 2005, about 101,100 ha of normally stocked THF became degraded resulting in a loss of 447,960 t C yr⁻¹ (Nakakaawa et al., 2011). About 136, 280 ha of THF forests were degraded in 2015 (MWE, 2017) resulting in approximately 2.21 MtCO₂ yr⁻¹.

Table 2-51: CO₂ emissions (Mt) from deforestation, 2000-2015

	Deforestation characteristics		CO ₂ emissions (Mt) through deforestation	
	Area deforested (ha)	Emission factors (tCO ₂ ha ⁻¹)	2000-2015	Average per year
Plantations	7063	260.3	1838540	122569
THF	124401	526.4	65484647	4365643
Woodlands	620739	91.0	56497181	3766479
Total			123,820,368	8,254,691

¹⁹⁵ a = FERL (2018), b = Obua et al. (2010), c = Nakakaawa et al. (2011), d = MWE (2018), e = MWE (2019a), f = MWE (2017)

¹⁹⁶ MWE, 2019a. Uganda's first biennial update report to the United Nations Framework Convention on Climate Change. Ministry of Water and Environment. Kampala, Uganda.

¹⁹⁷ Uganda, 2020. Draft national greenhouse gas inventory report for 2005-2015. Government of Uganda. Kampala, Uganda.

Table 2-52: values for above- and below-ground biomass

	Above-ground biomass	Annual above-ground biomass growth	Ratio of below-ground biomass to above-ground biomass	Carbon fraction of above-ground forest biomass	BCEFr
	(Mg d.m ha ⁻¹)	(Mg d.m ha ⁻¹ yr ⁻¹)	(R)	(CF; tonnes d.m ha ⁻¹ yr ⁻¹)	
THF	231	1,5	0,2	0,46	1
Woodlands	40	1,5	0,4	0,46	1
Plantations	114	12	0,24	0,47	1

2.14.2 Business as usual scenario

Drivers and assumptions

Key drivers of emissions from the FOLU sector include removal of trees for fuel and timber, conversion of forests to cropland and grassland and wildfires. All of these are, to an extent, driven by population growth. More information on these are below.

2.14.2.1.1 Wildfires

Wildfires are the largest sources of GHG emissions in Uganda, producing a total of 421 Mt CO₂e in 2015 (MWE, 2018). Over 96% of the emissions are caused by burning of forests while the remainder of the emissions are as a result of fires on non-forest land especially grasslands. Wildfires are the biggest cause of deforestation and forest degradation in the country (MWE, 2018). Emissions scenarios for wildfires were developed using (i) average deforestation rate (MWE, 2019a), (ii) population growth (3% per year) assuming that this would result in proportional land clearings to respond to various social demands, and (iii) wildfire incidences will remain constant until 2050 (MWE, 2018).

2.14.2.1.2 Fuelwood, poles and timber removals

Fuelwood was estimated using a power function ($1.57 \times 106 \times e^{0.0593x}$) built on the 2005-2015 historical data. It was estimated that 39% of all fuelwood was harvested from woodlands, 10% from THF and 1% forest plantations (MWE, 2019a). Fuelwood consumption in the country stands at 102 550 TJ per year (MWE, 2019a). Another fuelwood consumption scenario was developed using assumption that the fuelwood demand will increase similar to the national population growth of 3% using the same ratios between woodlands.

2.14.2.1.3 Croplands and grasslands

According to key category analysis, land converted to cropland is the third (after Forest land remaining Forest land, and enteric fermentation) largest source of emissions in Uganda.

CO₂ emissions attributable for conversion of forests to cropland and grassland are estimated at 36,830 Gg (MWE, 2019a).

2.14.2.1.4 Assumptions

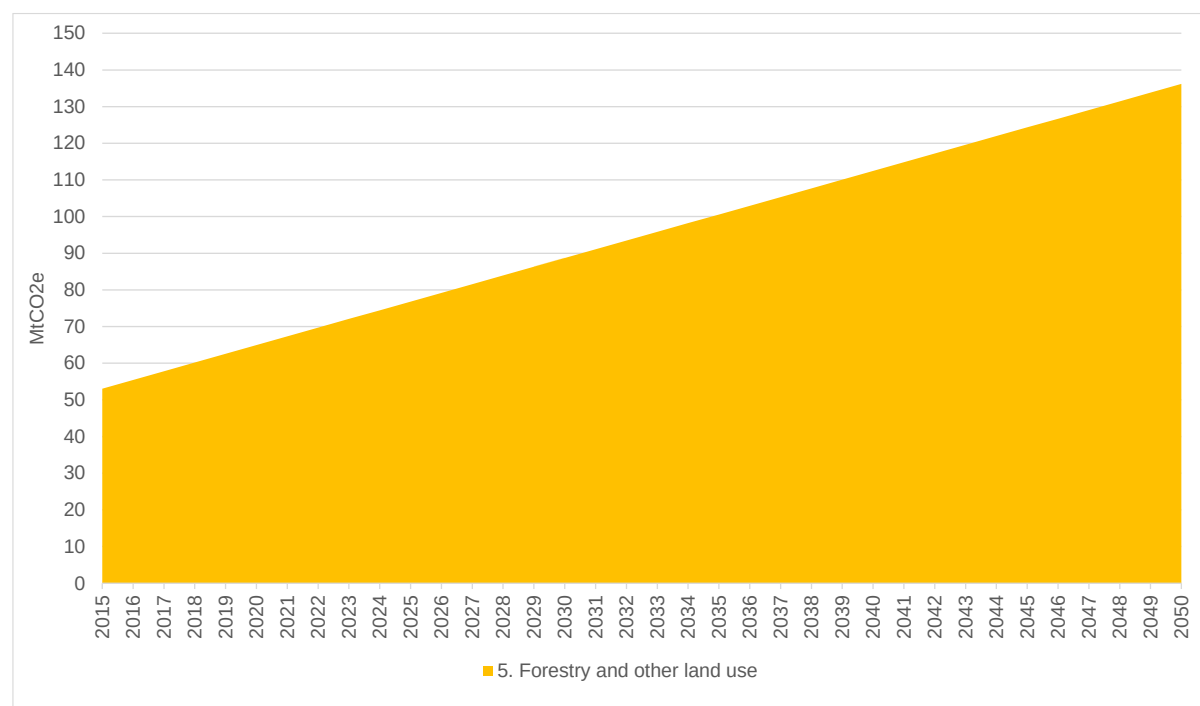
Emissions were calculated from changes in land use by applying default emissions factors. For the land use changes, cropland, grassland and other land use emissions were assumed constant through to

2050 based on information in the MWE (2019a). Emissions from forest land are the only ones changing as per the trend established based on three points; 2005, 2010, 2015 (MWE, 2019a).

GHG emissions projections

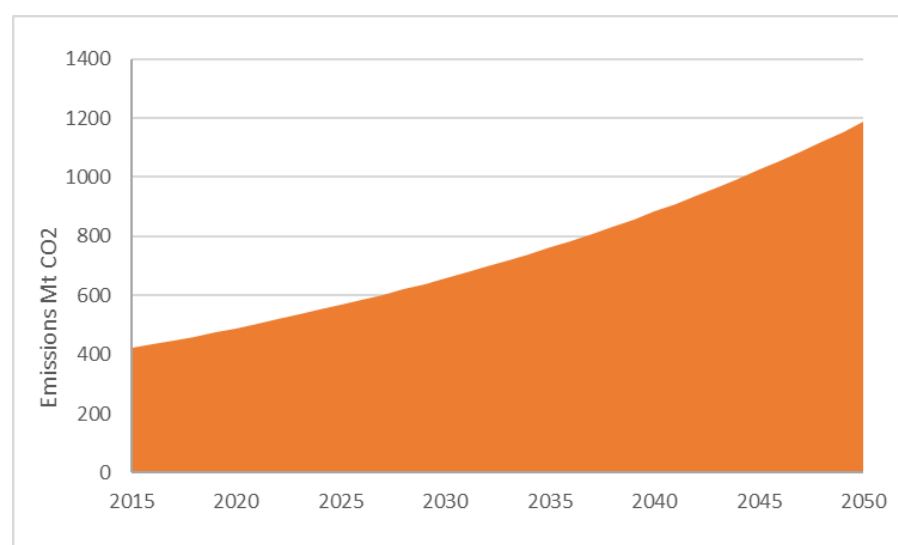
Following the assumptions outlined above, the business as usual emissions projection from land use are as shown in Figure 2-44 and Figure 2-45 below.

Figure 2-44: projected GHG emissions to 2050 from forest land remaining forest land (MtCO₂e)



Forest land remaining forest land (forest degradation) is projected to produce a net total of nearly 140 Mt CO₂ by 2050, up from 39 Mt CO₂ in 2015. This total consists of sources from woodlands (96 Mt CO₂) and THF (48 Mt CO₂). Plantations provide sinks of 6 Mt CO₂ in 2050.

Figure 2-45: projected GHG emissions from wildfires to 2050 (MtCO₂e)



Wildfires are the main sources of GHG emissions in Uganda. They are projected to grow their emissions from the current stable 421 Mt CO₂ to approximately 1200 Mt CO₂ in 2050. The projections are based

on the assumption that emissions will increase in proportion to human population growth rates and the influence of vegetation drying as a result of climate change.

Key uncertainties

Human population growth and future demand of services by people from forest lands are difficult to model. These affect future deforestation and forest degradation rates and their associated emissions.

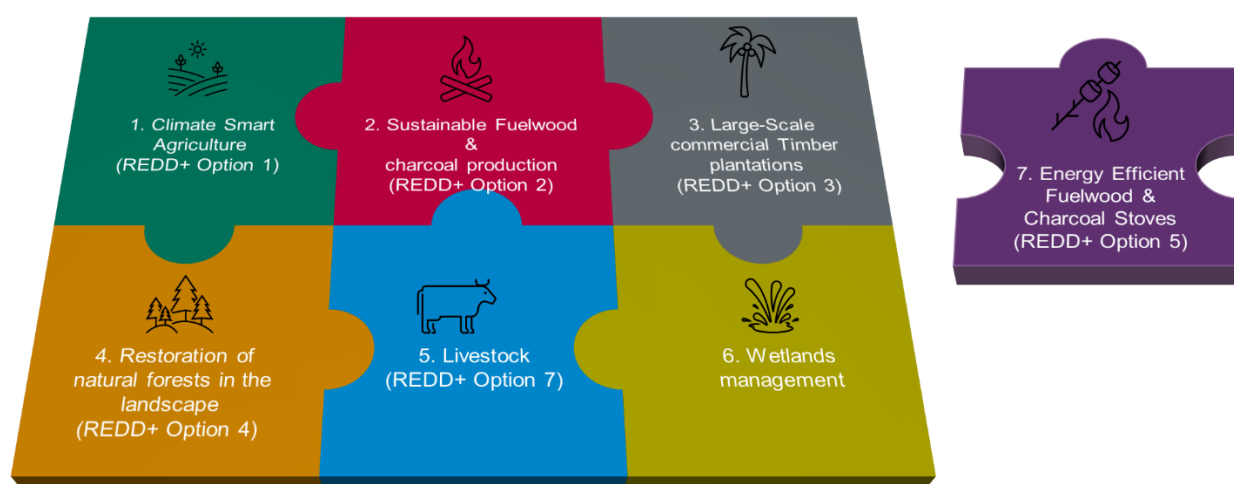
Known main causes of forest degradations are wildfires. Degradations that may be caused by drier conditions in the future are not modelled. Projected higher temperatures and drier conditions have a potential to cause forest degradations that have not been currently investigated.

Mitigation measures across the AFOLU sector have elements that are interlinked. Measures in the FOLU touch on some aspects of Agriculture (i.e. agroforestry and fodder productions). This causes uncertainties in relation to allocation and assessments/investigations of mitigation benefits.

2.14.3 Mitigation options

In the both the agriculture sector and the forestry and land-use sector, the main mitigation actions for limiting emissions will be those included with Uganda's REDD+ Strategy (Ministry of Water and Environment, 2017). The Government of Uganda has launched an annual 40 million trees a day programme which will serve as the main implementation programme for REDD+.

The figure below depicts the seven types of mitigation options that were modelled for the agriculture, forestry and other land use sector.



These measures and described below, along with the assumed savings that will be delivered by the 40 million trees a day programme. These measures concern both the agriculture and forestry sectors.

Climate smart agriculture

This measure includes three mitigation actions, which all aim to reduce the need for agricultural expansion to forest areas by intensifying and increasing agricultural production on existing agricultural land. This in turn reduces deforestation.

2.14.3.1.1 Sustainable Land Management/Use and Agroforestry Practices

It entails growing and managing a combination of fruit trees, fodder trees, fuelwood and construction wood species to address all kinds of tree product needs for the households, so that these would not need to be collected from either forests or non-forest areas.

45% of farming households in Uganda have already adopted these practices, so this action will target the remaining 55%.

2.14.3.1.2 Rainwater harvesting with collection tanks and drop irrigation

Crop yields are generally low in Uganda. Irrigation through rainwater harvesting is promoted in the country to increase crop production and improve food security. It is intended to provide water during dry spells and droughts and to prolong the growing season. Water storage technologies suitable to the local conditions have been considered and promoted. Due to the costs involved, targeted farmers are those that can afford upfront investments.

Irrigation will improve crop productivity and food security. Improved yields will reduce pressure to increase cropland in order to produce enough food for the people of Uganda. Expanding cropland to grow more food occurs at the expense of forest land which reduces national sinks.

The aim of this action is to enable the utilization of rainwater to prolong the two crop cultivation seasons in Uganda by storing rainwater for times when rainfall is not sufficient in crop cultivation season. It is expected that this will double the crop yield.

2.14.3.1.3 Vegetable production in greenhouses

Greenhouse structures will be built in the gardens to increase vegetable productivity. It will be promoted to a fraction of the wealthy households in the country. Greenhouses can conserve moisture, maintain higher temperatures and assist with improved retainment of crop residues after harvests. As a result, the use of greenhouses to increase yields of vegetables can reduce expansion of agricultural land at the expense of forest lands. Improved soil fertility can also reduce use of artificial fertilisers. Their benefits will be to reduce net GHG emissions. This measure will be targeted at 15% of the wealthiest farming households.

Sustainable fuelwood and commercial charcoal production

This measure has three interlinked mitigation actions, which aim to address the energy poverty in the context of climate change by promoting sustainable fuelwood and charcoal production, which in turn will reduce deforestation.

2.14.3.1.4 Commercial small-holder and community bioenergy woodlots

The aim of this action is to produce woodfuel in tree-based bioenergy woodlots all over Uganda. The focus here is on 20 % of the farmer households, which amounts to 866,246 households, each with about 1 ha of farmland. This will allow use of the one hectare of each household for energy wood, fodder and crop cultivation.

2.14.3.1.5 Commercial small-holder and community poles & timber plantations (with coffee agroforestry)

This action will focus on trees, giving the farmer households the option to focus on the pole or timber incomes or on agricultural crops growing in the shade of the trees.

If 3% of the trees planted through the programme go towards this action then it can potentially contribute 0.5 tCO₂e per annum to the country's NDC target by 2030.

2.14.3.1.6 Improved charcoal kilns linked to bioenergy woodlots

In this mitigation measure, one of the two previous measures (particularly the energy woodlot) is used to supply the improved charcoal kiln with a sufficient amount of sustainable wood to make this efficient and sustainable.

Large-scale commercial timber plantations

This strategic option contributes to an Inclusive Green Economy through promoting responsible investments in large-scale commercial transmission pole and timber plantations.

2.14.3.1.7 Commercial transmission pole and timber plantations

It is envisaged that 1.1% of the trees will go towards this initiative, abating approximately 0.2 MtCO₂e per annum by 2030.

2.14.3.1.8 Commercial pole and sawlog plantations

This measure will produce more wood products than is current practice in Uganda. The new products will stem from even selling small pieces of wood as small poles, fuelwood or charcoal.

2.14.3.1.9 Improved charcoal kilns & timber plantation sites

This is similar to the small-holder farmers' improved charcoal kiln action, except there will not be annual wood supply for this measure.

Restoration of natural forests in the landscape

This strategic option aims at restoring and rehabilitating natural forests within the context of climate-smart landscape.

2.14.3.1.10 Designated areas for natural forest regeneration

This measure looks into rehabilitation of natural forests in situations where this can be possible. The measure is designed to be done in collaboration with the communities who would use their own woodlands to obtain services while allowing for the regeneration of the primary forests. This measure will provide GHG mitigation through provision of GHG sinks.

2.14.3.1.11 Restoration of degraded protected natural forest

This measure will allow for degraded natural forests to evolve and regrow with time throughout Uganda. The restorations will be done on national parks, forest reserves and forests on privately owned land. This will be done with the involvement of the communities that live adjacent to the forest lands. The mitigation potential of this measure is in the provision of GHG sinks with time.

Energy efficient cooking stoves

This strategic option aims at promoting clean cooking solution and covers energy efficient fuelwood stoves as well as improved charcoal stoves. Both components target both households and institutions, including hospitals, prisons, cafeterias, etc. The former stove offers some 58% saving in households and about 45% savings in institutions, while that latter offers some 36% saving in households and approximately 45% in institutions.

Livestock rearing in the cattle corridor

This includes change to exotic cattle varieties, cross breeding, establishment of drinking waterdams, establishment of fodder agroforestry plantations, improvement of feed quality, feed supplementing and manure management.

Potential emission reductions estimated using data from Smith et al. (2014) - Breed feed improvements have a potential to reduce emissions by 5-15%, mid-value of 10% assumed. This is based on the fact that grasses in East Africa have good dry matter digestibility (DMD, generally > 65%, Ndungu et al., 2018) and improvement of fodder by creating pastures & agroforestry plantations can improve DMD to approx 70%.

Included in this measure is the Livestock NAMA to Improve feed quality, feed supplements and manure management.

Wetlands and peatlands management

Increase wetland coverage to 12% by 2030, from approximately 10.9% in 2014, through demarcation, gazettement and restoration of degraded wetlands. The proposed measures for wetland restoration will result in approximately 260,000 hectares of new or restored wetlands.

2.14.4 Mitigation scenarios

Key policies and strategies

The Government of Uganda has launched an annual 40 million trees a day programme which will serve as the main implementation programme for REDD+. The assumption below reflect what might be delivered by the programme

Assumptions for the mitigation scenarios

The table below describes the assumptions made for each FOLU measure for the three mitigation scenarios:

Sub-sector	Mitigation measure	WEM	CDP Scenario	WAM Scenario
			(Based on the 40 million trees per year programme = 102,000 ha/yr, up to 50% of the potential land identified in the REDD+ strategy)	(Double the 40 million trees challenge planting rate (i.e 204,000 ha/yr) upto 100% of the land potential identified in REDD+ strategy. Annual ratio ha planted per option remains constant)
Smart Agriculture [REDD+ SO 1]	1.1. Smart Agriculture - Agroforestry	Same as BAU	40% of 40 million annual trees planted; Equaling 43,200 ha planted annually, reaching 1,252,800 ha by 2050 (indigenous fruit and fodder trees)	Same as CDP
	1.2. Smart Agriculture - Rainwater harvesting & irrigation	Same as BAU	Implemented at the same penetration rate as SS 1.1. Implemented in increments of 35,340 ha or in 35,340 new households per annum	Same as CDP
	1.3. Smart Agriculture - vegetables production in greenhouses	Same as BAU	Implemented at the same penetration rate as SS 1.1. Implemented in increments of 11,781 new households per annum	Same as CDP
Sustainable fuelwood and commercial charcoal production [REDD+ SO 2]	2.1. Sustainable fuelwood & charcoal production - Bioenergy woodlots	Same as BAU	9% of 40 million annual trees planted; Equaling 9,600 ha planted annually. Full harvesting in year 10 and replanting on the same land in year 11. This means only using 96,000 ha of land	Same as CDP
	2.2. Sustainable fuelwood & charcoal production - Poles & Timber Plantations	Same as BAU	1% of 40 million annual trees planted; Equaling 1,200 ha planted annually. Full harvesting in year 10 and replanting on the same land in year 11. This means using 12,000 ha of land for this programme	Same as CDP
	SS 2.3. Sustainable fuelwood & charcoal production - Improved charcoal kilns	Same as BAU	Implemented at the rate of tree planting	Same as CDP
Large-scale commercial timber plantations [REDD+ SO 3]	3.1. LARGE SCALE TIMBER PLANTATIONS - Commercial transmission pole and timber plantations	Same as BAU	15% of 40 million annual trees planted; Equaling 16,200 ha planted annually & surpassing REDD+ target in 3 years. Harvesting in year 10 and replanting on the same land in year 11. This means only using 162,000 ha of land	Same as CDP

Sub-sector	Mitigation measure	WEM	CDP Scenario	WAM Scenario
			(Based on the 40 million trees per year programme = 102,000 ha/yr, up to 50% of the potential land identified in the REDD+ strategy)	(Double the 40 million trees challenge planting rate (i.e 204,000 ha/yr) upto 100% of the land potential identified in REDD+ strategy. Annual ratio ha planted per option remains constant)
	3.2. LARGE SCALE TIMBER PLANTATIONS - Commercial pole and sawlog plantations	Same as BAU	15% of 40 million annual trees planted; Equaling 16,200 ha planted annually & surpassing REDD+ target in 2 years. Harvesting in year 10 and replanting on the same land in year 11. This means only using 162,000 ha of land	Same as CDP
Restoration of natural forests in the landscape [REDD+ SO 4]	4.1. Designated areas for natural forest regeneration	Same as BAU	10% of 40 million annual trees planted; equalling 10,800 ha planted annually, surpassing the REDD+ ha target in 10 years, and reaching 313,200 ha by 2050	Same as CDP
	4.2. Restoration of degraded protected natural forest	Same as BAU	10% of 40 million annual trees planted; equalling 10,800 ha planted annually, surpassing the REDD+ ha target in 10 years, and reaching 313,200 ha by 2050	Same as CDP
Energy - Efficient Cooking stoves (REDD+ SO 5)	Energy efficient fuelwood stoves and improved charcoal stoves	Same as BAU	Implemented incrementally from 2.2 % of households & institutions which use wood & charcoal in 2022 reaching 15.7% and 34.5% by 2030 and 2050 respectively	Deploy at 1.7 the rate in CDP - reaching 27.3% and 60.1% by 2030 and 2050 respectively
Livestock (REDD+ Strategy)	Change to exotic cattle varieties and breeding	Same as BAU	No targets provided in the REDD+ strategy. Potential emission reductions estimated using data from Smith et al. (2014) - Breed feed improvements have a potential to reduce emissions by 5-15%, mid-value of 10% assumed. This is based on the fact that grasses in East Africa have good dry matter digestibility (DMD, generally > 65%, Ndungu et al., 2018) and improvement of fodder by creating pastures &	Same as CDP
	Establishment of fodder agroforestry plantations	Same as BAU		

Sub-sector	Mitigation measure	WEM	CDP Scenario	WAM Scenario
			(Based on the 40 million trees per year programme = 102,000 ha/yr, up to 50% of the potential land identified in the REDD+ strategy)	(Double the 40 million trees challenge planting rate (i.e 204,000 ha/yr) upto 100% of the land potential identified in REDD+ strategy. Annual ratio ha planted per option remains constant)
	Establishment of drinking water dams	Same as BAU	agroforestry plantations can improve DMD to approx 70%. Assumption here is that these measures will collectively reduce annual enteric fermentation emissions by 10% (including NAMA on Agriculture below)	
	Livestock NAMA - Improve feed quality, feed supplements and manure management	Same as BAU	Production of improved feed especially for dry seasons [this is supposed to improve feed dry matter digestibility (DMD) from current 65% to about of 70%) - Ndungu et al. (2018)].	Same as CDP
Wetlands and peatlands	Wetlands and peatlands management	Same as BAU	Between 0 and 0.8 MtCO ₂ e per year by 2030: For the aggregated indication, a mid-point value of 0.4 MtCO ₂ e is taken. Increase wetland coverage to 12% by 2030, from approximately 10.9% in 2014, through demarcation, gazettement and restoration of degraded wetlands. The proposed measures for wetland restoration will result in approximately 260,000 hectares of new or restored wetlands.	Same as CDP
Measure	BAU	WEM	CDP	WAM
All AFOLU sub-sectors	-	Same as BAU	Reduction of 30.4 MtCO₂e by 2030	Reduction of 35.514 MtCO₂e by 2030

Key uncertainties

Deforestation and forest degradation are major sources of emissions from FOLU. Reported data on deforestation and forest degradation rates are very variable between government sources and national researchers. FREL (MWE, 2018) reports 50,147 ha of forest which is small compared to the estimates 122,000 ha per year made by national experts (i.e. Kazungu, personal communication in 2021).

REDD+ strategy (MWE, 2017) has in it measures that may affect other non-forest production systems (ie. Fodder production). Mitigation that can be achieved by these other systems are not adequately described.

GHG mitigation projections

[To be added]

2.15 Emissions from managed soils (3C)

2.15.1 Baseline situation

Characteristics of the sector

Managed soils are all soils on land, including Forest Land, which is managed. Sources of emissions from managed soils are organic and inorganic fertilisers and crop residues. The current application rates of fertilisers in Uganda are very low with many farmers not using neither organic nor inorganic fertilisers at all, claiming their soils are fertile enough. Similar to other parts of Africa, fertiliser application rates and overall consumption in Uganda are low. Crop residues are not necessarily retained after harvests and ploughed back to the soil, but they are either used for household fuel or grazed by animals in the fields particularly during dry seasons.

GHG emissions

GHG emissions from managed soils include direct and indirect N₂O emissions and CO₂ emissions from liming (which can be used to reduce soil acidity and improve plant growth in managed systems, particularly agricultural lands and managed forests). Managed soils are largest sources of emissions from non-CO₂ emissions on land. They are followed by biomass burning and rice cultivation. There are, however, no reported emissions from lime and urea applications in the country.

Table 2-53: Estimated non-CO₂ GHG (Gg) emissions from land sources

Source category		2005	2006	2007	2008	2009	2010	2015
Emissions from biomass burning	CH ₄ & N ₂ O	1,101	959	816	674	531	389	
Direct emissions from Managed soils	N ₂ O	3,734	3,983	4,149	4,818	4,941	5,107	
Indirect emissions from managed soils	N ₂ O	1,235	1,329	1,371	1,592	1,616	1,668	
Indirect emissions from manure management	N ₂ O	70	68	48	133	137	-	
Rice cultivations	CH ₄	699	774	815	877	589	596	

2.15.2 Business as usual scenario

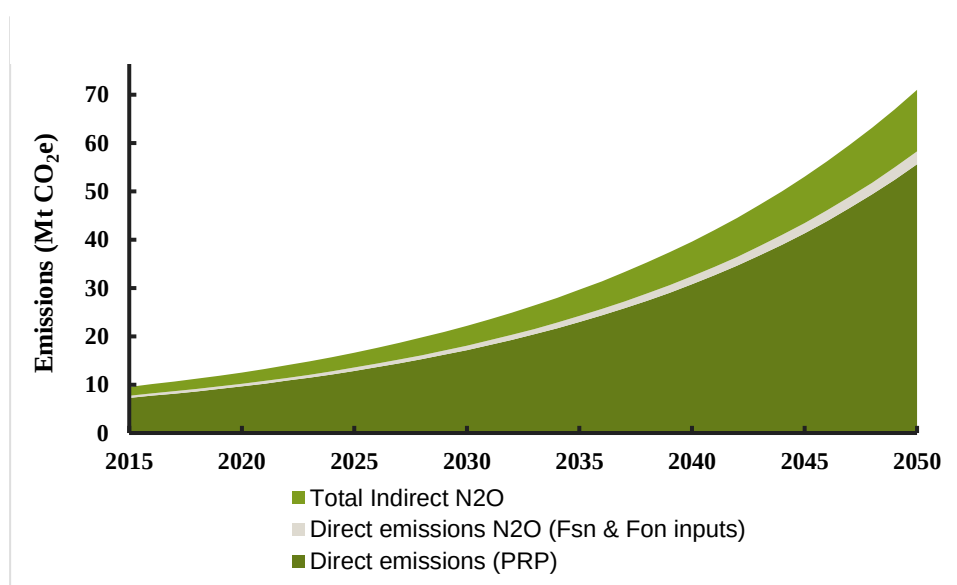
Drivers and assumptions

A key driver of emissions from the managed soils is population growth. The current situation of low application of fertilisers may change in the future, as soil loses fertility and Uganda intensifies crop production to meet growing demand due to population growth.

GHG emissions projections

Following the assumptions outlined above, the business as usual emissions projection from managed soils are as shown in the figure below. Direct N₂O emissions from pasture, range and paddock are the main sources of the emissions due to projected rapid growth of livestock population. The graph excludes future emissions from rice cultivation & non-CO₂ from biomass burning.

Figure 2-46: Business as usual emissions for non-CO₂ GHG sources on land



Key uncertainties

Livestock manure management systems are not fully documented. Data still rely much on expert judgement. This can affect types of GHG emissions as manure management systems influence the emissions produced. If undocumented properly, the growth of dairy sector in the country will increase uncertainties particularly with changes with feeding regimes and evolution of manure handling.

There is no data on the use of agricultural lime and urea in the sector, and there is limited information on the application of inorganic fertilisers. This can affect estimates of emissions from applications of minerals to soil.

There are no national estimates of N from crop residues retained in the fields after harvest. Relied on FAO estimates. The FAO statistics may not consider different uses of residues in the fields (i.e. grazing, burning, retained and ploughed back into the soil).

2.15.3 Mitigation options

No mitigation measures were identified for the managed soils sector.

2.15.4 Mitigation scenarios

Key policies and strategies

There are currently no policies, strategies or commitments to tackle GHG emissions from managed soils. The WEM, CDP and WAM scenarios are therefore the same as the BAU.

Although farmers are encouraged to apply more fertilisers to enhance their yields, there are no indications that fertiliser consumption in Uganda will increase in the near future. Recommendations in the country are for farmers to increase the application rates but affordability of manures remain a major hindrance. Current consumption rates are below recommended quantities for crop commodities. Increase in production is currently addressed by expanding cropland.

Although there are no specified mitigation measures directly related to managed soils, some measures identified for enteric fermentation will have impacts on emissions from managed soils. Reduction of traditional cattle with the promotion of exotic cattle is an example.

2.16 Waste (4)

2.16.1 Baseline situation

Characteristics of the sector

This sector covers solid waste and wastewater treatment. Waste composition is changing rapidly in Uganda particularly in urban areas. Waste generation rate in tonnes per day, is increasing annually in the cities because of growing urbanization. The country generates three types of waste in the form of domestic wastewater, industrial and solid waste. Despite challenges of poor data collection systems, it is estimated that the content of biodegradable material in the country decreased from 88.5% in 1990 to 77% in 2014. Content of plastics in the waste increased from 1.6% to 12.4% during the period. It is estimated that over 755 kilo tons of solid waste was generated in Uganda in 2017. The city of Kampala is responsible for the half of the waste generated in the country.

Volumes of wastewater are also increasing in the country especially in the cities. However, there are limited sewage systems with many households constructing their own septic tanks. The fraction of households that are connected to the municipal sewage system in the cities is estimated to be only 1%¹⁹⁸ with a projected growth to 30% in the next decade. In the absence of wastewater collection and treatment services, households often rely on onsite sanitation systems, such as pit latrines. In this case, GHG emissions depend on the individual use of the system, e.g. poor flush latrines vs. dry latrines, the quality and efficiency of faecal sludge management and the type of faecal sludge treatment.

GHG emissions

The total greenhouse emissions from the waste sector in 2015 were 2,080 Gg CO₂e, of which 70% was associated with Solid waste disposal (i.e. methane emissions from managed and unmanaged waste sites) and the remainder was from waste water treatment and discharge (Ministry of Water and Environment, 2019).

¹⁹⁸ Progress on household drinking water, sanitation and hygiene 2000-2020: five years into the SDGs. Geneva: World Health Organization (WHO) and the United Nations Children's Fund (UNICEF), 2021. Licence: CC BY-NC-SA 3.0 IGO

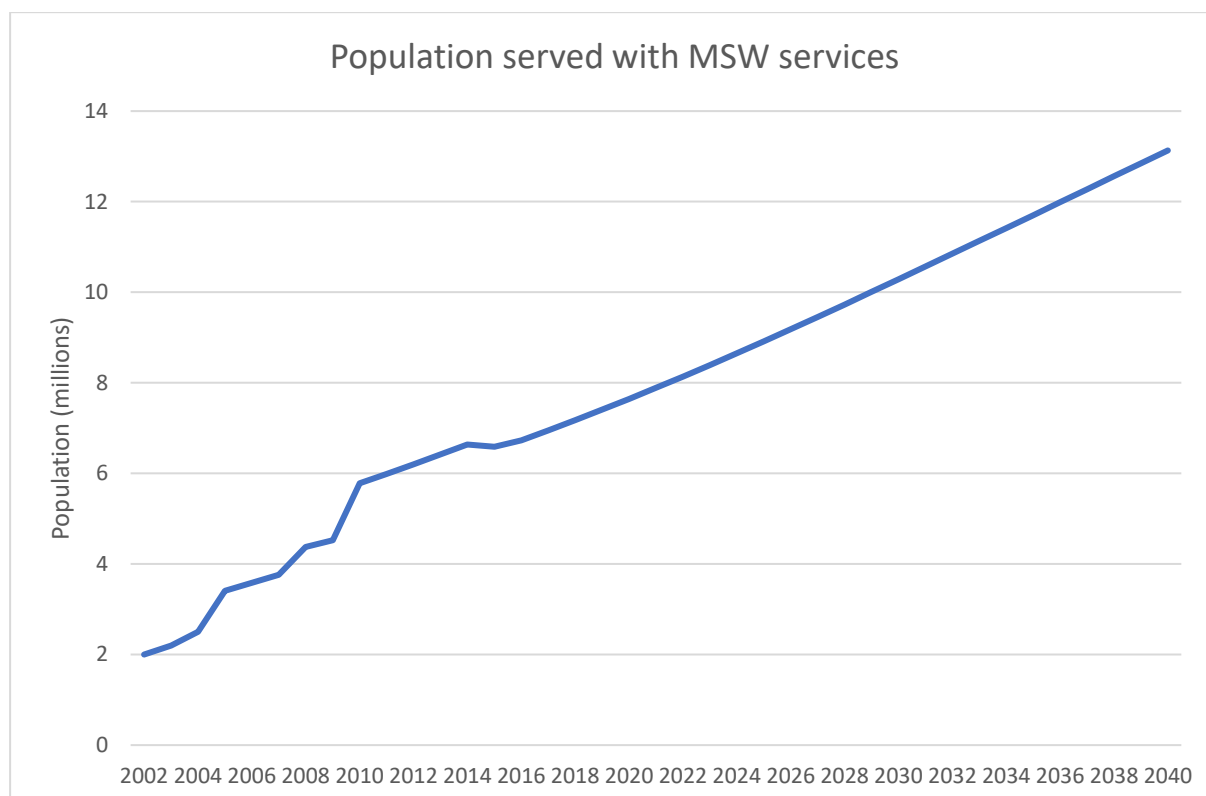
2.16.2 Business as usual scenario

Drivers and assumptions

The key drivers of GHG emissions from the waste sector are population, income levels and type of available waste management systems. More details on these are below.

2.16.2.1.1 Assumptions

It was assumed that the population serviced with waste management services grows in line with national population growth. See figure below.



It was assumed that consumption, waste generation and solid waste management practices remain the same over the projected period. Specific assumptions for solid waste disposal were:

- Waste management technologies remain constant from 2015 (40% unmanaged shallow & 60% managed anaerobic).
- Degradable organic carbon fraction remains constant for all waste types.
- Waste per capita remains constant at average 2002 - 2015 value (237 kg/cap/yr)
- % of waste to managed sites remains constant at 2015 value (60%)
- Waste categorization remains constant at 2015 values (38.7% Food; 34.4% garden; 10.6% paper; 1.3% wood; 0.5% textile; 0.5% nappies & 14% plastics)
- Sludge to managed sites remains constant at 2015 value (0%)
- Amount of methane recovered remains constant at 2015 value (0 kg)

For wastewater treatment and discharge, it was assumed that consumption, wastewater generation and treatment practices remain the same over the projected period, while wastewater generation grows in line with national population growth. Specific assumptions for wastewater treatment and discharge were:

- Degradable Organic component remains at default value.
- Types of treatment and discharge remains constant from 2015 to 2025 (mostly anaerobic deep lagoon; the rest latrine); from 2025 mostly septic tank and latrine, the rest lagoon

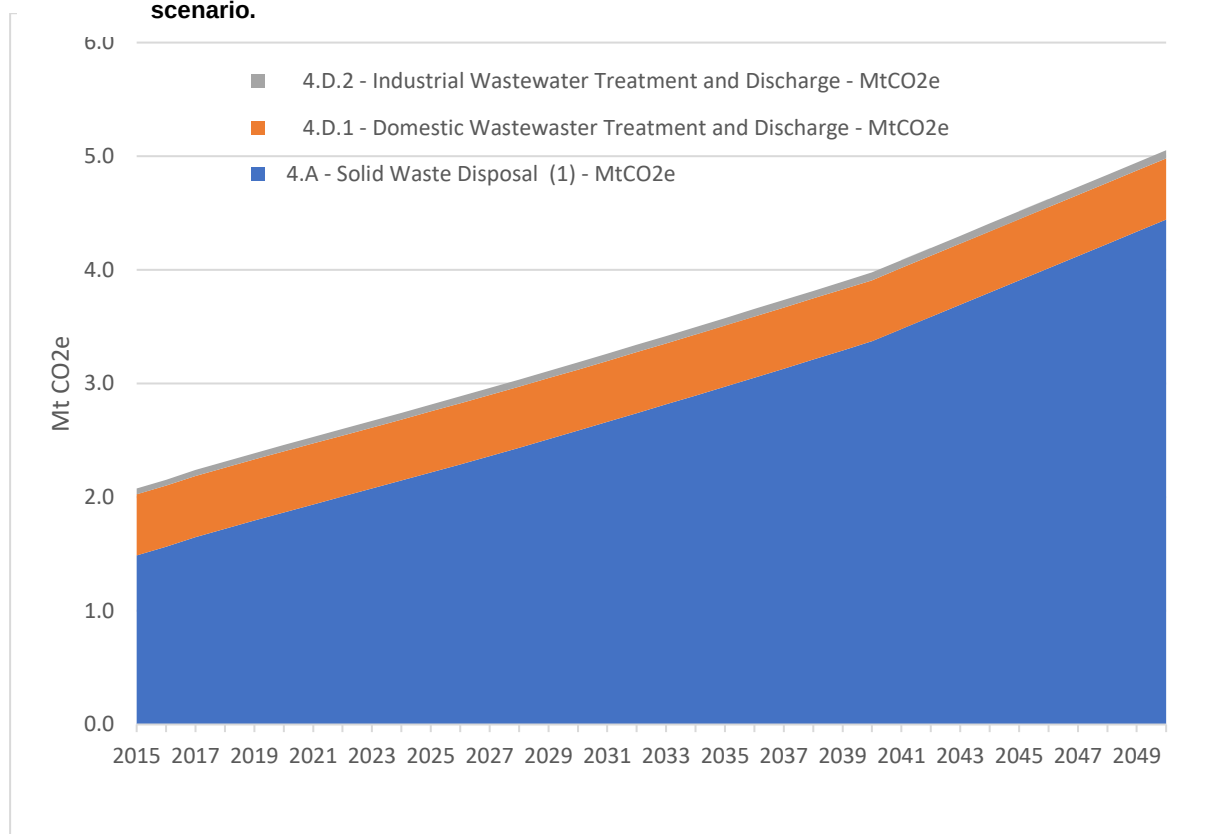
- Fraction of population using different treatments remains constant 2015 to 2025 (mostly anaerobic deep lagoon; the rest latrine); from 2025 mostly septic tank and latrine, the rest lagoon
- Degree of technology utilization remains constant 2015 to 2025 (mostly anaerobic deep lagoon; the rest latrine); from 2025 mostly septic tank and latrine, the rest lagoon
- Per capita protein consumption remains constant as of 2015 (17.888 kg/person/year).

For industrial wastewater treatment and discharge, future activity increases in line with the historical increase for the period 2005 - 2015.

GHG emissions projections

Following the assumptions outlined above, the business-as-usual emissions projection from waste are as shown in Figure 2-47 below.

Figure 2-47: GHG emissions (MtCO_{2e}) from waste sector in Uganda under the business-as-usual scenario.



It is projected that the waste sector emissions will reach 3.19 and 5.59 MtCO_{2e} by 2030 and 2050 respectively.

Key uncertainties

There is little data on national waste profiles. The emissions have been calculated relying heavily on waste characteristics of the waste in the city of Kampala. Waste generation was calculated using average waste per capita and IPCC default factors. The 2015 GHG inventory was used as the basis for projecting the BAU and that inventory contains a lot of gap and shortcomings which were carried over into the emission projections.

2.16.3 Mitigation options

Green cities waste management

This measure will target green development in the large municipal areas. It will involve supporting comprehensive physical planning and efficient waste management (both solid and wastewater) for at least five cities and 15 municipalities (National Planning Authority, 2017)¹⁹⁹. The water management elements (increased water harvesting and efficient water utilisation) are estimated to deliver savings equivalent to 1.1 MtCO_{2e} by 2030.

Wider co-benefits associated with the green cities includes include a clean healthier air to breathe in the cities, cleaner water for domestic consumption and economic use, among others (National Planning Authority, 2017)²⁰⁰.

School biolatrines

This measure is based on the relevant elements of a Nationally Appropriate Mitigation Action (NAMA) for improving cooking and sanitation conditions of schools. It involves the installation of biogas cookstoves fed by biogas from biolatrines with biodigesters (Ministry of Energy and Mineral Development, 2016)²⁰¹. The NAMA is based on 5% of schools, or 1,100 having the measures installed.

2.16.4 Mitigation scenarios

Key policies and strategies

The Green Cities Plan forms part of the Green Growth Development Strategy. Likewise, the school biolatrine measure is already captured within an existing NAMA.

Summary of scenarios

The table below describes the assumptions made for each FOLU measure for the three mitigation scenarios:

Measure	BAU	WEM	CDP	WAM	Notes
Green Cities Waste management	Waste generation based on population growth.	Assumed same as BAU	Support comprehensive physical planning and efficient waste management (solid and wastewater) for at least five cities and 15 municipalities.	Extend this to other municipalities that generate about 50% of the waste generated in the five cities and the 15 municipalities.	Green Growth Development Plan only includes CO ₂ savings estimate for the water management element, so saving will be underestimated.
School biolatrines	Waste generation based on per capita rates. Targeting schools.	Assumed same as BAU	1,100 schools with biolatrines with biodigesters.	Continuation of rate of installation to 2050	CO ₂ savings from the measures are not estimated in the NAMA.

¹⁹⁹ NPA (2017) The Uganda Green Growth Development Strategy 2017/18 – 2030/31

²⁰⁰ NPA (2017) The Uganda Green Growth Development Strategy 2017/18 – 2030/31

²⁰¹ Nationally Appropriate Mitigation Action on Integrated Sustainable Energy Systems for Schools in Uganda. Accessed from: <https://www.ndcs.undp.org/content/dam/LECB/docs/pubs-namas/undp-lecb-Uganda-Green-School-NAMA-2016.pdf?download>

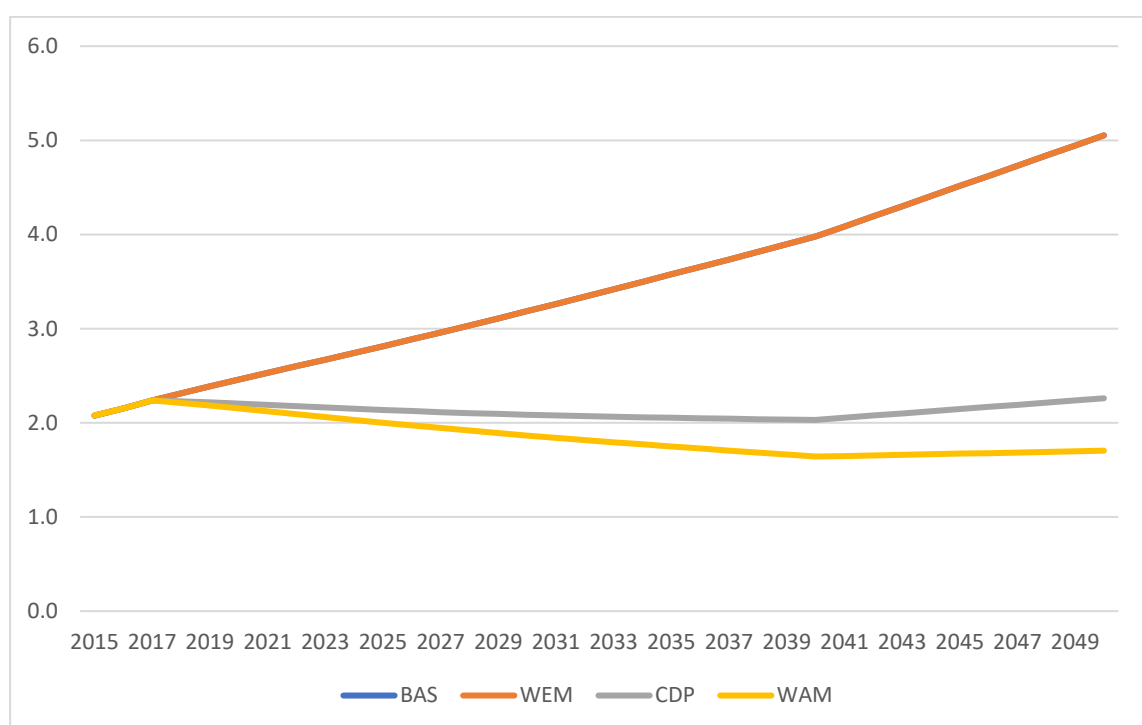
Key uncertainties

Uncertainties related to the growth assumptions, and also to the expected savings from the mitigation measures. Specifically, the green cities waste management option is expected to make a major contribution towards limiting the expected increase of the savings. However, this is uncertainty about how the CO₂ savings from this measure were derived. There is also uncertainty over the savings that might be delivered after 2030.

GHG mitigation projections

The resulting GHG emissions projections based on the assumptions described above are shown below. The business as usual scenario (BAU) is the same as the WEM. In both the CDP and WAM scenarios the mitigation measures are implemented in the 2015-2030 period, but since population continues to grow post 2030, so do the associated emissions.

Figure 2-48: Expected impact on GHG emissions of waste sector mitigation measures



2.17 Setting enhanced economy wide NDC Targets

2.17.1 Current emissions

Uganda's GHG emissions have slowly increased from 53 442 GgCO₂e in 2005 to 90 230 GgCO₂e in 2015. The Land Use and Land Use Change and Forestry (LULUCF) category was the dominant source of emissions, accounting for 59.5% (53 670 GgCO₂e) of the total emissions in 2015. The Land sector was estimated to be a source of emissions instead of a sink. Agriculture is the second largest source, contributing 26.9% in 2015, followed by energy (10.7%) and waste (2.3%). The Industrial processes and product use (IPPU) sector was insignificant in terms of emissions.

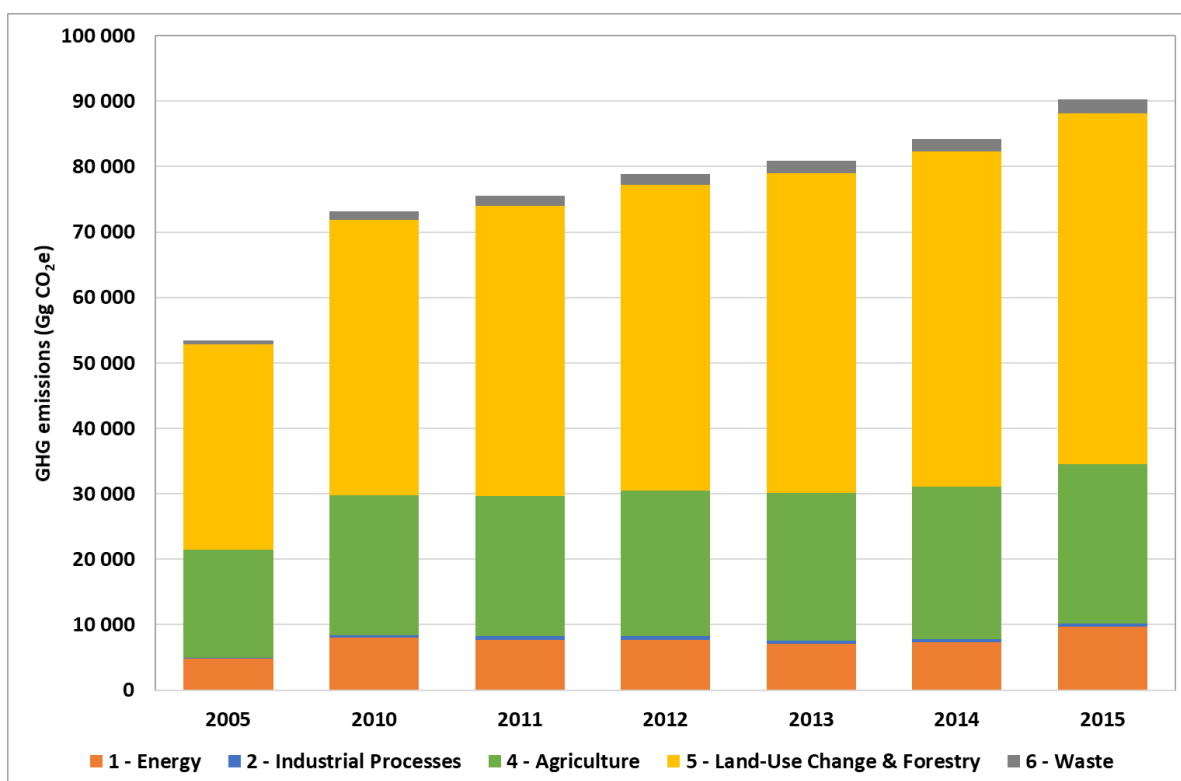


Figure 2-49: Sectoral trends in GHG emissions for Uganda (Source: Ministry of Water and Environment, 2019)

2.17.2 Future projections

The sectoral and sub-sector BAU projections presented in the preceding sections cumulatively make up Uganda's economy-wide BAU emissions. These are projected to increase from 90.1 MtCO₂e in 2015 to 148.8 MtCO₂e by 2030 and 235.7 MtCO₂e by 2050 under the Business-as-Usual (BAU) Scenario. This is illustrated in Figure 2-50 below.

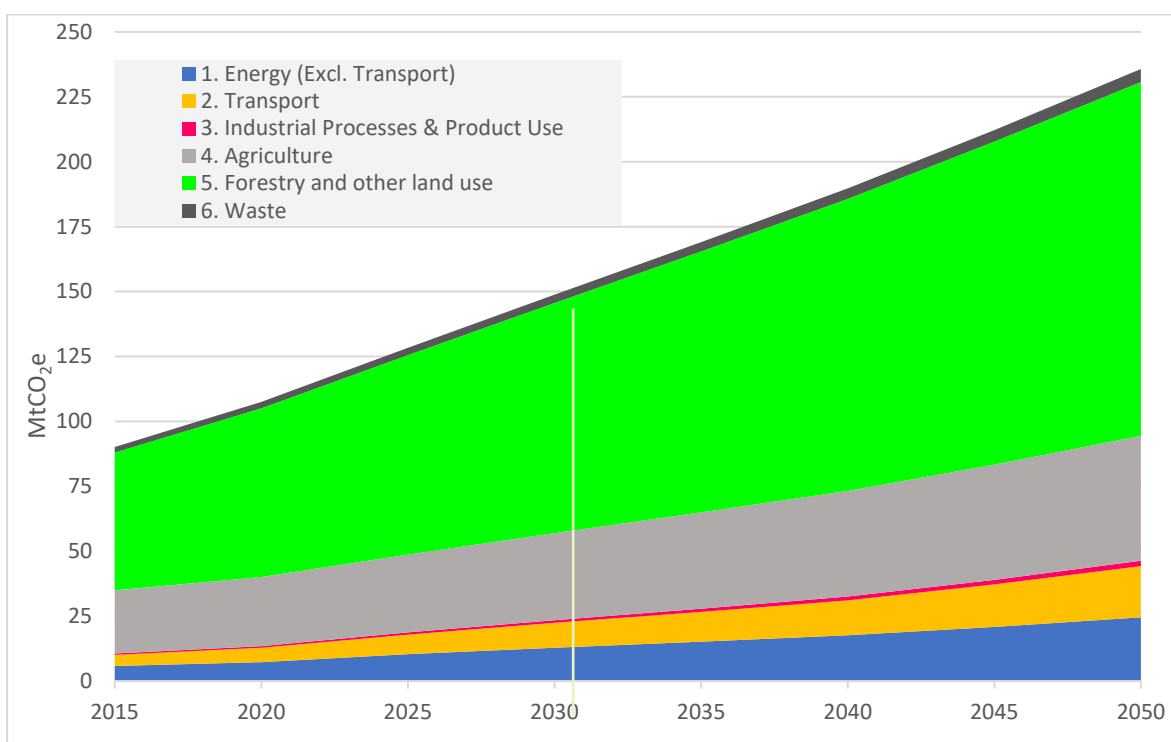


Figure 2-50: Uganda's BAU trajectory

Similarly, the economy-wide WEM, CDP and WAM scenarios are made up of the respective sectorial scenarios as shown in Figure 51 and Figure 52 below.

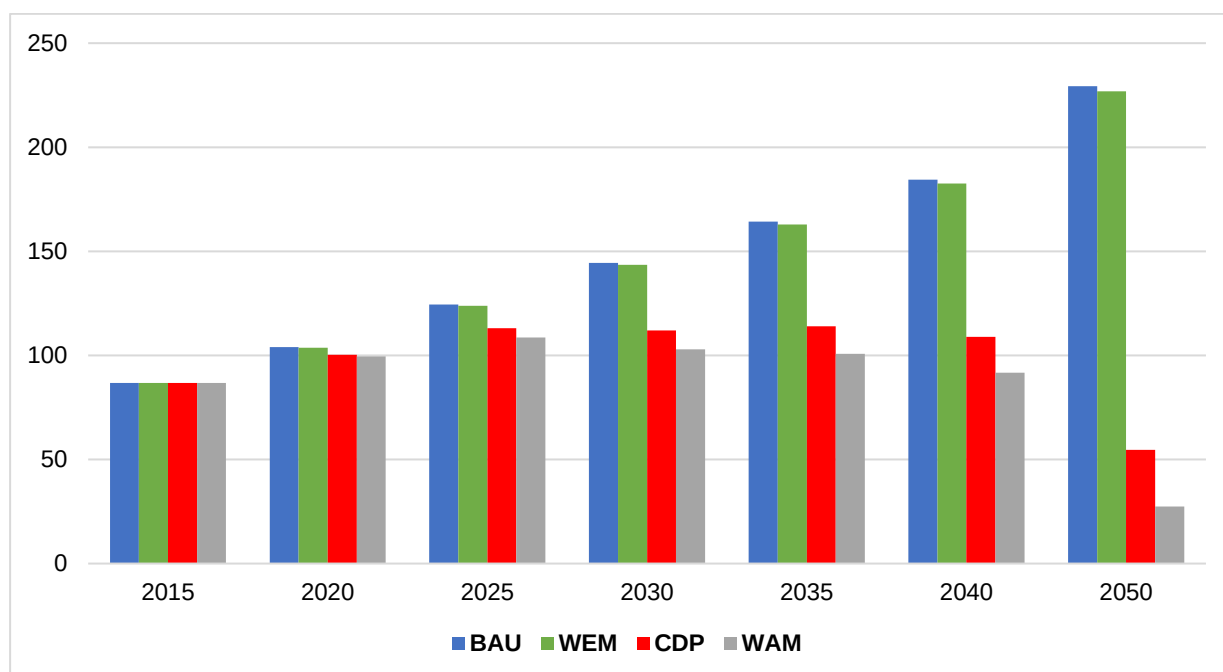


Figure 51: Economy-wide scenarios in 5 year intervals

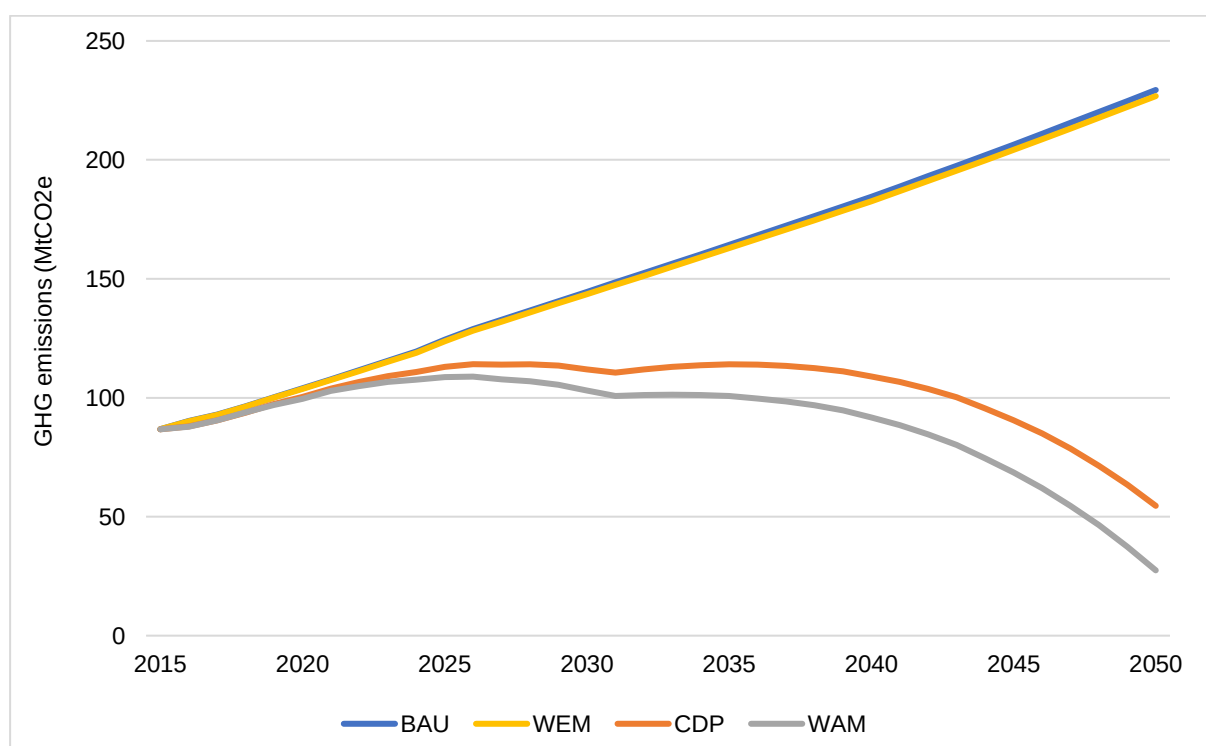


Figure 52: Complete economy-wide scenarios

The figures show that by 2030 emissions under the WEM, CDP and WAM scenarios are projected to be 0.64%, 24.6% and 29.74% below BAU scenario.

In order to ensure that the updated conditional NDC target is progressive compared to the country's first NDC target, while also ensuring that it is achievable, the CDP trajectory was adopted as the ideal trajectory upon which the updated NDC should be based.

However, because the metabolic assessment study carried out at the beginning of this project showed that all sectors will need to be included in the updated NDC, for IPPU, the WAM scenario was adopted because in this sector CDP is the same as BAU. The table below presents the scenarios adopted for the updated NDC for each of the sectors

Table 2-54: Uganda's Conditional and Unconditional Mitigation targets

Sector	Scenario adopted for updated NDC
Stationary Energy	Current Development Plans (CDP)
Transport	Current Development Plans (CDP)
Industrial Processes and Product Use (IPPU)	With Additional Measures (WAM)
Agriculture, Forestry and other land use (AFOLU)	Current Development Plans (CDP)
Waste	Current Development Plans (CDP)

When all the planned and intended mitigation actions under those scenarios have been implemented, it is envisaged that Uganda's net emissions will be reduced by 24.7% below the BAU trajectory by 2030 to 112.1 MtCO₂e (See **Error! Reference source not found.** below).

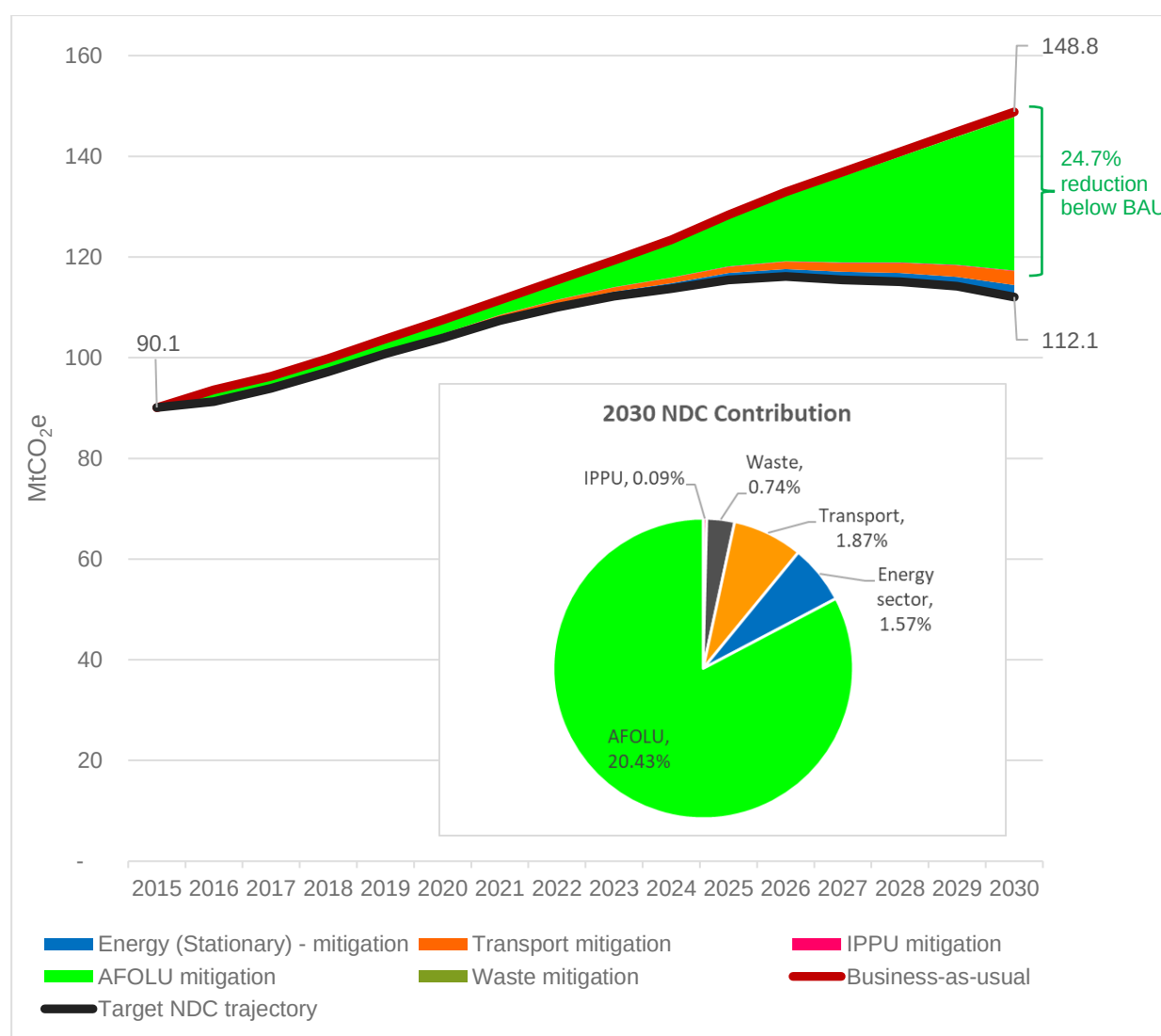


Figure 2-53: Illustration of Uganda's Conditional economy-wide mitigation contribution

The Figure shows that, consistent with the 2015 NDC, 22% reduction below BAU will come from AFOLU (20.43%) and energy sector (1.57%), while the newly added sub-sectors will contribute the additional 2.7%, made up of 1.87% from transport, 0.74% from waste and 0.09% from IPPU.

To determine the country's unconditional target, those mitigation measures that are already being implemented were identified from the various country documents, including the first Biennial Update Report under the UNFCCC. It was found that only four mitigation measures can be classified as already under implementation.

Table 2-55 below presents a summary of the mitigation actions to be implemented in each of these sectors, as well as their envisaged GHG reduction contributions under both conditional and unconditional economy-wide targets.

Table 2-55: Envisaged mitigation contribution under conditional and unconditional targets

No	Main Mitigation Actions	Conditional 2030 Mitigation Contribution (MtCO ₂ e)	Unconditional Contribution (MtCO ₂ e)
	Agriculture, Forestry and Land Use Sector	30.4 MtCO₂e	6.89 MtCO₂e
1	Climate Smart Agriculture – Sustainable land use and Agroforestry	1.18	
2	Climate Smart Agriculture – Rainwater harvesting and irrigation	1.55	
3	Climate Smart Agriculture – vegetables production in greenhouses	0.28	
4	Small-scale Bioenergy woodlots	2.94	
5	Small-scale Poles & Timber Plantations	0.37	
6	Small-scale fuel production – Improved charcoal kilns	3.37	
7	Large-scale commercial transmission pole and timber plantations	4.96	
8	Large-scale commercial pole and sawlog plantations	4.96	
9	Designated areas for natural forest regeneration	0.30	
10	Restoration of degraded protected natural forest	0.30	
11	Energy efficient fuelwood and charcoal stoves	6.89	6.89
12	Livestock management	2.93	
13	Wetlands and peatlands management	0.40	
	Energy Sector (Excl. Transport)	2.34 MtCO₂e	1.57 MtCO₂e
1	Renewable energy generation	0.0003	
2	Reduction in transmission and distribution losses	0.00001	
3	Improved efficiency of charcoal production	0.48	0.48
4	Industrial Energy efficiency	0.26	
5	Industrial Fuel switching	0.23	
6	Increased electricity access for households	0.28	
7	Lighting energy efficiency in households	0.003	
8	Cooking mitigation measures, incl. energy efficiency and fuel switch	1.09	1.09
	Transport Sector	2.78 MtCO₂e	0
1	Road transport fuel efficiency	1.58	
2	Alternative fuel switch	0.54	

No	Main Mitigation Actions	Conditional 2030 Mitigation Contribution (MtCO ₂ e)	Unconditional Contribution (MtCO ₂ e)
3	Modal shift of passengers to Mass Rapid & Non-Motorized Transport	0.66	
4	MGR – Meter Gauge Railway rehabilitation for freight transit	0.0005	
	Waste Sector	1.10 MtCO₂e	
1	Planned Green Cities Waste management	1.10	0.265
2	NAMA – Schools bio-latrines	0.0006	
	Industrial Processes and Product Use Sector	0.14 MtCO₂e	0
1	Clinker substitution in Cement production	0.14	
TOTAL		30.4 MtCO₂e	8.72 MtCO₂e

The unconditional mitigation measures lead to Uganda's net emissions being reduced by 5.9% below the BAU trajectory by 2030 to 140.1 MtCO₂e (See **Error! Reference source not found.** below).

Figure 2-54 and Table 2-56 table below present and compare the annual net emission targets for Uganda's Conditional and Unconditional NDC targets.

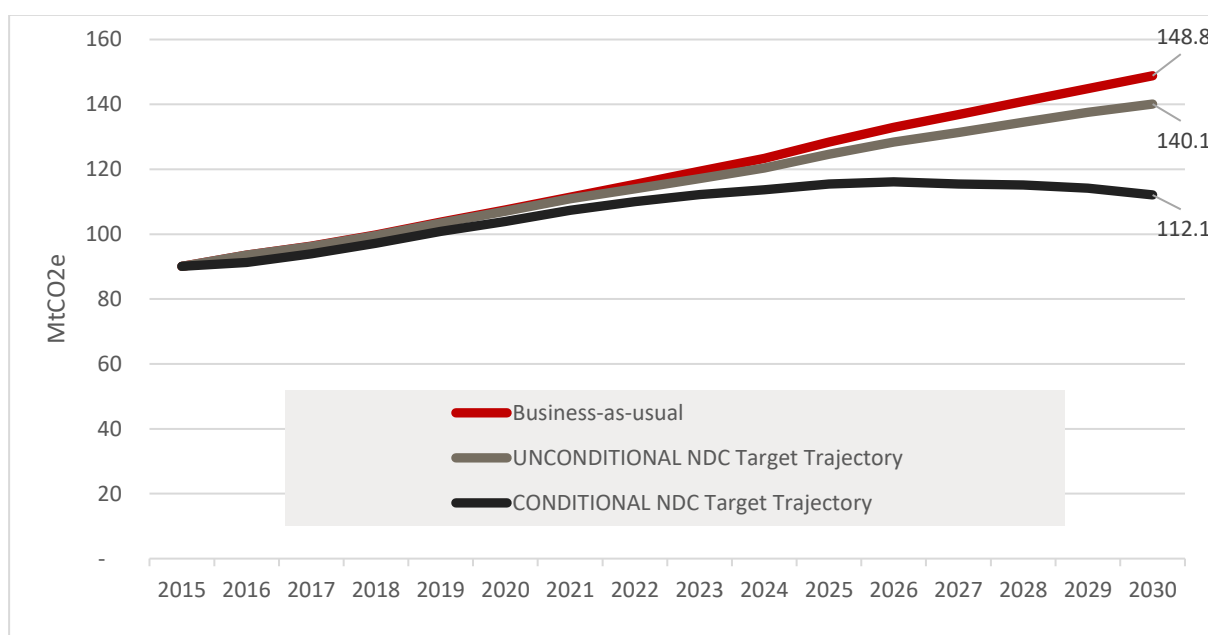


Figure 2-54: Comparison of Conditional and Unconditional NDC Targets

Table 2-56: Implied annual targets of Uganda's conditional and Unconditional NDC targets

Trajectory	Annual targets in MtCO ₂ e																
	Base year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	
Business-as-usual	90.1	93.6	96.3	99.8	103.7	107.5	111.4	115.4	119.4	120.4	128.4	132.9	136.9	140.9	144.9	148.8	
Un-Conditional NDC trajectory	90.1	93.5	96.2	99.6	103.5	107.1	110.9	114.0	117.2	120.4	124.6	128.3	131.4	134.5	137.6	140.1	
Conditional NDC trajectory	90.1	91.3	93.9	97.2	100.8	103.9	107.4	110.0	112.2	113.7	115.5	116.1	115.5	115.1	114.2	112.1	

Detailed descriptions of the envisaged mitigation actions, their respective mitigation impacts and the resulting annual conditional targets for each sector are outlined in the sub-sections below.

Table 2-57: Uganda's Conditional and Unconditional Mitigation targets

	2030 Conditional Target	2030 Unconditional Target
% Reduction	24.7%	5.9%
Reference point	Business As Usual trajectory (Figure 2-50)	Business As Usual trajectory (Figure 2-50)
Details	All planned and intended mitigation actions implemented	Some of the planned and intended mitigation actions implemented

2.17.3 Setting enhanced sectorial NDC Targets

2.17.4 Agriculture, Forestry and Land Use (AFOLU) Sector

Under Business-As-Usual conditions it is projected that AFOLU net emissions will reach 122.2 MtCO₂e by 2030, while with the implementation of all the main mitigation measures envisaged above, it is projected that the net emissions in this sector will be reduced by 24.9% to 91.8 MtCO₂e by 2030. This is illustrated in **Error! Reference source not found.** below.

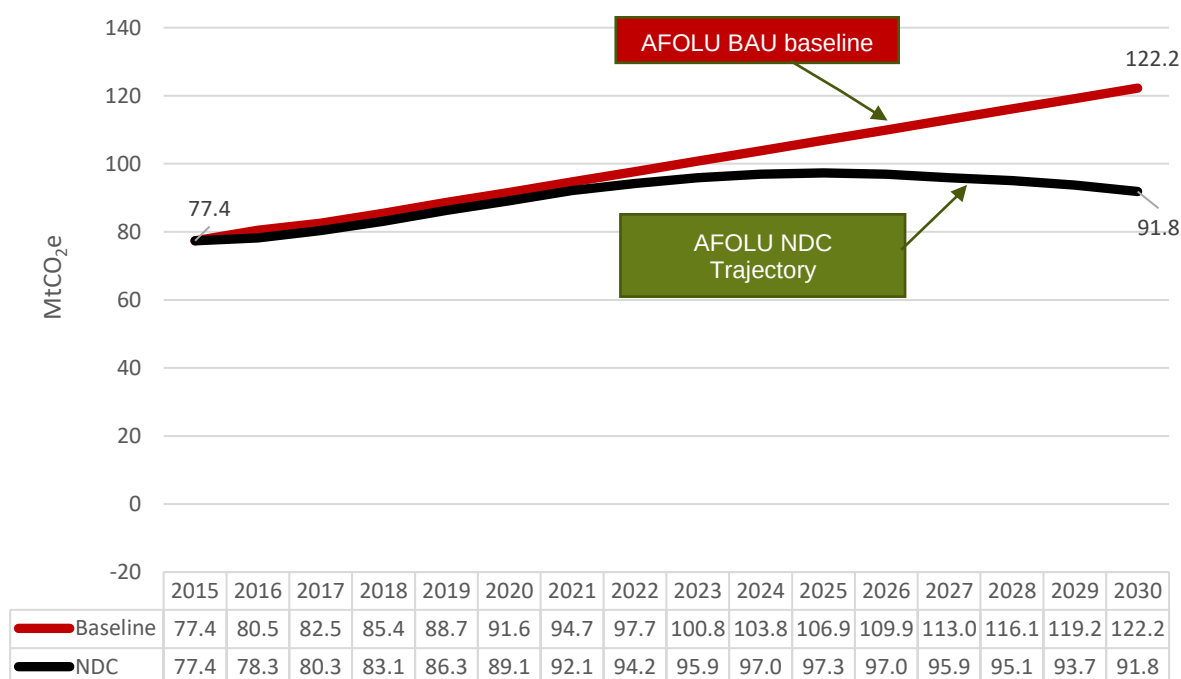


Figure 2-55: AFOLU baseline and annual targets under the conditional mitigation scenario

The figure also presents the annual mitigation targets for the AFOLU sector.

2.17.5 Energy Sector (Stationary)

The emissions under the Business-As-Usual scenario in the Energy sector (stationary) will increase from 5.7 MtCO₂e in 2015 to about 12.4 MtCO₂e by 2030. The main mitigation actions above have a potential to reduce GHG emissions in this sector by 18.8% to 10.1 MtCO₂e by 2030 as shown in the Figure below. The Figure also presents the sector's annual conditional mitigation targets.

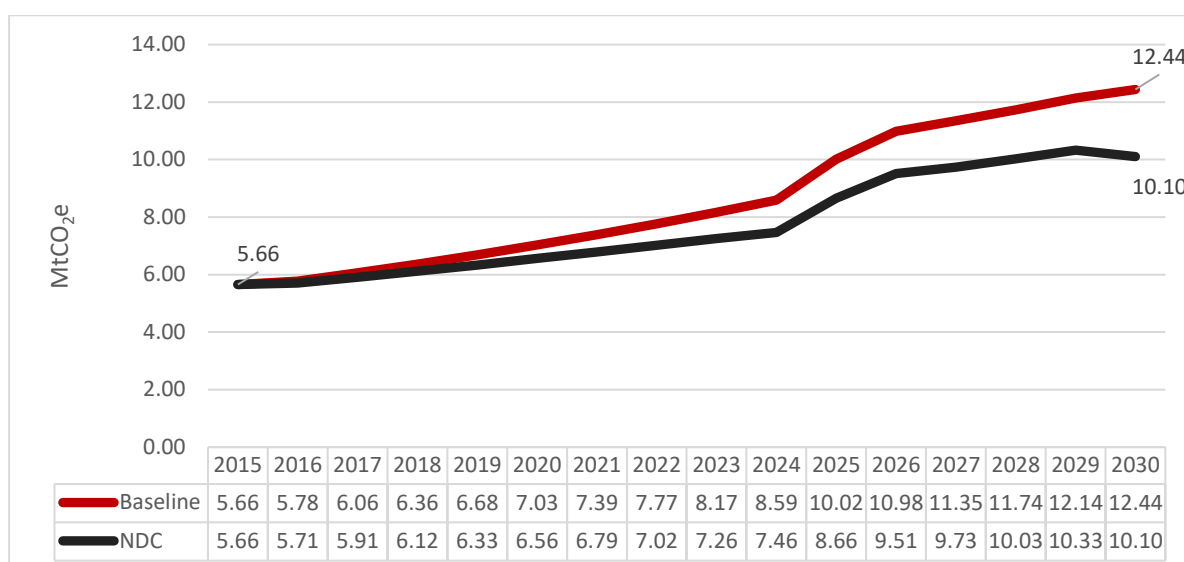


Figure 2-56: Energy sector baseline and annual targets under the conditional mitigation scenario

2.17.6 Transport Sector

Transport sector GHG emissions under Business-As-Usual conditions will more than double from 4.2 MtCO₂e in 2015 to 9.6 MtCO₂e in 2030. If all the main mitigation measures under the NDC scenario are implemented fully, they have the potential to limit the growth of the emissions by 29% to 6.8 MtCO₂e by 2030 as illustrated in the Figure below. The figure also presents annual targets under the conditional transport sector mitigation scenario.

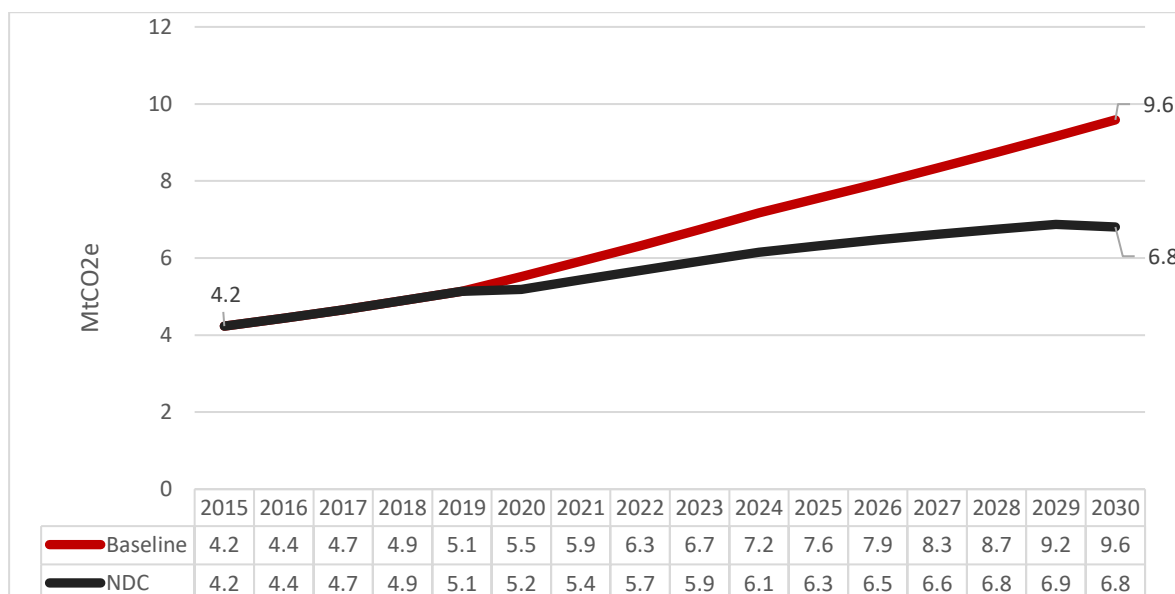


Figure 2-57: Transport baseline and annual targets under the conditional NDC scenario

2.17.7 Waste Sector

Under the BAU scenario waste sector emissions are projected to grow from 2.08 MtCO₂e in the base year to 3.19 MtCO₂e by 2030. If fully implemented, the main mitigation actions are projected to reduce the 2030 emissions by 34.8% to 2.09 MtCO₂e. This is illustrated in the Figure below.

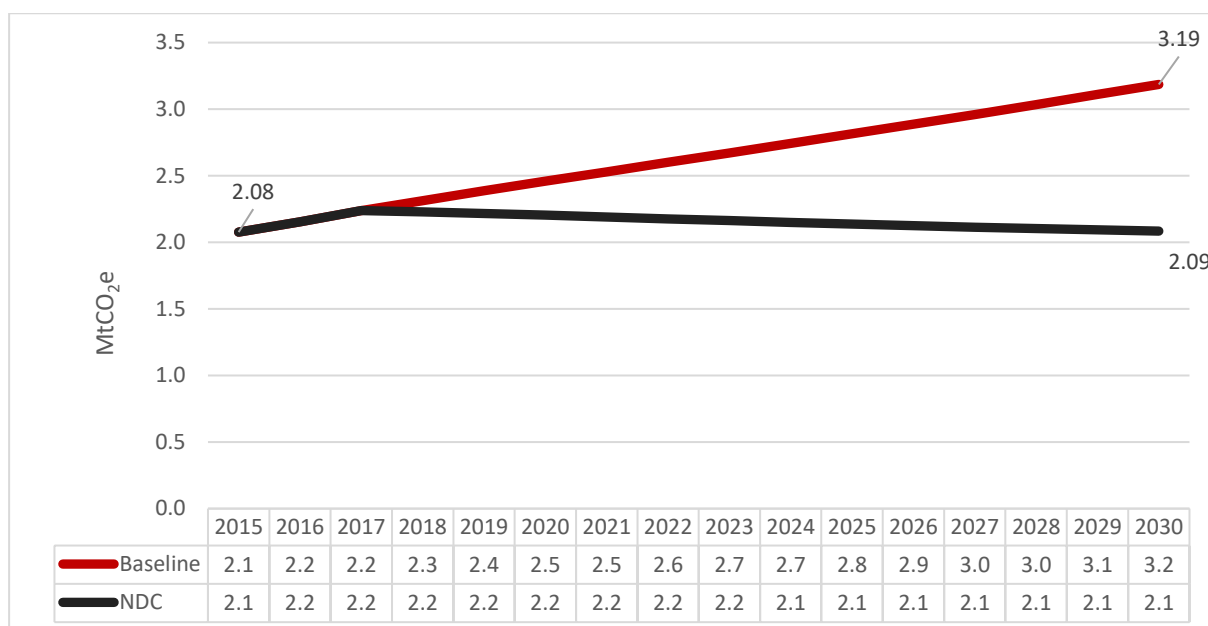


Figure 2-58: Waste sector baseline and annual targets under the conditional NDC scenario

The figure also presents the annual mitigation targets for the AFOLU sector. Planned green cities waste management has high mitigation potential over the schools biolatrines and can account for nearly all the current potential abatements in the NDC scenario.

2.17.8 Industrial Processes and Product Use

Under Business-as-Usual scenario, the industrial processes and product use sector emissions are projected to grow from 0.57 MtCO₂e in 2015 to 1.0 MtCO₂e by 2030. Implementation of the main mitigation action has the potential to reduce the emissions in this sector by 14% to 0.86 MtCO₂e by 2030 as shown in the Figure below. The figure also presents the annual mitigation targets for the IPPU sector.

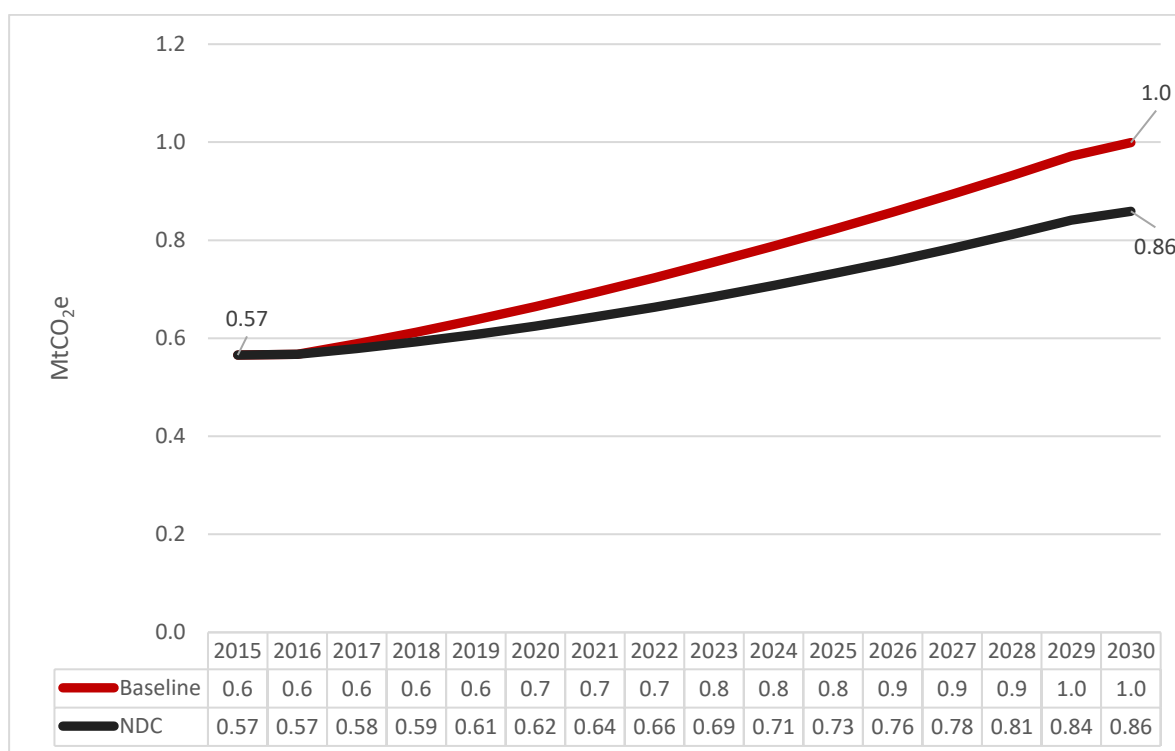


Figure 2-59: IPPU sector baseline and annual targets under the conditional NDC scenario

3 Chapter 3 Enhanced Adaptation Targets

3.1 Climate Change Risks for Uganda

It is recognised that climate change is a serious global challenge and that climate-related impacts may impede economic and social well-being and development efforts, particularly for developing countries such as Uganda. It is therefore essential to assess the relevant risks and vulnerabilities associated with a changing climate and the opportunities for adaptation, to ensure long-term resilience in Uganda.

A summary of some of the key climate change risks for different sectors is summarised below.

Table 3-1: Summary of the risks and vulnerabilities associated with a changing climate for Uganda

Sector	Risks and vulnerabilities
Water and Environment	The environment encompasses the land, vegetation, and biodiversity of Uganda. Under the influence of climate change there has been an increase in extreme climatic events i.e., floods and droughts. The impacts of flooding and drought on the environment and hydrological cycle include soil erosion, land degradation, gradual loss of biodiversity, displaced fauna, pollution (poor water quality) of rural and urban water courses and receiving water. Reduced streamflow, degrading wetlands, and lower dam levels not only is a threat to water supply but can also impact on the potential for hydropower generation. Sanitation concerns include damage and loss of infrastructure and services from floods and increased wastewater being transported into rivers.
Agriculture	The agricultural sector can experience major setbacks due to the increase in floods and droughts. Changing rainfall seasonality is already having an impact on farm crop selection and planting regimes (USAID, 2013). With more rain falling as heavy storm events, rainfall is less effective, there is increased erosion, increased streamflow, and an increased risk of flooding and greater environmental degradation. Higher evaporative demand offsets any benefits should rainfall possibly increase, also resulting in less effective rainfall. These changes also have societal impacts through crop yields. The agricultural sector experiences a loss of income with reduced crop yields, additional expenses to repair equipment and possible re-cultivation of the land.
Forestry	Forestry will be impacted by drought and increasing temperatures which could result in increasing fire risks. Deforestation is also a major contributor to the countries greenhouse gas emissions and impacts on water availability, increasing erosion, and environmental degradation. Higher temperatures are likely to increase the risk of fires. Reforestation and protection of forestry resources is also a critical focus area for reducing GHG emissions and also contributing to increased resilience particular through improved water security.
Fisheries	Climate change is likely to impact on fisheries through changes in lake levels and river flows as well as associated impacts on water quality including temperature, eutrophication, and stratification. Fluctuating lake levels will impact on the critical infrastructure, ports and jetties, necessary to support the fishing community.
Energy	Increased temperatures will increase the risk of forest fires and reduced livelihood options will exert more pressure on forest products contributing to deforestation and associated impacts for water security, biodiversity and land. Increased variability of rainfall could lead to lower lake levels and associated loss of hydropower potential resulting in more frequent power outages and reduce capacity for energy exports from Uganda. Increasing temperatures will increase the requirement for cooling (refrigeration and air conditioning) appliances and this will increase the energy demand in addition to the risk of ozone depleting substances. Increased temperatures also decrease the efficiency of the energy supply infrastructure, i.e., electric generation, transmission and distribution.

Sector	Risks and vulnerabilities
Health	The people of Uganda, in both rural and urban areas, are at risk to various health issues under climate change conditions. Climate variables control the geographical distribution of pests and diseases; therefore, it is possible for diseases to move into new areas as the climate changes. The environment and living standards during a flood or a drought increase people's susceptibility to diseases such as malaria and cholera. Furthermore, water scarcity and food insecurity have an impact on the overall nutrition of the population.
Transport	Depending on the magnitude and frequency, extreme climatic events have the potential to deteriorate, and cause major damage to transportation infrastructure, mainly roads and bridges, and the overall built environment. Industries in Uganda which rely on transport, could potentially be limited in their ability to travel to various parts of the country. Specific risks including increasing temperatures which require updated design guidelines for pavement design, increasing flood and landslide risks and increased variability in lake levels and storms.
Cities and the Built Environment	Uganda is one of the fastest urbanizing countries in Africa and as a result it is critical to address the specific climate change risks associated with urban areas. These include increasing temperatures due to the urban heat island effect as well as increased risk of urban pluvial flooding due to increasing rainfall intensities and insufficient capacity of existing urban drainage systems. Climate change is also likely to have a greater impact in rural areas potentially contributing to a further increase in urban migration and demand for services.
Manufacturing, Industrial Processing and Mining	Floods in Uganda regularly destroy and damage manufacturing and industrial infrastructure worth billions of Shillings. Other climate change related risks pertaining to the manufacturing sector include, fires, hailstorms, increased electricity prices and disruptions. The probability of these risks resulting in material financial impacts to manufacturing companies will increase with climate change over the coming decades. Food and beverage industries, which are dependent on raw materials from agricultural supply chains, are exposed to risks for higher input prices due to reduced supply of input materials from more extreme drought and extreme weather events.
Tourism	Tourism is a sensitive industry which is likely to be impacted by increasing climate change risks. This would be due to damage to infrastructure, the loss of biodiversity and degraded ecosystems. In addition, tourism is one of the main economic sectors that is directly affected by the access to water supply.
Gender	For women and girls across the country, the reality of an increased incidence and intensity of climate related disruptions to basic services such as water supply, have life-changing implications. This includes more time spent searching for safe drinking water, which interferes with school and work attendance and exposes them to a greater chance of encountering violence. Women, particularly in the rural areas, are also likely to be disproportionately impacted due to increasing climate change risks affecting subsistence agriculture and have the most to benefit from the adoption of Climate Smart Agriculture (CSA) as well as improved access to water and sanitation services and better roads.

3.2 The Economic Impacts of Climate Change

The projected impacts of climate change will undermine further advancements in critical development areas such as food security, water resources management, health and economic growth (USAID, 2012). An assessment of the economic impacts of climate change on the agriculture sector in Uganda concluded that while the potential impact on food crops would not be significant (i.e., less than 0.2% of GDP per year, there could be significant impact on export crops (USAID, 2012). It was estimated that climate induced losses for coffee could be in the order of 50 – 75% by 2050, because of a combination

of yield reductions and loss of areas where coffee can be grown. These would represent a major impact on the economy which is currently deriving 18% of its export earnings from coffee. Similar impacts would be likely for both tea and cotton which are also major exports and critical to the economy.

Although generally considered to have a negative impact, there is the potential that climate change could have a positive economic benefit at least in some sectors of the economy. The World Bank study (Cervigni et al, 2015) on Enhancing the Climate Resilience of Africa's Infrastructure (ECRAI) showed that on the whole climate change contribute to up to a 50% increase in hydropower revenue from the Nile Equatorial Lakes region, including for Uganda due to higher lake levels.

An assessment of the potential economic impacts of climate change for Uganda (MWE, 2015) show that if no adaptive action is taken, annual costs could be in the range of US\$3.2 - 5.9 billion within a decade, with the biggest impacts being on water, followed by energy, agriculture, and infrastructure.

Table 3-2: Summary of costs (US\$Mn.) of inaction to climate variability and change (Source: MWE, 2015)

	2025 LB	UB	2050 LB	UB	Total 2010 - 2050 LB	UB
Agriculture	293	513	1,401	2,458	22,200	38,300
Water	2,437	4,499	5,538	10,255	120,356	222,225
Energy	338	338	10,433	10,433	123,600	123,600
Infrastructure*	94	505	581	3,857	6,478	52,747
Total Costs to Uganda	3,162	5,855	17,963	26,983	272,634	436,872
As percent of PPPGDP	3.01	5.57	2.31	3.48	2.80	4.49

Notes:

LB/UB: Lower/Upper bound estimates obtained from projections under SSP1/RCP4.5 and SSP5/RCP8.5 scenarios respectively
Damages to infrastructure are expected values based on expected frequency of extreme events. GDP projections are based on SSP1, which has annual growth rates for Uganda from 2010-2050 of 7.8% per annum

*Loss in the capacity of infrastructure to withstand climate change

Even if there were no further increases in climate impacts, the cost of inaction would rise over time because of an increase in population. If we assume a growth path close to the Uganda Vision 2040, the economy will expand enormously by 2050 – to about 20 times its size in 2010. Under this scenario, the potential damages due to climate change also increase at a similar rate and the estimates amount to approximately 2-4% of cumulated GDP over the period. If the economy does not see this level of growth, damages will be slightly less, but as a percentage of GDP they will still rise.

The same study by MWE (2015) determined that the total spending required on adaptation actions from 2015 to 2020, based on selected adaptation options from the Costed Implementation Strategy, is estimated at about US\$406 million. On an annual basis this amounts to about 5% of net official assistance received by the country and 3.2% of total government revenues (excluding grants). In future the adaptation budgets will rise significantly: to US\$644 million for 2021-2025 and US\$596 million for 2026-2030, and it is likely that, as a share of government revenues in those years, the demands for adaptation will rise. However, the cost of inaction is much greater, estimated at around US\$3.1-5.9 billion a year by 2025, a range that is around 24-46 times greater than the proposed adaptation budget. These costs combine current climate variability with future climate change, and therefore some of these costs will occur regardless of the expected climate impacts. Many of the proposed adaptation options also have significant co-benefits that cannot easily be quantified.

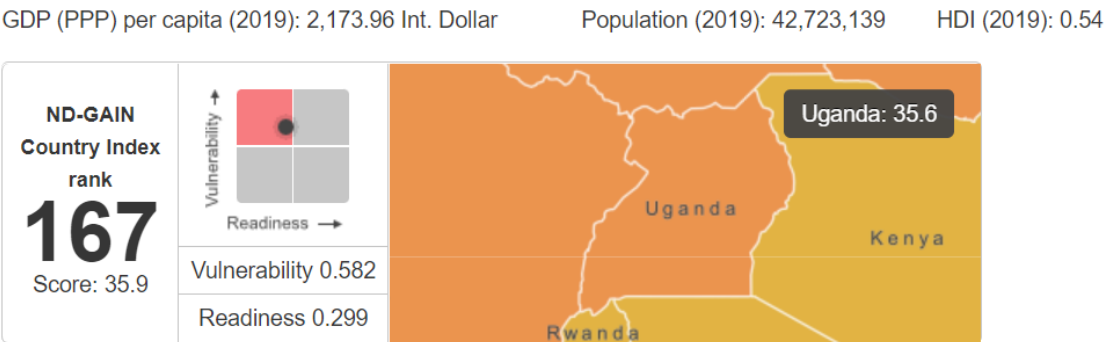
3.3 Climate Change Risks and Vulnerabilities

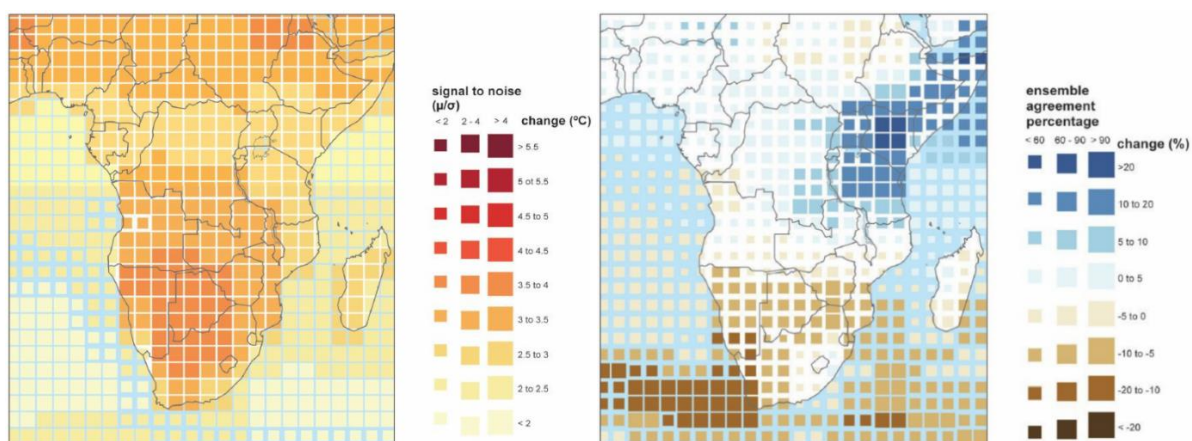
According to the Notre Dame Global Adaptation Initiative (ND-GAIN), Uganda is the 10th most vulnerable country and the 37th least ready country. The high vulnerability score and low readiness score of Uganda places it in the upper-left quadrant of the ND-GAIN Matrix. It has both a great need for investment and innovations to improve readiness and a great urgency for action.

A comprehensive climate change risk and vulnerability assessment was undertaken as part of the Long-Term Scenarios project (MWE, 2020). The outcomes of this study, and previous studies are described

in the following sections. In addition, these results were presented at a series of regional workshops where stakeholders contributed to the assessment of specific risks, vulnerability and the identification of priority actions addressing both climate change mitigation and adaption.

Uganda





*From 1960-1990 baseline climate, percentage change in average annual precipitation, averaged over 21 Climate Model Intercomparison Project 3 (CMIP3) models

Figure 3-2: Percentage change in average annual temperature (left) and rainfall (right) for Africa by 2100 from 1960-1990 baseline climate (Source: IPCC AR5 Report)

Table 3-3: Summary scenario 1 (RCP 2.6) (Source: LTS Report, 2020)

Parameter	Projected impacts
Reports reviewed	(UNFCCC/ The Republic of Uganda, In draft) only.
National	
Temperature	Increase in national mean annual temperature of ~0.5 to 1°C by 2050 and of ~1 to 1.5°C by 2100 . All parts of the country will get warmer. Increases in mean temperature are broadly consistent annually and seasonally.
Rainfall	Increase in mean annual rainfall of ~0-10mm by 2040 and of ~10mm by 2100 . Seasonally, the long rains (MAM) are projected to decrease, whereas JJA, SON and DJF rains are projected to increase.
Regional	
Central	Temperature: Least increase in mean annual temperature over Lake Victoria area. Rainfall: Least decrease in mean annual rainfall over Lake Victoria area.
Western	Temperature: Greatest increase in mean annual temperature.

Table 3-4: Summary scenario 1 (RCP 4.5) (Source: LTS Report, 2020)

Parameter	Projected impacts
Reports reviewed	(Baastel/ FAO, 2016) (Ministry of Water and Environment, 2015) (UNFCCC/ The Republic of Uganda, In draft) (World Bank, 2015)
National	
Temperature	All reports indicate an increase in mean annual temperature of ~1.4 to 2°C by 2050 and of ~1.5 to 2.5°C by 2100 . Fewer colder periods and more warmer periods. Increase in extreme daily temperature events is likely. Projected warming is greatest in Uganda's coolest seasons (MAM, JJA). (Ministry of Water and Environment, 2015) projects higher mean annual temperatures across Uganda than the other reports.
Rainfall	Decrease in mean annual rainfall of ~0-10mm by 2040 and of ~10mm by 2100 . Little change to current conditions. There is a small possibility of increased periods of drought but, in general, limited change overall to rainfall characteristics. Increase in extreme daily rainfall events is considered unlikely. (Ministry of Water and Environment, 2015) projects a decrease in mean annual rainfall throughout Uganda, while (UNFCCC/ The Republic of Uganda, In draft) projects an increase throughout most of the country. (World Bank, 2015) shows an increase in projected rainfall over the Northern Region and a decrease over the Central and Western Regions.

Parameter	Projected impacts
	Increase in seasonal rainfall for the DJF season (up to 100% greater than present), indicates a longer wet season that extends from SON towards DJF.
Regional	
Northern	Temperature: North Eastern Drylands will experience the greatest increase in mean annual temperature (Baastel/ FAO, 2016) Rainfall: Higher intensity of rainfall in the north than in the south (World Bank, 2015). Overall decrease in mean annual rainfall levels (Baastel/ FAO, 2016).
Eastern	Temperature: Less increase in mean annual temperature around Lake Kyoga/ Lake Kwanja (Baastel/ FAO, 2016) Rainfall: Largest decrease in mean annual rainfall on the low-lying Kyoga plains (Baastel/ FAO, 2016)
Central	Temperature: Lowest increase in mean annual temperature increase over the Victoria area (Ministry of Water and Environment, 2015) (UNFCCC/ The Republic of Uganda, In draft) (World Bank, 2015) Rainfall: Decrease in mean annual rainfall, largest change in the Lake Victoria area (Ministry of Water and Environment, 2015) (UNFCCC/ The Republic of Uganda, In draft) (World Bank, 2015)
Western	Temperature: As national Rainfall: As national

Table 3-5: Summary scenario 1 (RCP 8.5) (Source: LTS Report, 2020)

Parameter	Projected impacts
Reports reviewed	(Baastel/ FAO, 2016) (Ministry of Water and Environment, 2015) (UNFCCC/ The Republic of Uganda, In draft) (World Bank, 2015) (World Bank, 2020)
National	
Temperature	Increase in mean annual temperature of ~1.8 to 3°C by 2050 and of ~3 to 5°C by 2100. The percentage of cooler days will decrease significantly Likely increase in extreme daily temperature events During JJA (currently the coolest season) temperatures will rise by up to 5.5°C. (Ministry of Water and Environment, 2015) projects higher mean annual temperatures than the other reports.
Rainfall	One report projects an increase in mean annual rainfall due to large seasonal changes in rainfall and an increase in rainfall intensity on wet days (UNFCCC/ The Republic of Uganda, In draft). Others project a decrease in mean annual rainfall (Baastel/ FAO, 2016) (Ministry of Water and Environment, 2015). Where a decrease is projected, mean annual rainfall is expected to reduce by ~10mm by 2040 and by ~20mm by 2100. Decreases in seasonal rainfall will be greater than decreases in mean annual rainfall. A significant increase in seasonal rainfall for the DJF season (up to 100% greater than present) indicates a longer wet season that extends from SON towards DJF. An increase in extreme daily rainfall events is unlikely.
Regional	
Northern	Temperature: By 2100, the highest relative increase in mean annual temperature is in the Eastern regions/ North Eastern Drylands (Baastel/ FAO, 2016) Rainfall: A decrease in mean annual rainfall is projected in the north, although the intensity of rain on wet days will increase. (World Bank, 2015) (Baastel/ FAO, 2016)
Eastern	Temperature: Low relative increase in mean annual temperature in the Low-lying Kyoga plains and around Lake Kyoga/ Lake Kwanja (Baastel/ FAO, 2016) Rainfall: An increase in the intensity of rain on wet days by 2100 (World Bank, 2015) but an overall decrease in mean annual rainfall (Baastel/ FAO, 2016). The Low-lying Kyoga plains will experience the greatest decrease in mean annual rainfall.
Central	Temperature: Least increase in mean annual temperature over the Victoria area. Rainfall: A large decrease in mean annual rainfall around Lake Victoria (-20%) (Ministry of Water and Environment, 2015). All reports project reduced mean annual rainfall over the Central Region and Lake Victoria except for (World Bank, 2020), which projects increased mean annual rainfall over almost all of Uganda.

Parameter	Projected impacts
Western	Temperature: The Albert area will experience the greatest increase in mean annual temperature. Rainfall: As national.

In August 2021, the latest report from the International Panel on Climate Change (IPCC) on the future climate change risks was released. This report confirmed the likelihood of increasing climate change risks, particularly for Africa where the increase since the previous studies was the greatest. The latest scenarios for annual temperatures, precipitation and rainfall intensity for Africa are shown in Figure 3-3.

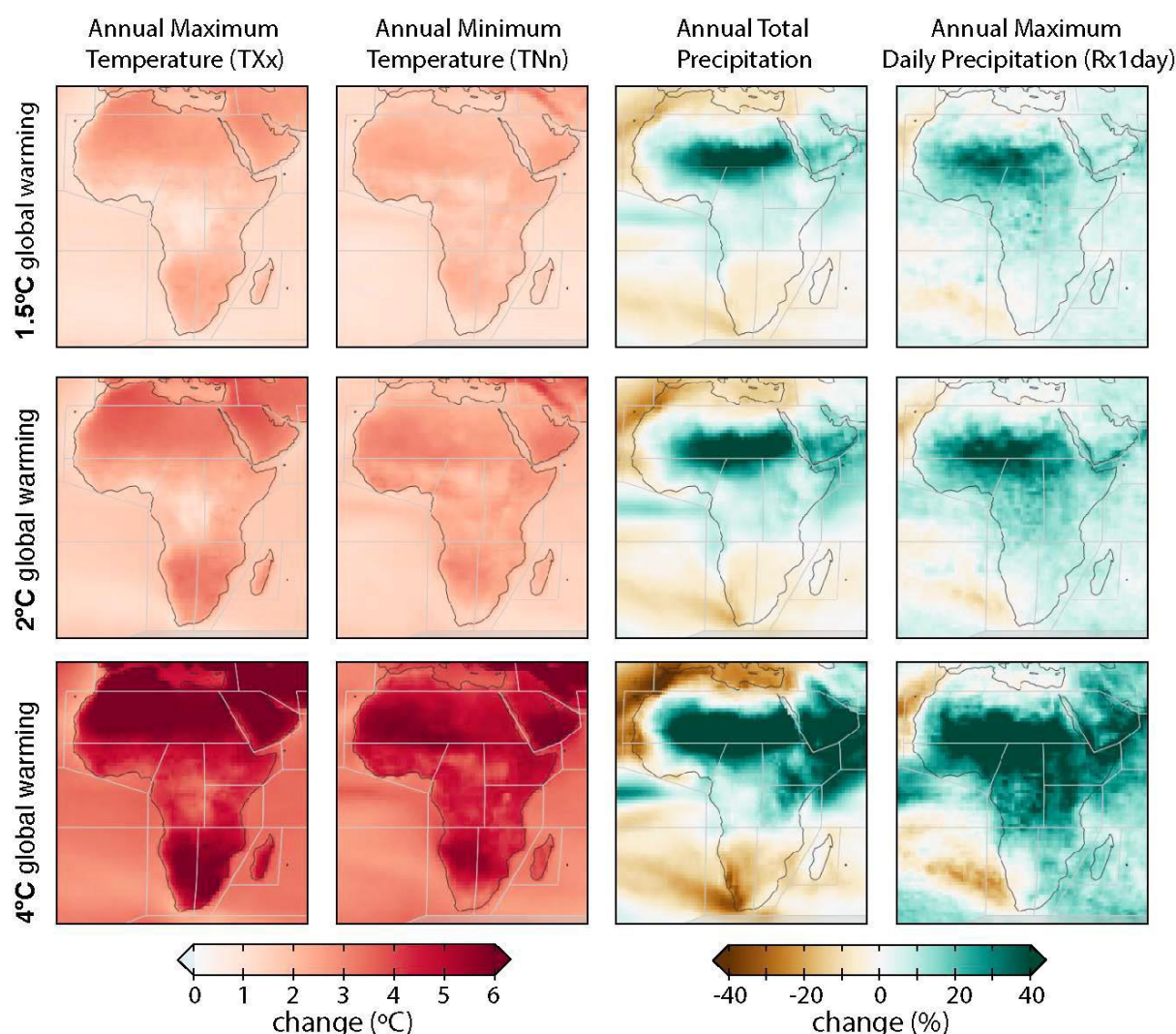


Figure 3-3: Future climate change scenarios for Africa – median impacts by 2100 (IPCC, 2021)

One of the main conclusions from the AR6 report is that the latest scenarios show even greater likelihood for increasing temperatures for Africa, and that this increase is greater for Africa than for any other region. The results also confirm that much of East Africa is likely to get wetter by the end of the century, with significant increases in the maximum daily rainfall, although there is also likely to be increased variability, which could also contribute to increasing risks of both floods and droughts.

3.5 Risk, Vulnerability and Priority Sectors for Adaptation

As a basis for identifying adaptation actions in the LTS study, a sectoral vulnerability and risk assessment was carried out. The vulnerability and risk assessment and subsequent deliverables make systematic use of the definitions of terms from the Intergovernmental Panel' Fifth Assessment Report (AR5) (IPCC, 2014) to assess climate vulnerabilities and potential impacts. As such, it defines:

- Climate vulnerability as a function of climate sensitivities and adaptive capacities
- Climate impact as a function of climate vulnerability and exposure to hazards
- Climate risks potential as a function of climate impacts and the likelihood of hazardous events,
- Adaptation as the process of reducing climate sensitivities, and/or increasing adaptive capacities, and/or reducing exposure to hazards.

These key terms are defined in Table 3-6 below.

Table 3-6: IPCC AR5 Definitions and Key Terms

Term	Definitions	
Vulnerability	“Propensity or predisposition to be adversely affected. Vulnerability encompasses... sensitivity or susceptibility to harm and lack of capacity to cope and adapt.”	
	Climate sensitivity	“Degree to which a system or species is affected, adversely or beneficially, directly or indirectly, by climate variability or change.”
	Adaptive capacity	“Ability of systems, institutions, humans, and other organisms to adjust to potential damage, take advantage of opportunities, or respond to consequences”
Impact	“Effect on a natural or human system... the interaction of climate changes or hazardous climate events occurring within a specific time period and the vulnerability of an exposed society or system.”	
	Hazard	“Potential occurrence of a natural or human-induced physical event or trend or physical impact that may cause loss of life, injury, or other health impacts, as well as damage and loss to property, infrastructure, livelihoods, service provision, ecosystems, and environmental resources.”
	Exposure to hazards	“The presence of people; livelihoods; species or ecosystems; environmental functions, services, and resources; infrastructure; or economic, social, or cultural assets in places and settings that could be adversely affected.”
Risk	“The potential, when the outcome is uncertain, for adverse consequences on lives, livelihoods, health, ecosystems and species, economic, social and cultural assets, services (including environmental services) and infra-structure.”	
	Likelihood	“The chance of a specific outcome occurring, where this might be estimated probabilistically “
Adaptation	“The process of adjustment to actual or expected climate and its effects. In human systems, adaptation seeks to moderate or avoid harm or exploit beneficial opportunities.”	

The vulnerability and risk assessment was performed at both a national and sectoral level. The following sectors were assessed: Agriculture; Built infrastructure; Business; Energy; Fisheries; Forestry; Health; Manufacturing; Mining and Quarrying; Transport; Water; Ecosystems. Where relevant, the assessment was further detailed into sub-sector and performed at a regional level.

For each sector a vulnerability and impact rating was assigned based on extensive desk research and the expert judgement of the national and international adaptation experts in the project team.

To assess risks, “likelihood” was estimated based on a synthesis of climate change scenarios. The scenarios that were used are described in Section 3.2. Two timescales were considered: 2040/2050, as this is the time horizon addressed by Uganda’s LTS; and 2080/2100 (long-term), in order to enable consideration of longer-term risks, as requested by the Ugandan Government.

Ratings and their interactions are described in Figure 3-4 below.

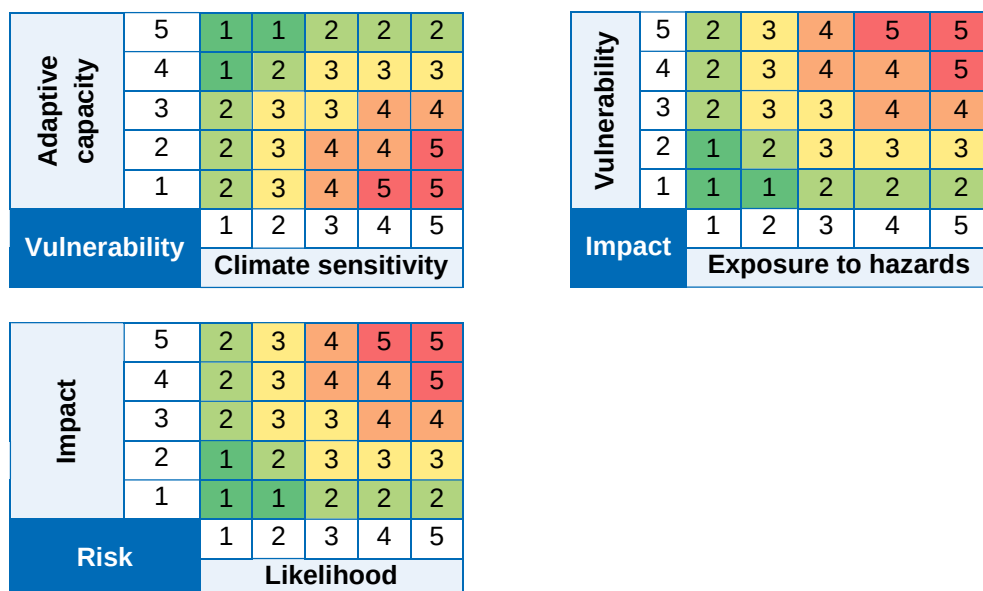


Figure 3-4: Rating of vulnerability and impact in relation to climate variability and change

Ratings were attributed to each sector, sub-sector and region based on the definitions described in Table 3-7. Final ratings for individual sectors and sub-sectors are interpreted as described in Table 3-8.

Table 3-7: Ratings of climate sensitivities and adaptive capacities

Climate sensitivity rating	
1	Insensitive to climate
2	May be sensitive to climate
3	Climate sensitive
4	Very climate sensitive
5	Extremely climate sensitive
Adaptive capacity rating	
1	Major challenges to adjust or respond within existing climate limits
2	Minor challenges to adjust or respond within existing climate limits
3	Able to adjust or respond within existing climate limits but major challenges beyond them
4	Able to adjust or respond within existing climate limits but minor challenges beyond them
5	Able to adjust or respond regardless of climate
Exposure to hazard	
1	Asset with few or no unit with little to no value is not exposed to the hazards.
2	Asset with some or few units of moderate value has some exposure to the hazards.
3	Asset with many units of moderate value has some exposure to the hazards.
4	Asset with some or few units of important value is highly exposed to the hazards.
5	Asset with many units of important value is highly exposed to the hazards.
Likelihood	
1	Very low likelihood

2	Low likelihood
3	Medium likelihood
4	High likelihood
5	Very high likelihood

Table 3-8: Interpretation of the vulnerability and impact ratings

Vulnerability rating		Impact rating	
1	Highly resilient	1	Very low impact
2	Resilient	2	Low impact
3	Potentially vulnerable	3	Medium impact
4	Vulnerable	4	High impact
5	Highly vulnerable	5	Very high impact
Risk rating			
1	Very low impact, very low likelihood		
1	Very low impact, low likelihood		
1	Low impact, very low likelihood		
2	Medium impact, very low likelihood		
2	Low impact, low likelihood		
2	Medium likelihood, very low impact		
3	Medium impact, high likelihood		
3	High impact, medium likelihood		
4	Medium impact, high likelihood		
4	High impact, medium likelihood		
5	Very high impact, high likelihood		

As a result of the vulnerability and risk assessment, sectors were provided a vulnerability and a risk score from 1 to 5 (least to most vulnerable, and least to most at risk). The resulting sectoral analysis is presented in **Error! Reference source not found.** below. Further detailed results by subsectors and regions are provided in the full LTS *Risk and Vulnerability Assessment Report* (REF), and also in an interactive Excel workbook.

As part of the Long-Term Scenarios (LTS) study, a sectoral vulnerability and risk assessment was carried out at both a national and sectoral levels. Sectors were provided a vulnerability and a risk score from 1 to 5 (least to most vulnerable, and least to most at risk). The resulting sectoral analysis is presented in **Error! Reference source not found.** below.

The Risk and Vulnerability Assessment shows that the priority sectors for adaptation in Uganda remain as Water, Environment, Agriculture and Forestry (**Error! Reference source not found.**). These were also identified as priority sectors in the initial NDC. However, there is also significant risk in other sectors, particularly in terms of the general business sector, mining and quarrying, transport, and the built environment. With a high dependency on hydropower, the energy sector is also potentially at risk due to climate change as it is also closely linked to the water and the environmental sector.

For all sectors, a significant cross-cutting issue remains disaster risk sector, which has also been identified as a priority area for adaptation globally. It has also been noted that due to Uganda's rapid rate of urbanisation, that there should be greater focus on the urban sector. This would help to address the increasing risk and vulnerability in the business, transport and built environment sectors.

Table 3-9: Results from the vulnerability and risk assessment, by sector

Sector	Vulnerability rating	Risk rating
Agriculture	4.07	4.18
Built infrastructure	3.27	3.83
Business	3.75	4.00
Energy	3.40	3.40
Fisheries	3.00	4.00
Forestry	3.25	4.25
Health	3.67	3.33
Manufacturing	3.00	3.00
Mining and quarrying	3.00	4.00
Transport	3.20	3.80
Water	4.63	4.75
Ecosystems	5.00	5.00

Table 3-10: Priority sectors for adaptation based on LTS risk and vulnerability ratings.

Vulnerability Rating	Risk Rating	Sector	Supporting Strategies and Plans
5.00	5.00	Ecosystems	Protection of critical ecosystems and Nature Based Solutions (NBS) (UNDP Report on NBS for Uganda)
4.63	4.75	Water	Improved Water Security. Strategic Investment Plan for the Water and Environment Sector, Uganda (2018)
4.07	4.18	Agriculture	Climate Smart Agriculture (CSA) practices. National Adaptation Plan for the Agricultural Sector (2018)
3.25	4.25	Forestry	Preventing deforestation, rehabilitation of natural forests, and support for Nature Based Solutions (NBS). REDD+ Strategy and Plan (2017)
Cross-cutting		Disaster Risk Reduction	Investments in Early Warning Systems, Flood Protection, Disaster Risk Reduction, Insurance Systems and Recovery.
New Cross-cutting		Urban Areas	Future Cities Africa Uganda Second Cities Report

In terms of identifying the priority sectors for adaptation, the water and environment sectors were combined as they are closely linked and are both covered by the same Ministry. Addition priority sectors that should also be retained from the original NDC are Health, Energy and Transport.

A comparison of the priority sectors for adaption in the original NDC compared to the recommended priority sectors for adaptation as part of the updated NDC are given in **Error! Reference source not found..** Specific adaptation actions have been identified under each of these sectors and are described in a later section.

Table 3-11: Updated Priority Sectors for Adaptation

Priority Sectors in NDC 2015	Priority Sectors in NDC 2021
Agriculture	Water and Environment (including Sanitation)
Forestry	Agriculture
Water	Forestry
Infrastructure	Fisheries
Energy	Energy
Health	Health
Risk Management	Urban, Transport and the Built Environment
	Disaster Risk Reduction (Cross-cutting)

3.6 Adaptation Pillars

As part of the LTS Study, thirteen adaptation “pillars” were identified in response to the risk and vulnerability assessment. These twelve pillars are given below, and a series of associated adaptation actions were also identified under each. These have been taken into consideration in updating the NDC, however they represent more of a long-term vision, while the NDC is focused on the short term.

Table 3-12: Adaptation Pillars identified in the Long-Term Scenarios (LTS) study

Pillar	Description
1	Promote climate resilient and low carbon agricultural production and value chains
2	Promote and apply land management practices that support sustainable and productive use
3	Promote climate-resilient water supply systems and increase water supply capacity
4	Promote sustainable management of ecosystems and the use of nature-based solutions, including through community engagement
5	Strengthen climate information services through improved data collection and sharing infrastructure
6	Develop and promote a clean and resilient energy system
7	Promote climate resilient and low-carbon urban planning and development
8	Increase availability and promote access to finance for climate-resilience, low-carbon investments and climate impacts recovery
9	Promote education on climate change science, and sharing of indigenous knowledge
10	Mainstream and institutionalise climate change responses in policies, plans, programmes and budgets at all levels of governance
11	Promote a multimodal shift to low carbon mobility and create climate-resilient transport infrastructure to support economic growth
12	Promote a resource-efficient circular economy
13	Further develop a resilient and “fit for future” health systems

3.7 Policy alignment for Adaptation Actions

Many of the identified adaptation actions have been recommended in multiple policy documents. The priority adaptation actions in terms of policy alignment are shown in Table 3-13. These have also been linked to the Adaptation Pillars from the LTS showing that they cover multiple aspects of adaptation.

Table 3-13: Priority adaptation actions in terms of policy alignment and linked to the LTS adaptation pillars.

Sector	Program or Action	Policy Alignments	LTS Adaptation Pillar
Forestry	Encouraging agro-forestry	9	1
Agriculture	Expanding Climate Smart Agriculture (CSA) practices	8	1
Agriculture	Expanding value addition, efficient harvesting, post-harvest handling and storage and access to markets, including micro-finances	7	1
Water and Environment	Ensuring increased access to water supply to key economic sectors, especially agriculture	7	3
Water and Environment	Improving water catchment protection	7	4
Forestry	Promotion of reforestation, afforestation and sustainable management of forests (including in urban areas)	7	4
Disaster Risk Reduction	Expanding climate information and early warning systems	6	5
Transport and the Built Environment	Develop standards and guidelines for climate proofing of urban built infrastructure, through revision of design codes and land use so as to embed resilience thinking.	5	8

3.8 Prioritization of Adaptation Actions and Targets

A multi-criteria assessment based on the criteria in the table below was used to prioritize the actions and targets.

Criterion	Brief description
Adaptation	Role of the action is reducing climate sensitivity, increasing adaptive capacity or reducing exposure.
Mitigation	Level of emission reduced or increased through the action

Impact	Criterion	Brief description
Economic	Job-creation	Employment created or lost through the action
Economic	Competitiveness	Competitiveness gained or lost through the action
Social	Social inclusion	Increases or reductions in social justice through the action, and integration of local communities in its development and implementation process (E.g. youth)
Social	Gender	Increases or reductions in gender equality through the action

Impact	Criterion	Brief description
Environmental	Ecosystems	Damages, protection or restoration of ecosystems caused by the action

Criterion	Brief description
Coherence	Role of the action is helping to achieve the objectives identified as part of national and sectoral development plans

Criterion	Brief description
Costs	Amount of investment required to complete the action, and the extent to which external financing is needed

A scoring of -5 to +5 was used to come up with the priority adaptation actions and targets presented in the section below.

3.9 Priority Adaptation Actions and Targets

In terms of adaptation, Uganda has an overall objective, enshrined in the country's National Development Plan III, to promote inclusive climate resilient development at all levels. In the original INDC (2015), while priority adaptation sectors and actions were identified, there were no targets set for these sectors. In this updated NDC, the country's economy-wide and sectorial adaptation targets and related prioritised actions have been presented. Adaptation actions were identified from the existing NDC as well as the National Development Plan (NDPIII) and other relevant strategies and plans such as the Adaptation Action Plan for Agriculture and the Water and Environment Investment Strategy. The identified actions were grouped under the relevant sector and aligned with the LTS Adaptation Pillars.

A summary of the identified actions and sub-actions for each sector are given in Table 3-14.

Table 3-14: Summary of actions and sub-actions per sector and links to the LTS adaptation pillars

Sector	Number of Actions	Number of Sub-actions	Adaptation Pillars
Environment and Ecosystems	7	5	2, 4 and 5
Water and Sanitation	9	11	3
Agriculture	3	38	1, 3 and 5
Forestry	7	0	1, 2, 4 and 6
Fisheries	2	25	2
Energy	5	0	6 and 12
Transport	6	1	11
Manufacturing, Industrial Processes and Mining	5	1	1 and 6
Cities and the Built Environment	3	0	2, 7
Tourism	2	0	4
Education	3	0	9
Health	7	0	13
Disaster Risk Reduction	6	3	5 and 8

The table below presents Uganda's top thirteen economy-wide priority actions identified for adaptation based on the extent of policy alignment, particularly the National Development Plan III.

Table 3-15: Top Ranking Economy-wide Adaptation Actions based on Uganda's policies

No	Sector	Priority Actions	Policy Alignment	Adaptation Pillars
1	Agriculture	Expanding Climate Smart Agriculture (CSA) practices	9	1
2	Forestry	Encouraging agro-forestry	9	1
3	Agriculture	Expanding value addition, efficient harvesting, post-harvest handling and storage and access to markets, including micro-finance	8	1
3	Disaster Risk Management	Expanding climate information and building more effective early warning systems for natural disasters such as droughts, flooding and landslides	7	5
4	Water and Sanitation	Ensuring resilient access to water supply both for domestic and productive purposes, especially in areas most exposed to climate hazards	7	3
5	Environment and Ecosystems	Improving water catchment protection	7	4
6	Forestry	Promotion of reforestation, afforestation and sustainable management of forests (including in urban areas)	7	4
7	Environment and Ecosystems	Expanding rangeland management	6	4
8	Environment and Ecosystems	Promote soil conservation and management at catchment level	5	4
9	Water and Sanitation	Promotion of water conservation, efficiency, and reuse practices in domestic, commercial, institutional and industrial water use	5	3
10	Environment and Ecosystems	Improved wetlands and lake management	5	4
11	Fisheries	Promote climate resilient fisheries sector and sustainable fisheries resource management	5	2
12	Energy	Increasing the efficiency in the use of biomass in the traditional energy sector	5	6
13	Energy	Promotion of use of alternative/renewable energy sources and promotion of energy efficient technologies to reduce electricity demand	5	6

Error! Reference source not found. presents the country's overarching objective on adaptation as well as the annual targets.

Table 3-16: Annual performance targets set for Objective 5 of the National Development Plan III

Outcome	Indicator	Performance Targets					
		Base year	Baseline	2021/22	2022/23	2023/24	2024/25
Objective 5: Promote inclusive climate resilient and low emissions development at all levels							
Climate Change Responsive Development Pathway.	Changes in Climate Vulnerability Index.	2.5	3	3.5	4	4.5	5

The tables below present Uganda's 10 economy-wide priority actions for adaptation inclusive and exclusive of cost considerations.

Table 3-17: Summary of Identified Adaptation Actions and Sub-actions

Sector	Number of identified specific actions areas or programs. (with identified sub-actions).	Primary Source
Water and Environment	12 (5)	INDC (2015) Investment Plan for Water and Environment (2018), NDP III.
Agriculture	5 (41)	INDC (2015). National Adaptation Plan for Agriculture (2018)
Forestry	10	INDC (2015), REDD++ (2020)
Fisheries	2 (25)	INDC (2015). National Adaptation Plan for Agriculture (2018)
Energy	5	INDC (2015)
Health	8	INDC (2015)
Transport and the Built Environment	9 (2)	INDC (2015), CRMAS for the Transport Sector (2012)
Disaster Risk Reduction	5 (2)	INDC (2015). National Policy for Disaster Preparedness (2010)
TOTAL	54 (75)	

In total there were 54 adaptation actions or programs identified with an additional 75 sub-actions. The list of identified actions was also shared with the LTS study in terms of the long-term considerations.

The tables below present Uganda's 10 economy-wide priority actions for adaptation based on the Multi-Criteria Analysis (MCA) inclusive and exclusive of cost considerations.

Table 3-18: Top Ranking Economy-wide Adaptation Actions (including cost considerations)

No	Sector	Action	Adaptation Pillars
1	Water and Sanitation	Ensuring resilient, safe and reliable access to water supply both for domestic and productive purposes, especially in areas most exposed to climate hazards	3
2	Environment and Ecosystems	Improving water catchment protection	4

No	Sector	Action	Adaptation Pillars
3	Water Sanitation and	Promotion of water conservation, efficiency, and reuse practices in domestic, commercial, institutional and industrial water use	3
4	Agriculture	Expanding Climate Smart Agriculture (CSA) practices	1
5	Disaster Risk Management	Expanding climate information and building more effective early warning systems for natural disasters such as droughts, flooding and landslides	5
6	Water Sanitation and	Increasing access to reliable and safe water and sanitation (including wastewater treatment) infrastructure and services, especially in districts of Uganda where climate risks overlap with lower access levels	3
7	Water Sanitation and	Managing water resource systems, including wetlands, particularly in cities, in such a way that floods are prevented, and existing resources conserved (through the establishment of an Integrated Water Resources Management system)	4
8	Health	Strengthening public health infrastructure and systems by building hospitals (including regional referral hospitals) and supplying them with medicine, equipment and well-trained personnel	13
9	Energy	Promotion of use of alternative/renewable energy sources and promotion of energy efficient technologies to reduce electricity demand	6

Table 3-19: Top Ranking Economy-wide Adaptation Actions (excluding cost)

No	Sector	Action	Adaptation Pillars
1	Environment and Ecosystems	Improving water catchment protection	4
2	Cities and the Built Environment	Develop standards and guidelines for climate proofing of urban built infrastructure, through revision of design codes and land use to embed resilience thinking in infrastructure design	7
3	Transport	Building resilience of road and bridge infrastructure to climate change, through design codes and guidelines (climate proof standards for road designs construction and maintenance)	11
4	Water and Sanitation	Promotion of water conservation, efficiency, and reuse practices in domestic, commercial, institutional and industrial water use	3
5	Agriculture	Expanding Climate Smart Agriculture (CSA) practices	1
6	Disaster Risk Reduction	Expanding climate information and building more effective early warning systems for natural disasters such as droughts, flooding and landslides	5
7	Energy	Promotion of use of alternative/renewable energy sources and promotion of energy efficient technologies to reduce electricity demand	6
8	Disaster Risk Reduction	Developing vulnerability risk mapping based on better data on climate change impacts at sectoral and regional level	5

No	Sector	Action	Adaptation Pillars
9	Energy	Increase access to energy efficient cook stoves and other appliances.	6

For each adaptation priority sector, the adaptation targets and related prioritized actions have been outlined in the sections below.

1.1.2 Environment and Ecosystems Sector

The table below outlines the 2030 targets for the Environment and Ecosystems Sector

Outcomes	Sub-sector	Indicators	Baseline	2030 Target Conditional (i.e., MOD)
Ecosystem Protection	Wetlands Management	1. Wetlands coverage	8.6%	8%

Table 3-20: Adaptation actions and sub-actions for the Environment and Ecosystems Sector

Priority Actions and Sub-Actions	Adaptation Pillars
1. Promote soil conservation and management at catchment level	4
2. Improved wetlands and lake management	4
a. Develop and implement wetland and forest management plans	4
b. Demarcate and gazette conserved and degraded wetlands	4
3. Improving water catchment protection	4
a. Integration of climate change in country wide catchment management planning to enhance sustainable use and management of water resources	4
b. Create awareness amongst communities of the upstream and downstream impacts of climate change throughout the basin	4
4. Enhancing adaptation to climate change through strengthening of weather, climate and hydrological monitoring capabilities (e.g., expansion of the weather/meteorological network)	5
5. Expanding rangeland management	4
a. Developing rangeland resources governance systems including water committees	4
6. Promoting biodiversity & watershed conservation (including re-establishment of wildlife corridors)	4
7. Implement national targets on threatened/endangered species, restoration of natural habitats, management of invasive alien species with support and participation of local communities and indigenous peoples	4

1.1.3 Water and Sanitation Sector

The table below outlines the 2030 targets for adaptation in the Water and Sanitation Sector, based the Water and Environment Sector Investment Plan (WESIP) (MWE, 2018). These targets are based on the moderate funding scenario which assumes a 50% increase in current funding for this sector.

Table 3-21: Baseline and Enhanced adaptation targets for the Water and Sanitation Sector

Outcomes	Sub-sector	Indicators	Baseline	2030 Target Conditional (i.e., MOD)
Clean Water Supply	Rural water and sanitation	2. Village water supply	66%	100%
		3. Functional rural water sources	85%	93%
	Rural water and sanitation/urban water and sewerage	4. Improved drinking water	70%	80%
		5. Safely managed drinking water	7%	16%
	Urban Water and Sewerage	6. Urban water service functionality	92%	100%
Sanitation and Public Health	Sanitation and Hygiene	7. Solid waste disposal	68%	81%
		8. Improved sanitation	19%	68%
		9. Safely managed sanitation	10%	19%
		10. Handwashing at home	37%	49%
		11. Handwashing at school	35%	66%
Water for Food Production	Water for Production	12. Irrigation	0.49%	1%
Water for Economic Activity		13. WfP functionality	85%	100%
		14. Water Storage Capacity	24%	48%
Water Quality and Quantity	Water Resource Management	15. Compliance with water standards	61%	80%
		16. Water permit compliance	71%	90%
	WRM/Sanitation and Hygiene	17. Wastewater treatment	20%	18%
	Water Resource Management	18. Ambient water quality	0	16%

Notes: baseline figures are from the 2017 Water and Environment Sector Performance Report. 2030 targets are based on the moderate investment funding scenario of the WESIP.

The following table lists the priority actions and sub-actions to be implemented towards achieving those targets for the Water and Sanitation Sector and associated LTS Adaptation Pillars.

Table 3-22: Adaptation actions and sub-actions for the Water and Sanitation sector

Priority Actions and Sub-actions	Adaptation Pillars
1. Ensuring resilient access to water supply both for domestic and productive purposes, especially in areas most exposed to climate hazards	3
2. Promotion of water conservation, efficiency, and reuse practices in domestic, commercial, institutional and industrial water use	3
a. Community sensitisation and advocacy on water efficiency in cities and urban areas	3
b. Develop policy briefs to advocate for updating policies to emphasize water use efficiency improvements in response to climate change	3
c. Adaptation Strategy and Action Plan for the Water Resources Sector	3
3. Develop and implement integrated catchment management plans for water resources catchment areas	3
4. Managing water resource systems, including wetlands, particularly in cities, in such a way that floods are prevented, and existing resources conserved (through the establishment of an Integrated Water Resources Management system)	4

5. Increasing access to reliable and safe water and sanitation (including wastewater treatment) infrastructure and services, especially in districts of Uganda where climate risks overlap with lower access levels	3
a. Develop a situation analysis to strengthen the resilience of sanitation interventions and increase access to climate proofed sanitation facilities	3
b. Map communities with an overlap of high exposure to floods and the practice of open defecation to identify where behavioural change approaches and access to safely managed sanitation needs to be prioritised by developing hazard profiles of the different targeted districts, identifying areas of high flood risk	3
c. Conduct a review of previous sanitation projects and programmes in the country to understand how climate hazards (i.e., floods and drought) have affected desired sanitation outcomes and impacts	3
d. Conduct an analysis on the drivers of demand for sanitation, including the influence of hazards (i.e., floods) in people's long-term willingness and ability to (re)invest in sanitation, which products they would prefer and how much they would be willing to pay	3
e. Conduct a market survey to provide information on the potential suppliers of sanitation goods and services (toilets, pit emptying services etc) in the areas of intervention – assessing ease of access, affordability and the presence of any existing locally adapted products which could be scaled up.	3
f. Based on analysis undertaken, support the construction of climate smart toilets facilities in prioritised rural, peri-urban communities, and in institutions to increase coverage	3
g. Review sanitation designs manuals and construction procedures, including in institutions (e.g., schools and healthcare centres) to ensure that they address climate impacts	3
Additional Actions and Sub-actions	
6. Promotion of water conservation, efficiency, and reuse practices in domestic, commercial, institutional and industrial water use:	3
a. Community sensitisation and advocacy on water efficiency in cities and urban areas	3
b. Develop policy briefs to advocate for updating policies to emphasize water use efficiency improvements in response to climate change	3
c. Adaptation Strategy and Action Plan for the Water Resources Sector	3
7. Extending electricity or expanding use of off-grid solar system, new technologies and biogas to support a low carbon, net Zero, water supply and sanitation system	3
8. Building resilience to climate change in communities in drought and flood prone areas through ecosystem-based water shed management and rainwater harvesting	4
a. Conduct an analysis that considers a range of financing options and intermediaries to support households with reinvestment and re-building after flooding events	4
9. Promoting appropriate and sustainable water harvesting, storage and utilisation technologies	3

1.1.4 Agriculture Sector

Error! Reference source not found. below outlines Uganda's enhanced adaptation targets for the agriculture sector. These are derived primarily from the REDD+ strategy. While specific actions were

identified in the National Adaptation Plan for Agriculture (NAPA), specific targets till need to be set for these priority actions.

Table 3-23: Enhanced adaptation targets for the Agriculture, Forestry and Fisheries Sector

Outcome	Indicators	Baseline (2020)	2030 Target	Target aligned with
Climate Smart Agriculture (CSA) Practices	Land utilizing sustainable land management & agroforestry practices (ha)	1,650,000 ha (45% of farming households)	3,600,000 ha (100% of farming households)	REDD+ Strategy NAPA
	Agricultural land equipped with rainwater harvesting (ha)		1,950,000 ha	
	Land utilized for greenhouse cultivation of vegetables (ha)		10.4 ha	
	Soil and water conservation bunds/terraces (km)	Un-known	1,000	

The tables below outlines the identified priority actions and sub-actions for the Agriculture sector.

Table 3-24: Priority adaptation actions and sub-actions for the Agriculture sector

Priority Actions and Sub-actions	Adaptation Pillars
1. Expanding value addition, efficient harvesting, post-harvest handling and storage and access to markets, including micro-finances	1
a. Invest in improved storage facilities and establish public-private partnerships management for long term food/grain storage and supply	1
b. Train producers, processors and marketers in post-harvest management, preservation and long-term storage of food and seed	1
c. Provide regular market information to improve distribution of agricultural commodities and food stuffs	1
d. Facilitate establishment of marketing centres in rural areas, with the appropriate infrastructure	1
e. Strengthen the warehousing receipt system and link smallholder farmers to warehousing receipt system in the grain supply chain	1
f. Increase timely access to quality agricultural inputs and their efficient use, including inorganic fertilizers	1
g. Establish linkages with the private sector on development and transfer of affordable technologies	1
2. Undertake comprehensive management needs assessment of existing large-scale irrigation schemes	3
a. Build capacity of water committees in watersheds to manage water for irrigation and other production purposes	3
b. Train extension workers on irrigation and water management technologies and impart skills to enable them undertake irrigation extension	3
c. Strengthen water harvesting and irrigation farming to build resilience to droughts and floods by identifying suitable sites for rainwater harvesting, training farmers/ household members in water harvesting and agricultural water management technologies and facilitating the construction of water harvesting structures at household and community levels	3
d. Conduct studies on the irrigation potential and identify sites in various river floodplains and underground water sources for micro-irrigation systems	3
e. Develop appropriate efficient small-scale irrigation technologies and packages	3

Priority Actions and Sub-actions	Adaptation Pillars
f. Establish links to input and output markets and service providers of irrigation technologies	3
3. Expanding Climate Smart Agriculture (CSA) practices	1
a. Promote and encourage highly adaptive and productive livestock breeds	1
b. Promoting pests and disease surveillance	1
c. Adoption of weather-index based agricultural insurance schemes and micro credit schemes for agricultural and pastoral production.	1
d. Promotion of tree biomass production in targeted landscapes	1
e. Promote and encourage highly adaptive and productive crop varieties and cultivars in drought-prone, flood-prone and rain-fed crop farming systems	1
f. Conduct a crop vulnerability and suitability assessment in all agro-ecological zones of Uganda	1
g. Conduct studies on climate resilient crop varieties and cultivars (early maturing and drought tolerant) in the different agro-ecological zones	1
h. Build capacity of seed producers and nursery operators in all agro-ecological zones and support them to expand and improve quality of seedlings	1
i. Scale up awareness and access to climate resilient crop varieties by farmers and communities in different agro-ecological zones	1
j. Support farmer-researcher linkages for adaptive research, multiplication of improved varieties and technology transfer	1
k. Promote and encourage conservation agriculture and ecologically compatible cropping systems to increase resilience to the impacts of climate change	1
l. Identify and conduct field trials on climate resilient cropping practices in different agro-ecological zones of Uganda using conservation agriculture and related practices	1
m. Conduct demonstrations of climate resilient cropping patterns in association with water management systems	1
n. Identify and establish trials for high value, multipurpose tree crops and agroforestry systems	1
o. Promote and scale up conservation agriculture practices such as agroforestry and sustainable land management	1
p. Support innovative paddy rice production techniques that can increase productivity, better manage water resources, and reduce GHG emissions	1
q. Promote diversification and integration of agricultural enterprises at all levels to spread climate risks	1
r. Promote diversification of livelihoods through supporting of alternative and non-weather dependent enterprises and employment	1
s. Support community-based adaptation strategies through expanded and climate smart extension services	1
t. Mainstream climate change into agricultural extension and community management	1
u. Train and build the capacity of extension and community development workers to deliver climate smart agricultural technologies and practices.	1
v. Strengthen platforms, through which small scale farmers can access agricultural information and extension services and promote innovation platforms that build on indigenous knowledge and partnerships along the commodity value chains	1
w. Strengthen partnerships and networks to enhance a common approach to climate resilient agriculture	1
x. Promote integrated crop-livestock systems to enhance community resilience to a changing climate	1
y. Empower communities to wisely use early warning information to enhance resilience	1

1.1.5 Forestry Sector

Current national deforestation rates stand at 2% annually with an average of 122,000 hectares lost every year. Uganda has committed to halting deforestation through various smart-agriculture and sustainable energy initiatives, and to increase forest cover from an estimated 12.5% in 2020 to 21% in 2030 and 24% by 2040 through reforestation and forest rehabilitation. Already a 40 million tree campaign was launched on 2 March 2021, focussing on forest restoration using indigenous trees. The proposed target for improved forestry cover for Uganda is given in Table 3-25 as derived from the NDP III report.

The identified priority actions and sub-actions for the Forestry sector are given in Table 3-26.

Table 3-25: Current baseline and Targets for the forestry sector in Uganda.

Outcome	Indicators	Baseline (2020)	2030 Target	Target aligned with
Reforestation	Forest cover (percentage of total land area)	Approximately 12.5%	21% (18% by 2025)	NDP III

Table 3-26: Adaptation actions for the Forestry sector

Priority Actions	Adaptation Pillars
1. Promotion of reforestation, afforestation and sustainable management of forests (including in urban areas)	4
2. Encouraging agro-forestry	1
3. Encouraging efficient biomass energy production and utilization technologies	6
4. Reduction of pressure on forests through provision of alternatives to wood fuel and construction materials	2
5. Encouraging cultivation of forest medicinal and edible plants outside protected areas	4
Additional Actions	
6. Integration of climate change issues into forest sectoral planning and implementation	4
7. Promote use of trees in demarcation of protected areas such as national parks and game reserves	4

1.1.6 Fisheries Sector

No specific targets have been set for the fisheries sector, but the identified actions and sub-actions and their associated LTS Adaptation Pillars are listed in Table 3-27.

Table 3-27: Adaptation actions and sub-actions for the Fisheries sector

Priority Actions and Sub-actions	Adaptation Pillars
1. Promote climate resilient fisheries sector and sustainable fisheries resource management	2
a. Undertake risk and vulnerability assessment of the fisheries sub-sector and value chains	2
b. Enhance capacity of the fisheries sub-sector to address the impacts of climate change on fisheries (training, avail appropriated fishing technologies, aquaculture inputs and value addition technology)	2
c. Promotion of zonation and protection of fish breeding grounds along shoreline	2
d. Promotion of renewable energy / energy-saving technologies for postharvest fish handling, environmentally friendly artisanal processing of small pelagic fish species	2

Priority Actions and Sub-actions	Adaptation Pillars
e. Enhance capacity of fish quality assurance (cold chain, post-harvest handling infrastructure); initiate development of standards and their enforcement	2
f. Strengthen operation and maintenance of existing infrastructure	2
g. Strengthen management and control of invasive water weeds	2
h. Put in place economic incentives to diversify livelihood options and reduce dependence on climate-sensitive fisheries resources	2
i. Strengthen monitoring capacity and the capability to prevent overfishing and unauthorized exploitation of water bodies	2
j. Up-scale climate resilient strategies/technologies in fisheries and climate resilient fish varieties	2
k. Establish, restore and maintain the bio-physical health of water bodies	2
2. Promote aquaculture to ensure climate resilient fisheries resources	2
a. Strengthen capacity of aquaculture infrastructures/structures (hatcheries, feed production and farmer organisations)	2
b. Enhance capacity of the aquaculture sub-sector to address the impacts of climate change	2
c. Strengthen both pond or cage culture	2
d. Identify suitable sites for setup of aquaculture parks	2
e. Encourage climate change resilient practices, including commercial fish farming, training in building fishponds and making fish fodder	2
f. Address challenges of quality and quantity of fish seed and feed	2
g. Promote the breeding of climate resilient fish breeds/species, appropriate to particular ecosystems and communities	2
h. Train farmers/ household members in semi-intensive and intensive aquaculture farming	2
i. Enhance capacity for integrated agriculture-aquaculture mixed farming systems to increase efficient use of water resources and adaptability to changing climate	2
j. Amplification of incentives available in government for private sector investment in climate-change suitable farming systems	2
k. Train extension workers on aquaculture production technologies in the face of changing climatic conditions	2
l. Undertake comprehensive management needs assessment of existing large-scale farms	2
m. Promotion of aquaculture in wetlands, small lakes and reservoirs and dams through stocking and restocking programs	2
n. Undertake studies of viability of climate-smart aquaculture enterprises	2

1.1.7 Energy Sector

The electricity sector in Uganda is particularly vulnerable to climate change impacts due to the dependency on hydropower but increasing temperatures will also contribute to reduced efficiencies of electricity supply infrastructure, i.e., generation, transmission and distribution.

There will also be more pressure exerted on the forest cover and biodiversity in search of biomass for thermal purpose especially cooking. Increase in temperatures will also increase the demand for energy for cooling (refrigeration and air conditioning) purposes and this has related impacts to ozone depleting.

The enhance adaptation (and mitigation) targets for the energy sector are given in Table 3-28.

Table 3-28: Enhanced adaptation targets for the Energy sector

Outcome	Indicators	Baseline	2030 Target	Target aligned with
Sustainable energy for cooking	Share of biomass energy used for cooking	from 85% in 2018/19	40% (50% in 2024/25)	NDP III
	share of clean energy used for cooking	15% in 2017/18	65% (50% in 2025/25)	
Sustainable fuel wood and charcoal production	Land utilized for bioenergy woodlots (ha)	Unknown?	866,000 ha	REDD+ Strategy
	Number of improved charcoal kilns on Commercial small-holder and community bioenergy woodlots	Unknown?	100,000	
	Number of improved charcoal kilns on large-scale commercial plantations (per 1000 ha)	Unknown?	100	
	% Biomass coming from sustainable sources (e.g., bagasse)		40%	Biomass Energy Strategy
Energy Efficient practices and technologies	Minimum Energy Performance Standards implemented	Voluntary programme in place, 12 standards in place, 1 testing equipment	Mandatory MEPS programme in place	NDP III, Energy Efficiency Roadmap for Uganda
	Energy efficiency in buildings programme in place	Regulations for energy efficiency in buildings under drafting	Mandatory regulations enforced for energy efficiency in buildings	
	Clean energy for transport programme in place	0 charging infrastructure to support electric mobility, No legal framework to support e-mobility	50 charging stations in place, Legal framework in place to support e-mobility	
	Energy Service Company (ESCO) Programme for the industrial sector in place	0	An operational ESCO programme in place	
	Minimum Energy Performance Standards implemented	Voluntary programme in place, 12 standards in place, 1 testing equipment	Mandatory MEPS programme in place	
Access and utilization of electricity from sustainable sources.	Proportion of the population with access to electricity	24% in 2018/19	75% (60% in 2025)	NDPIII
	Per capita electricity consumption	100 kWh in 2018/19	578 kWh	
	Transmission capacity (in km of	from 2,354 km in 2018/19	6,300 km (4,354 km in 2025)	

Outcome	Indicators	Baseline	2030 Target	Target aligned with
	high voltage ⁷ transmission lines)			

It is envisaged that these targets will be achieved through the following adaptation actions.

Table 3-29: Adaptation actions for the Energy sector

Priority Actions	Adaptation Pillars
1. Increasing the efficiency in the use of biomass in the traditional energy sector	6
2. Promotion of use of alternative/renewable energy sources (other than hydropower) and promotion of energy efficient technologies to reduce electricity demand	6
3. Ensuring the best use of hydropower by careful management of the water resources	3
4. Climate proofing investments in electricity power sector	8
5. Increasing access to clean cooking technologies	6
6. Develop and enforce minimum energy performance standards and regulations for energy efficiency	6
7. Implement incentives to promote uptake of energy efficiency technologies	6
8. Support the building code to promote energy efficiency in buildings	6
9. Promote establishment of energy service companies (ESCOs)	6
10. Promote the productive use of energy	6

1.1.8 Transport Sector

For the Transport sector, Table 3-30 details the enhanced adaptation targets for 2030, aligned to various national policies, strategies and plans. To achieve these targets, the actions in Table 3-31 will be undertaken.

Table 3-30: Enhanced adaptation targets for the Transport sector

Outcome	Indicators	Baseline	2030 Target	Target aligned with
Increased stock of resilient transport infrastructure	i. Kms of paved National Roads	4 971	7 500	National Development Plan III
	ii. Kms of paved urban roads	1 248	1 748	
	iii. Permanent way /railway road (Km)	262	462	
	iv. Construction of Roll-on Roll-off vessels (international)	5	7	
	v. Construction of domestic (Ro 'Pax) passenger ferries	10	15	
	vi. Ferry crossings	9	13	
Increased average infrastructure life span	1. Tarmac roads – years	Unknown	20 years	National Development Plan III
	First class murram	Unknown	2 years	

Table 3-31: Adaptation actions and sub-actions for the Transport sector

Priority Actions and Sub-actions	Adaptation Pillars
1. Building resilience of road and bridge infrastructure to climate change, through design codes and guidelines (climate proof standards for road designs construction and maintenance) and where cost effective, future proof existing infrastructure	7
a. Promote thermally resilient road and rail infrastructure using heat resistant materials	7
2. Promote modes of transport that consider greenhouse gas emissions reduction such as use of non-motorised transport and increased access to high-volume public transport options to replace single-car commuter trips in cities and urban areas	11
3. Updating transport codes and regulations and implementing measures to ensure compliance with them	11
Additional Actions	
4. Updating of transport risk assessment guidelines	11
5. Promote the development, approval, and effective implementation of a long-term national transport policy and plan that will take greenhouse gas mitigation concerns into account	11
6. Effect a gradual shift to the use of less carbon-intensive fuels (including compressed natural gas, ethanol, and liquefied petroleum gas) in vehicles, instead of relying heavily on gasoline and diesel fuels	11

1.1.9 Manufacturing, Industrial Processes and Mining Sector

No specific targets have been set for the Manufacturing, Industrial Process and Mining sector. The table below presents the identified adaptation actions for this sector.

Table 3-32: Adaptation actions and sub-actions for the Manufacturing, Industrial Processes and Mining sector

Priority Actions and Sub-actions	Adaptation Pillars
1. Switching to energy efficiency technologies in the mineral value chain	6
a. Integration of renewable energy in the mining sector	6
2. Forest smart mining with land scape management	2
3. Adoption of a circular economy for low-carbon minerals	12
4. Organise, formalise and regulate the artisanal and small-scale miners	2
5. Provide incentives for acquisition of appropriate and clean technology	6

1.1.10 Cities and the Built Environment Sector

For the Cities and the Built Environment sector, Table 3-33 details the enhanced adaptation targets for 2030, aligned to various existing national policies, strategies and plans. To achieve these targets, the actions given in Table 3-34 will be undertaken.

Table 3-33: Enhanced adaptation targets for the Cities and the Built Environment sector

Outcome	Indicators	Baseline	2030 Target	Target aligned with
Green Resilient Cities	Number of green cities	0	5 Cities:	Green Growth Dev. Strategy 2017/18 – 2030/3, 2017
	i. Urban unemployment rate	14.4%	7%	
	ii. Acute housing deficit	2.2 million	1.76 million	

Outcome	Indicators	Baseline	2030 Target	Target aligned with
Sustainable urbanization and housing	iii. Percentage of urban dwellers living in slums and informal settlements	60%	40%	National Development Plan III
	iv. Average travel time per km in GKMA	4.1 min/km	3.5 min/km	
	v. Proportion of tarmacked roads in the total urban road network	1229.7 km (6.1 percent)	2459.4 km (12.2 percent)	
	vi. Proportion of surveyed land	21 percent (2020/21)	50% (40 percent by 2024/25);	
	vii. Efficiency of solid waste collection	30%	70% (50% by 2025)	

Table 3-34: Adaptation actions for the Cities and the Built Environment sector

Priority Actions	Adaptation Pillars
1. Develop standards and guidelines for climate proofing of urban built infrastructure, through revision of design codes and land use to embed resilience thinking in infrastructure design	8
2. Investing in making existing and new buildings more climate resilient and promoting climate resilient urban planning including Green Buildings standards and guidelines	8
3. Rehabilitation and expansion of urban drainage and flood management systems in flood prone cities including Water Sensitive Urban Design (WSUD) principles and guidelines	8
4. Benchmarking cities as Water Sensitive Cities	3
5. Develop, expand, and maintain greenbelts within cities	2

1.1.11 Tourism Sector

The Tourism sector is considered highly sensitive to climate change risks however, no specific targets have been set for this sector. The table below lists the current proposed adaptation actions for the Tourism sector.

Table 3-35: Adaptation actions for the Tourism sector

Priority Actions	Adaptation Pillars
1. Promote natural and cultural/heritage conservation	4
2. Enhance and maintain ecological integrity of wildlife conservation	4

1.1.12 Education Sector

Education, particularly related to the potential impacts of climate change, is critical in terms of adaptation. Specific targets have, however not been set for this sector as part of the NDC. These will need to be determined as part of the implementation phase.

The table below presents the current proposed adaptation actions for the Education sector.

Table 3-36: Adaptation actions for the Education sector

Priority Actions	Adaptation Pillars
1. Improve education, awareness raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning	9
2. Integrating climate change information into school syllabus	9

Priority Actions	Adaptation Pillars
3. Investing in making existing and new school buildings more climate resilient (Links to the Cities and the Built Environment sector adaptation actions)	7

1.1.13 Health Sector

Hotter and wetter climate scenarios are expected to increase the risk of diseases such as malaria which will have a significant impact on the health and productivity of the people of Uganda. A study of the potential economic impacts of climate change determined that the cost of malaria alone may rise by around a factor of three from a range of USD 8.7 million to US 221 million in 2010, to a range of USD 20.1 million to USD 560 million in 2050 in just one district, Tororo. A failure to provide basic water and sanitation services will also have a significant impact on human health and productivity.

Table 3-37 presents Uganda's targets related to adaption in the health sector given in NDPIII. These targets will be achieved through the priority adaptation actions listed in Table 3-38.

Table 3-37: Enhanced adaptation targets for the Health sector

Outcome	Indicators	Baseline	2030 Target	Target aligned with
Access to health care	Proportion of the population accessing universal health care	44% in 2018	70% (60% in 2025)	National Development Plan III
Access to sanitation	Proportion of population with access to basic sanitation	19%	50% (40% by 2025)	
Access to safe water supply	Proportion of population with access to safe water	70% rural 74% urban	85% rural 100% urban	

Table 3-38: Adaptation actions for the Health sector

Priority Actions	Adaptation Pillars
1. Conducting vulnerability assessments of the health sector to climate change impacts	13
2. Assessing the impacts of climate change on human health and well-being	13
3. Putting in place contingency plans to develop climate change-resilient health systems	13
4. Strengthening public health infrastructure and systems by building hospitals (including regional referral hospitals) and supplying them with medicine, equipment and well-trained personnel	13
5. Making provision for a safe water chain and sanitation facilities to limit outbreaks of water-borne diseases and implement strong public awareness programmes to promote better hygiene	3
6. Enhance early warning systems, strengthen adaptive mechanisms and provide adequate preparedness for dealing with climate change related diseases	5
7. Enhance community resilience to malaria and other diseases caused by climate change (increasing the distribution of mosquito nets, household initiatives and community led initiatives)	13

1.1.14 Disaster Risk Reduction Sector

International experience has shown that investing in improved disaster risk reduction measures and in particular the development of improved monitoring and early warning systems has the greatest benefits in terms of adaptation to the impacts of climate change and improved resilience.

As most climate scenarios indicate that Uganda is likely to get wetter and to experience more intense rainfall events, it is critical that priority be given to improved flood risk management and including both the development of physical infrastructure and improved flood risk management. The strengthening of early warning systems is also critical in adapting to increasing risk from droughts.

The table below presents the cross-cutting adaptation actions aimed at disaster risk reduction:

Table 3-39: Adaptation actions and sub-actions for the Disaster Risk Reduction sector

Priority Actions	Adaptation Pillars
1. Promote continuous integration of climate change and disaster risk reduction in planning, budgeting and reporting	5
2. Developing vulnerability risk mapping based on climate change impacts at sectoral and regional level and improve regulation of development in high-risk areas	5
3. Preparation of drainage and flood risk management plans for flood prone cities and regions	5
4. Expanding climate information and building more effective early warning systems for natural disasters such as droughts, flooding and landslides	5
a. Increase coverage of automated weather monitoring stations country-wide and integrate use of indigenous knowledge into community early warning systems	5
b. Put in place community information platforms, to facilitate and disseminate climate information early warnings	5
c. Build capacity to manager, maintain, interpret, and use monitoring and early warning systems	5
5. Improve capacity to anticipate events and emergency related institutions and establishing a contingency fund to take care of emergency needs following an extreme climate event	8
6. Improved access to climate related information to affected communities	9

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Appendix title

Appendix heading

Appendix heading 1

In diversity there is beauty and there is strength.

MAYA ANGELOU

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